Imperial College London

Fundamental Studies on Solid-state Stamp Forming of Unidirectional Fibre Reinforced Thermoplastics for Automobile Components

By

Hongyan Wang

A thesis submitted for the degree of Doctor of Philosophy of Imperial College London and the Diploma of Imperial College London

Dyson School of Design Engineering

Imperial College London

Sept 2023

Declaration of Authorship

I, Hongyan Wang, declare that the work presented in this thesis titled 'Fundamental Studies on Solid-state Stamp Forming of Unidirectional Fibre Reinforced Thermoplastics for Automobile Components' is my own, all else is appropriately referenced.

Declaration of Copyright

The copyright of this thesis rests with the author. Unless otherwise indicated, its contents are licensed under a Creative Commons Attribution-Non-Commercial 4.0 International Licence (CC BY-NC).

Under this licence, you may copy and redistribute the material in any medium or format. You may also create and distribute modified versions of the work. This is on the condition that: you credit the author and do not use it, or any derivative works, for a commercial purpose.

When reusing or sharing this work, ensure you make the licence terms clear to others by naming the licence and linking to the licence text. Where a work has been adapted, you should indicate that the work has been changed and describe those changes.

Please seek permission from the copyright holder for uses of this work that are not included in this licence or permitted under UK Copyright Law.

Abstract

Climate change has led to extensive research on lightweight materials for transportation industries, and unidirectional fibre reinforced thermoplastics (UD FRTPs) stand out due to their exceptional properties. To incorporate UD FRTPs into vehicles, a cost-effective manufacturing method suitable for mass production is necessary. In this regard, solid-state stamp forming offers potential due to its short cycle time and energy-saving capabilities. Despite its potential, challenges arise from insufficient lab experiments, limited understanding of deformation behaviours, and lack of material models.

This thesis aims to address the challenges of solid-state stamp forming for complex-shaped UD FRTPs by encompassing four major facets:

1) Experimental characterisation and mechanism analysis of thermoplastics, with a particular focus on polyamide 6 (PA6) and polyether ether ketone (PEEK). It confirms the feasibility of solid-state stamp forming and identifies initial processing windows for FRTP materials.

2) Experimental characterisation and mechanism analysis of UD FRTPs. Specifically, an automotive-grade material, unidirectional carbon fibre reinforced polyamide 6 (UD CF/PA6) was studied. The developed knowledge and understanding helps to refine the forming windows for UD FRTPs and provides crucial insights for the process design.

3) Constitutive model development and numerical simulation, by creating two sets of unified physically based constitutive models to describe the deformation of thermoplastics and UD FRTPs, respectively. These models are subsequently implemented into commercial Finite Element Analysis (FEA) software for process simulations of solid-state stamp forming of UD FRTPs, enabling the analysis of potential defects and optimal processing parameters.

4) Stamp forming experiments and post-form analysis via forming dome-shaped demonstrator components. The post-form quality is assessed using various methods, which validates the findings from earlier studies. The successful production of stamp-formed demonstrators indicates the promising potential of using UD FRTPs in the automotive industry.

Acknowledgements

In the early stages of my doctoral journey, I drew inspiration from the poem 'Ithaka' by Constantine P. Cavafy. The words, *'As you set out for Ithaka / hope your road is a long one / full of adventure, full of discovery'* resonated with my academic aspirations, and provided a profound reassurance that this challenging expedition would be fulfilling. Through the trials of a global pandemic, personal struggles with mental health, and life's varying circumstances, these verses have offered a steadfast source of strength.

I would like to express my profound gratitude, first and foremost, to my supervisor, Dr. Nan Li. Her consistent guidance and unwavering support have been invaluable throughout my academic journey. Her high standards for scholarly investigation, professional diligence, and extensive expertise in our research field have not only shaped my professional growth but have also had a profound impact on my personal evolution. My gratitude extends to Prof. Peter Childs and Prof. Jianguo Lin, co-supervisors who provided valuable insights during our group discussions. I am also indebted to Dr. Haibao Liu, who generously shared his expertise during numerous discussions and the preparation of research papers.

I would like to acknowledge the contributions of our technical staff: Mr. Suresh Viswanathan Chettiar and Dr. Ruth Brooker from the Department of Mechanical Engineering, and Miss Ingrid Logan from Design Engineering, at Imperial College London. Their invaluable support greatly contributed to the successful execution of my experimental work. Additionally, I am profoundly thankful to Dr. Caroline Hargreaves and Dr. John Pinney from Graduate School, for their generous support during the writing of my thesis.

The financial support from Imperial College London and China Scholarship Council scholarship is gratefully acknowledged.

My colleague and friend, Zerong Ding, deserves special recognition. We have been teammates for several years, and his professional acumen, coupled with the warmth of friendship, has enriched this journey. I am also grateful for the camaraderie and support from my colleagues: Dr. Lei Zhu, Famin Tian, Haosu Zhou, Lianna Zhao, Sijin Sun, Xiao He, and Xi Hua, especially the sisterhood support from Dr. Shiya Li, Dr. Xianyan Zhou and Dr. Qi Rong. I extend my heartfelt thanks to my long-time friend, Minyan Guo, who has been a constant source of emotional support since our childhood. I also express my gratitude to my sister, who has been a sounding board for my life choices. Additional thanks are due to my friends, Wen Shi, Dr.

Muhao Chen, and Guang Li, whose support and companionship have been much appreciated throughout this journey.

Lastly, but by no means least, my gratitude extends to a comforting and furry presence, the most adorable cat in the world, who is curled up beside me even now, as I pen these acknowledgments.

Contents

Declaration of Authorship ..... **II**

Declaration of Copyright ..... **II**

Abstract ..... **III**

Acknowledgements ..... **IV**

Contents ..... **VI**

List of abbreviations ..... **XI**

Nomenclature ..... **XIII**

List of Figures ..... **XX**

List of Tables ..... **XXVIII**

CHAPTER 1 Introduction **1**

1.1. Industrial drivers: the Net-Zero target and automotive lightweighting 1

1.2. Unidirectional Fibre Reinforced Thermoplastics (UD FRTPs) 2

1.3. Solid-state stamp forming 5

1.4. Aim and objectives 7

1.5. Thesis structure 8

CHAPTER 2 Manufacturing Technologies and Fundamental studies on UD FRTPs **11**

2.1. Manufacturing technologies for FRTP components 11

2.1.1. Conventional methods 11

2.1.2. Automated methods 16

2.2. The state of the art in stamp forming technologies for UD FRTPs 20

2.2.1. UD FRTP demonstrators manufactured through stamp forming 20

2.2.2. Common defects from UD FRTPs stamp forming 22

2.2.3. Other innovations in FRTPs stamp forming 26

2.3. Deformation mechanisms and characterisation methods 27
2.3.1. Thermoplastics 28
2.3.2. UD FRTPs 28
2.4. The thermomechanical properties of thermoplastics and UD FRTPs 33
2.4.1. Thermoplastics 33
2.4.2. UD FRTPs 36
2.5. Constitutive models for thermoplastics and UD FRTPs 38
2.5.1. Thermoplastics 39
2.5.2. UD FRTPs 40
2.6. Numerical simulation for FRTPs stamp forming process 42
2.7. Summary 48
CHAPTER 3 Investigation of the Thermomechanical Properties of Thermoplastics 49
3.1. Introduction 49
3.2. Experimental design for uniaxial tensile tests 50
3.2.1. Design of specimen and test conditions 50
3.2.2. Experimental setup 51
3.2.3. In-situ strain measurement 53
3.3. Experimental results and analysis from the uniaxial tensile tests 55
3.3.1. Uniaxial tensile test results of thermoplastics 55
3.3.2. Key-features of the stress-strain curves 57
3.3.3. Analysis of the temperature effects 59
3.3.4. Analysis of strain rate effects 61
3.4. Fundamental studies on the V-Notched Rail (VNR) shear tests 63
3.4.1. Introduction to the VNR shear tests 63
3.4.2. Experimental investigation 65
3.4.3. A detailed explanation of VNR shear tests for stress-strain characterisation problem 69
3.4.4. Numerical simulations 77
3.4.5. Analysis of the numerical simulation results 81

3.4.6. A new two-step correction method 84
3.4.7. Validation of the developed correction method 89
3.5. Summary 92
CHAPTER 4 Investigation of the Thermomechanical Properties of UD FRTPs 93
4.1. Introduction 93
4.2. Fabrication of UD FRTP laminates 93
4.2.1. Hot press and moulding rig design 93
4.2.2. Laminate fabrication and quality assessment 96
4.3. Characterisation of the transverse tensile and longitudinal shear properties of UD FRTPs 98
4.3.1. Specimen design 98
4.3.2. Experimental setup 99
4.3.3. Strain measurement and data postprocessing 100
4.4. Experimental results and analysis 102
4.4.1. Transverse tensile properties of UD FRTPs 102
4.4.2. Longitudinal shear properties of UD FRTPs 103
4.4.3. Analysis of the intra-ply deformation properties 104
4.5. Summary 108
CHAPTER 5 Development and Implementation of Constitutive Models 109
5.1. Introduction 109
5.2. Development of a constitutive model for thermoplastics 109
5.2.1. The intermolecular resistance (I) 113
5.2.2. The network resistance (N) 115
5.2.3. The heat generation during deformation 117
5.2.4. Discussion of the proposed constitutive model and associated deformation mechanisms 118
5.2.5. Model calibration and performance 121
5.3. Numerical implementation of the constitutive model for thermoplastics 124

5.4. Development of a constitutive model for UD FRTPs 128
5.4.1. The contribution of the polymer constituent reinforced by fibres (PrF) 130
5.4.2. The contribution of the polymer-fibre interface (P-F) 134
5.4.3. Model calibration and performance 139
5.5. Numerical implementation of the UD FRTP model 142
5.6. Finite element (FE) simulation and analysis of the solid-state stamp forming process 144
5.6.1. Numerical model construction 144
5.6.2. Analysis of the numerical simulation results 147
5.7. Summary 154
CHAPTER 6 Forming Experiments and Quality Assessment 155
6.1. Introduction 155
6.2. Experimental design for the solid-state stamp forming process 155
6.2.1. Setup of the stamp forming machine and tools 155
6.2.2. Stamp forming specimen preparation 157
6.2.3. Solid-state stamping conditions and procedure 157
6.3. Solid-state stamp forming trials and analysis 159
6.3.1. The effect of blank shape 159
6.3.2. The effect of layer blocking 160
6.3.3. The effect of forming temperature 161
6.4. Quality assessment of the formed components 164
6.5. Comparison of the numerical simulation and experimental results 167
6.6. Summary 169
CHAPTER 7 Conclusions and Suggestions for Future Work 170
7.1. Conclusions 170
7.1.1. Thermomechanical properties of thermoplastics 170
7.1.2. Thermomechanical properties of UD FRTPs 171
7.1.3. Constitutive models and the implementation for numerical simulations 171

7.1.4. Experimental investigations on the solid-state stamp forming process 172

7.2. Suggestions for future work..... 172

References 175

List of publications 198

List of abbreviations

1D	One dimensional
2D	Two dimensional
3D	Three dimensional
8HS	8 Harness satin
AFP	Automated fibre placement
ATP	Automated tape placement
CAGR	Compound annual growth rate
CF	Carbon fibre
CO_2	Carbon dioxide
CT	Computer tomography
DFFIM	Direct fibre feeding injection moulding
DIC	Digital Image Correlation
DKT	Discrete Kirchhoff Triangle
DMA	Dynamic Mechanical Analysis
FEA	Finite element analysis
fps	Frames-per-second
FRP	Fibre reinforced polymer
FRTP	Fibre reinforced thermoplastic
FRTS	Fibre reinforced thermoset
GF	Glass fibre
IEA	International Energy Agency
IFRM	Ideal fibre reinforced material
IR	Infrared
JC	Johnson-Cook
LFT	Long fibre reinforced thermoplastic
PA6	Polyamide 6
PBC	Periodic boundary conditions
PEEK	Polyether Ether Ketone
P-F	Polymer-fibre interface

PI	Polyimide
PID	Proportional-Integral-Derivative
PJN	Pin-joint-net
PMMA	Poly methyl methacrylate
PrF	Polymer constituent reinforced by fibres
PP	Polypropylene
PPS	Polyphenylene sulfide
ROI	Region of interest
RT	Room temperature
RVE	Representative volume element
SFT	Short fibre reinforced thermoplastics
UD	Unidirectional
VARTM	Vacuum assisted resin transfer moulding
VBO	Vacuum-bag-only
VNR	V-notched rail

Nomenclature

Greek letters	
$\dot{\gamma}_{01}$	The material constant relating to the dashpot in the intermolecular viscoelastic part
$\dot{\gamma}_{02}$	The material constant relating to the dashpot in the network resistance
$\gamma_{m,s}$,$\gamma_{m,p}$	The maximum shear strain under simple shear and pure shear deformation
γ_{xy}	The in-plane maximum shear strain
$\gamma_{m,c}$	The maximum shear strain of the central point on the VNR shear specimen
$\bar{\gamma}_{m,ROI}$	The mean maximum shear strain values of all the nodes in the ROI
$\varepsilon_{Ie}, \varepsilon_{Iv}, \varepsilon_{Ip}$	The equivalent (true) strains of the intermolecular elastic, viscous, and plastic parts
$\varepsilon_{Ne}, \varepsilon_{Nv}$	The equivalent (true) strains of the network elastic and viscous parts
$\dot{\varepsilon}$	the total equivalent (true) strain rate
$\dot{\varepsilon}_{Ie}$, $\dot{\varepsilon}_{Iv}, \dot{\varepsilon}_{Ip}$	The equivalent (true) strain rates of the intermolecular elastic, viscous and plastic parts
$\dot{\varepsilon}_{Ne}$, $\dot{\varepsilon}_{Nv}$	The equivalent (true) strain rates of the network elastic and viscous parts
$\dot{\varepsilon}_{0p}$	The material constant relating to the dashpot in the viscoplastic part
$\bar{\varepsilon}$	The Von-Mises effective strain and effective stress
ε_t	The tensile strain of the transverse tensile test results
ε_s	The shear strain of the longitudinal shear test results
$\bar{\varepsilon}_{pl}$	The plastic strain in Johnson-cook model
$\varepsilon_{12}, \bar{\varepsilon}_{12}$	The in-plane shear strain on the longitudinal shear test specimen, the arithmetic mean value of the in-plane shear strains within the ROI
ε_{22}^{RVE}, ε_{12}^{RVE}	The transverse tensile strain, the longitudinal shear strain of an RVE
ε_{PrF}	The effective strain of PrF
$\varepsilon_{equ-i}^{polymer}$, $\bar{\varepsilon}_{equ}^{polymer}$	The equivalent strain of a polymer element and the averaged equivalent strain of all polymer elements in RVE
ε_C, ε_{PrF}, ε_{P-F}	The strains of composite, the PrF and P-F parts
$\varepsilon_1, \varepsilon_2$	The strain of the turning point in the cohesive model, the strain when the cohesive model curve approaches zero

$\varepsilon_{10}, \varepsilon_{20}$	The material constants in the cohesive model that related to the temperature dependency
ε_{ROI}^{eng} , ε_t	The Mean value of maximum principal strains within the ROI in engineering strain and true strain form.
ε_1, ε_2, ε_3	The maximum, middle, minimum principal strains
$\varepsilon_{+45°}$, $\varepsilon_{-45°}$	The strains measured from the +45° and -45° strain gauges
$\varepsilon_x, \varepsilon_y$	The in-plane strain components of the analysed element at the specimen centre
$\bar{\varepsilon}_c$	The Von-Mises effective strain of a central point on the VNR shear specimen
ε_{SG}	The effective strain calculated from strain gauge measurement
ε_{DIC}	The effective strain calculated from DIC measurement
ε_{ROI}	The effective strain of the first correction target, i.e., all nodes in the ROI
$\bar{\varepsilon}_{m,\mathrm{ROI}}$	The averaged value of maximum principal strains of all nodes in the ROI
η_{Iv}	The coefficient of viscosity of the dashpot in the intermolecular viscoelastic part
η_{Iv0}	The material constants relating to η_{Iv}
η_{Nv}	The coefficient of viscosity of the dashpot in the network resistance
η_{Nv0}	The material constants relating to η_{Nv}
$\theta, \theta_{ref}, \theta_g, \theta_m$	The current, reference, glass transition, and melting temperatures (absolute temperature)
θ_c	The corrected temperature during the deformation when the adiabatic heat comes into effect (absolute temperature)
$\dot{\theta}$	The temperature evolution (absolute temperature)
θ_0	The controlling temperature of PA6 in Johnson-Cook model (absolute temperature)
λ_{Ie}	The intermolecular elastic stretch ratio
$\bar{\lambda}_{Ne}$	The average network elastic stretch ratio
$\bar{\lambda}_L$	The locking stretch
ν	The Poisson's ratio
ν_{12}, ν_{13}, ν_{23}	The material constants for the orthotropic elasticity in ABAQUS
ξ_{SG} , ξ_{DIC}	The calculation errors for strain gauge and DIC measurement

ρ, ρ_c, ρ_a	The densities of the thermoplastic, the pure crystal phase, and the pure amorphous phase
σ	The total effective stress / The flow stress in Johnson-cook model
σ_{lim}	The stress limit set on the viscoplastic friction element
σ_{TO}	The material constants relating to σ_{lim}
σ_{Ip}	The equivalent stress carried by the dashpot in the viscoplastic part
σ_{Iv},σ_{Ip}, σ_{Nv}	The equivalent stress carried by the dashpot in the intermolecular viscoelastic part, the viscoplastic part and the network resistance
σ_c	The corrected total effective stress considering thermal softening by adiabatic heat
σ_t	The tensile stress of the transverse tensile test results
σ_s	The shear stress of the longitudinal shear test results
σ_C, σ_{PrF},σ_{P-F}	The stress responses of the composite, the PrF, and the P-F parts
σ_1	The stress of the turning point in the cohesive model
$\sigma_{P-F}(\varepsilon_C)$	The cohesive stress of the cohesive model
$\partial\sigma_{P-F}(\varepsilon_C)$ / $\partial\varepsilon_C$	The first-order derivative of the cohesive model
σ_{10}	The material constant in the cohesive model that related to the temperature dependency
$\bar{\sigma}$	The Von-Mises effective stress
σ_1, σ_2, σ_3	The maximum, middle, minimum principal stresses
σ_{11}, σ_{22}, σ_{12}	The stress components of a basic deformation element
$\bar{\sigma}_c$	The Von-Mises effective stress of a central point on the VNR shear specimen
σ_{SG}	The effective stress calculated from strain gauge measurement
σ_{DIC}	The effective stress calculated from DIC measurement
σ_{ROI}	The effective stress of the first correction target, i.e., all nodes in the ROI
$\bar{\sigma}_{m,ROI}$	The averaged value of maximum principal stresses of all nodes in the ROI
σ_{SG}^{corr1} , σ_{DIC}^{corr1}	The effective stresses after the first step correction of strain gauge measurement and DIC measurement
σ_{SG}^{corr2} , σ_{DIC}^{corr2}	The effective stresses after the second step correction of strain gauge measurement and DIC measurement
χ_c	The degree of crystallinity

ω	The ratio of energy that is converted to temperature change to the total plastic dissipation
$\omega_{tg}, \omega_{SG}, \omega_{DIC}$	The rotational angle for the first correction target curve, using strain gauges or DIC measurement
ω_c	The rotational angle for the second correction target curve
	English symbols
A_0	The initial cross-section area of the transverse tensile or longitudinal shear test specimen
A, B, n, m	The material constants for the Johnson-Cook model
C_1	The temperature-dependent parameter in the intermolecular resistance
C_p	The temperature-dependent parameter relating to the dashpot in the viscoplastic part
C_{p1} , m_{cp}	The material constants relating to C_p
C_R	The temperature-dependent modulus of the network resistance
$C_{R_{ref}}, C_{r1}$	The material constants relating to C_R
$\bar{C}$	The temperature-dependent specific heat capacity per unit volume
$\bar{C}_{ref}, \bar{C}_1$	The material constants relating to $\bar{C}$
E	The temperature-dependent elastic modulus
E_{ref}	The temperature-dependent elastic modulus at the reference temperature
E_1	The material constants relating to E
E_1, E_2, E_3	The material constants for the orthotropic elasticity in ABAQUS
F	The load measured from the tensile testing machine
G_{12}, G_{13}, G_{23}	The material constants for the orthotropic elasticity in ABAQUS
J_{Ne}	The Jacobian determinant associated with the network elastic deformation gradient
$\mathcal{L}^{-1}$	The inverse of the Langevin function
T_g, T_m	The glass transition temperature, the melting temperature (Celsius temperature)
V_{RVE}, V_p	The total volume of the RVE, the total volume of polymer elements inside the RVE
V_{F_p}, V_{F_f}	The volume fractions of polymer constituent and fibres inside the composite material

X_i^o, X_i^U, X_i^d $(i = 1,2,3)$	The coordinates of the original, stretched, and deformed configurations of an analysed element
b	A notation for making the equation simpler
$b_{\varepsilon 2}$	The material constant in the cohesive model that related to the temperature dependency
d	The change of the length within the gauge length on the transverse tensile test specimen
dv	Local material volume
f_{SG}^1, f_{DIC}^1	The first correction factors for strain gauge and DIC measurement
f_{SG}^2, f_{DIC}^2	The second correction factors for strain gauge and DIC measurement
$k^{map}, k_{\varepsilon_{22}}^{map}, k_{\gamma_{13}}^{map}$	The strain mapping constant, the strain mapping constants for transverse tensile, longitudinal shear deformation
$k_{\varepsilon 1}, k_{\sigma 1}, k_{\varepsilon 2}$	The material constants in the cohesive model that related to the temperature dependency
l_0	The initial gauge length of the transverse tensile or longitudinal shear test specimen
m_{rI}, m_{Iv}	The material constants relating to the dashpot in the intermolecular viscoelastic part
$m_{\theta I}$	The material constant relating to η_{Iv}
m_E	The material constant relating to E
m_σ	The material constant relating to σ_{lim}
m_{Ip}	The material constant relating to the dashpot in the viscoplastic part
m_{rN}, m_{Nv}	The material constants relating to the dashpot in the network resistance
$m_{\theta N}$	The material constant relating to η_{Nv}
m_h	The material constant representing the softening parameter
p_1, p_3	The positions of the 1st and 3rd marked lines along elongation throughout the test
t_0	The initial thickness at the gauge section of the tensile specimen
w_0	The initial width at the gauge section of the tensile specimen
Tensors and vectors	
$\boldsymbol{\sigma}$	The Cauchy (true) stress tensor
$\boldsymbol{\sigma_I}, \boldsymbol{\sigma_N}$	The stress tensors of intermolecular and the network resistance

$\boldsymbol{\sigma_P}, \boldsymbol{\sigma_I}, \boldsymbol{\sigma_N}$	The stress tensors of the polymer, the intermolecular branch, and the network branch
$\boldsymbol{\sigma_C}, \ \boldsymbol{\sigma_{PrF}}, \boldsymbol{\sigma_{P-F}}$	The stress tensors of the composites, the PrF part, and the P-F part
$\boldsymbol{\sigma_{ts}}, \boldsymbol{\sigma_{pre-ts}}$	The stress tensors within the tensile and the pre-tensioned springs in the P-F part
$\boldsymbol{\theta_p^o}$,$\boldsymbol{\theta_p^d}$	The maximum principal strain directions of pure shear under original and deformed configurations
$\boldsymbol{\theta_s^o}$, $\boldsymbol{\theta_s^U}$, $\boldsymbol{\theta_s^d}$	The maximum principal strain directions of simple shear under the original, stretched, and deformed configurations
$\boldsymbol{\theta_s^{Ud}}$	The counterclockwise rotational angle of the analysed element (i.e., the rigid body rotation)
$\boldsymbol{\theta_s^{od}}$	The rotational angle of the principal strain axes of an analysed element
$\widehat{\boldsymbol{v}}_{\boldsymbol{1}}$	The first eigenvector of the stretch tensor $\boldsymbol{U}$
$\boldsymbol{\Omega_0}, \bar{\boldsymbol{\Omega}}, \bar{\bar{\boldsymbol{\Omega}}}, \breve{\boldsymbol{\Omega}}, \boldsymbol{\Omega}$	The reference, intermedia plastic, intermolecular intermedia relaxed, network intermedia relaxed, and current configurations
$\nabla_{\boldsymbol{X}}$	The material gradient
$\boldsymbol{B_{Ie}}$	The intermedia elastic left Cauchy-Green deformation tensor
$\boldsymbol{B_{Ne}^*}$	The deviatoric network elastic left Cauchy-Green deformation tensor
$\bar{\bar{\boldsymbol{D}}}_{\boldsymbol{Iv}}, \bar{\boldsymbol{D}}_{\boldsymbol{Ip}}, \breve{\boldsymbol{D}}_{\boldsymbol{Nv}}$	The intermolecular elastic, viscous, and plastic deformation rate tensors
$\boldsymbol{F_P}, \boldsymbol{F_I}, \boldsymbol{F_N}$	The deformation gradient tensors of the polymer, the intermolecular and the network branch
$\boldsymbol{F_{Ie}}, \boldsymbol{F_{Iv}}, \boldsymbol{F_{Ip}}$	The intermolecular elastic, viscous, and plastic deformation gradient tensors
$\boldsymbol{F_{Ne}}, \boldsymbol{F_{Nv}}$	The network elastic and viscous deformation gradient tensors
$\boldsymbol{F_{ts}}, \boldsymbol{F_{pre-ts}}$	The deformation gradient tensors of the tensile and the pre-tensioned springs in the P-F part
$\boldsymbol{F_{Ne}^*}$	The deviatoric part of $\boldsymbol{F_{Ne}}$
$\boldsymbol{F_C}, \boldsymbol{F_{PrF}}, \boldsymbol{F_{P-F}}$	The deformation gradient tensors of the composites, the PrF part, and the P-F part
$\boldsymbol{F}, \boldsymbol{F_I}, \boldsymbol{F_N}$	Deformation, intermolecular deformation, network deformation gradient tensors
$\boldsymbol{F_{Ie}}, \boldsymbol{F_{Iv}}, \boldsymbol{F_{Ip}}$	The intermolecular elastic, viscous, and plastic deformation gradient tensors
$\boldsymbol{F_{Ne}}, \boldsymbol{F_{Nv}}$	The network elastic and viscous deformation gradient tensors

$\boldsymbol{F_P}$,$\boldsymbol{F_s}$	The deformation gradient tensors of pure shear and simple shear deformation
$\boldsymbol{I}$	The identity matrix
$\boldsymbol{M_{Iv}}, \boldsymbol{M_{Ip}}, \boldsymbol{M_{Nv}}$	The Mandel stress tensors for the intermolecular viscous, plastic and the network viscous configurations
$\boldsymbol{R}$	The rotation tensor
$\boldsymbol{R_e}, \boldsymbol{R_v}$	The rotation tensors for the elastic and viscous deformation
$\boldsymbol{U}$	The stretch tensor
$\boldsymbol{U_e}, \boldsymbol{U_v}$	The stretch tensors for the elastic and viscous deformation
$\boldsymbol{U_{Ie}}, \boldsymbol{U_{Iv}}, \boldsymbol{U_{Ip}}$	The intermolecular elastic, viscous, and plastic stretch tensors
$\boldsymbol{X^o}$, $\boldsymbol{X^d}$	The original and deformed configurations of the analysed element
$\boldsymbol{r_{Iv}}, \boldsymbol{r_{Ip}}$,$\boldsymbol{r_{Nv}}$	The intermolecular viscous, plastic and network viscous deformation potential tensors

List of Figures

Figure 1-1. Breakdown of CO_2 emissions in the transportation sector worldwide 2021. Data obtained from [1]. 1
Figure 1-2. a) Weight reduction by using fibre reinforced polymers (FRPs) and b) FRP components, for an example car body [2]..... 2
Figure 1-3. Several types of fibre reinforcements [9] 3
Figure 1-4. Mechanical properties of FRPs [6, 10]..... 3
Figure 1-5. a) The production cost composition when the production volume reaches 50,000; b) The trends and forecast of FRP production in 2012-2022; c) Global Market value of organosheet and semi-finished thermoplastic UD-tape laminate [12-14]. 5
Figure 1-6. Illustration of the FRTPs stamp forming process. 6
Figure 1-7. Comparison of solid-state and melt-state stamp forming for FRTPs 7
Figure 1-8. The flowchart of the thesis structure 10
Figure 2-1. Schematic of the compression moulding used for FRTPs. Figure adapted from [24]..... 12
Figure 2-2. Schematic of the DFFIM process FRTPs. Figure adapted from [25, 26]..... 13
Figure 2-3. Schematic of a) VBO process [27] and b) VARTM [28]. 14
Figure 2-4. Schematic of a) pultrusion and b) extrusion process for FRTPs [30, 32]. 16
Figure 2-5. Schematic of AFP process [36]. 17
Figure 2-6. The schematic of the filament winding process [22]..... 18
Figure 2-7. Example demonstrators that formed from UD CFRTPs. 20
Figure 2-8. Common defects during the stamp forming of UD FRTPs 23
Figure 2-9. Recent innovations in stamp forming technologies of FRTPs. a) Frictional blank holder; b) Clamping frame with tape-preform covering tool with FRTP sheets; c) Construction of stamp forming set up with diaphragm and samples exploded; d) Induction heating assisted tooling and the sandwich FRTP sheet design; e) Variothermal tooling installed in 950 kN laboratory press [10, 48, 70, 71, 73]. 27
Figure 2-10. Deformation mechanisms in the thermoplastic constituent in FRTPs during stamp forming 28
Figure 2-11. Deformation mechanism in one layer of UD FRTPs under stamp forming. 29

Figure 2-12. Characterisation methods for the intra-ply shear properties of UD FRTPs and their corresponding example pictures. 30
Figure 2-13. Reviews of the characterisation results on a) PA6 and b) PEEK. Data of PA6 extracted from [91-96]; Data of PEEK extracted from [74, 97-100]. (The figures are in log-log scales. The T_g, T_m are respectively 47 ℃ and 230 ℃ for PA6, and 143 ℃ and 343 ℃ for PEEK.) 34
Figure 2-14. Typical stress–strain curves of polymers tested at different temperatures [103] , curves (a) – (d) representing an increasing trend of temperatures. 35
Figure 2-15. Thermomechanical properties of two typical UD CF/PEEK materials (Solvay APC and Toray TC1225) characterised by bias extension test. [107] 37
Figure 2-16. Various numerical simulation focuses during the FRTPs stamp forming process. 43
Figure 2-17. Bending behaviour of the a) fibrous material and b) classic continuous material [147] 45
Figure 2-18. Model strategies for decoupling the intra-ply and bending behaviours in FRTPs. a) New shell element composed of DKT element and membrane element; b) Combination of shell element and membrane element; c) Combination of DKT element with linear shape function shell element used for Aniform [5, 8, 15]. 46
Figure 3-1. Uniaxial tensile test specimens: a) the design and dimensions; photographs of b) PA6 and c) PEEK tensile test specimens. (unit: mm) 51
Figure 3-2. Experimental setup. a) Test setup on low-rate INTRON; b) Test setup on high-rate INSTRON; ①Load cell; ②Fan; ③Environmental chamber; ④Low-speed camera; ⑤Light; ⑥Thermocouple and data-logger; ⑦Tensile test grip; ⑧Tensile test specimen; ⑨High-speed camera; ⑩Laptop and software to capture the image from the high-speed camera. 53
Figure 3-3. Tensile strain captured using an image processing method. a) An example frame of the deformation pictures; b) Identifying the position of marked lines through pixel values in the example frame; c) The tracked positions of three marked lines throughout a uniaxial tensile test 54
Figure 3-4. Experimental true stress – true strain curves from uniaxial tensile tests for PA6 under different temperatures (160 - 220 ℃) and strain rates (0.01 - 1 /s). a) Temperature

effects at the strain rate of 0.1 /s; strain rate effects at temperatures of b) 180 ℃ and c) 220 ℃ 56
Figure 3-5. Experimental true stress – true strain curves from uniaxial tensile tests for PEEK at different temperatures (180 – 310 ℃) and strain rates (0.001-50 /s). a) Temperature effects at a strain rate of 0.1 /s. b) Strain rate effects at a temperature of 240 ℃..... 57
Figure 3-6. The analysis of the key features of a representative stress-strain curve: a) three stress response regions; b) the identification of the yield point and elastic modulus; c) the definition of the pre-strain hardening modulus and the post-strain hardening modulus..... 58
Figure 3-7. Temperature effects on the a) yield stresses and b) elastic moduli of PA6 and PEEK at different temperatures (160 – 220 ℃ for PA6 and 180 – 310 ℃ for PEEK) and a strain rate of 0.1 /s..... 60
Figure 3-8. Temperature effects on the a) pre-strain hardening modulus and b) post-strain hardening of PA6 61
Figure 3-9. Stress analysis at strain levels of 0.1 and 0.7 for a) PA6 at a temperature of 180 ℃, and b) PEEK at a temperature of 240 ℃, at various strain rates (0.001 – 50 /s). (The figures are in log-log scales)..... 62
Figure 3-10. An example of deformation images of PEEK at a temperature of 240 ℃, and at a strain rate of a) 0.001 /s (no necking observed) and b) 50 /s (necking observed), at the same deformation stage (ε_e = 0.14) 62
Figure 3-11. The experimental setup and specimen designs for VNR shear tests. a) Design and dimensions (unit: mm) of the specimen. b) An example picture of the specimen and grips used for the VNR shear test. c) Experimental setup: ①VNR shear test specimen; ②VNR shear test grips; ③Instron 5584 load frame; ④Crosshead embedded with a load cell; ⑤Environmental chamber; ⑥Light; ⑦Thermocouples and data-logger; ⑧⑨Fans; ⑩Camera 66
Figure 3-12. The DIC measurement methods for the VNR shear test 67
Figure 3-13. The effective stress-strain curves obtained from the tensile and the VNR shear tests on PEEK at a temperature of 180 ℃ and at a strain rate of 0.1 /s..... 69
Figure 3-14. Deformation mechanism of the VNR shear specimen. a) Simple shear deformation; b) Simplification of the deformation mechanism at small deformation: pure shear deformation + rigid body rotation. (Terms $\sigma_{11}, \sigma_{22}, \sigma_{12}$ are the stress

components of the analysed element. Terms γ_m, s and γ_m, p are the maximum shear strains under simple shear and pure shear deformation, respectively.) 70
Figure 3-15. Deformation mechanisms of a) Pure shear and b) Simple shear deformation. (The terms $X_{io}, X_{iU}\ X_{id}$ $(i = 1,2,3)$ are the coordinates of the original, stretched, and deformed configurations, respectively. The arrows, $\boldsymbol{\theta_{po}}$,$\boldsymbol{\theta_{pd}}$ represent the maximum principal strain directions of pure shear under original and deformed configuration, respectively; $\boldsymbol{\theta_{so}}$, $\boldsymbol{\theta_{sU}}$, $\boldsymbol{\theta_{sd}}$ are the maximum principal strain directions of simple shear under the original, stretched, and deformed configurations, respectively. Note: $\boldsymbol{\theta_{po}}$, $\boldsymbol{\theta_{so}}$ are two virtual directions for the purpose of angle calculations.)..... 71
Figure 3-16. The shear strain measurement methods in the VNR shear test. a) Using strain gauge; b) Using DIC. 75
Figure 3-17. Flow stress comparison between the JC numerical modelling (solid lines) and the tensile test (dash lines) results of PEEK at different temperatures (180 – 300 °C) and at a strain rate of 0.1 /s. 78
Figure 3-18. Schematic illustration of the boundary conditions of the FE model 79
Figure 3-19. Validation of the numerical model by comparing a) strain distribution along the ligament at four displacement stages (Solid lines are the numerical results; Dot lines are the experimental results); strain fields of b) experimental and c) numerical results at Stage 3; d) vertical loads obtained from experimental and numerical results. 80
Figure 3-20. The deformation state of two representing nodes, Node 1 and Node 2, on the test specimen: a) illustration of the analysed nodes, and b) deformation state of the two analysed nodes in the forming limit diagram..... 81
Figure 3-21. The ground truth validation: effective stress-strain curves of the central point (Node 2 in Figure 3-20) and the input constitutive model (the JC model). 82
Figure 3-22. Comparison between the results of strain gauge and DIC measurements and the ground truth of material properties: a) effective strains and b) effective stresses. 84
Figure 3-23. Two-step correction for the effective stress-strain curves. a) The first correction step: a)-1, The relationship between effective stresses and rotational angle; a)-2, The relationship between the calculation errors and rotational angle; a)-3, The first step correction effect; b) The second correction step: b)-1, The relationship between effective stresses and rotational angle; b)-2, The relationship between the calculation errors and rotational angle; b)-3, The second step correction effect. 85

Figure 3-24. Comparison of the effective stress-strain curves of strain gauge and DIC methods before and after the two-step correction. (BC means before correction, and AC means after correction)..... 88
Figure 3-25. The effective stress-strain curves of the VNR shear tests a) before and b) after correction on PEEK at a temperature of 180 ℃ and at a strain rate of 0.1 /s..... 88
Figure 3-26. Flow stress-plastic strain curves obtained from the JC model predicted results and the VNR shear experimental results at a temperature of 180 ℃ and a strain rate of 0.1/s. (BC means before correction, AC means after correction. Exp refers to the post-processed experimental results)..... 89
Figure 3-27. Comparison between experimental and numerical results using shear test results as model input at 180 ℃: strain field comparison at four displacement stages using a) model fitted using uncorrected shear test data and b) model fitted using corrected shear test data. (The solid lines are the numerical results, and the dot lines are the experimental results.)..... 90
Figure 3-28. Verification of the correction method by a) changing the test condition of the same material (PEEK); and b) changing the material (PA6). (BC means before correction, and AC means after correction)..... 92
Figure 4-1. The hot press mounted on the biaxial machine, and its electrical unit for controlling the heating function..... 94
Figure 4-2. The two designs of consolidation tool: a) Picture frame tool and b) Metal mould tool. The example pictures of fabricated laminates via different consolidation tools (c-e). 95
Figure 4-3. Microstructural observation of the laminate quality 97
Figure 4-4. Design of experimental specimen for a) transverse tensile and b) VNR shear tests. (Unit: mm) 99
Figure 4-5. Experimental setup. ①Instron 3369 tensile testing machine with environmental chamber; ②PC and BLUEHILL software; ③Camera; ④Lamp; ⑤Thermocouples and datalogger; ⑥Tensile test rig; ⑦Transverse tensile test specimen; ⑧VNR shear test rig; ⑨VNR shear test specimen. 99
Figure 4-6. Experimental results of transverse tensile tests on UD CF/PA6 under a) temperature ranges in 180 – 220 ℃, at a strain rate of 0.01 /s and b) a strain rate ranges in 0.001 – 0.25 /s, at a temperature of 200 ℃. 102

Figure 4-7. Experimental results of VNR shear tests on UD CF/PA6 under a) temperature ranges in 180 – 220 ℃, and at a strain rate of 0.04 /s and b) strain rate ranges in 0.004 – 0.4 /s at a temperature of 200 ℃. 103
Figure 4-8. The tangent modulus versus strain at different temperatures (180 – 220 ℃) under a) transverse tensile at a strain rate of 0.01 /s, only the data within a range of ±1000 are shown, and b) VNR shear tests at a strain rate of 0.04 /s, only the data within a range of ±100 are shown. 105
Figure 4-9. Localised deformation analysis of VNR shear tests under the temperature range of 180 – 220 ℃. a) Shear strain evolution versus the crosshead displacement, and shear strain distribution at the displacement of b) 0.2 mm and c) 0.7 mm at different temperatures. 106
Figure 4-10. Stress-strain curves of UD CF/PA6 under longitudinal shear compared to PA6 under uniaxial tension, under various temperatures (180 – 220 ℃). 107
Figure 5-1. The proposed new constitutive model. (a) The rheology scheme; (b) The kinematics ($\boldsymbol{F}$ is the deformation gradient, $\boldsymbol{F_I}, \boldsymbol{F_N}$ represent the intermolecular deformation and the network deformation gradients, respectively. $\boldsymbol{F_{Ie}}$,$\boldsymbol{F_{Iv}}$,$\boldsymbol{F_{Ip}}$ represent the intermolecular elastic, viscous, and plastic deformation gradients, respectively. $\boldsymbol{F_{Ne}}$, $\boldsymbol{F_{Nv}}$ represent the network elastic and viscous deformation gradients, respectively.). 111
Figure 5-2. Schematic illustration of microstructural evolution and the governing parts of the constitutive model, corresponding to the stress-strain response at different stages, during the tensile deformation of semicrystalline thermoplastics [106, 196, 197]. 119
Figure 5-3. Comparison between experimental (symbols) and modelling (solid lines) results at different temperatures and strain rates for PA6. a) PA6 at different temperatures (160 – 220 ℃), at a strain rate of 0.1 /s; b) PA6 at different strain rates (0.01 – 1 /s), at a temperature of 180 ℃. 123
Figure 5-4. Comparison between experimental (symbols) and modelling (solid lines) results at different temperatures and strain rates for PEEK. a) PEEK at different temperatures (180 – 310 ℃), at a strain rate of 0.1 /s; b) PEEK at different strain rates (0.001 – 50 /s), at a temperature of 240 ℃. 123
Figure 5-5. The flow chart of the ABAQUS subroutine 124
Figure 5-6. Kinematics of viscoelastic deformation 125

Figure 5-7. The validation of the ABAQUS subroutine via simulation of the VNR shear deformation. a) Comparison of the maximum principal strain distribution at four deformation stages from the simulation results (solid lines) and experimental results (circle lines); the maximum principal strain distribution of b) experimental and c) simulation results at deformation stage 2; d) the comparison of the load from the experimental and simulation results. (The studied case is the PEEK material at 240 ℃). 128
Figure 5-8. Rheology of the constitutive model form of UD CF/PA6. 129
Figure 5-9. Analysis of the mapping relationships between the composite deformations and the effective deformations of PrF using RVE. a) ε_{PrF} vs. ε_{22}^{RVE} under transverse tensile at various temperatures (180 – 220 ℃), and b) the equivalent strain distribution in polymer elements when $\varepsilon_{22}^{RVE} = 0.036$, at 200 ℃. c) ε_{PrF} vs. ε_{12}^{RVE} under longitudinal shear at various temperatures (180 – 220 ℃) and d) the equivalent strain distribution in polymer elements when $\varepsilon_{12}^{RVE} = 0.188$, at 200 ℃. 133
Figure 5-10. a) The calculation method for obtaining the expected cohesive stress curve. The Expected cohesive stress at different temperatures (180 – 220 ℃) under b) transverse tension and c) longitudinal shear deformation (circles representing the turning point of each curve). 135
Figure 5-11. The cohesive model. 137
Figure 5-12. The expected controlling parameters for cohesive stress curves under a) transverse tension and b) longitudinal shear deformation. 138
Figure 5-13. The comparison between the constitutive model predictions (solid lines) and the experimental results (circles) of transverse tensile tests. 142
Figure 5-14. The comparison between the constitutive model predictions (solid lines) and the experimental results (circles) of VNR shear tests. 142
Figure 5-15. The comparison between the FEA and experimental results of a bias-extension test. a) Comparison of the in-plane shear strain distribution at the displacement of 2.7 mm (1 is the fibre direction); b) The comparison of the history of the averaged in-plane shear strain across the middle section of the specimen. 144
Figure 5-16. The model assembly and loading conditions of the FEA model. 145
Figure 5-17. Comparison of the numerical simulation results a) with (using solid elements) and b) without bending decoupling (superimposed shell and membrane elements) 146

Figure 5-18. Design of blank shapes (unit: mm)..... 148
Figure 5-19. Comparison of the blank shape effects via the in-plane shear strain ($\varepsilon 12$) distributions from FEA simulation results. ① The excessive material on the flange area. ② the relative flat area on the flange. ③ the incompatible deformation between layers and wrinkles. 149
Figure 5-20. Analysis of the layer blocking effects reflected via a) the maximum in-plane stress (σ_{max}) and b) the cutting view of deformed components. 151
Figure 5-21. Analysis of the temperature effects reflected via the in-plane shear strain (ε_{12}) distribution at temperatures between 180 – 210 ℃. 152
Figure 5-22. Analysis of the temperature effects reflected via the modified mean curvature (κ) at temperatures between 180 – 210 ℃. 153
Figure 6-1. Experimental setup of the solid-state stamp forming process of CF/PA6 sheet. a) Stamp forming equipment setup b) Stamp forming tool set; c) Cross section view of the stamp forming tool set shown in a CAD model; d) Critical dimensions of the dome forming tools..... 156
Figure 6-2. A temperature profile example of forming at 205 ℃..... 158
Figure 6-3. The effect of blank shape. ① the excessive material on the flange area; ②③④⑤⑥ wrinkles ; ⑦ the limited area on the flange area. 160
Figure 6-4. The effect of layer-blocking at a) three layers blocked (0.42 mm for a single layer) and b) unblocked (0.14 mm for a single layer) configurations. 161
Figure 6-5. The effect of forming temperature (180 – 260 ℃). 163
Figure 6-6. Optical microscope observation of the as-formed part trial 1 (forming temperature at 208 ℃) 165
Figure 6-7. CT scanning results of a) forming trial 2 (forming temperature at 205 ℃) and b) forming trial 6 (forming temperature at 195 ℃) 166
Figure 6-8. Comparison of the numerical results and the experimental results on a) as-formed part dimensions; b) comparison of wrinkles on the outer surface; c) comparison of wrinkles on the inner surface. ① Wrinkle area on the simulation results. ②Wrinkle area on the experimental results. ③Wrinkle area observed from CT scan on the experimental results. ④⑤Wrinkle area on the inner surface of the simulation results. ⑥⑦Wrinkle area on the inner surface of the experimental results. 168

List of Tables

Table 1-1 Comparison of the advantages and disadvantages of using UD FRTPs compared to other materials. 4
Table 2-1. Summary of the FRTPs processing metrics [7, 22, 38] 19
Table 2-2. Existing characterisation research on the thermo-mechanical properties of UD FRTPs. 37
Table 2-3. Comparing features for the FEA software used for stamp forming applications. 47
Table 3-1. Mechanical and thermal properties of PA6 and PEEK. 50
Table 3-2. Uniaxial tensile test matrices 52
Table 3-3. The detailed effective stress-strain calculation for the VNR shear tests 68
Table 3-4. Input parameters fitted from tensile test results of PEEK at 180 – 300 ℃ 79
Table 3-5. Details of the two-step correction method 87
Table 3-6. Input parameters fitted from the VNR test results of PEEK at 180 ℃, before and after the correction 89
Table 3-7. Input parameters fitted from tensile test results of PA6 at 160 ℃ 91
Table 4-1. The grinding and polishing process for UD CF/PA6 97
Table 4-2. Test matrices for transverse tensile and VNR shear tests 100
Table 5-1. Material constants for PA6 and PEEK. 122
Table 5-2. Material constants for carbon fibre and matrix material input in ABAQUS 130
Table 5-3. Summary of the unified constitutive model. 140
Table 5-4. Summary of all the material constants in the unified constitutive model 141
Table 5-5. Model setup for numerical simulation 147
Table 6-1. Stamp forming condition matrices. 159

CHAPTER 1 Introduction

1.1. Industrial drivers: the Net-Zero target and automotive lightweighting

Climate change has driven various industry sectors to seek innovative solutions to reduce the greenhouse gas emissions. According to the International Energy Agency (IEA), the transportation sector, as one of the largest contributors, accounted for 17 % of the global greenhouse gas emission in 2021, with an emission of around 7.7 billion tons [1]. Figure 1-1 illustrates the breakdown of carbon dioxide (CO_2) emissions in the transportation sector worldwide in 2021, highlighting the significant contribution from passenger cars [1]. These substantial greenhouse gas emissions have prompted intensive studies, especially on lightweight materials and their applications in automotive industries, to improve operational efficiency and contribute towards achieving the net-zero target.

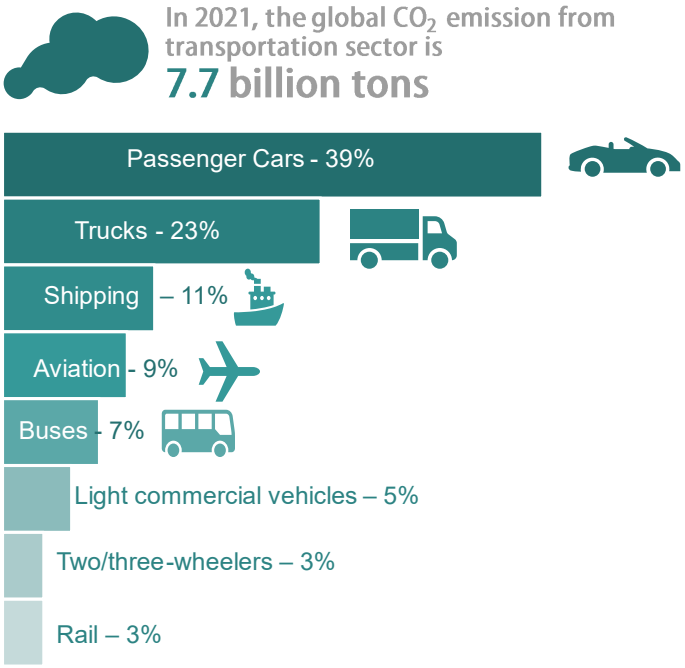


Figure 1-1. Breakdown of CO_2 emissions in the transportation sector worldwide 2021. Data obtained from [1].

Among the various lightweight materials available, fibre reinforced polymers (FRPs) are considered one of the most promising solutions due to their superior strength-to-weight ratio. Figure 1-2a shows the potential for weight reduction in cars by incorporating FRPs compared to traditional cars. By replacing certain steel components with FRPs, a significant 30 % reduction of body weight can be easily achieved. FRPs have versatile applications in automobiles, including interiors, exteriors, chassis, powertrain, etc. An example of a car body that already utilises FRP materials is depicted in Figure 1-2b, showing the successful implementation of FRPs in the automotive industry.

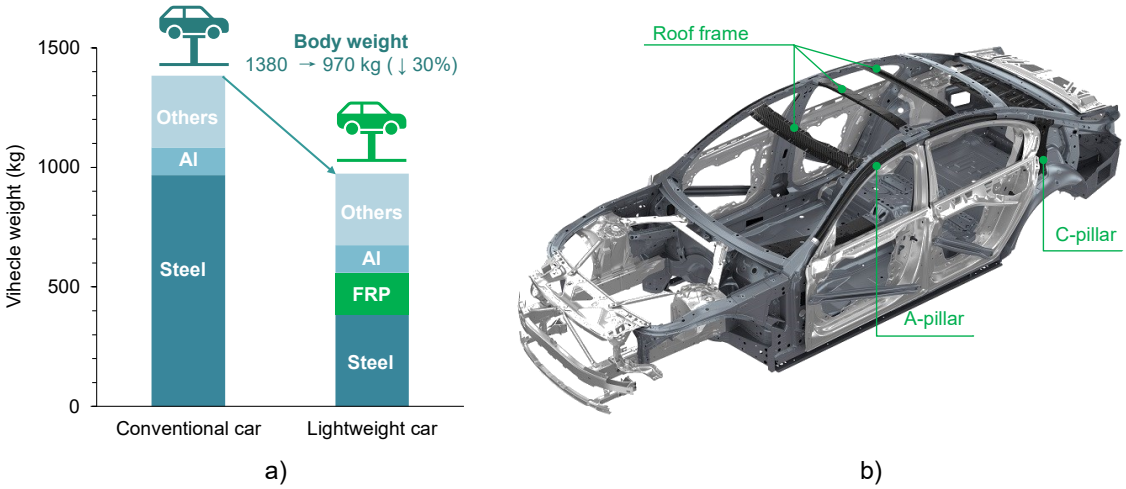


Figure 1-2. a) Weight reduction by using fibre reinforced polymers (FRPs) and b) FRP components, for an example car body [2]

1.2. Unidirectional Fibre Reinforced Thermoplastics (UD FRTPs)

With the great potential of FRP to be used in cars, there is a various selection of polymer matrices and fibre reinforcement. For the polymer matrix, there are thermoset and thermoplastics, which differ in the intermolecular bonds within the polymer chains. In the thermoplastics, weak intermolecular forces exist and there are no cross-lines between chains, making the thermoplastics soften when heated [3]. In contrast, thermosets retain their structure when heated. This unique characteristic of thermoplastics suggests a promising future for fibre reinforced thermoplastics (FRTPs) to be used in automobiles. Compared to fibre reinforced thermosets (FRTSs), FRTPs have better recyclability, weldability, and good impact resistance [4, 5]. With regard to the reinforcement types, there are continuous and discontinuous fibres as

illustrated in Figure 1-3, and the mechanical properties of various fibre types are compared in Figure 1-4. It can be concluded that, when it comes to the reinforcement types, the unidirectional (UD) fibre has always shown the best properties, with the ability to embedded a higher fibre volume fraction, and superior mechanical properties [6, 7]. Moreover, the unidirectional fibre reinforced thermoplastics (UD FRTPs) are usual comes in the form of prepreg tape. This format enables the flexibility of tailoring lay-ups in terms of orientation and thickness, opening up a wider possibility for design optimisation [8]. Therefore, this research recognises the significant potential of UD FRTPs and focuses on studying these materials for automotive applications.

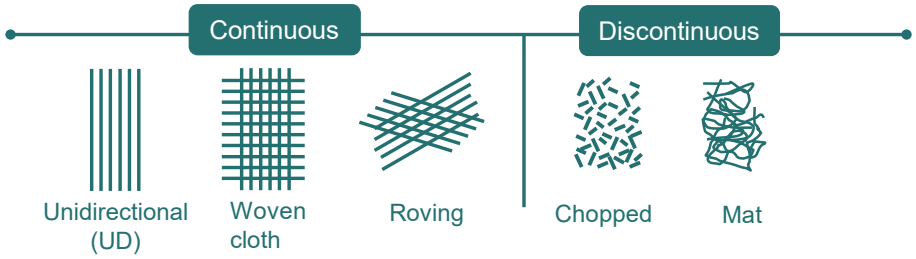


Figure 1-3. Several types of fibre reinforcements [9]

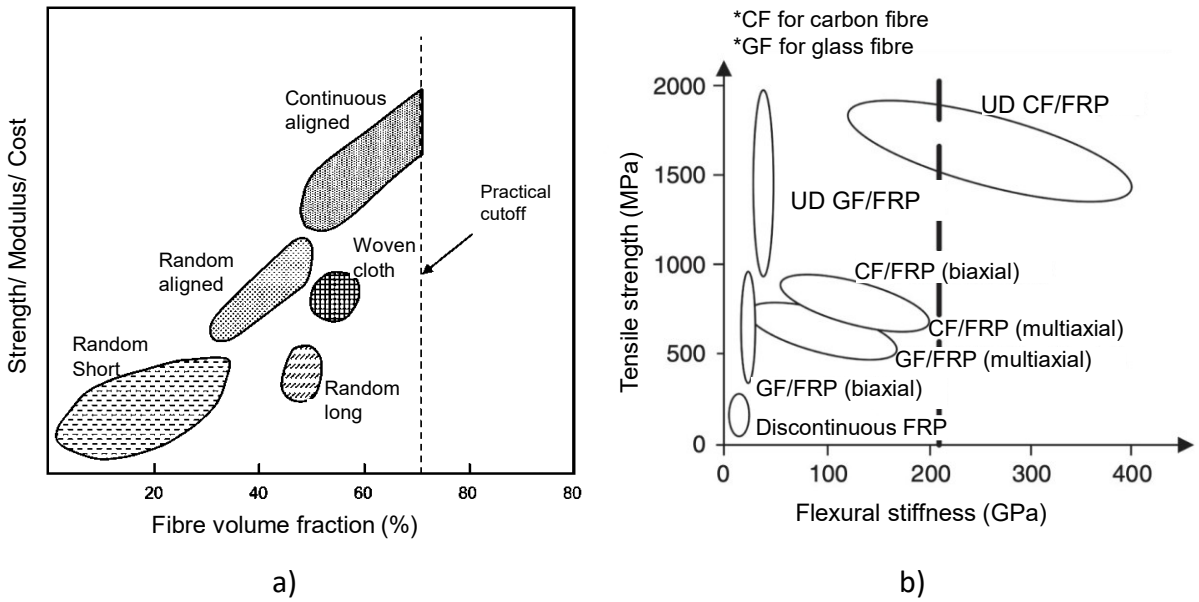


Figure 1-4. Mechanical properties of FRPs [6, 10]

A detailed comparison of the advantages and disadvantages of using UD FRTPs in automobiles is listed in Table 1-1.

Table 1-1 Comparison of the advantages and disadvantages of using UD FRTPs compared to other materials.

Advantages
• High strength-to-weight ratio compared to woven FRTPs [6, 7] • Higher fibre volume fraction • Less fibre waviness compared to other UD composite, such as UD non-crip fabric (NCF) • Suitable for automated tape placement to create tailored blanks [8]
Disadvantages
• Poor formability compared to woven FRTPs • High material cost • High potential of defects because of no constraint on transverse direction

Admittedly, the high cost of raw UD FRTPs and their manufacturing cost are inhibiting their wider application in automobiles. Reports indicate that the cost of adopting UD FRTPs in cars can be more than five times that of traditional steels [11]. However, this situation is gradually changing as this material becomes more widely utilised. As illustrated in Figure 1-5a and b, Shama et al. [12] presented that when the production volume reaches 50,000, the major investment comes from the material cost, and the cost of FRP per kilogram (kg) in USD is projected to have dropped by 67 % between 2012 and 2020. Meanwhile, the consumption of FRP is consistently increasing. In terms of the compound annual growth rate (CAGR), it is expected to be 9.0 % in 2025. In the aspect of UD FRTP tapes, Figure 1-5c depicts the global market size from 2019 to 2022, with a projected CAGR of 10.92 % until 2031. These statistics demonstrate the significant market and potential of UD FRTPs to be used in the automotive industry in the near future. However, the development of a cost-effective manufacturing method suitable for mass production is a prerequisite for wider adoption.

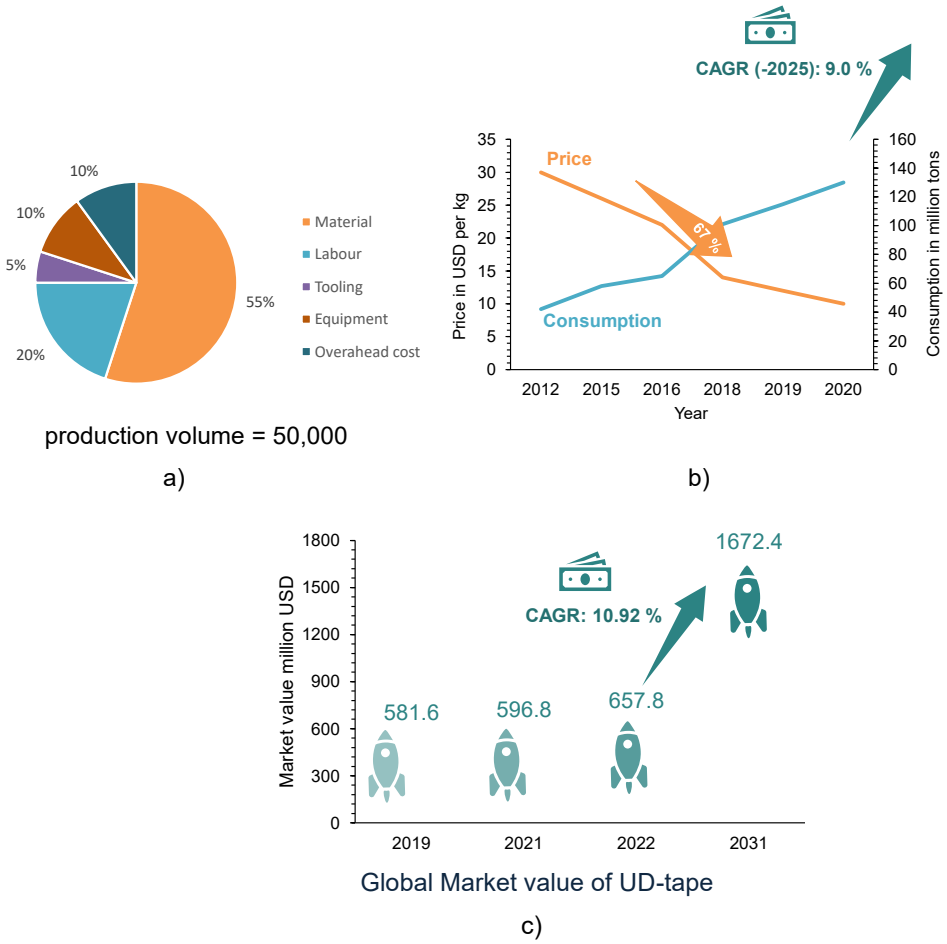


Figure 1-5. a) The production cost composition when the production volume reaches 50,000; b) The trends and forecast of FRP production in 2012-2022; c) Global Market value of organosheet and semi-finished thermoplastic UD-tape laminate [12-14].

1.3. Solid-state stamp forming

Among the available techniques, stamp forming technology has emerged as the most viable option for the manufacturing of UD FRTPs, mainly due to its rapid cycle time of approximately 10 to 60 seconds. The process consists of four distinct steps, as illustrated in Figure 1-6: heating the FRTP laminate to a higher temperature; transferring and positioning the FRTP laminate on to the forming die; stamp forming the composite sheet into mould cavity using a metallic punch; and finally releasing the as-formed parts [15]. For industrial applications, stamp forming equipment generally consists of a press machine, a set of forming dies, and an integrated heating system. This configuration allows for seamless integration into pre-existing manufacturing infrastructure. This compatibility is particularly advantageous for large-scale automotive

component production, as it minimises transition times and upholds operational efficiency. Moreover, stamp forming machines offer a comparatively low cost per unit in high-volume production scenarios, making them the most suitable option for the automotive sector.

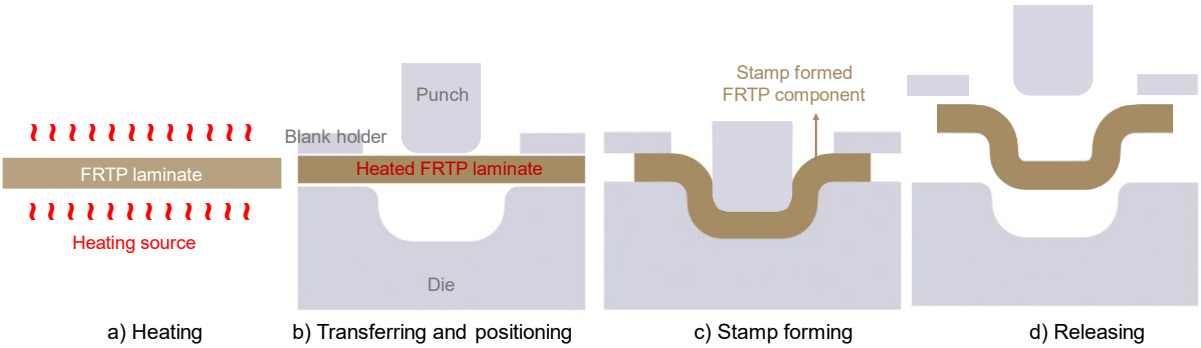


Figure 1-6. Illustration of the FRTPs stamp forming process.

Traditional manufacturing technologies for FRTP components, such as compression moulding or autoclave forming, typically process material at melt state. These technologies require a long cycle time and involve high costs. In contrast, solid-state stamp forming of FRTPs reduce the cycle time and improve the flexibility of the manufacturing process [16]. Additionally, under solid state, the fibres remain embedded within the matric surface, resulting in a better surface finish of the as-formed part [17]. Moreover, solid-state stamp forming can effectively reduce the risk of flattening defects in the corner area [18]. Research conducted by Brands et al. [19] when studying the forming limit of FRTPs demonstrated that, the maximum strain is higher when the material is under solid state. Figure 1-7 provides a summary comparison of solid-state and melt-state stamp forming for FRTPs.

Solid-state stamp forming	Melt-state stamp forming
Forming condition • Below the melting point (T_m)	**Forming condition** • Above the melting point (T_m)
Research status and properties • Limited research data or knowledge • Poor inter-ply formability • Shorter cycle time • Longer tool lifetime • Less energy consumption • Potential to improve surface finish • Potential to reduce the risk of flattening corner • Potential to improved forming limit in transverse direction • Reduced risk of material degradation	**Research status and properties** • More existing research data and knowledge • Better inter-ply formability • Longer cycle time • More energy consumption • Shorter tool lifetime • Dry fibre/cracks may appear on the surface • Corners may flatten • Potential risk of material degradation

Figure 1-7. Comparison of solid-state and melt-state stamp forming for FRTPs

Despite these many potential advantages of solid-state stamp forming of UD FRTPs [8], noticeable research gaps exist, as currently, the majority of existing stamp forming studies concentrate on the melt state [20] and woven FRTPs [20, 21]. There are no comprehensive studies on solid-state stamp forming of UD FRTPs, resulting in insufficient lab experiments, limited understanding of deformation behaviours, and a lack of material models. To address these research gaps, the novelty of this thesis lies in its pioneering exploration of the thermomechanical properties of UD FRTPs under solid-state stamp forming conditions and providing optimal processing windows for manufacturing UD FRTP components.

1.4. Aim and objectives

The aim of this research is to develop solid-state stamp forming of UD FRTPs for complex-shaped components, by developing a comprehensive understanding of the deformation behaviours of thermoplastic matrices and UD FRTPs under corresponding conditions. This research has a particular focus on automotive-grade FRTPs, using unidirectional carbon fibre reinforced polyamide 6 (UD CF/PA6) as the studied material. The successful implementation of this process will significantly reduce cycle time, energy consumption, and thereby costs,

promoting wider application of FRTPs in the automotive industry. To achieve this aim, the following objectives have been outlined:

1) Characterise and develop a thorough understanding of the primary deformable constituent of UD FRTPs, i.e., thermoplastics, under the conditions of solid-state stamp forming.
2) Analyse the principal deformation mechanism of UD FRTPs, taking into account their material anisotropy, and identify the most effective characterisation method. The thermomechanical response of UD FRTPs will be comprehensively characterised.
3) Establish unified, physically based, viscoelastic-viscoplastic material constitutive models for thermoplastics and their corresponding UD FRTPs. These models will be capable of describing the deformation behaviours of FRTPs under solid-state stamp forming conditions. These models will be integrated into a finite element analysis (FEA) software package for the simulation and optimisation of stamp forming of FRTPs.
4) Identify the optimal processing windows for stamp forming of FRTPs in a controlled setting with minimal defects. Demonstrator components will be formed, followed by an analysis of forming parameters and an evaluation of the quality of the as-formed components. The effectiveness of the FEA model will also be validated.

1.5. Thesis structure

The structure and interconnections between each chapter of this thesis are illustrated in Figure 1-8. A summary of each chapter is as below:

Chapter 2 presents a systematic literature review of the processing technologies of FRTPs, with special attention to the mass production of UD FRTPs using solid-state stamp forming. It introduces the deformation mechanisms, characterisation methods, thermomechanical properties, constitutive modelling, and numerical simulation of the UD FRTPs. The research gaps and research methods have been identified.

Chapter 3 conducts the characterisation studies on the thermoplastic constituent. The thermomechanical properties of two semi-crystalline thermoplastics are characterised and analysed under solid-state stamp forming conditions. Besides, an alternative characterisation method which involves simple shear deformation, is studied and a post-processing method is proposed for obtaining the real stress-strain properties.

Chapter 4 introduces the fabrication of UD FRTP laminates and the mechanical characterisation of their intra-ply transverse tensile and longitudinal shear properties. The thermomechanical properties of UD FRTPs are systematically analysed under the solid-state stamp forming conditions and the possible phenomena during the solid-state stamp forming process are analysed.

Chapter 5 focuses on the numerical studies of the thermoplastics and UD FRTPs. The constitutive model of thermoplastics is built based on the experimental studies from Chapter 3, and the constitutive model of UD FRTPs is built based on the constitutive model of thermoplastics and the experimental studies of Chapter 4. These constitutive models are written in a Fortran VUMAT subroutine and embedded in FEA software, ABAQUS, for simulating the solid-state stamp forming process. The processing defects are predicted from the numerical simulation and initial processing windows are proposed.

Chapter 6 manufactures UD FRTP dome-shaped demonstrators and the processing parameters in terms of initial blank shape, fibre-configuration, and temperature are investigated. The quality of the stamp-formed demonstrators is examined, and the optimised processing parameters are identified. The conclusions from the numerical simulations conducted in Chapter 5 are validated. Besides, the numerical simulation results are compared to the demonstrators in dimensions and wrinkle formation, which proves the validity of these constitutive models and FEA models.

Chapter 7 concludes the thesis by summarising the major findings. It also provides recommendations for future research in this area. The achieved aim and objectives of the work are discussed, highlighting the potential of solid-state stamp forming technology for UD FRTPs. The material constitutive models and FEA models developed in this thesis can be adopted for other types of UD FRTPs, enabling their wider application.

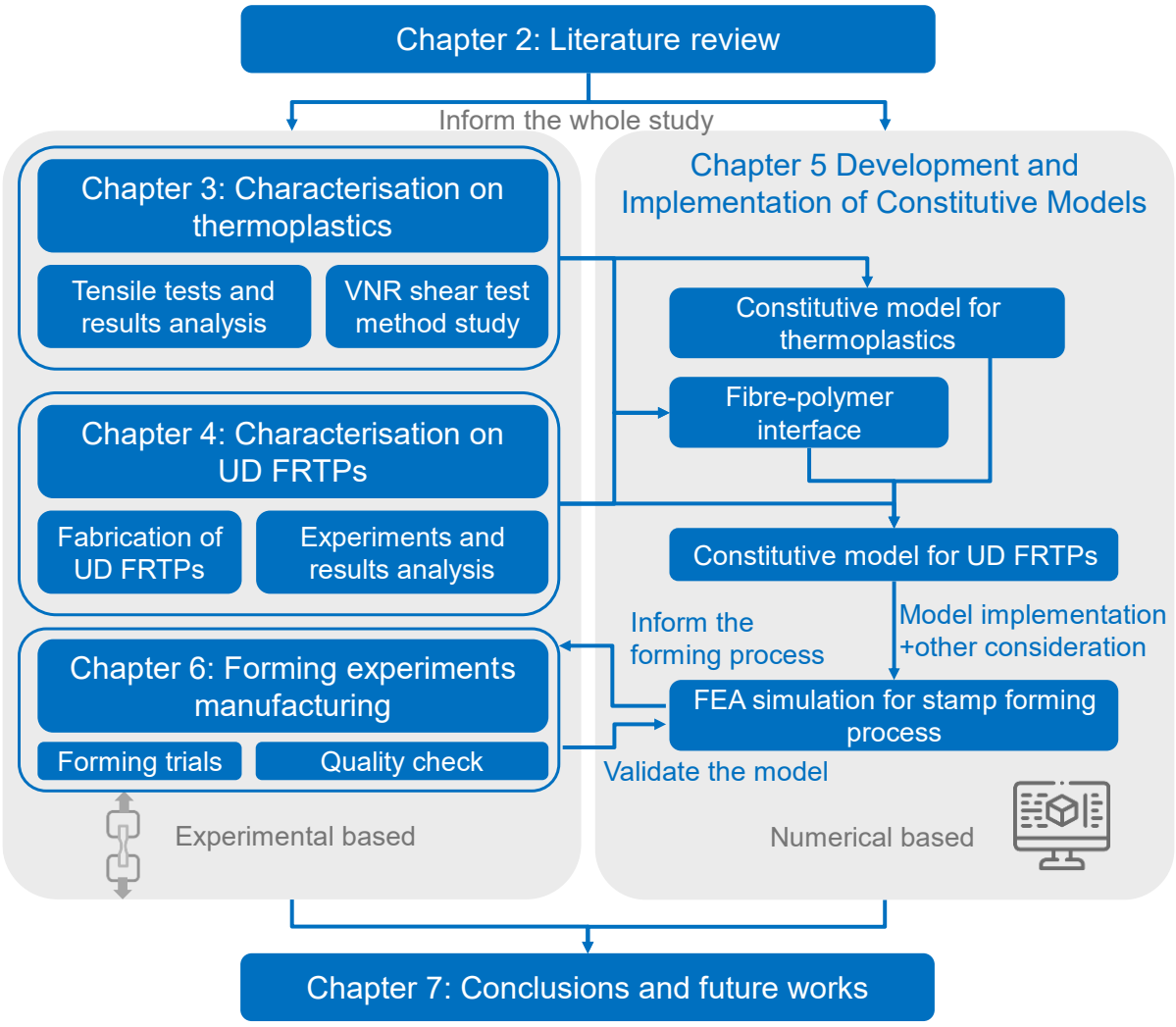


Figure 1-8. The flowchart of the thesis structure

CHAPTER 2 Manufacturing Technologies and Fundamental studies on UD FRTPs

2.1. Manufacturing technologies for FRTP components

Over the past three decades, manufacturing technologies for FRTP components have been intensively studied, with respect to different reinforcement forms, component shapes, applications, and quality needs. Therefore, it is essential to review the commonly used manufacturing methods to identify the most promising ones for automotive applications.

From a processing point of view, the FRTP manufacturing technologies can be classified into the conventional methods and the automated methods [22]. The conventional methods refer to those that involve heating up the matrices of FRTPs to deform them into shapes [7]. With the development of automation in the manufacturing process, automated manufacturing methods have also become areas of research interests, as they require little or no human intervention [22].

2.1.1. Conventional methods

- Autoclave forming

In industrial manufacturing, the autoclave represents a vital tool, providing a high-pressure and high-temperature environment ideal for the production of FRTPs [23]. By processing these materials within the autoclave, the as-formed parts have superior surface quality and reduced void content, which in turn bolsters the dependability of their mechanical properties. Importantly, the autoclave method is uniquely proficiency at handling large, intricately structured components. However, the long cycle time required for autoclave forming, approximately 5 hours, constrains the overall production rate and could potentially impact the operational efficiency. Furthermore, the high costs associated with the high-pressure and high-temperature requirements of autoclave equipment pose a significant financial burden.

- Compression moulding

The compression moulding process shapes molten FRTPs by moulding them into a specific cavity. Figure 2-1 shows the schematic of the compression moulding process. The compression moulding process involves three major steps: placing the bulk material, moulding through compression, and demoulding. This process can be used for manufacturing large basic shapes or complex shapes to a certain level. It is widely used for composites due to its high precision quality and relative low cost when mass-produced. The cycle time for compression moulding is about 1-2 minutes. However, the requirements for temperature and pressure present challenges in tool design to maintain consistent production quality. Given the complexity of the equipment setup process, this method may not be suitable for low-volume production.

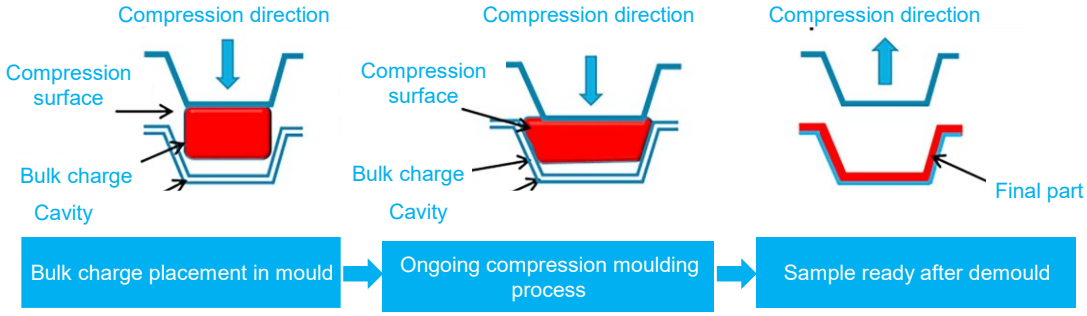


Figure 2-1. Schematic of the compression moulding used for FRTPs. Figure adapted from [24]

- Injection moulding

The injection moulding technology has been adopted for producing plastic products since the 18th century [25]. When applied to FRTPs, injection moulding is generally used for short fibre reinforced thermoplastics (SFT) and long fibre reinforced thermoplastic (LFT) pellets. In the process, short fibre mixed with molten polymers is injected rapidly into a mould cavity with anticipated shape [7]. It is a cost-effective method typically used for high-volume production. However, some defects include gas-burn marks due to moisture toxic gases within the molten polymer solution. Besides, there is additional cost from the pellet drying process. Recently, an improved injection moulding method known as Direct Fibre Feeding Injection Moulding (DFFIM) technology has emerged, which streamlines the process of making and drying fibre pellet. The process of DFFIM is illustrated in Figure 2-2, which shows that the continuous fibre

roving strands are directly guided into the vent hole of the injection moulding machine, and then fed into the molten material by shearing motion of the screw [25]. The quality of products formed through injection moulding technology is highly dependent on the control of pressure and temperature during the injection and cooling steps [7]. Defects such as voids, jetting or flash, fibre exposure due to difference in shrinkage, and weld lines can occur. Controlling fibre length and orientation remains a challenge in this manufacturing technology.

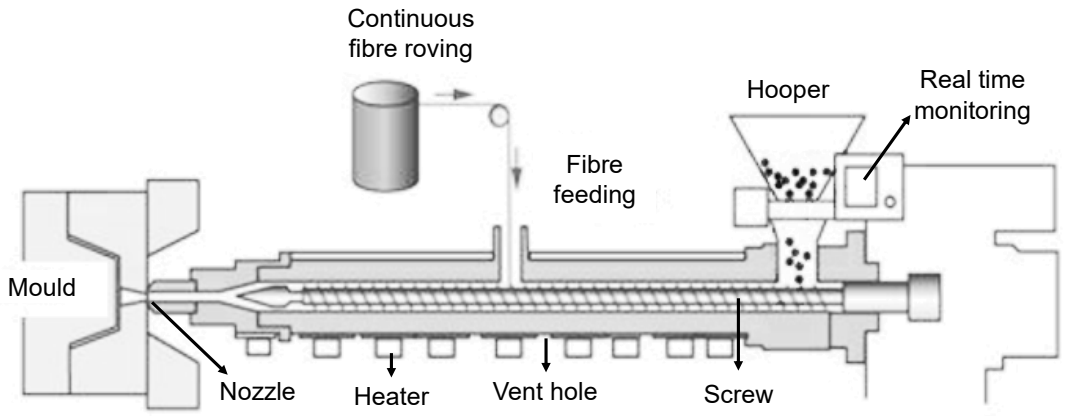


Figure 2-2. Schematic of the DFFIM process FRTPs. Figure adapted from [25, 26].

- Vacuum bagging and Vacuum Assisted Resin Transfer Moulding (VARTM)

The vacuum bagging method, typically used for composite manufacturing, removes air from the laminate, often with the assistance of heating and preforming [22]. The vacuum-bag-only (VBO) process for FRTPs is illustrated in Figure 2-3a. There are three major steps. First, the thermoplastic film and reinforcement plies are stacked on a plate, with the spacers, caul plate and vacuum bag in place; then the air within the system bag is vacuumed using a pump. Afterwards, the assembly is heated in an oven to above the melting temperature of thermoplastics for consolidation [27]. This method of manufacturing requires a relatively low equipment investment. However, the cycle time for per component is relatively long and it cannot be used to manufacture sophisticated shapes.

There is another manufacturing method involving vacuum bagging, namely Vacuum Assisted Resin Transfer moulding (VARTM), that has proved to be a cost effective process [28]. The schematic of the VARTM is illustrated in Figure 2-3b. A single rigid mould is used in the VARTM, for the dry fabrics to drape on (also called preform process). The vacuum part of this

process is similar to other vacuum bagging system. A pressure differential, generated by the vacuum in the preformed fabrics and the resin tank, transfers the resin into the mould cavity, enabling the resin to permeate the entire preformed part. This VARTM method can fabricate more complex components compared to the VBO process. However, challenges persist with high temperature processing (≥ 177 °C) due to the need for higher pressures to accommodate high-viscosity thermoplastics [28].

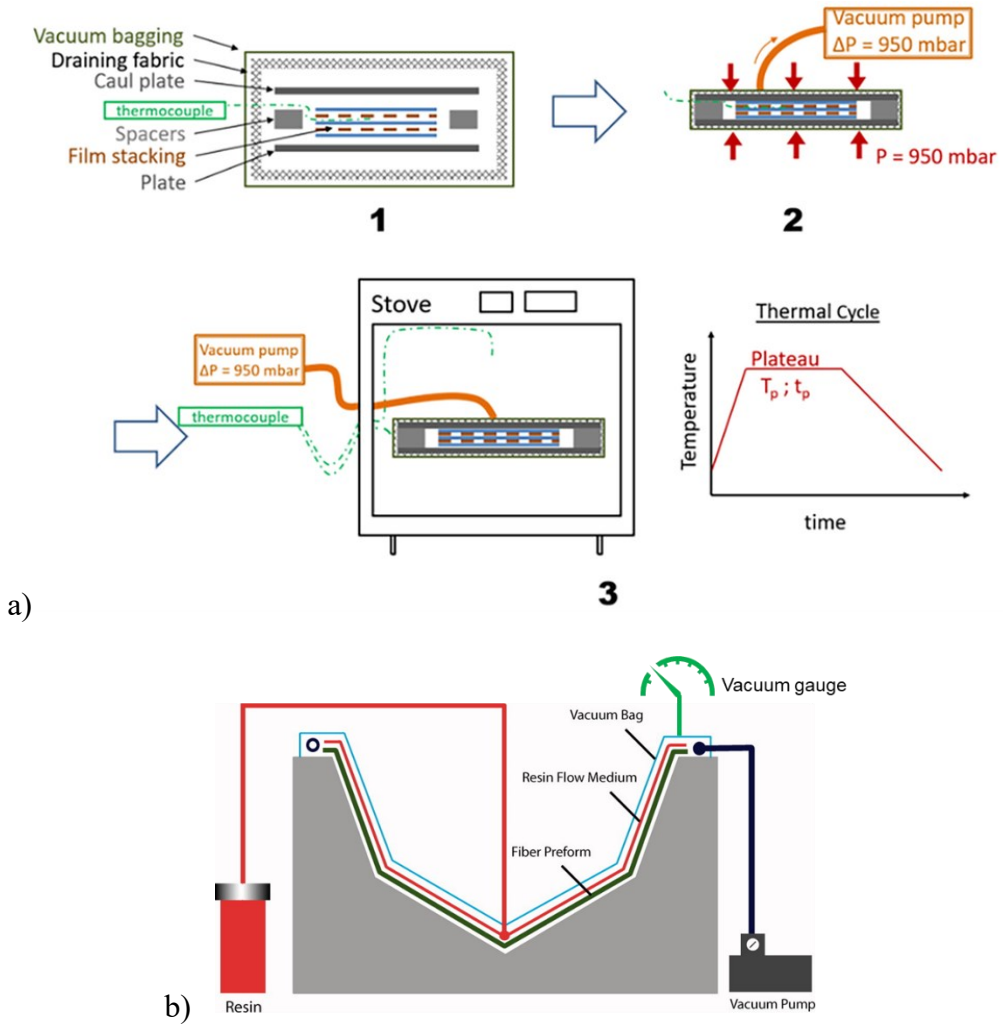


Figure 2-3. Schematic of a) VBO process [27] and b) VARTM [28].

- Pultrusion and extrusion

Pultrusion process is a continuous and automated manufacturing technology for producing constant section and rectilinear products. Originally invented for fibre reinforced thermoset (FRTS) composites, it has gradually gained more interest for manufacturing FRTPs. The

pulsion process of FRTPs is illustrated in Figure 2-4a. first, the pre-impregnated FRTPs are pulled into the pre-heating zone through the guidance system, where they are heated to the required temperature and consolidated to the desired size. Once solidified in the cooling zone, the formed components are extracted and cut from the system [29]. The pultrusion process requires low labour input and processes a high raw material conversion efficiency [30]. The pultrusion process is also highly dependent on the processing parameters. For instance, rapid pulling speeds may result in poor impregnation, while slow pulling speeds could yield low productivity. In general, the productivity of FRTPs pultrusion process is still unsatisfactory compared to FRTSs pultrusion, and further studies in this filed are expected to make this technology more mature [31].

Extrusion can be used to mix the filler and the matrix using mechanical stirring. The schematic of extrusion process is illustrated in Figure 2-4b. During the extrusion process, the FRTPs are continuously heated until melted and then transported to the die via the screw. The extruded components are then cooled down in the cooling chambre [32]. This process is particular suitable for the LFT for pipes, sheet and films. There are also manufacturing methods that combine the extrusion process with another process such as injection moulding or hot pressing [22]. For example, Li et al. [33] manufactured a testing specimen of short carbon fibre reinforced polyethersulphone (CF/PES) using a combination of extrusion compounding and injection moulding.

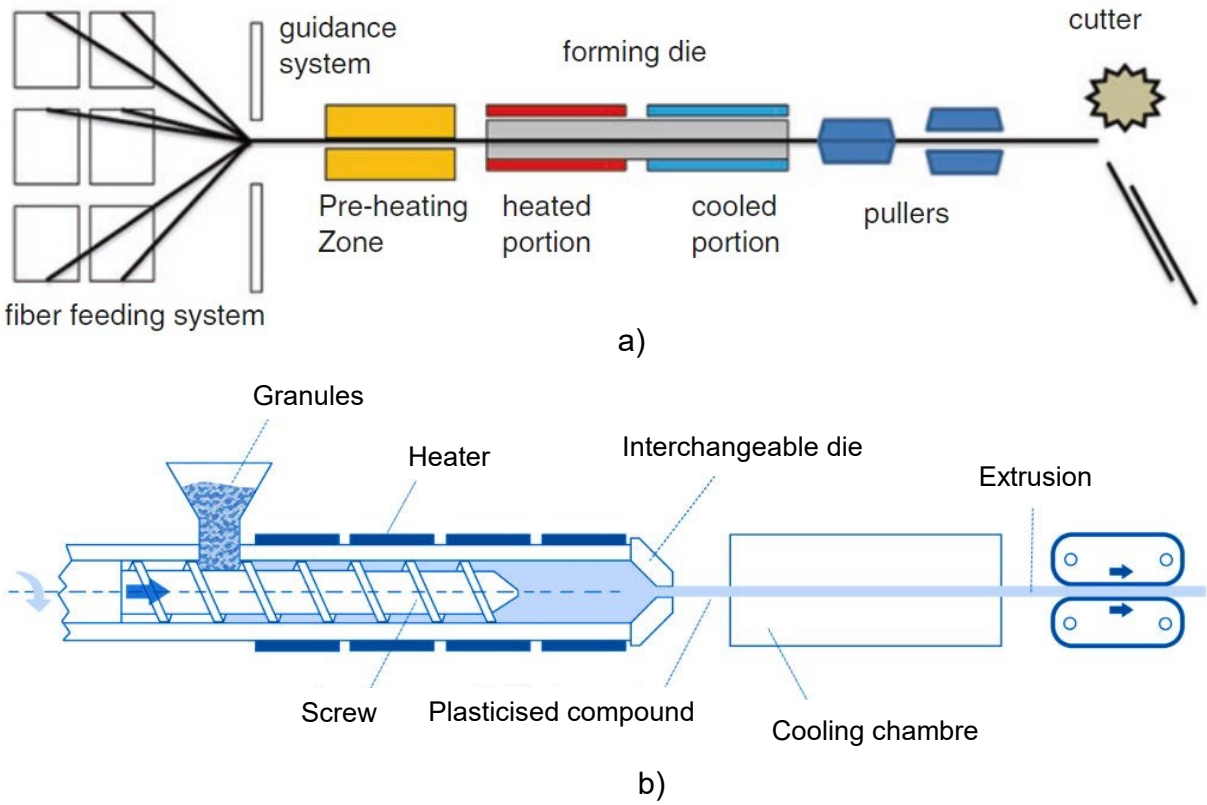


Figure 2-4. Schematic of a) pultrusion and b) extrusion process for FRTPs [30, 32].

- Stamp forming

The stamp forming process has been introduced in section 1.3. and will be elaborated in Section 2.2.

2.1.2. Automated methods

The automated FRTPs manufacturing processes, facilitated by advancements in robotics, sensor technology, and computer-aided design, have the potential to significantly enhance production efficiency, precision, and repeatability, while reducing labour-intensive tasks and production costs. Currently, the manufacturing processes that can be fully automated include automated fibre/ tape placement (AFP / ATP), filament winding and additive manufacturing [22].

- AFP/ ATP

The AFP process, schematically illustrated in Figure 2-5, involves placing thin impregnated fibre strips under a certain controlled program using a robot arm [34]. In the ATP process, this

fibre can be wider as tape for large-size applications. After the fibre is placed, the whole thermoplastic composite part is consolidated under high temperature and pressure added by a roller [34]. Compared to traditional manufacturing process, the AFP/ATP techniques can produce the components with precise and consistent fibre orientation control [35], and the flexibility and high degree of design freedom of the process provide the possibility of manufacturing complex geometries. However, the processing procedure is time-consuming, and the initial equipment invest is rather high. Certain features, such as radii and sharp corners, are highly dependent on the size of the roller and steering control algorithm.

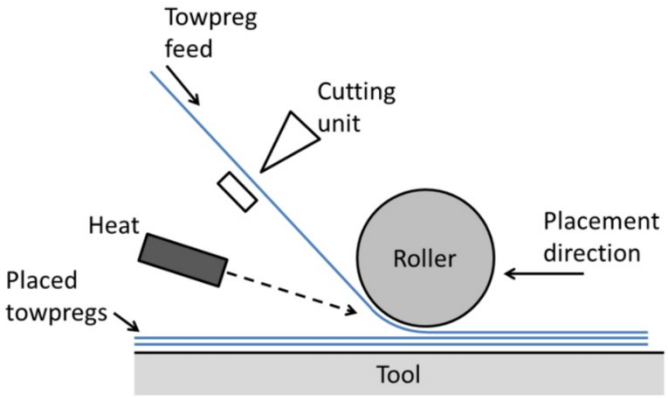


Figure 2-5. Schematic of AFP process [36].

- Filament winding

Filament winding is another FRTPs manufacturing technique that can be fully automated. This technique utilises the pre-heated, impregnated, continuous fibres that are directed by the carriage and precisely wound onto a rotating mandrel to form both symmetric and non-symmetric parts such as pipes, tanks, or pressure vessels, as illustrated in Figure 2-6 [37]. This process builds up to continuous fibre reinforced lay-ups with well-defined fibre orientation and accurate positioning of the fibre. It is one of the most cost-effective processes used for composite manufacturing [7]. While this process can be used for manufacturing parts with a winding core, it is not suitable for complex shapes. During the filament winding process, it is essential to ensure the adequate fibre impregnation and properly maintain the fibre tension, which requires the high stability of the filament winding machine.

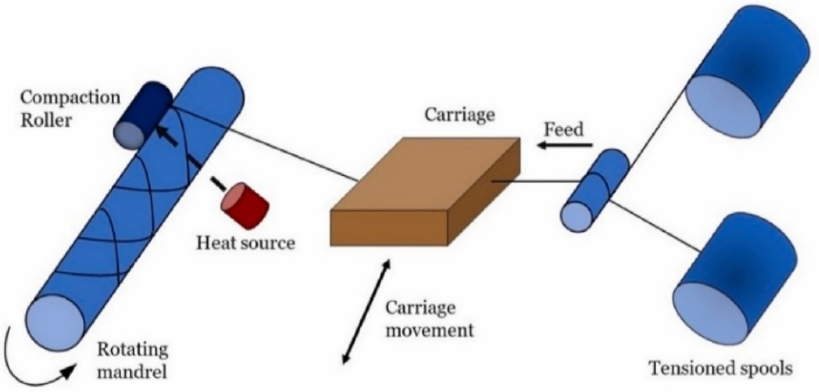


Figure 2-6. The schematic of the filament winding process [22].

- Additive manufacturing

The additive manufacturing process, often referred to as three dimensional (3D) printing, has been intensively studied for manufacturing metal, polymer and FRTPs [35]. In this process, the FRTPs components are created by the precise, layer-by-layer deposition of a thermoplastic matrix material embedded with continuous or short fibre reinforcements [36]. This approach is highly suitable for customised products, offering flexible geometry and reduced waste. Nevertheless, there are still challenges such as achieving uniform fibre distribution to avoid cracks, managing the thermal stress and optimising the printing parameters for better mechanical properties [22].

The manufacturing processes presented are selectively summarised in terms of their application, maturity and cost, as detailed in Table 2-1. For the automobile components, the ability for mass production with affordable time and equipment cost is the key. With all these manufacturing processes compared, the stamp forming process shows the most advantages.

Table 2-1. Summary of the FRTPs processing metrics [7, 22, 38]

Processing method	Reinforcement form	Reinforcement length scale	Process maturity[1]	Process modelling maturity	Design combined with process	Geometry complexity[2]	Cost matric[3]	Cycle time
Compression moulding	Powders, particles, fibres, films	Nano to continuous	8	7	8	8	5	Medium
Injection moulding	Powders, flakes, particles, short fibres	Nano to 6 mm	8	8	7	9	9	Medium
Vacuum bagging	Woven fabrics, films, sheets	Continuous	4	7	6	4	4	Medium
Diaphragm forming	Woven fabrics, films, sheets	Continuous	9	7	7	9	8	Long
Extrusion	Short and long fibres	3 – 25 mm	6	6	7	8	6	Medium
Pultrusion	Tapes, sheets	Continuous	4	5	3	5	6	Medium
Liquid moulding (VARTM, RTM)	Woven fabrics	Continuous	3	7	1	4	6	Long
Stamp forming	Woven fabrics, films, sheets	Continuous	5	6	5	5	6	**Short**
Automated fibre placement	Tows, tapes, rods	Continuous	4	3	5	6	7	Medium
Filament winding	Tows, tapes	Continuous	4	4	4	5	6	Long

[1] Maturity refers to how well the process has been developed and being adopted by industry, with 1 refers to least mature and 10 refers to most mature.

[2] Complexity refers to the features in the as-formed parts, with 1 refers to the simplest structures and 10 refers to the most complex structures.

[3] Cost is estimated based on tooling and manufacturing fees.

2.2. The state of the art in stamp forming technologies for UD FRTPs

2.2.1. UD FRTP demonstrators manufactured through stamp forming

For the stamp forming of UD FRTPs, several studies have developed demonstrators ranging from relatively simple shapes to complex geometries with multiple features [39]. Figure 2-7 provides examples of these demonstrators, with researchers addressing various challenges.



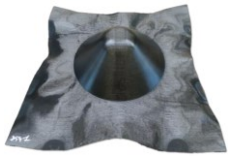
a) U shape bending

UD CF/PEEK [40]



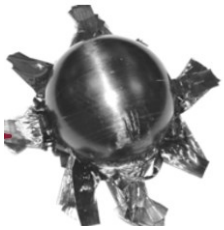
b) Rib with double curvature

UD CF/PEEK [41]



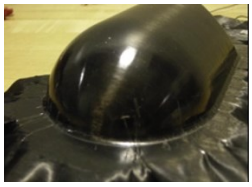
c) Cone shape

UD CF/PEEK [5]



d) Dome shape

UD CF/PA6 [42]



e) Double dome shape

UD CF/PEEK [43]



f) Complex-shaped

(Dome, trapezoid, tetrahedron)

UD CF/PA6 [5]



g) Complex-shaped

(Case corners and beads)

UD CF/PA6 [44]

Figure 2-7. Example demonstrators that formed from UD CFRTPs.

Figure 2-7a shows a U-shaped part stamp formed from UD carbon fibre reinforced poly-ether-ether-ketone (CF/PEEK) under melt state, with a focus on optimising the process to minimise the corner thinning, edge-squeezed-out, wrinkles and fibre breakages [40]. Processing parameters (such as tool temperatures, consolidation pressures and forming speeds) and tools improvements (such as replacing the metal punch with a rubber punch and adding tension springs on the clamp holder) were investigated. It was pointed out that forming a defect-free component is challenging but feasible through various explorations within the design space.

Figure 2-7b demonstrates a rib with double curvatures studied by Haanappel et al. [41], used in aircraft, with several embedded bending and shallow features. The rib was stamp formed from melt-state UD CF/PEEK and woven glass fibre (eight harness satin weave, 8HS) reinforced polyphenylene sulfide (8HSs GF/PPS). It was concluded that the formability of UD CF/PEEK was inferior to that of 8HS GF/PPS. The UD CF/PEEK blanks were very susceptible to wrinkles near these double curvature areas, which were difficult to eliminate.

Figure 2-7c displays a cone-shaped component that was stamp formed from UD CF/PEEK, with the primary objective of investigating the effects of stamp forming from unconsolidated spot-welded UD CF/PEEK layups at melt state of matrix [5]. It was found that unconsolidated prepregs with welding spots were inadequate for producing flawless parts without prior consolidation. Alternative methods, such as vacuum setups with polyimide (PI) film or vacuum setups with polyamide 6 (PA6) film, improved the quality of the as-formed parts. Although this method could be valuable in terms of reducing the pre-consolidation process, it still involved complex preparation process, and there will be PI or PA6 film remaining on the final part.

Figure 2-7d exhibits the dome-shaped component that was stamp formed from UD CF/PA6 tailored laminate by Burkhart et al. [42]. Although it was a successful trial demonstrating the potential of defect-free components formed from UD CF/PA6, wrinkles were still observed on the component.

Figure 2-7e presents the UD CF/PEEK double-dome-shaped component that was stamp formed under melt state [43]. The researchers employed the Taguchi method to investigate the effect of four operational parameters (preheating temperature, mould temperature, transferring time, stamping pressure) on the part thickness, interlaminar shear strength, and the degree of crystallinity. It was found that the mould temperature and stamping pressure had the most significant effects on the part consolidation quality. It was also found that thinner parts

exhibited higher interlaminar shear strength and a higher degree of crystallinity compared to thicker parts. To enhance the quality of the as-formed part, PI film was added on both sides of the laminate before forming.

Figure 2-7f shows a relatively complex-shaped component formed by Shug et al. [45], featuring a combination of dome, trapezoid and tetrahedron shapes. The component was formed from UD CF/PEEK at melt state. Forming parameters, such as tool temperature, pre-heating temperature, and forming speed, were studied to optimise the quality of the formed component. It was determined that a higher tool temperature could improve the as-formed part quality. Nevertheless, severe fibre waviness, fibre splits and wrinkles persisted even after these improvements.

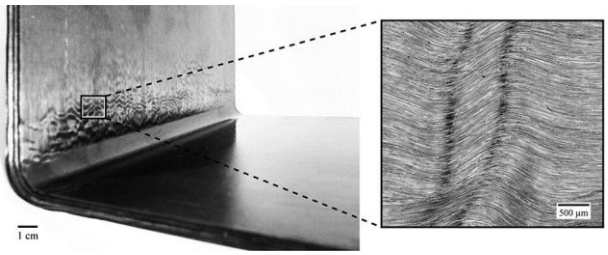
Figure 2-7g displays a complex-shaped UD CF/PA6 component with case corners and beads manufactured by Dorr et al. [44] in its melt state. The focus of this study was on the simulation of the forming process. The in-plane membrane deformation was modelled using either a Voigt-Kelvin or a generalized Maxwell approach and implemented in a user-element subroutine within the commercially available FEA solver, ABAQUS. It was found that only the generalized Maxwell approach can represent the whole material characteristics. Regarding the stamp forming quality, although not particularly studied, the authors have mentioned the existence of wrinkles on the formed demonstrators.

As can be concluded from the above analysis, there are studies on the melt-state stamp forming process of UD FRTPs. However, unresolved defects persist in components. In recent years, solid-state stamp forming of fabric reinforced composites has gained interest [46] and findings suggest that the formability of solid-state woven CF/PEEK laminates is primarily influenced by reinforcements and resins, with forming temperature playing a significant role [16]. In the case of UD FRTP materials, Uriya et al. [47-49] systematically investigated the solid-state stamping formability of unidirectional carbon fibre/epoxy sheets using various tests, demonstrating the feasibility of solid-state stamp forming. Based on the insights gained from literature and the thermomechanical and constitutive characterisation [50], it is posited that solid-state stamp forming could yield improved formability for UD FRTPs.

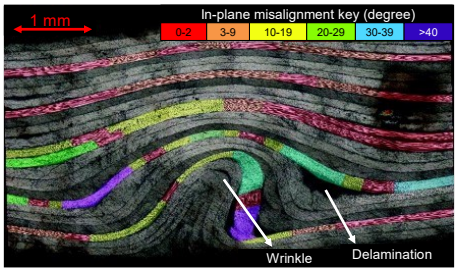
2.2.2. Common defects from UD FRTPs stamp forming

There are several forming defects in the stamp forming process, including fibre waviness, wrinkles, delamination, shape distortion and incompatible deformation. The example pictures

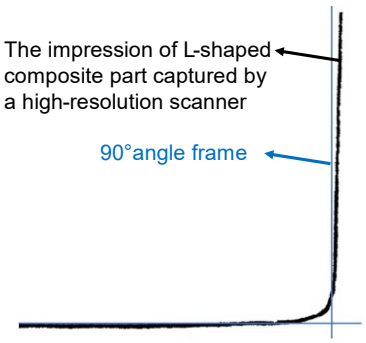
of the defects are illustrated in Figure 2-8. It is essential to understand the generation of these forming defects, so that the forming parameters can be better optimised for the stamp forming process.



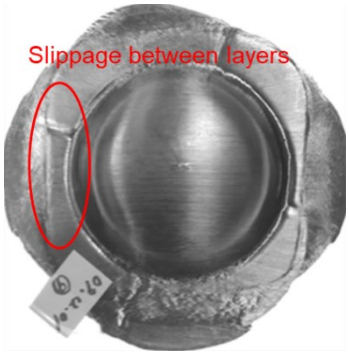
a) Fibre waviness [51]



b) Wrinkle and delamination [52]



c) Spring-back [53];



d) Excessive inter-ply slippage [54] ;



e) Fibre split, gap [5].

Figure 2-8. Common defects during the stamp forming of UD FRTPs

In Figure 2-8a, the in-plane fibre waviness observed in the stamp formed V-shaped component made of unidirectional (UD) carbon fibre reinforced polyether ether ketone (CF/PEEK) is demonstrated by Sachs [51], illustrating how the UD fibres distort and deviate from their original straight path. Other studies have used different terms to describe this in-plane waviness, including fibre misalignment, wrinkling, folding, marcel and undulation [52, 55-57]. Sitohang et al. [58] conducted experiments to evaluate the impact of in-plane waviness on the mechanical properties of FRTP components. They found that components with waviness experienced a significant reduction in compressive strength. The main factor contributing to the formation of

in-plane waviness is the axial compressive stress on the fibres. Under such loading conditions, the fibres in the piles buckle and the transverse material around the fibre fails to provide sufficient support [59]. Consequently, enhancing the transverse strength of the material can help alleviate the formation of in-plane waviness.

Wrinkling is the most prominent defect occurred during the forming process of FRTPs, and extensive research has been conducted to understand this defect [60]. Figure 2-8b illustrates the wrinkles observed by Lightfoot et al. [52] when stamping CF/Epoxy into a double curvature shape. The formation of wrinkles can lead to local variation in FRTP laminate thickness, which in turn can result in poor consolidation quality, delamination, and fibre cracks. Hallander et al. [61] identified three distinct defects occurring during the forming of UD prepregs, namely, variations in fibre angle, in-plane waviness, and spreading or tightening of the fibre tow. These defects arise from inter-ply friction and combination of fibre orientations. They concluded that these defects are caused by excessive material compression in the deformation area. To mitigate the formation of wrinkles, Hallander et al. [62] found that altering the local friction or reducing the number of difficult fibre angle combinations can slightly improve the quality of the as-formed parts. Haanappel [8] investigated the effects of bending parameters on the stamp formed UD CF/PEEK part using numerical predictions. The study found that smaller bending stiffness led to more pronounced wrinkling, as it induced greater intra-ply shear deformation.

Shape distortion, spring-back and spring-in are common defects that can occur from the stamp forming of FRTPs. These defects are typically attributed to residual stresses in the forming parts, which arise from inter-ply slip, anisotropic thermal expansion and polymer crystallisation [63]. Figure 2-8c shows an L-shape part after forming[53], where the bending angle clearly deviates from the intended shape. Shape distortion can result in deviations from the intended design, thereby affecting the performance and functionality of the final product [53]. The occurrence of shape distortions, spring-back and spring-in is affected by various process parameters, including velocity, pressure, temperature, holding time, and part geometry [63, 64]. Optimising these process parameters is crucial to mitigate these defects. For example, ensuring sufficient cooling time to fully solidify the matrix before the end of forming process, suppressing the inter-ply slippage by reducing the forming temperature, etc. [63]. Other approaches such as tool compensation can also help with obtaining the desired components [64].

Excessive inter-ply slippage is another potential defect that can occur during the stamp forming process. Chen et al. [65] mentioned that a certain degree of inter-ply slippage may be desired to reduce residual stress or wrinkles, however, when inter-ply slippage become excessive, it can lead to inter-ply defects such as wrinkles, fibre folds and delamination, and eventually compromising the mechanical properties [66]. Figure 2-8d shows a stamp formed dome-shaped component formed from UD CF/PA6, which exhibits pronounced inter-ply slippage between layers [54]. The strong anisotropy of the material has led to unbalanced distribution of material in the flange area of this dome shape. To ensure the quality of as-formed components, it is recommended to control and minimise inter-ply slippage during the stamp forming process. This can be achieved through various means, such as surface treatment, adhesive bonding, or reducing the processing temperature and pressure [65, 67]. Additionally, using laminates with good consolidation quality before stamp forming can also help with minimising inter-ply slippage [67].

Last but not least, fibre splits or gaps on the top layer of as-formed parts are commonly observed in UD CFRTPs. Figure 2-8e shows the occurrence of fibre splits after stamp forming of UD CF/PEEK [5]. Due to the weak in-plane transverse tensile strength, gaps between fibres on the surface can be easily developed through the transverse sliding of the fibres, even with small deformation [68].

In summary, to prevent defects during the stamp forming process of UD CFRTPs, it is crucial to have a thorough understanding of the deformation mechanisms and to carefully control the stamp forming parameters. Based on the analysis above, strategies to alleviate in-plane waviness and wrinkles, which are indicative of higher in-plane transverse and bending stiffness, suggest the use of UD CFRTP laminates in a solid state. Similarly, approaches to address distortion and excessive inter-ply slippage suggest the adoption of lower forming temperatures. These findings indicate that solid-state stamp forming has the potential to yield higher quality components. Recent research has demonstrated success in solid-state stamp forming of woven CF/PEEK laminates [46]. Based on the insights gained from literatures and thermomechanical constitutive characterisation [50], it is proposed that solid-state stamp forming could improve the formability of UD CFRTPs. However, there is currently a lack of research on the solid-state stamp forming technologies of UD CFRTPs. Therefore, further investigations incorporating numerical and experimental studies are necessary to fully explore the potential of this forming process.

2.2.3. Other innovations in FRTPs stamp forming

In recent years, other innovations are also developed related to stamp forming process of FRTP materials, Figure 2-9 illustrated these innovations.

In terms of the holding method the friction-based holding and spring with gripper-based system are the two main holding methods[69], as shown in Figure 2-9a and b. Khan et al. [70] used the frictional-based holding system for the preforming of woven fabrics, and for the stamp forming of UD FRTPs, researchers [5, 43, 71] tended to adopt the spring with gripper-based holding system. Harrison et al. [72] compared these two holding methods and found that the spring with gripper-based holding system can mitigate the formation of wrinkles during stamping. However, concentrated holding force in the spring with gripper-based holding system can also induce defects such as the in-plane waviness in local areas. Careful design of the number of springs and spring force within each spring is required for further studies of the stamp forming technique [72]. As for the heating method, as illustrated in Figure 2-9b-e, several techniques have been adopted, such as the infrared (IR)-radiator [71], induction heating [10, 48], and variothermal (cartridge heaters and compressed air) [73] for achieving fast heating ability. Furthermore, Figure 2-9c and d show the methods for improving the formability of UD FRTPs. Harrison et al. [10] combined the incremental matched-die and diaphragm forming methods, and introducing the molten tin inter-layers, heated rapidly and efficiently via induction heating. Their research has effectively improved the wrinkle formation during the multi-cavity geometries, although there are still limitations for application, as it is difficult to completely remove the residual tin inside the as-formed part. Besides, for stamp forming of UD FRTSs, to enhance the formability, Uriya et al. [48] incorporated dummy metal sheet on both sides of UD FRTSs, and stamp form the sandwich sheets under cold or warm tooling conditions. Their approach yielded promising results. However, due to the necessity of removing the dummy sheet post-processing, this method necessitates a simplistic shape design for practical adoption.

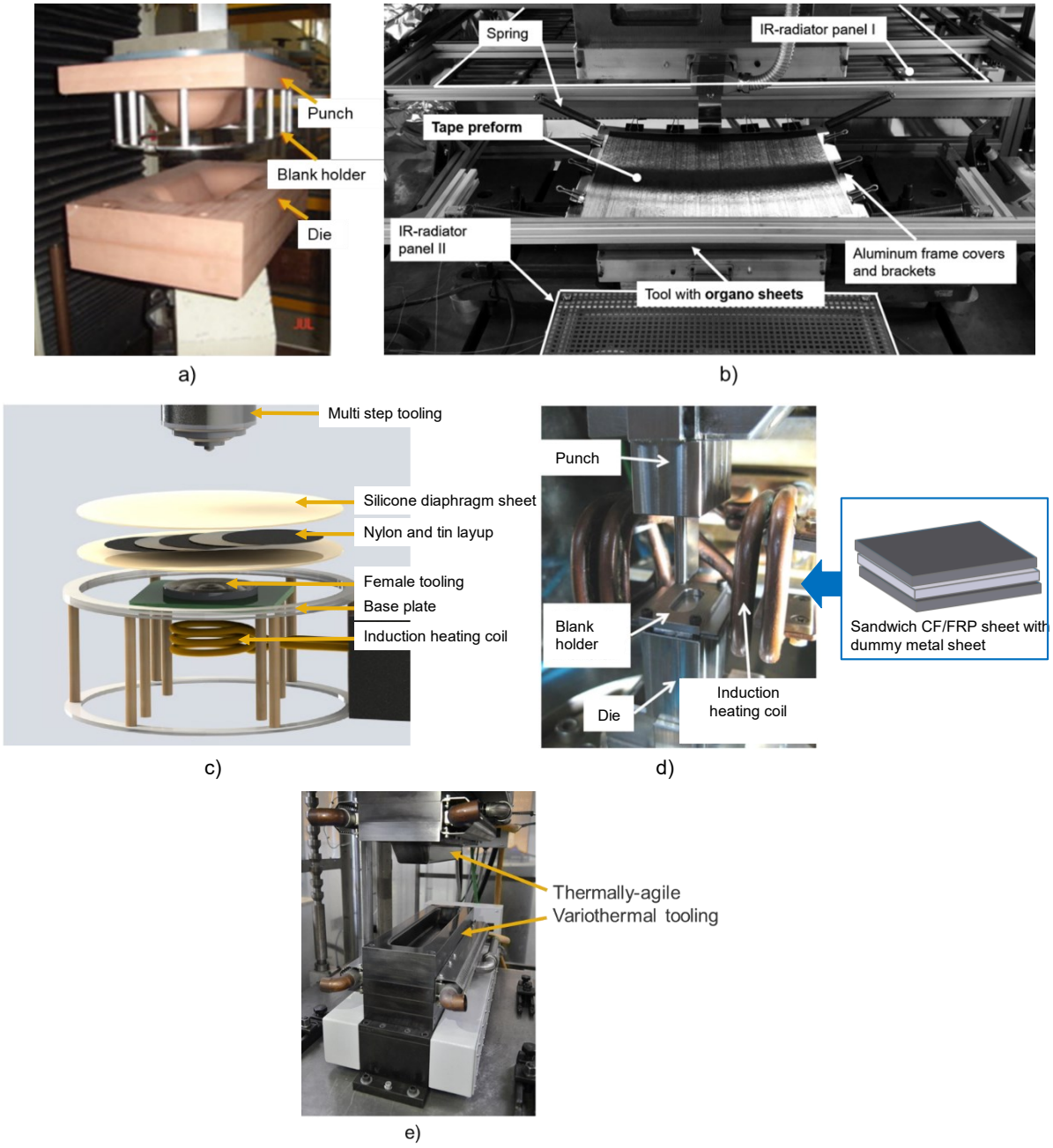


Figure 2-9. Recent innovations in stamp forming technologies of FRTPs. a) Frictional blank holder; b) Clamping frame with tape-preform covering tool with FRTP sheets; c) Construction of stamp forming set up with diaphragm and samples exploded; d) Induction heating assisted tooling and the sandwich FRTP sheet design; e) Variothermal tooling installed in 950 kN laboratory press [10, 48, 70, 71, 73].

2.3. Deformation mechanisms and characterisation methods

As introduced above, for better improving the stamp forming process and reducing the forming defects, a thorough analysis of the formation mechanism and identifying accurate

characterisation methods are essential. The studies regarding thermoplastics and UD CFRTPs are presented as follows.

2.3.1. Thermoplastics

Given that polymer matrices constitute the primary component undergoing deformation during the forming process, it is crucial to investigate the thermomechanical properties of thermoplastics under solid-state stamp forming conditions. Such investigation is essential for a deeper understanding of the stamp forming process of thermoplastics and FRTPs. Figure 2-10 illustrates the deformation mechanisms of thermoplastics within various types of FRTPs, underscoring the necessity for accurately characterising these mechanisms. It is evident that, regardless of the type of FRTP, thermoplastics always undergo either tensile or shear deformation during the deformation process. Given the isotropic nature of thermoplastics, their thermomechanical properties are typically characterised via tensile or compression tests under various conditions. Emphasis is particularly placed on tensile properties for stamp forming applications [74].

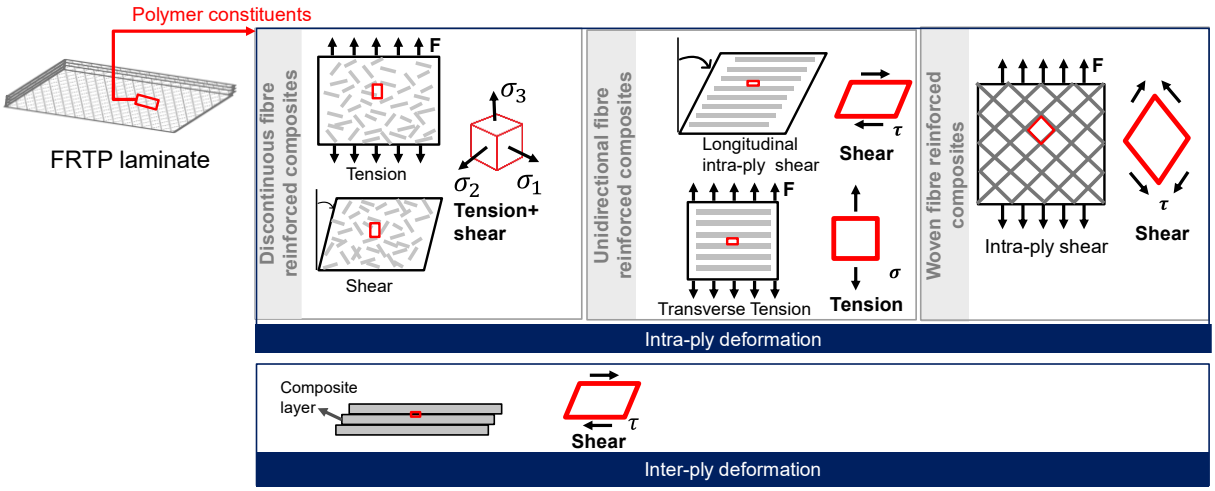


Figure 2-10. Deformation mechanisms in the thermoplastic constituent in FRTPs during stamp forming

2.3.2. UD FRTPs

For UD FRTPs, the stamp forming process involves both intra-ply and inter-ply deformation mechanisms [44, 67, 75, 76], as analysed in Figure 2-11. Bussetta et al. [76] concluded that intra-ply deformation mechanisms are the most studied mechanisms during the numerical

forming of composites. Figure 2-11 shows the intra-ply tensile and shear deformation mechanisms during the stamp forming of UD FRTPs. Since UD FRTPs are considered to be sheet materials, deformation along the thickness direction (i.e., transverse shear) is less significant. Additionally, as fibres are typically considered to be inextensible [8], longitudinal extension is trivial. Therefore, the transverse tension and intra-ply longitudinal shear are identified as the most dominant deformation mechanisms, which will be the focus of this thesis. Transverse tension refers to the tensile deformation that occurs when the prepreg layer is stretched to separate fibres from their original positions [8], while intra-ply longitudinal shear refers to the shear deformation that occurs when fibres slide parallel to each other.

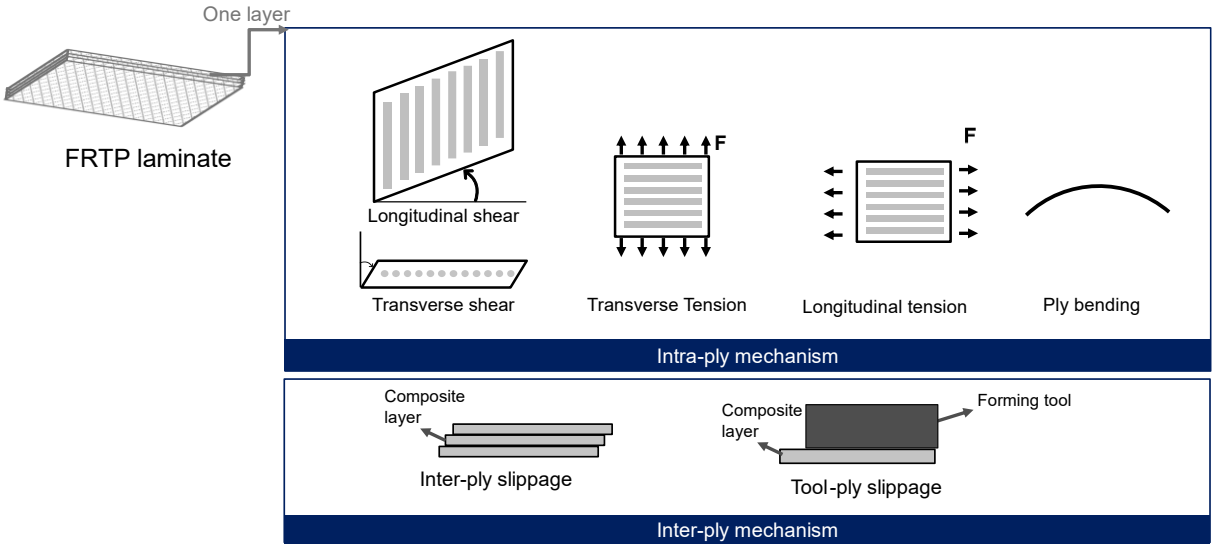
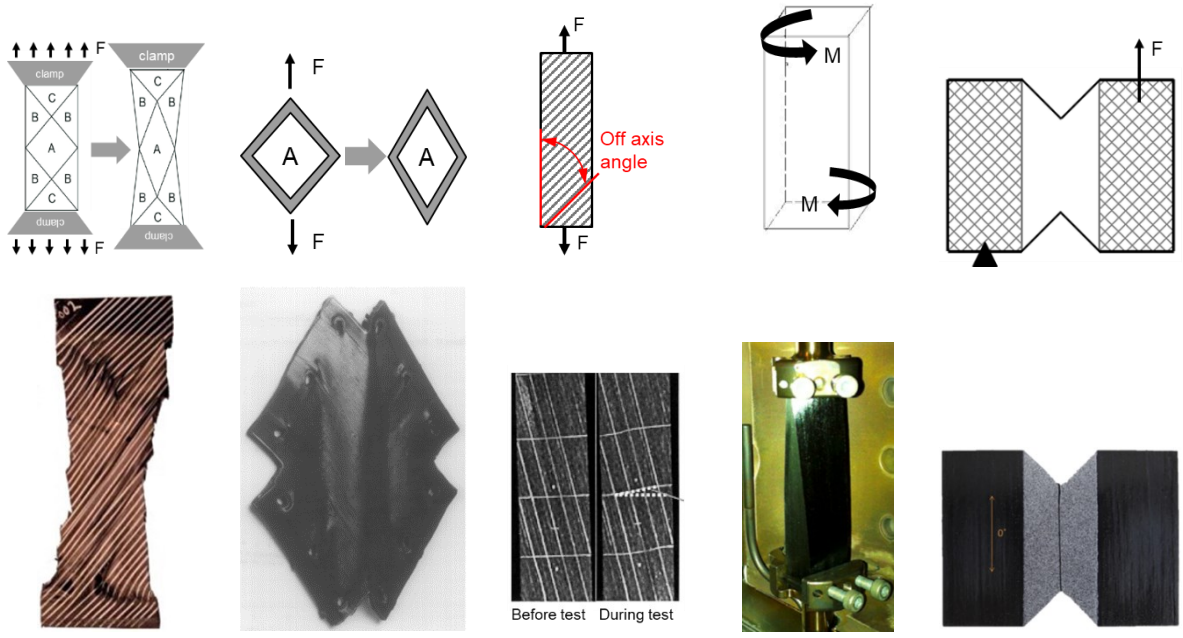


Figure 2-11. Deformation mechanism in one layer of UD FRTPs under stamp forming.

For the characterisation of the intra-ply longitudinal shear properties, many researchers have done extensive research and reviews [67, 77-79] on the characterisation methods of the shear properties. Figure 2-12 summarises the characterisation methods for obtaining the intra-ply shear properties of UD composites as employed in previous studies. The most popular methods are the bias-extension test and the picture frame test, both of which were originally invented to characterise textiles [80].



[67] a) Bias-extension test

[77] b) Picture frame test

[68] c) Off axis test

[67] d) Torsional test

[78] e) VNR shear test

Figure 2-12. Characterisation methods for the intra-ply shear properties of UD FRTPs and their corresponding example pictures.

Figure 2-12a shows the schematic illustrating the deformation mechanism of bias-extension. Regarding its application to UD composites, Potter [79] explored the response of cross-plied (at ±45°) UD carbon fibre reinforced epoxy (CF/epoxy) via bias-extension test at room temperature (RT). However, the test revealed complex deformation mechanisms, including significant fibre wrinkling and compaction loss. It was concluded that the overall deformation is sensitive to misaligned fibres, which can induce compressive loads and thus fibre buckling. Later, Larberg et al. [81, 82] also conducted the bias-extension test on two types of cross-plied CF/epoxy, namely HexPly® T700/M21 and Cycom® HTS/977-2, to characterise the in-plane deformation at different stacking sequences, i.e., $[45/-45/90]_s$ and $[45/90/-45]_s$ (90° layers are the layers transverse to the pulling direction). Although they gained some effective results for the in-plane shear properties of HexPly® T700/M21, the behaviour of the other material, Cycom® HTS/977-2, revealed that the inter-ply friction played an important role after the initial in-plane shear dominant deformation. In addition, their research also showed a significant

effect of prepreg stacking directions on the deformation response. To check the validity of bias-extension tests on molten UD FRTPs, Haanappel et al. [67] conducted bias-extension on UD CF/PEEK, and the picture in Figure 2-12a shows a typical result. Due to the weak integrity of the laminate at the melt state, there was severe material buckling.

Figure 2-12b shows the schematic illustrating the deformation mechanism of the picture frame test method. McGuinness et al. [77] conducted this test on cross-plied CF/PEEK laminates with layups of $[0_4/90_4]_{2s}$, under various processing temperatures and strain rates. Although it was concluded that the method was effective in characterising the intra-ply shear properties, there were severe diagonal folds or buckles on the specimens when the enclosed shear angle reached 15-18°, as shown in the picture in Figure 2-12b. Haanappel et al. [67] conducted picture frame test on UD CF/PEEK prepregs, and it was observed that ply splitting was easily induced by even small fibre misalignments in the prepregs. Besides, for the picture frame test, an oversized specimen is required to obtain the uniformity in-plane shear deformation. Consequently, a large prepreg size is required, and an equally large environmental chamber is required for the high-temperature characterisation tests.

Figure 2-12c illustrates the off axis extension test which is another potential useful method for characterising shear properties. Potter [68] performed off axis extension tests on UD CF/epoxy at various off axis angles (15 – 90° with 15° intervals) to determine both transverse and intra-ply shear deformation properties at RT and 50 ℃. The specimen before and after deformation is presented in the figure with the white lines drawn along the fibre direction to monitor the fibre rotation. Although the in-plane deformation properties could be obtained through the off axis extension test, the deformation at various off axis angles was coupled by both shear and transverse deformation, including transverse tension or compression. Therefore, the coupling effect prevents accurate characterisation of the intra-ply shear properties.

Figure 2-12d shows the torsional test proposed by Haanappel et al. [67] for characterising the shear deformation of molten UD FRTPs. The specimen for this torsional test was a prismatic bar with a constant rectangular cross-section, where the fibres were aligned in the direction of the torsional axis. Torque M was applied to one end of the specimen, while the other end was fixed to introduce the longitudinal shear. The experiments were conducted under a rheometer with an oscillating load applied within a small strain regime to characterise viscoelastic properties. This torsional experiment setup was also adopted by Dörr et al. [44] on the melt UD

CF/PA6. In contrast to the experiments by Haanappel et al. [67], a transient load under a single deflection outside the small strain regime was applied [44]. These studies found that both fibre tension and bending were involved in this characterising method, especially in the higher strain regime. Additionally, there was a high requirement for the clamping fixtures to avoid slippage between the fixture and the specimen, which added difficulties to the experiments.

Lastly, Figure 2-12e depicts the V-notched rail (VNR) shear test, as defined by ASTM 7078 [83], which is another experimental method for characterising the properties of UD FRTPs. As illustrated, there is a vertical load F applied on one side of the specimen and a constraining force on the other side, making the central ligament deform in the manner of simple shear. Tan et al. [78] studied the VNR shear test on UD CF/PEEK at RT with various fibre configurations, and an example picture of the specimen at $[0]_{24}$ is shown in the figure . It can be observed from the fracture of the specimen that the in-plane shear deformation was concentrated on the ligament, without having any out-of-plane deformation, as also concluded by the authors [78]. Similarly, Totry et al. [84, 85] utilised the VNR shear test to study the in-plane shear properties of UD glass fibre reinforced epoxy (GF/epoxy) [85] and UD CF/epoxy [84] at room temperature. Although not widely employed, the VNR shear test method has demonstrated advantages over other methods, such as generating a relatively uniform shear stress state along the ligament and eliminating edge crushing.

For the characterisation of the transverse tensile properties of UD FRTPs, as discussed earlier, off axis extension test characterises transverse tension for certain off axis angles (such as 30 – 90 º) [68], but it was difficult to decouple the in-plane transverse tension and shear properties. Wang et al. [86] tested the transverse tensile strength of a UD CF/epoxy specimen at RT using a dog-bone specimen on a tensile test machine, but only the maximum force was captured. Margossian et al. [87] designed experiments to directly measure the transverse tensile properties of melt-state UD CF/PA6. They initially explored the option of tensile test on a tensile test machine with an environmental chamber. However, there were difficulties in obtaining results with good reproducibility, mainly due to the thermal expansion of the metallic clamps and their interactions with the specimen. Then, a Dynamic mechanical analysis (DMA) system with a constant loading rate on the specimen was used and satisfactory results were obtained for melt-state UD CF/PA6. To the authors' best knowledge, this is the only research that characterised the pure transverse tensile properties of UD FRTPs for the application of thermoforming processes, although it was conducted only at the melt state of the matrix. In order to simulate

the forming process, Dörr et al. [88] utilized the virtual characterisation results derived from representative volume element (RVE) studied by Roesner et al. [89] to obtain solid-state properties.

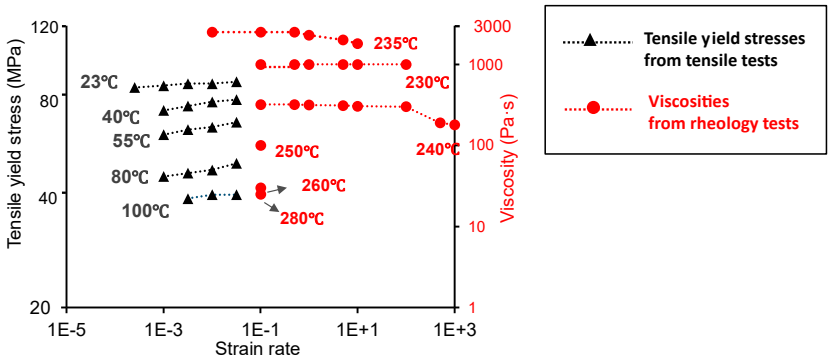
To summarise, there has been a fair number of studies on characterising the intra-ply deformation properties of UD FRTPs. However, most of the characterisation methods have issues in isolating the intra-ply tension and shear deformation mechanisms. Besides, the majority of experiments were performed either at RT or above the melting temperature (T_m) of the matrices, and no experimental results have been reported at the conditions required for solid-state stamp forming. From a literature analysis, it has been determined that the VNR shear test is suitable for obtaining satisfactory shear properties for UD FRTPs tested under temperatures corresponding to solid-state stamp forming conditions. Additionally, traditional tensile test can be used to characterise the transverse tensile properties, provided that the fixture thermal expansion problem is effectively addressed.

2.4. The thermomechanical properties of thermoplastics and UD FRTPs

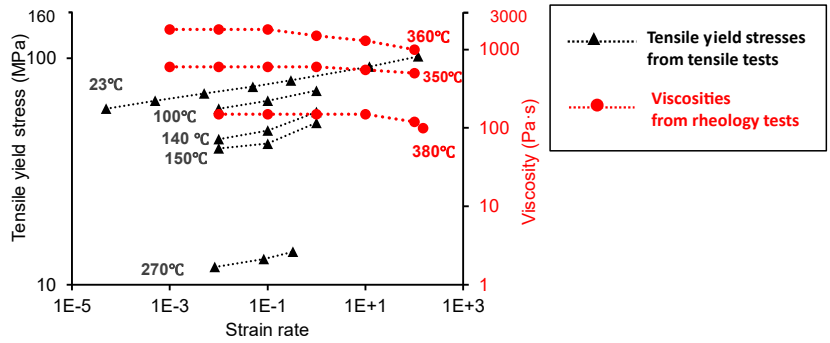
2.4.1. Thermoplastics

PA6 and PEEK are two types of preferred thermoplastics in the automotive and aerospace industries, respectively [16, 90]. They can serve as either pure polymers or as matrices for FRTP components. Both applications have been attracting increasing research and industrial interest, especially in relation to their manufacturing technologies.

An overview of the existing research regarding PA6 and PEEK in terms of yield stresses (characterised under solid state) or viscosities (characterised under melt state) is presented in Figure 2-13. It can be concluded that the mechanical properties of PA6 and PEEK are both high temperature- and strain rate-dependent. Besides, existing research focuses on the temperatures either under the glass transition temperature (T_g) or above the melting temperature (T_m) of the thermoplastics, leaving gaps between the T_g and T_m of the thermoplastics, which are the ideal solid-state stamp forming temperatures.



a) PA6



b) PEEK

Figure 2-13. Reviews of the characterisation results on a) PA6 and b) PEEK. Data of PA6 extracted from [91-96]; Data of PEEK extracted from [74, 97-100]. (The figures are in log-log scales. The T_g, T_m are respectively 47 ℃ and 230 ℃ for PA6, and 143 ℃ and 343 ℃ for PEEK.)

For stamp forming applications, specifically those involving solid-state stamp forming, it is critical to consider various stress-strain characteristics of the material. These include viscoelasticity, viscoplasticity, strain hardening, and strain softening. Brinson et al. [101] presented a comprehensive summary of typical stress-strain curves of thermoplastics, revealing that these curves display nonlinear responses, which are a combination of viscoelastic and viscoplastic behaviours. These behaviours signify the time, strain rate, temperature, and pressure dependencies of thermoplastics. Among these parameters, temperature and strain rate are particularly dominant and have garnered significant research interest [102]. Figure 2-14 depicts typical stress-strain curves of polymers, with curves (a) – (d) representing an increasing trend of temperatures. For semi-crystalline thermoplastics, they typically exhibit behaviour

similar to curve 'c', demonstrating strain softening, strain hardening, and post-strain hardening properties.

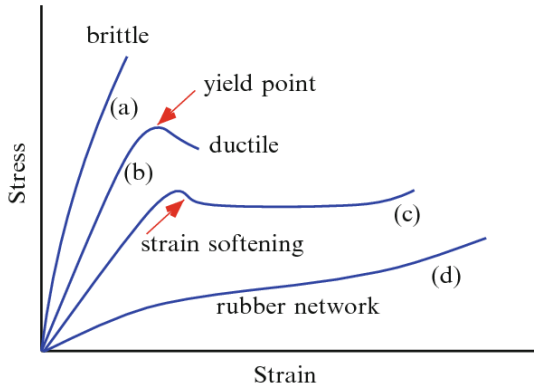


Figure 2-14. Typical stress–strain curves of polymers tested at different temperatures [103] , curves (a) – (d) representing an increasing trend of temperatures.

For PA6, there are only a few studies on its thermomechanical properties. Tensile tests at a range of temperatures (23, 40, 60, 80, and 120 °C) were conducted by Parodi et al. [104] at a strain rate range from 10^{-4} to 10^{-1} /s. It was found that the tensile stress-strain curves changed noticeably with the temperature increase from RT to an elevated temperature. With increasing temperature, the elastic modulus and yield stress decreased, while the ductility increased. Further, at lower temperatures (23 and 40 °C), there was strain softening followed by strain hardening at large deformation, and the strain softening behaviour gradually disappeared at higher temperatures (60, 80, and 120 °C), at which only strain hardening was observed. The strain rate effects on the stress responses, however, were not significant within the studied conditions. For PEEK, the thermomechanical properties have been more extensively studied. Rae et al. [105] investigated the temperature effects on the tensile properties at temperatures between 50 – 150 °C, and a strain rate of 1.7 x 10^{-4} /s. With increasing temperature, the ductility increased remarkably while the yield stress and elastic modulus dropped, and this was especially significant when the temperature was above the glass transition temperature, T_g. In addition, their study showed that there was strain softening after material yielding at temperatures below 100 °C and no strain hardening behaviour was observed; whereas at higher temperatures (140, 150 °C), it was found that the strain softening behaviour disappeared and the strain hardening behaviour appeared. Zheng et al. [74] studied the temperature effects on PEEK at a temperature range between T_g and the melting temperature, T_m, and found that there

was no obvious yield point at temperatures above T_g. However, their study only presented a limited strain level of the deformation. The strain rate effects of PEEK at RT were systematically studied by Alberola at al. [99] via tensile tests at a strain rate range from 10^{-5} to 300 /s. The strain rate effects on the stress responses were found to be marginal at the strain rate range from 10^{-5} to 10^{-1} /s. At higher strain rates (10-1 -300 /s), however, it was reported that there were appreciable increases in yield stress. The strain rate effects at elevated temperatures were studied by Chen et al. [100], and they found that there were no obvious strain rate effects at temperatures of 100, 140, and 150 ℃. The strain rate effects for PEEK above T_g were studied at 240 ℃ at strain rates ranging from 10^{-3} to 10^{-1} /s by Barba et al.[106]. Although the strain rate effects were not significant at the studied range, there were non-negligible adiabatic heat effects. To summarise, there are studies that have investigated the thermomechanical properties of PA6 and PEEK polymers at various conditions. However, the material behaviour between the T_g and T_m (the T_g and T_m are respectively 47 ℃ and 230 ℃ for PA6, and 143 ℃ and 343 ℃ for PEEK), where the forming temperature is normally set to obtain the optimal formability for solid-state stamp forming [16], is still unclear, especially when the temperature is approaching T_m. Therefore, there is a demand for more comprehensive studies on the tensile behaviour between the T_g and T_m.

2.4.2. UD FRTPs

As summarised in section 2.3.2. intra-ply transverse tension and longitudinal shear are the two major intra-ply deformation properties identified, and there are intensive studies to characterise these intra-ply properties under melt state. For UD FRTPs, typical testing responses are shown in Figure 2-15, studied by Brands et al. [107]. They characterised the thermos-mechanical properties of two typical UD CF/PEEK materials via bias-extension test, under various temperatures (all under melt state) and strain rate. From the force-shear angle curves, a s-shape curve is shown, indicating the strong temperature- and strain rate- dependent. Besides, although these curves are not in the forms of stress-strain curves, the strain hardening effects can be inferred. Considering the potential similar response of UD FRTPs and UD FRTSs, other characterisation studies of these materials and their major conclusions are summarised together and listed in Table 2-2. From the testing conditions and testing methods, it can be concluded that, for the thermos-mechanical characterisation of UD FRTPs, further research is required for these materials deforming under solid-state stamp forming conditions using a more accurate

characterisation method that is suitable for the testing condition. Besides, most studies only obtain the force-shear angle results, which require further analysis of the actual deformation under the stamp forming process.

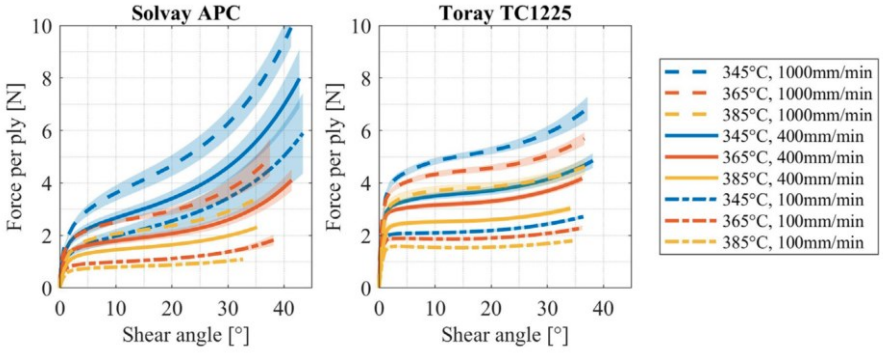


Figure 2-15. Thermomechanical properties of two typical UD CF/PEEK materials (Solvay APC and Toray TC1225) characterised by bias extension test. [107]

Table 2-2. Existing characterisation research on the thermo-mechanical properties of UD FRTPs.

Reference/ Year/ Material	Conditions	Testing method	Major conclusions
Martin et al. [108], 1995 UD glass fibre reinforced polypropylene GF/PP	Melt-state 170 ℃	V-bending test Fibre ⊥ bending line	Non-linear viscoelasticity were studied.
McGuinness et al. [77], 1998 UD CF/PEEK	Melt-state 350-380 ℃	Picture frame test	Longitudinal and transverse viscosities were evaluated; An anisotropic power-law model was employed to model the experimental results.
Potter [79], 2002 UD CF/Epoxy	RT	Bias extension	'Locking angle' is essentially meaningless for cross-plied prepreg; The longitudinal and transverse strains are broadly in line with the Pin-joint-net (PJN) model.
Potter [68], 2002 UD CF/Epoxy	RT	Off-axis tension	The material deforms non-uniform under these experimental conditions;

			The shear deformation is very much stiffer than the transverse deformation model.
Harrison et al. [109], 2005 UD GF/PP	Melt-state 180 ℃	Squeeze flow; Picture frame test	Longitudinal and transverse viscosities were evaluated; A power-low model was able to analyse the properties.
Haanappel et al. [67], 2014 UD CF/PEEK	Melt-state 365 ℃/ 390 ℃	Torsional test	Both linear and non-linear responses were observed, more focus was paid on the linear small strain responses.
Margossian et al. [87], 2016 UD CF/PA6	Melt-state 280 ℃	Transverse tensile test	Tensile test conducted on DMA; Strain rate effects were studied
Larberg et al. [110], 2007 UD CF/Epoxy	20 ℃, 70℃.	Bias extension	No or little slip was seen before wrinkling in the RT tests; At 70 ℃, a larger amount of slippage was seen than in the tests at RT.
Larberg et al. [111], 2012 UD CF/Epoxy Three types were studied	RT and 40/70℃.	Bias extension	Deformation occurred in the form of bands with no further shearing of the material inside the bands.
Brands et al. [112], 2021 UD CF/PEEK	Melt-state 385 ℃	Bias extension	The strain localisation noticed. Similar response with woven fabrics, but a high intra-ply viscosity is observed.
Brands et al. [107], 2022 UD CF/PEEK	Melt-state 345 - 385 ℃	Bias extension	A s-shape viscoelastic behaviour observed with both temperature- and rate- dependent. This study provides reliable shear test results.

2.5. Constitutive models for thermoplastics and UD FRTPs

With the thermomechanical properties of thermoplastics and UD FRTPs characterised, the next critical step in conducting FEA is the establishment of a suitable constitutive model. Constitutive models play a pivotal role in FEA as they define the material's stress-strain relationship with the consideration of temperature and strain rate effects.

2.5.1. Thermoplastics

As previously introduced and illustrated in Figure 2-14 curve 'c', the typical stress-strain curves of thermoplastics are shown as a s-shape curve, which shows the complex deformation features. There are two approaches to establish constitutive models of thermoplastics: the phenomenological approach, based on the macroscopic observation of the stress-strain response, and the physical approach, based on the physical interactions of the microstructures. Many phenomenological models are adopted from models developed for metal forming, such as the Johnson-Cook (JC) model. For example, Garcia-Gonzalez et al. [113] adopted the JC model to represent the viscoplastic behaviour of PEEK while simulating PEEK components under impact. Chen et al. [100] proposed a modified JC model based on mechanical test results to improve the JC model prediction under various temperatures and strain rates. In ABAQUS, there are several hyperelastic phenomenological models, which are defined as total stress-strain relationship, that are specially designed for polymers, such as the New-Hookean form, the Mooney-Rivlin form, and Ogden form [114]. However, the phenomenological models are generally used for engineering applications where the strain level is limited, as these models are not capable of capturing the viscoelastic and viscoplastic behaviours of thermoplastics at large deformation [115].

The physically based models are built based on multiple entropic sources forces associated with molecular movement [113]: the separation, reorientation, and stretching of the molecular chains. From the microstructural point of view, both PA6 and PEEK are semicrystalline thermoplastics comprising well-arranged crystals and amorphous molecular chains. However, many existing physical models are built for amorphous polymers, such as the Boyce-Arruda model [116, 117], based on an eight-chain assumption, and applied to amorphous thermoplastics. Xiao et al. [118] employed a back stress tensor for molecular orientations to improve the strain hardening prediction capability for amorphous polymers. To understand the macro-yield process in amorphous polymers, Jiang et al. [119] established the constitutive model via studying two molecular chain microstructures, i.e., sub-entanglement and topological entanglement. In their following study [120] on polymethyl methacrylate (PMMA), they illustrated that these two microstructures could reasonably be used to interpret the rate- and temperature- dependent macro-yield behaviour of amorphous polymers. Compared to amorphous polymers, the microstructural interactions are more complicated in semicrystalline thermoplastics since there are not only interactions between multiple microstructure phases, but

also phase transformation during deformation [106]. Some studies on the physically based models for semicrystalline thermoplastics have been conducted in recent years. For example, to describe the stress responses of semicrystalline thermoplastics at room temperature, Polanco-Loria et al. [121] built a hyperelastic-viscoplastic constitutive model by considering the pressure sensitivity, volume plastic strain, and strain rate sensitivity under isothermal conditions. Abdul-Hameed et al. [122] made progress by revising Boyce et al. [117]'s model, combining the influence of degree of crystallinity to study the strain rate effects on semicrystalline thermoplastics. Berezvai et al. [123] further studied the strain rate effects on thermoplastics by establishing a constitutive model based on the contributions of crystalline, amorphous, and rigid amorphous fractions. To understand the thermal effects, Garcia-Gonzalez et al. [102] improved Polanco-Loria's model [121] with consideration of thermal-mechanical coupling, including the effects of thermal softening, thermal expansion, and the adiabatic heat caused by plastic dissipation during deformation. For considering the material behaviour of thermoplastics at temperatures above T_g, Barba et al. [106] studied a broader application of the model of Garcia-Gonzalez et al. [102] with experimental data on PEEK corresponding to three thermal regimes, which were below, at, and above T_g, so that the reorientation mechanism of the crystal phase and the amorphous phase could be better understood. However, the model predictions showed noticeable discrepancies compared to experimental data. To conclude, the constitutive models aiming to predict the thermomechanical behaviour at temperatures between T_g and T_m are still not well studied. Therefore, there is a demand for a physically based constitutive model with comprehensive consideration of temperature and strain rate effects to accurately predict the thermomechanical behaviour.

2.5.2. UD FRTPs

In the aspect of the constitutive models for UD FRTPs, it is crucial to account for the multiscale nature of the material, which involves different physics that occur at different scales, i.e., the macro-scale (the scale of multilayers in a laminate), the meso-scale (the scale of one prepreg layer) and the micro-scale (the scale of fibres) [76]. Understanding this multiscale nature is essential for developing accurate constitutive models for UD FRTPs. With this in mind, the choice of modelling scale depends on a balance of model accuracy and computational cost. Normally, macro-scale models for woven FRPs and meso-scale models for UD FRPs are used for numerical simulations of stamp forming processes [76]. Both cases require constitutive laws

that consider the unique mechanical behaviour of the fibre reinforcements, including the anisotropic behaviour of the warp and weft frames that evolve during transformation due to in-plane shear. Additionally, other behaviours, such as geometric nonlinearities, quasi-inextensibility of the fibres, and strong nonlinear in-plane shear deformation, must be accounted for [20]. To address these specificities, several categories of constitutive models are being studied, including the non-orthogonal elastic model, hypoelastic model, hyperelastic model and the "Ideal Fibre Reinforced Material (IFRM)" model [69]. The non-orthogonal elastic model utilizes the elastic stress-strain relationship for the material to examine the impact of fibre strength variations and orientations on the anisotropy during the preforming or fabric draping processes [69, 124-126]. Based on the basic frame of the non-orthogonal elastic model, Peng et al. [127] proposed a non-orthogonal hypoelastic model, which was defined by the relationship between the stress rate and strain rate, for the application of large deformation of woven glass fabrics. In this hypoelastic mode, the global constitutive relation for woven glass fabrics is derived by transforming the contravariant and covariant components to the Cartesian components of the stress and strain tensors. However, the major drawback with the hypoelastic model is that the loading path is unrecoverable. And the hyperelastic models derived from the potential energy function could effectively overcome this drawback [69]. With regards to the formulation of the energy function, Peng et al. [128] decomposed the strain energy function into fibre stretches and yarn cross-over shearing to propose a hyperelastic model that could analyse the large deformation of dry fabric. Later, Gong et al. [129] further refined Peng et al. [128]'s work by proposing an anisotropic hyperelastic model for woven thermoplastic composite prepregs that takes into account the fibres, matrix, and fibre-matrix interaction. They demonstrated its ability to predict the behaviour of woven prepregs during thermoforming processes. Although the above-mentioned constitutive modelling techniques have been used to simulate the stamp forming process of woven FRPs [130], research on the constitutive models of UD FRTPs is still limited. For constitutive models of UD FRTPs, several researchers have employed IFRM for stamp forming applications. The IFRM assumes that the material is incompressible and inextensible in the fibre direction and exhibits Newtonian viscous behaviour for all intra-ply behaviour. This model strategy has proven to be particularly useful since current characterisation studies for stamp forming applications are conducted under the melt state. The IFRM model was firstly introduced by Rogers et al. [131] and later adopted by Haanappel [8] to simulate the stamp forming process of molten UD CF/PEEK. Similarly, Dörr

et al. [88] also adopted the IFRM model for molten UD CF/PA6 by assigning rate-dependent hyperelastic models to the intra-ply shear properties, following either a Voigt-Kelvin approach or a generalised Maxwell approach. The simulation results showed that the Voigt-Kelvin approach was more suitable for predicting the forming behaviour. In addition, the implementation of the IFRM strategy into the commercial software Aniform [8, 132] has consequently increased its popularity [5]. However, as indicated by some material characterisation studies [16, 50, 74], the thermomechanical properties of UD FRTPs at temperatures close to the T_m of the matrices are more complicated. Therefore, simple elastic, hyperelastic or even viscous-hyperelastic models might not be adequate to describe the complicated temperature-dependent strain hardening or softening material behaviours. To tackle this issue, it is necessary to consider certain physical factors, such as the temperature and strain rate dependence of the matrices, and the properties of the polymer-fibre interface, in order to accurately describe the thermomechanical behaviour of UD FRTPs below and close to T_m. Therefore, a constitutive model that is tailored to the unique characteristics of UD FRTPs under solid-state stamp forming conditions is required.

2.6. Numerical simulation for FRTPs stamp forming process

Conducting numerical simulations of the stamp forming process is an integral component for researching and enhancing this manufacturing technique, offering a practical solution to avoid the expensive 'trial and error' methods. As previously outlined, the stamp forming process comprises four stages, each necessitating specific considerations during numerical simulations, as demonstrated in Figure 2-16.

Whilst some researchers have conducted comprehensive process simulations - for instance, Lamers [133] simulated both mechanical deformation and thermal distortion in his thesis, and Pascon et al. [134] evaluated the stamp forming process, residual stress, and crystallisation - the predominant focus within numerical simulation research remains on the localised process of stamp forming [69]. This is primarily attributable to the fact that deformation is the dominant factor contributing to the phenomena and defect formation witnessed in stamp forming, such as wrinkles. Additionally, this stage necessitates a heightened level of understanding to propose an optimised process. Consequently, this thesis prioritises the local stamp forming process and its numerical simulation strategies.

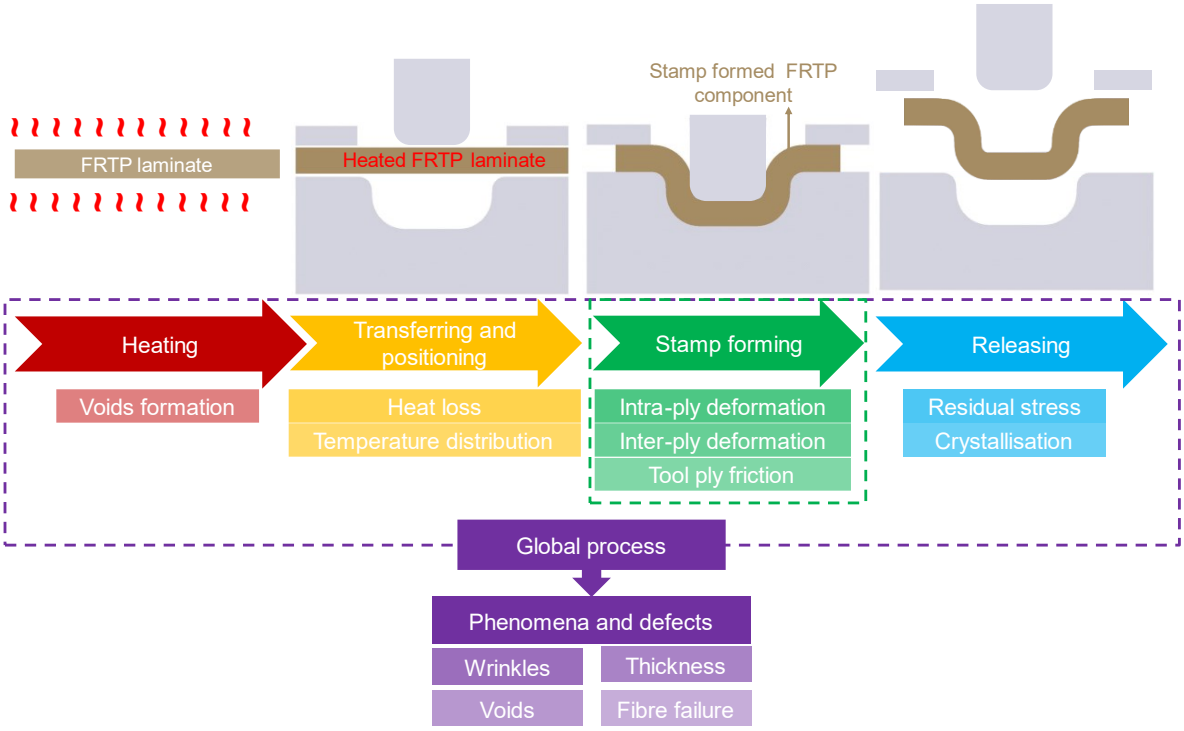


Figure 2-16. Various numerical simulation focuses during the FRTPs stamp forming process.

With regard to simulating the stamp forming process, three approaches are discrete approach, semi-discrete approach and continuum approach [76]. The choice of these approaches depends on the balance of cost and reliability.

Among the aforementioned methods, the discrete approach is typically employed to model the meso-scale of woven or knitted fabrics, taking into account the interactions between fibres and yarns. This method can be utilised to simulate the deformation of FRTPs during manufacturing processes. For example, Skordos et al. [135] modelled the pre-impregnated woven composites as a truss of tow and shear elements within a simplified finite element model. The unit cell of this model consists of a network of pin-jointed bars, where stiff elements signify the tows of the woven FRTP. By considering factors such as wrinkling and strain rate effects, they successfully predicted the forming behaviour of a semi-spherical component. Other strategies include modelling the material as a trellis of truss [136], beam [137], spring [138], shell [139], or membrane [140]. These methods have also been employed to predict the mechanical properties of composite structures. However, it should be noted that the discrete approach carries a significant numerical cost and is generally limited to smaller structures.

The semi-discrete approach is the intermedium approach that costs less resource but models a coarser material architecture [15]. It is a hybrid of the discrete approach and continuous approach [76], i.e., some components of the material are modelled as a continuum (using a continuous approach), while others are modelled discretely (using a discrete approach). Boisse et al. [141] firstly introduced the woven unite cell concept to simulate the preform process of woven fabrics, focusing only on biaxial tensile behaviour. Later, Boisse et al. [142] further improved the model with consideration of the in-plane tension and bending behaviour. Despite its common use in woven fabric materials, for FRTPs application, Liang et al. [143] and Wang et al. [144] both utilised woven unit cell to study the forming process of woven FRTP prepregs, and the wrinkles could be accurately predicted.

The continuous approach regards the FRTP material as a continuum, rather than individually modelling the matrix and fibre. This continuous approach requires a material constitutive model, such as the hyperelastic model or hypoelastic model [76]. As the continuous approach models the FRTP material at macro scale, the simulation cost in terms of time and computing resources is significantly reduced, making it more suitable for simulating stamp forming problems [69]. The relatively low out-of-plane bending flexural stiffness of FRTPs led many early studies to focus predominantly on its tensile and in-plane shearing behaviours during numerical simulations [145]. However, subsequent research revealed that bending stiffness significantly influences the quantity and size of wrinkles generated during the thermo-stamping process [60, 146]. As illustrated in Figure 2-17, in traditional continuum materials, bending deformation normally follows the Kirchhoff laws, whereby the normal direction of the midplane remains perpendicular to the middle plane. However, in fibrous materials, slippage between fibres and the inextensibility of fibres causes the normal direction of the midplane to deviate from this perpendicularity [147]. This unique bending behaviour of fibrous materials results in considerably lower bending stiffness compared to traditional continuous materials. Consequently, the assumption that tension stiffness determines bending stiffness in traditional continuum materials is insufficient to describe the deformation of composites. Therefore, special techniques that can decouple the bending and in-plane behaviour are required to accurately model the deformation of fibrous materials.

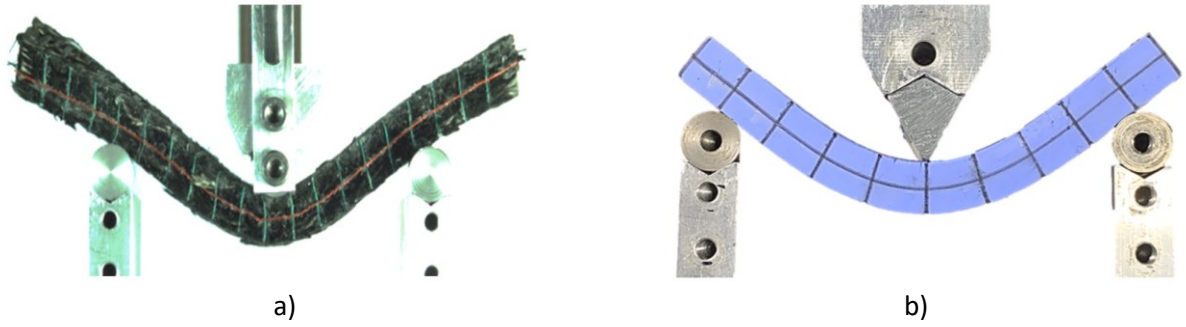


Figure 2-17. Bending behaviour of the a) fibrous material and b) classic continuous material [147]

Researchers have employed various approaches to decouple the bending behaviour of textiles and other composite materials. Döbrich et al. [148] considered the bending of textile material according to the laminate theory. Jauffrès et al. [149] utilised the discrete mesoscopic modelling method, combining one dimensional (1-D) beam elements and two dimensional (2-D) membrane elements to characterise the deformation of woven fabric materials. In the context of FRTPs, Figure 2-18 shows several strategies that have been successfully used for the numerical simulation of stamp forming process. For simulating the stamp forming behaviour of UD CF/PEEK prepreg, Haanappel [8] decoupled the in-plane deformation and bending deformation by modelling these behaviours using membrane element and Discrete Kirchhoff Triangle (DKT) element, respectively, as depicted in Figure 2-18a. Similarly, Nishi et al. [150] and Dörr et al. [151] applied a superimposed element method, comprising a shell element for bending and a membrane element for in-plane behaviour, as illustrated in Figure 2-18b. In the commercial FEA software, Aniform, the decoupling strategy is depicted in Figure 2-18c, which is composed of shell element with linear shape functions (LTR3D) and DKT elements. These elements share the same nodes and deform mutually. More recently, Schäfer et al. [152] used a solid-shell element, to account not only for the in-plane and bending behaviour of woven fabrics, but also for the stresses and strains in thickness directions. Their studies using these solid-shell elements, demonstrated the typical characteristics of the membrane-bending interdependency in coupon tests as well as in forming simulations.

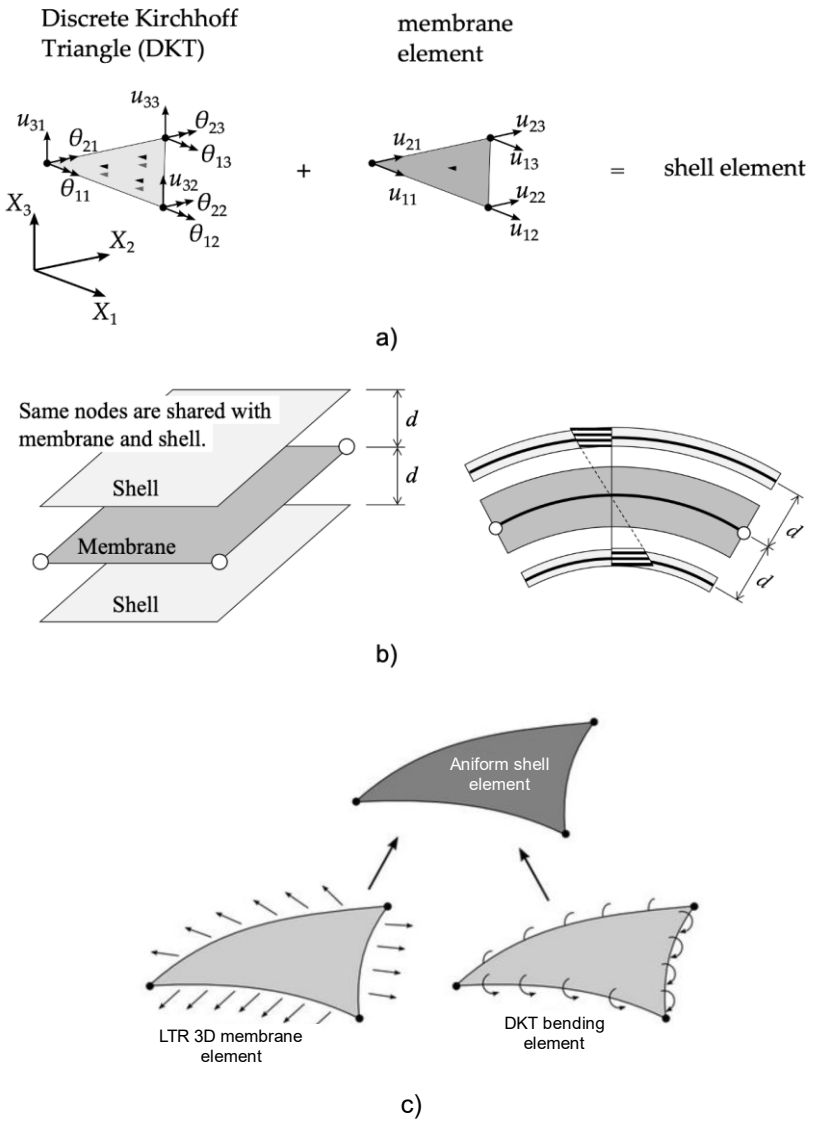


Figure 2-18. Model strategies for decoupling the intra-ply and bending behaviours in FRTPs. a) New shell element composed of DKT element and membrane element; b) Combination of shell element and membrane element; c) Combination of DKT element with linear shape function shell element used for Aniform [5, 8, 15].

Regarding simulation platform, several commercially available FEA software packages are frequently adopted. To evaluate the capabilities and features of these commercially used FEA software, Dörr et al. [132] conducted research by simulating a benchmark UD FRTP component on platforms including PAM-FORM and Aniform, as well as two approaches implemented in the multi-purpose FE solver LS-Dyna and ABAQUS. These simulation results were compared

to experimental results by examining the deformed surface and outer contour. The primary findings of the study included a comparison of the available features and functionalities of the different FE codes, as listed in Table 2-3, as well as an assessment of their quality of forming predictions. It is worth mentioning that both PAM-FORM and AniForm possess specially developed material model applicable for FRTPs. For ABAQUS, however, although all of these features can be achieved, most features require a further user development, such as the use of subroutine. Some features such as the fully decoupled bending behaviour, the rate-dependent membrane behaviour, is not accessible in LS-Dyna, even with customised development.

Table 2-3. Comparing features for the FEA software used for stamp forming applications.

Comparing feature	PAM-FORM	AniForm	LS-Dyna	ABAQUS*
Fully decoupled bending behaviour	Yes	Yes	No	Yes*
Assignment of characteristic curves	Yes	No	Yes	No*
Deformation described by constitutive models	No	Yes	No	Yes*
Rate-dependent membrane behaviour	Yes	Yes	No	Yes*
Rate-dependent bending behaviour	No	Yes	No	Yes*
Rate-dependent interface mechanisms	Yes	Yes	Yes	Yes*
Thermomechanical modelling	Yes	Yes	Yes	No*
Modelling of grippers / clamp holders	Yes	Yes	Yes	Yes*
Tailoring determination Export fibre orientation	Yes	Yes*	No	Yes*
Export fibre orientation	Yes	Yes*	Yes*	Yes*

*Only available by customised developments.

2.7. Summary

This section provides a systematic review regarding the technology advancements of the manufacturing technologies of FRTPs. From this review and comparison of various manufacturing technologies, the stamp forming technology has been identified as the most promising technology for the mass production of automobile components. After analysing the common defects of the stamp forming technology, it has been confirmed that solid-state stamp forming can provide better component quality. Subsequently, the deformation mechanism and characterisation methods of the FRTPs, and the major deformable constituent, the matrix thermoplastics, are reviewed. In addition, the thermomechanical properties of thermoplastics and UD FRTPs are critically examined from existing research. Furthermore, the material constitutive models and numerical simulation strategies with regard to the solid-state stamp forming are reviewed. Through this chapter, the research gaps and experimental methods have been identified.

CHAPTER 3 Investigation of the Thermomechanical Properties of Thermoplastics

*This Chapter is reproduced from publication **A1** and **A4** in the Publication list.

3.1. Introduction

In this chapter, the thermomechanical properties of two typical semicrystalline thermoplastics, PA6 and PEEK, under solid-state stamp forming conditions are comprehensively characterised via the uniaxial tensile tests on, under a temperature range between T_g and T_m (roughly, 0.7 – 0.9 T_m at absolute temperature) and a strain rate range from 0.001 to 50 /s, which were defined corresponding to typical solid-state stamp forming conditions. The temperature and strain rate effects on the stress-strain properties of these thermoplastics are analysed, so that the further constitutive model could be built.

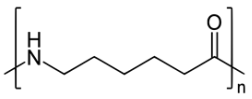
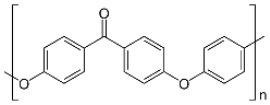
Except the thorough characterisation on the thermoplastics via uniaxial tensile tests, an alternative shear test method is also studied and analysed, based on the property benchmarks that have been established via the uniaxial tensile test. The reason for a further studies on shear test is that the stress-strain responses under shear loading have been attracting great interests [153] but there is no universal way of conducting shear tests on polymers. And as mentioned by several researchers [101, 154, 155], deviatoric stresses that generated under shear loading have more influence on the mechanical behaviour of polymers. In this regard, the VNR shear test according to ASTM 7078 [83] is selected and the fundamental deformation mechanism, measurement and postprocessing errors are analysed. A two-step correction method is proposed, and its validity is confirmed via further numerical studies.

3.2. Experimental design for uniaxial tensile tests

3.2.1. Design of specimen and test conditions

The PA6 and PEEK materials were supplied by Direct Plastics and Ai Engineering Plastics & Laminates, respectively, in the UK. The mechanical and thermal properties of the investigated PA6 and PEEK are provided in Table 3-1.

Table 3-1. Mechanical and thermal properties of PA6 and PEEK.

Properties	PA6	PEEK
Elastic modulus at 20 ℃ (GPa)	3.5	4.3
Yield stress at 20 ℃ (MPa)	80	115
Elongation at yield at 20 ℃ (%)	4	5
Elongation at break at 20 ℃ (%)	55	7
Density (kg/m3)	1150	1310
Density of crystal phase and amorphous phase (kg/m3)	$\rho_c = 1210$ $\rho_a = 1090$	$\rho_c = 1400$ $\rho_a = 1260$
Glass transition temperature T_g (℃)	47	143
Melting temperature T_m (℃)	230	343
Coefficient of thermal expansion (/K)	80 – 90 x 10^{-6}	50 – 70 x 10^{-6}
Molecular weight (g/mol)	113.16	288.31
Repeat unit		

Since both the PA6 and PEEK polymers are semicrystalline thermoplastics, the degree of crystallinity in mass fraction, χ_c, can be estimated according to [113, 156, 157]:

$$\chi_c = \frac{\rho_c(\rho - \rho_a)}{\rho(\rho_c - \rho_a)} \tag{3-1}$$

where, ρ is the density of the thermoplastic and ρ_c, ρ_a are the densities of the pure crystal phase and the pure amorphous phase, respectively. The degrees of crystallinity of PA6 and PEEK are estimated to be approximately 52 % and 30 %, respectively.

The uniaxial tensile tests were conducted based on ASTM 638 [158] which is a standard tensile testing method for plastics. Considering the large elongation of thermoplastics at elevated temperatures, the dimensions of the tensile test specimen design were determined and are shown

in Figure 3-1a. Two holes were added on the specimen at the two ends for mounting onto the grips. The gauge length was defined as 7.62 mm according to the testing standard. The thicknesses of PA6 and PEEK were 2 mm and 5.5 mm, respectively. The PA6 and PEEK specimens were manufactured by laser cutting and water jet cutting, respectively, and the dimensions were inspected. The manufactured tensile test specimens of PA6 and PEEK are shown in Figure 3-1b and Figure 3-1c, respectively.

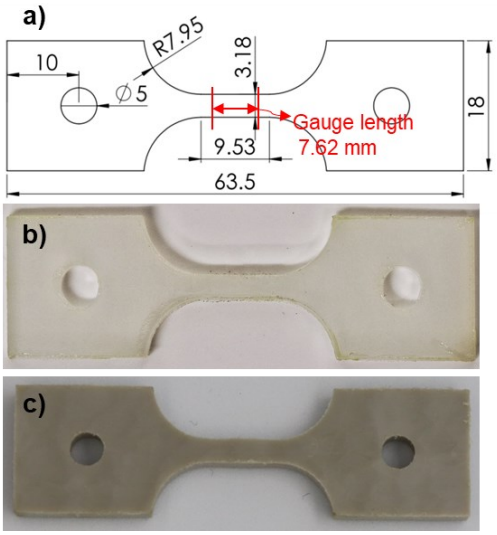


Figure 3-1. Uniaxial tensile test specimens: a) the design and dimensions; photographs of b) PA6 and c) PEEK tensile test specimens. (unit: mm).

3.2.2. Experimental setup

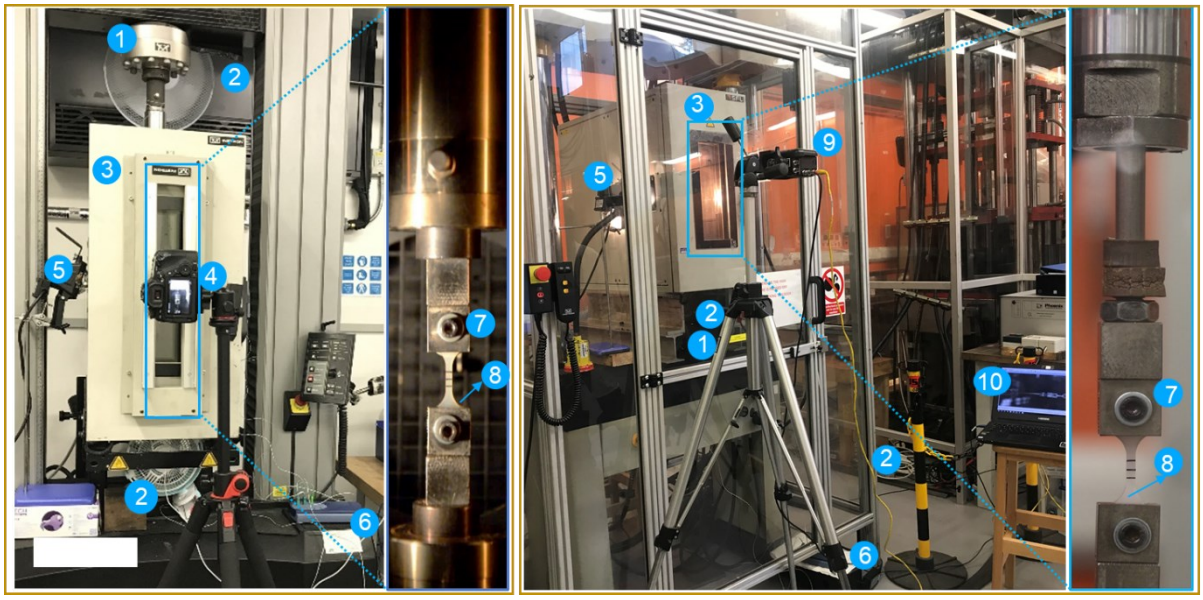
For both types of thermoplastics, the tested temperature range was around 0.7 – 0.9 T_m on the absolute temperature scale, in which range the solid-state polymer materials are more deformable compared to room temperature; the tested strain rate range was set at 0.001 – 50 /s which covers stamp forming conditions [159]. The test conditions for PA6 and PEEK are listed in Table 3-2. In this table, the strain rates are referring to the initial strain rates of the testing material. For each condition, tests were repeated three or four times to check their repeatability.

Figure 3-2 shows two sets of experimental setups for a) low-rate and b) high-rate tests, respectively. All of the low-rate tests were performed on an INSTRON 5584 tensile testing machine, shown in Figure 3-2a, and all of the high-rate tests were performed on an INSTRON

VHS servo-hydraulic machine, shown in Figure 3-2b. An environmental chamber was mounted to each machine. The average heating rate was around 4 ℃/min. The tested temperature on the surface of the specimens was monitored by using thermocouples connected to a data-logger. To achieve a uniform temperature distribution, the specimens were soaked at testing temperatures for 10 minutes before the tests started. Fans were used to prevent the load cell and the machine from overheating. A 60 frames-per-second (fps) low-speed camera and a 500 fps high-speed camera were used to capture the deformation at low-rate and high-rate tests, respectively. A light was used on each setup to assist in capturing the deformation images. Bespoke lightweight tensile test grips were used to reduce the inertia effects during the high-rate tests.

Table 3-2. Uniaxial tensile test matrices

Test equipment	Low-rate INSTRON			High-rate INSTRON		
Strain rate (/s) / Temperature (℃)	0.001	0.01	0.1	1	10	50
PA6						
160			✓			
180		✓	✓	✓	✓	
200			✓			
220		✓	✓	✓	✓	
PEEK						
180			✓			
210			✓			
240	✓	✓	✓	✓	✓	✓
270			✓			
300			✓			
310			✓			



a) b)

Figure 3-2. Experimental setup. a) Test setup on low-rate INTRON; b) Test setup on high-rate INSTRON; ①Load cell; ②Fan; ③Environmental chamber; ④Low-speed camera; ⑤Light; ⑥Thermocouple and data-logger; ⑦Tensile test grip; ⑧Tensile test specimen; ⑨High-speed camera; ⑩Laptop and software to capture the image from the high-speed camera.

3.2.3. In-situ strain measurement

In order to measure the extra-large tensile strains of the thermoplastics at elevated temperatures, an in-house Digital Image Correlation (DIC) script was developed in MATLAB. As shown in Figure 3-3, the specimen was marked with three lines, where the 1st and 3rd lines marked the two ends of the gauge length (7.62 mm), and the 2nd line marked the middle of the gauge length. The script recognised and tracked the three marked lines by detecting the differences in pixel values along the specimen using a sliding window schematically marked in Figure 3-3a. After detecting one frame of images, the differences in pixel values along the elongation could be obtained, shown as Figure 3-3b. The three marked lines' pixel positions along the elongation were extracted by locating the pixel positions of the three peaks in Figure 3-3b. By conducting the same process on all the frames, the pixel positions of the three marked lines along the elongation were obtained as shown in Figure 3-3c. Therefore, the evolution of the average strain within the gauge length could be calculated from the distance change between the 1st and 3rd lines. As illustrated by the example in Figure 3-3c, the 2nd line is always in the middle of the

1st and 3rd line, demonstrating a uniform deformation within the gauge length throughout the test. It is worth mentioning that not all test results were under uniform deformation until fracture. Necking behaviour was observed at several high-rate or low-temperature tests.

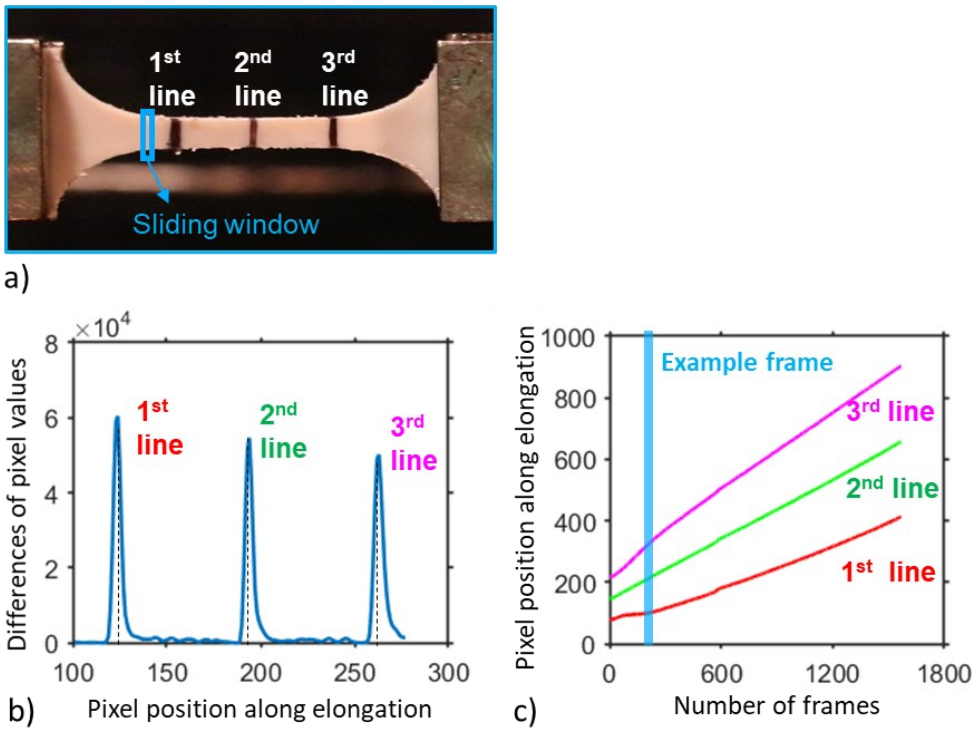


Figure 3-3. Tensile strain captured using an image processing method. a) An example frame of the deformation pictures; b) Identifying the position of marked lines through pixel values in the example frame; c) The tracked positions of three marked lines throughout a uniaxial tensile test

The true tensile strain and true tensile stress were calculated according to Eqs. (3-2)-(3-5).

$$\varepsilon_e = \frac{p_3 - p_1 - l_0}{l_0} \tag{3-2}$$

$$\varepsilon_t = \ln(1 + \varepsilon_e) \tag{3-3}$$

$$\sigma_e = \frac{F}{w_0 t_0} \tag{3-4}$$

$$\sigma_t = \sigma_e(1 + \varepsilon_e) \tag{3-5}$$

where, ε_e and ε_t are the experimentally measured engineering tensile strain, and true tensile strain, respectively. $p_3,\ p_1$ are the positions of the 3rd line and the 1st line along elongation throughout the test as shown in Figure 3-3, and l_0 is the initial distance between the 3rd line and the 1st line (i.e., gauge length). σ_e and σ_t are the experimentally measured engineering tensile stress and true tensile stress, respectively. F is the measured load from the load cell on the tensile test machine. w_0 and t_0 are the initial width and initial thickness, respectively, at the gauge section measured from each tested specimen.

3.3. Experimental results and analysis from the uniaxial tensile tests

3.3.1. Uniaxial tensile test results of thermoplastics

Figure 3-4 shows the uniaxial tensile test results of PA6. The true stress – true strain curves at different temperatures and the same strain rate of 0.1 /s are shown in Figure 3-4a. There is a noticeable decrease in overall stress level with an increase in temperature. The initial elastic deformation part shows nonlinear properties, and no obvious yield points can be detected in the tested material. At larger deformation stages, there is a significant strain hardening behaviour when PA6 deforms at temperatures below or equal to 200 ℃, whereas the strain hardening effect is much reduced at 220 ℃.

Figure 3-4b and c show the tensile responses of PA6 under various strain rates at 180 ℃ and 220 ℃, respectively. In general, for both temperatures, it can be observed that the strain rate effects, within the tested range of 0.01 – 10 /s, are not as obvious as the observed temperature effects. There is no significant difference in stress responses when the true tensile strain is less than 0.6. The effect of the strain rate in stress response at a larger tensile strain becomes more obvious. Interestingly, it is found that the stresses are higher at lower strain rates (0.01, 0.1 /s) when the tensile strain is higher than 0.6. Please note that the vertical scale in Figure 3-4c is different with that in Figure 3-4a and Figure 3-4b since the stresses are relatively small.

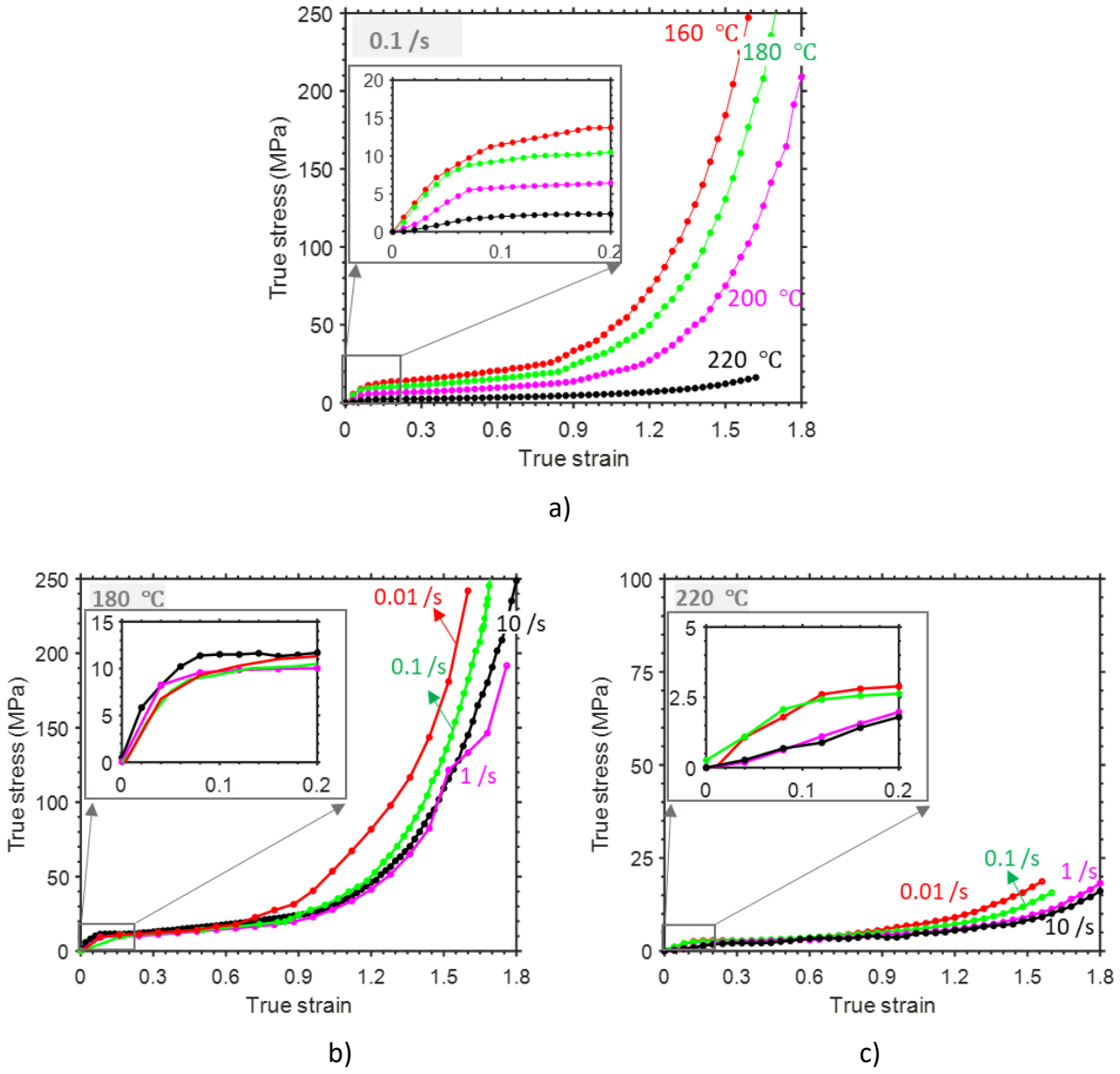


Figure 3-4. Experimental true stress – true strain curves from uniaxial tensile tests for PA6 under different temperatures (160 - 220 °C) and strain rates (0.01 - 1 /s). a) Temperature effects at the strain rate of 0.1 /s; strain rate effects at temperatures of b) 180 °C and c) 220 °C

Figure 3-5 shows the uniaxial tensile test results of PEEK. The true stress – true strain curves at different temperatures and the same strain rate of 0.1 /s are shown in Figure 3-5a. Similar to the temperature effects on PA6, there is a considerable decrease in the overall stress level with increasing temperature. At the initial stage of deformation, the material also shows nonlinear behaviour within the studied temperature range. At 180 °C, there is a region showing slight strain softening after the initial elastic deformation, which is believed to be related to the

reorientation of crystallites and the fragmentation of larger crystalline phases [160]. This behaviour disappears at higher temperatures. The strain hardening phenomena appear at larger stages of deformation, when the temperature is lower than 310 ℃. At 310 ℃, the strain hardening behaviour is much reduced and is not obvious.

The strain rate effects on PEEK can be inferred from Figure 3-5b. In general, the stress responses of PEEK at different strain rates, at 240 ℃, are similar compared with those of PA6 at different strain rates (see Figure 3-4b and c). The tensile stress is found to be higher at a higher strain rate within a low tensile strain level, and the opposite trend is observed when the tensile strain is higher than 0.6.

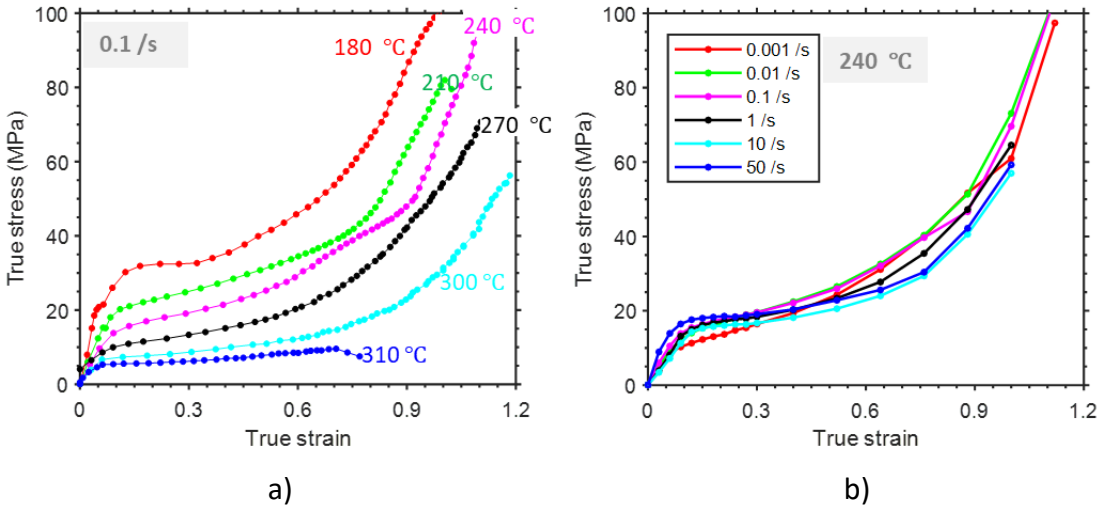


Figure 3-5. Experimental true stress – true strain curves from uniaxial tensile tests for PEEK at different temperatures (180 – 310 ℃) and strain rates (0.001-50 /s). a) Temperature effects at a strain rate of 0.1 /s. b) Strain rate effects at a temperature of 240 ℃

3.3.2. Key-features of the stress-strain curves

Based on the tensile test results from Figure 3-4 and Figure 3-5, in order to conduct a more in-depth analysis of the temperature and strain rate effects, three stress response regions are qualitatively defined for each true stress – true strain curve [74], as illustrated in Figure 3-6a. Region I is the initial nonlinear elastic region where the stress increases elastically with the increasing strain; region II is a pre-strain hardening region where the stress increases steadily with increasing strain; region III is a post-strain hardening region where the stress increases

exponentially with increasing strain. Key features for the three regions are further characterised and discussed.

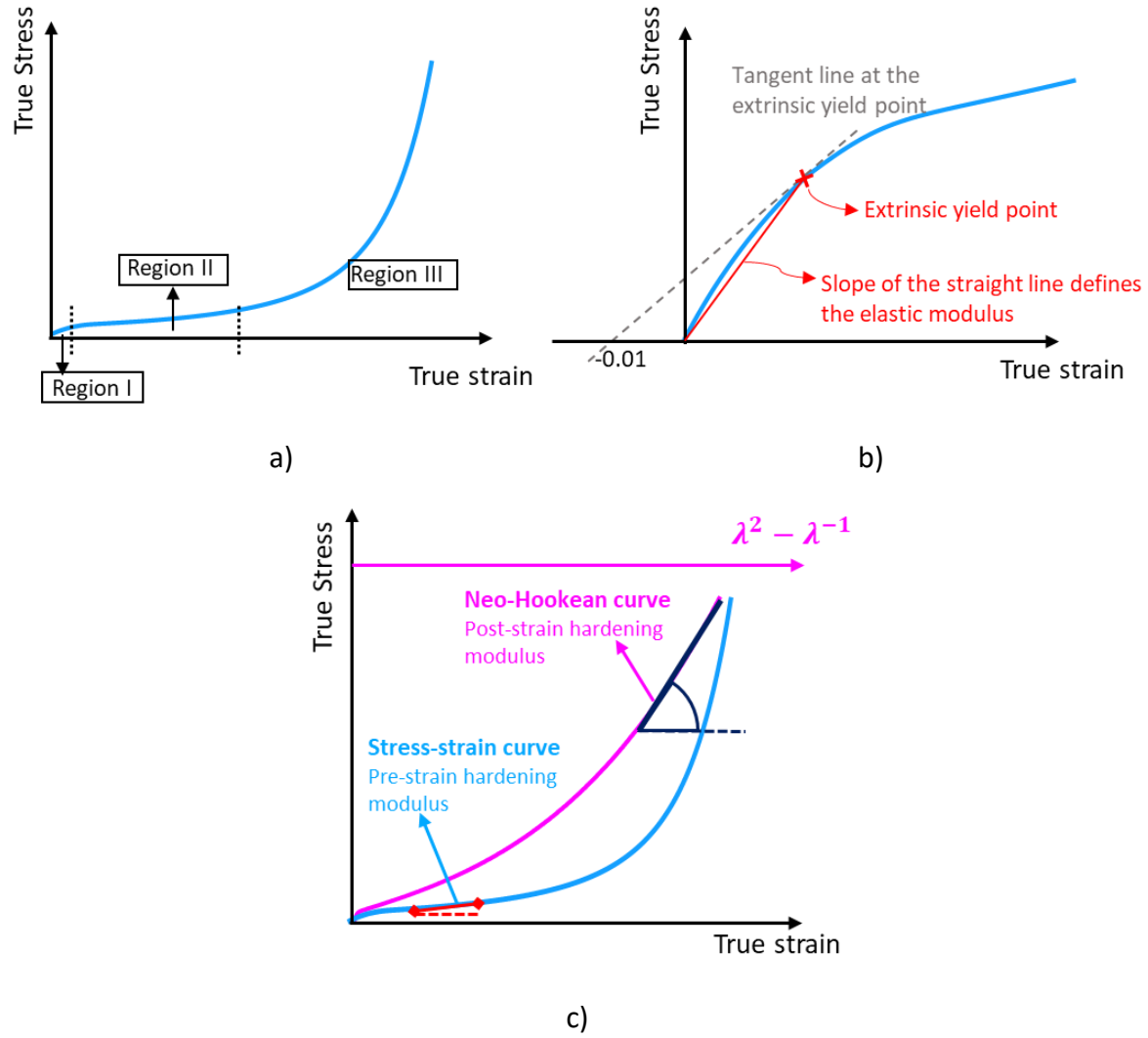


Figure 3-6. The analysis of the key features of a representative stress-strain curve: a) three stress response regions; b) the identification of the yield point and elastic modulus; c) the definition of the pre-strain hardening modulus and the post-strain hardening modulus

For region I, the elastic modulus and yield stress are analysed. As has been mentioned, there was no obvious yield point observed for the thermoplastics at the tested temperatures. Thus, it is difficult to locate the elastic limit from the true stress – true strain curves. Therefore, the extrinsic yield point is used in this study, as mentioned by Brinson et al. [101]. As shown in Figure 3-6b, the extrinsic yield point is captured by drawing the tangent lines from points on

the stress-strain curve and locating the point when the intersection of the tangent line and the horizontal axis is -0.01 [101]. The elastic modulus is defined by the slope of the line connecting the undeformed point (i.e., origin) and the extrinsic yield point.

In region II, the strain hardening behaviour appears. Although there is slight strain softening for PEEK at 180 ℃, it lasts for only a short deformation stage and is also followed by a steady strain hardening stage. Therefore, Region II is identified as the pre-strain hardening stage and is described by a pre-strain hardening modulus. As can be observed from Figure 3-4 and Figure 3-5, at a strain range of 0.3 – 0.5, there is a steady strain hardening behaviour for both materials at all test conditions. Therefore, the pre-strain hardening modulus is calculated by the average gradient of the true stress – true strain curve at the strain range of 0.3 – 0.5, as shown in Figure 3-6c.

For region III, defined as the post-strain hardening region, the stress response could be described by a Neo-Hookean curve according to Melick et al. [161], shown as the pink line in Figure 3-6c. The Neo-Hookean curve is defined by the relationship between the true stress and the term $(\lambda^2 - \lambda^{-1})$, where $\lambda = \varepsilon_e + 1$ is the stretch ratio, representing the ratio between the final and the initial gauge length of the tested material. The post-strain hardening modulus is determined as the slope of the last portion of the Neo-Hookean curve, which is linear. In this study, the last 20 % of the Neo-Hookean curve is used to calculate the post-strain hardening modulus.

3.3.3. Analysis of the temperature effects

The temperature effects on the stress response are analysed in detail as below.

1) The elastic modulus and the yield stress

The yield stresses and elastic moduli of PA6 and PEEK at different temperatures and the strain rate of 0.1 /s are shown in Figure 3-7. It is clear that, with increasing temperature, the elastic modulus and the yield stress decrease nonlinearly for both materials.

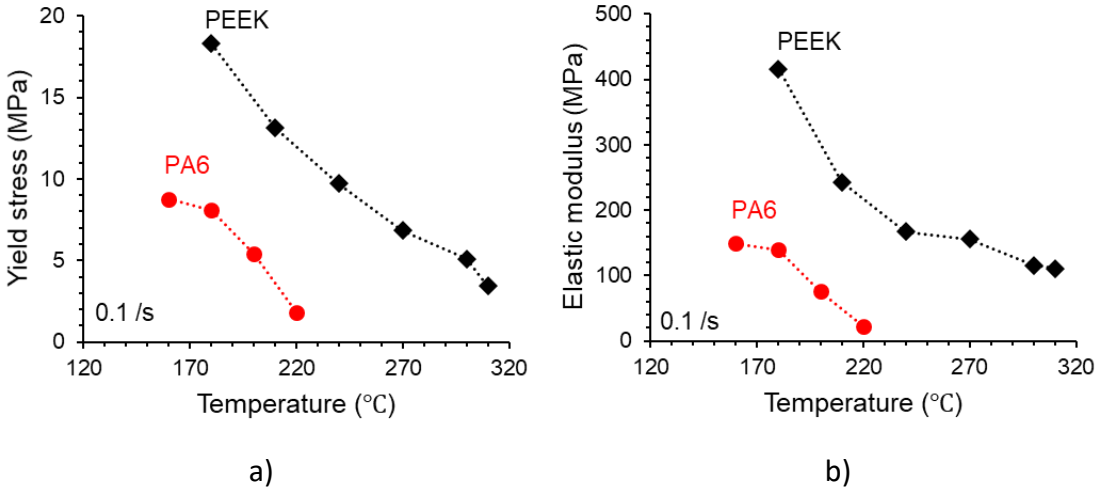


Figure 3-7. Temperature effects on the a) yield stresses and b) elastic moduli of PA6 and PEEK at different temperatures (160 – 220 ℃ for PA6 and 180 – 310 ℃ for PEEK) and a strain rate of 0.1 /s

Figure 3-8a shows the pre-strain hardening moduli of tensile deformation at different temperatures and a strain rate of 0.1 /s for both materials. As can be observed, with increasing temperature, the pre-strain hardening modulus decreases linearly. The extent of the temperature dependence of the pre-strain hardening modulus can be defined by the slope of each trendline, and the slopes for both materials are almost equal.

Figure 3-8b shows the post-strain hardening moduli of tensile deformation at different temperatures and a strain rate of 0.1 /s for both materials. As can be observed, there are also linear relationships between the post-strain hardening modulus and the temperature for both materials in the lower temperature ranges (160 – 200 ℃ for PA6, 180 – 300 ℃ for PEEK). The slopes of the trendlines for PA6 and PEEK represent the temperature dependencies of post-strain hardening, and the slope for PA6 is higher than the slope for PEEK. At temperatures of 220 ℃ for PA6 and 310 ℃ for PEEK, the post-strain hardening moduli are near zero. This corresponds to Figure 3-4a and Figure 3-5a, where the post hardening disappeared at the temperatures of 220 ℃ and 310 ℃ for PA6 and PEEK, respectively. This phenomenon indicates that 220 ℃ and 310 ℃ are near the critical temperatures approaching T_m, for PA6 and PEEK, respectively.

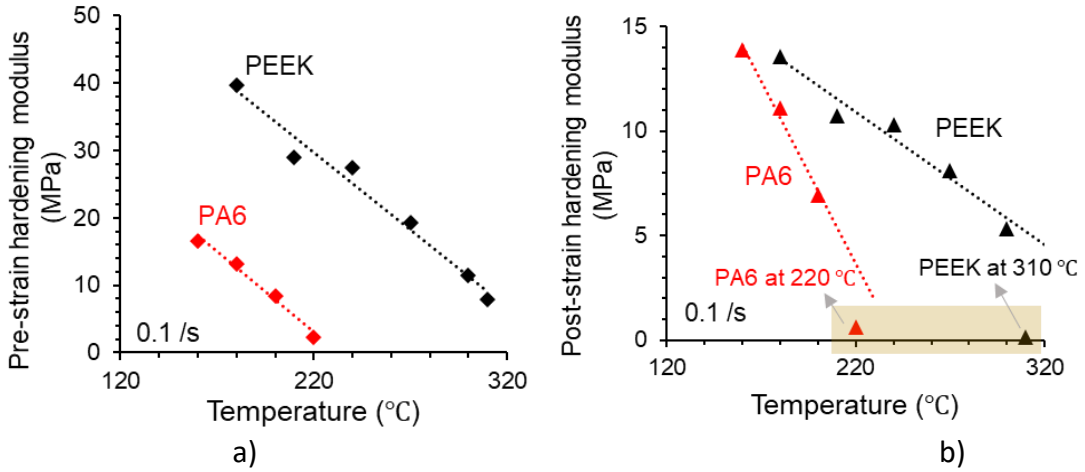


Figure 3-8. Temperature effects on the a) pre-strain hardening modulus and b) post-strain hardening of PA6

3.3.4. Analysis of strain rate effects

Understanding the strain rate effect on thermoplastics' thermomechanical properties is pivotal. When materials are subjected to dynamic loading conditions, such as in crash scenarios or during high-speed operations, their response can vary significantly based on the rate at which they are deformed. This is referred to as the strain rate sensitivity. For thermoplastics, which are frequently used in applications ranging from automotive parts to aerospace components, capturing this sensitivity is essential. It allows engineers and designers to make informed decisions about the material's suitability for specific applications, ensuring safety, longevity, and optimal performance. In this study, identifying any variations in material behaviour under different strain rates can influence processing techniques and tooling designs. From the tensile test results shown in Figure 3-4b,c and Figure 3-5b, the strain rate effects are not as obvious as the temperature effects. The elastic moduli, yield stresses, pre-strain hardening, and post-strain hardening moduli are almost the same at different strain rates for both PA6 and PEEK. However, there is a different trend at the early stage and the later stage of the deformation. Therefore, to demonstrate this phenomenon, the stress levels corresponding to two strain levels are analysed. Figure 3-9 shows true stresses of PA6 tested at 180 ℃ and PEEK tested at 240 ℃ at the strain levels of 0.1 and 0.7, corresponding to region II and region III respectively, at various strain rates. It can be observed that, at a small strain level of 0.1, for PA6 at 180 ℃ or PEEK at 240 ℃, the stress remains unchanged (for PA6) or increases (for PEEK) with increasing strain rate. On the contrary, at a larger strain level (i.e., 0.7), the stress slightly decreases with an increase in strain rate. The shift of strain rate effects on stress responses at

different deformation stages indicates that the true stress level is not a monotonic function of strain rate for PA6 at 180 ℃ or PEEK at 240 ℃, which can be attributed to the adiabatic heat at higher strain rates [162, 163]. At higher strain rates, there is not sufficient time for the plastic deformation energy to dissipate. Therefore, fast deformation results in heat accumulation and temperature increase in the deformed area [106], which softens the material in the deformed area at the later stage of deformation. The existence of adiabatic heat can be proven by the necking behaviour observed in the tensile tests of PEEK, with an example shown in Figure 3-10. In the tensile tests at various strain rates for PEEK at 240 ℃, necking behaviour was found in the tests at the highest strain rate (50 /s), but not in a strain rate of 0.001 /s throughout the whole test. This adiabatic heat effect will be further discussed in the constitutive modelling part.

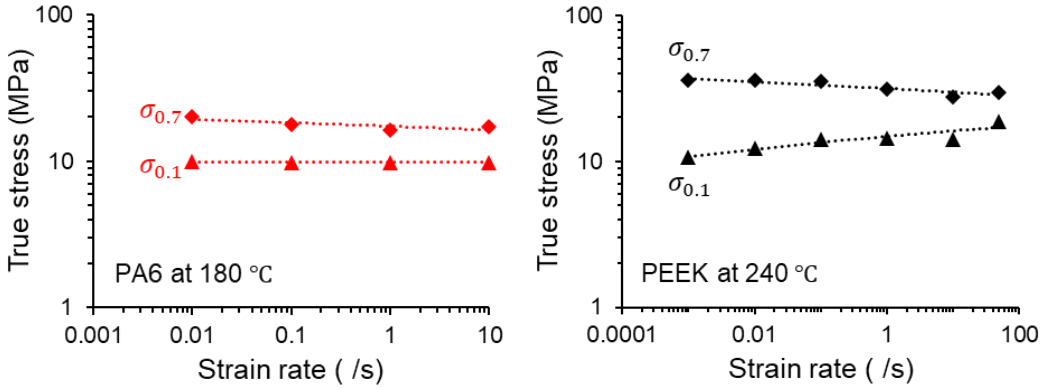


Figure 3-9. Stress analysis at strain levels of 0.1 and 0.7 for a) PA6 at a temperature of 180 ℃, and b) PEEK at a temperature of 240 ℃, at various strain rates (0.001 – 50 /s). (The figures are in log-log scales)

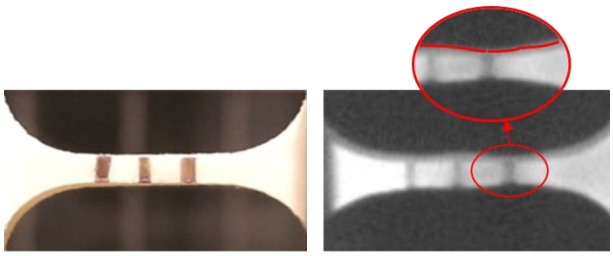


Figure 3-10. An example of deformation images of PEEK at a temperature of 240 ℃, and at a strain rate of a) 0.001 /s (no necking observed) and b) 50 /s (necking observed), at the same deformation stage (ε_e = 0.14)

Based on the experimental studies on the thermomechanical behaviour of solid-state thermoplastics, certain implications of the solid-state stamp forming parameters can be established. With regard to temperature, for stamp forming of pure thermoplastics and discontinuous FRTPs, temperature ranges of 180 – 200 °C and 240 – 300 °C are recommended for PA6 and PEEK, respectively, by considering the elongation and strain hardening characteristics of the materials and the energy cost of manufacturing. For FRTPs reinforced with continuous woven fibres, the large deformation is attributed to the fibre reorientation and fibre bending, and there is more local shear deformation in the polymer constituent. This means strain hardening, which is a key characteristic to enhance drawability [164] in stretch dominated deformation, is no longer the most important consideration. Instead, minimum shear resistance is preferred to guarantee the fibre motions within polymers. Therefore, processing temperatures of ≥ 220 °C and ≥ 310 °C are suggested for composites with PA6 and PEEK matrices, respectively. With regard to the strain rate, since there are no obvious differences in material responses with the varying strain rates in the studied range, high forming speeds within the relevant machine capacities may be adopted to increase productivity increase productivity and to facilitate the use of cold forming tools with minimal heat loss.

3.4. Fundamental studies on the V-Notched Rail (VNR) shear tests

3.4.1. Introduction to the VNR shear tests

With the thermomechanical properties of thermoplastics obtained, the benchmark has been established for furthering investigation of another experimental method, the VNR shear test, for characterising the effective stress-strain properties of thermoplastics. Please note that it is assumed the thermoplastics are isotropic in this section.

In shear deformation interpretation, it is crucial to distinguish between pure shear and simple shear as two distinct concepts when evaluating the mechanical behaviour from shear tests [155]. Commonly used shear test methods on sheet materials, such as the V-Notched Rail (VNR) shear test according to ASTM 7078 [83], the two rail and three rail tests according to ASTM D4255 [165, 166], and the Iosipescu test according to ASTM D5379 [167, 168], are all based on simple shear deformation. While many researchers have considered simple shear as a combination of pure shear and rigid-body rotation within small deformation [169, 170], this simplification is not applicable to large deformations [153] and may cause calculation errors. Therefore, it is necessary to study the calculation errors and corresponding correction methods to gain a deeper

understanding, especially for polymers, which have increasingly found applications in polymers and Polymer Matrix Composites (PMCs) for lightweight panel components, both of which involve large deformation [171]. Furthermore, as highlighted by several researchers [101, 154, 155], deviatoric stresses play a more significant role in the mechanical behaviour of polymers, emphasizing the importance of accurately characterising the stress-strain response of polymers under shear loading.

For a fundamental understanding of the simple shear deformation, Thiel et al. [155] explained the incompatibility between Cauchy pure shear stress and simple shear deformation. They also introduced the notion of an idealised finite simple shear form that can be applied to certain classes of hyperelastic materials under plane stress conditions without pressure, such as the Mooney-Rivlin material. Belytschko et al. [172] presented solutions to simple shear problems by introducing the objective stress rates for hyperelastic materials. However, the equations for solving the stress state under simple shear deformation were complicated, and different results would be obtained by using various objective stress rates. To study the deformation of a cylinder without axial load, Rajagopal et al. [173] established the shear strain field of the simple shear deformation in the form of stretch ratios along the principal axes of the nonlinear isotropic elastic material. Moreira et al. [153] conducted a comparison between the simple shear and the pure shear on a hyperelastic material under large deformation. In their research, the simple shear deformation was achieved by a single lap joint test, and the pure shear deformation was obtained using the planar tension test. Although the above studies have provided insights into the deformation mechanisms of simple shear and identified the differences between simple shear and pure shear, the stress and strain states of materials under simple shear are still unclear, and a unified post-processing method is lacking.

Several error sources have been reported for stress-strain behaviour calculated from simple shear tests. G'Sell et al. [174] and Dayan et al. [170] reported the rotation of principal strains under simple shear deformation, which causes the directions of the principal stresses to rotate continuously during the simple shear test. This rotation leads to inaccurate effective stress calculations based on pure shear deformation. In addition to the error due to rotation, Hedner et al. [175] reported non-uniform distribution of shear stresses when simulating the simple shear deformation of polymers. This non-uniform deformation could affect the effective stress-strain calculation results if stress localisation is not well addressed. Melin et al. [176] also mentioned the calculation errors generated by the non-uniform deformation. To achieve a more uniform

deformation, they redesigned the Iosipescu test specimen by rescaling the opening angle of the test specimen according to the orthotropic ratio, which is the ratio of the two in-plane principal Young's moduli. To improve the accuracy of calculations, several methods have been proposed by giving correction factors during the testing data post-processing process. For example, Oh et al. [177] defined a correction factor, denoted as the ratio of the average strain over the central area to the central point strain of the testing specimen, based on the Finite Element Analysis (FEA) results, to obtain a more precise shear modulus when measuring the shear modulus from the Iosipescu test on Carbon/Epoxy composites. Taheri-Behrooz et al. [178] also used similar correction factors to correct non-uniformity-induced shear stresses. However, these correction factors are usually fixed numbers and can only be applied to specific testing materials and conditions. In summary, although several studies have pointed out error sources of simple shear tests and proposed correct solutions, none have proposed a universal correction method suitable for a variety of materials under a simple shear test method.

The aim of this section is to demonstrate the errors that occur during VNR shear tests on solid-state polymers, provide an in-depth analysis of the sources of error, and introduce a new universal method to correct the effective stress-strain properties.

3.4.2. Experimental investigation

The VNR shear tests were conducted, and experimental results were post-processed using a traditional calculation method as defined in reference [101]. To identify errors, the obtained effective stress-strain properties were compared with the results obtained through the uniaxial tensile test [101], which is a common material characterisation practice.

- Experimental design

For the specimen design, since distortion was observed when conducting the VNR shear test on standard test specimen defined in ASTM 7078 [83] in this study, a modification was made on the specimen design by reducing the ligament length. The dimensions of the modified VNR shear specimen are shown in Figure 3-11a. Two deep notches were machined on the standard specimens. The ligament length of the modified specimen is 15.68 mm. The modified shear test specimen with the shear test grips is shown in Figure 3-11b. While mounting the specimen to the fixtures during the VNR shear experiment, a pair of space blocks were used to guarantee the good alignment between the specimen ligament and the loading axis. Besides, special care was taken while tightening the bolts in the VNR shear grips to avoid any out-of-plane twisting.

The VNR shear tests were performed on an INSTRON 5584 universal tensile testing machine shown in Figure 3-11c. An environmental chamber was mounted on the machine to heat the test specimen to the target temperatures. The temperature of the specimen was monitored by thermocouples connected to a data-logger during the experiments. A camera was used to record the testing images for strain measurements, at a recording speed of 60 frame-per-second (fps). A light source was also used to assist the image recording. Fans were used to prevent the overheating of the load cell and the machine itself. To achieve a large deformation in the shear experiments, the tests were conducted at 180 ℃, with the crosshead speed set as 1.57 mm/s. The consistency of experimental results was examined by repeating the tests three times.

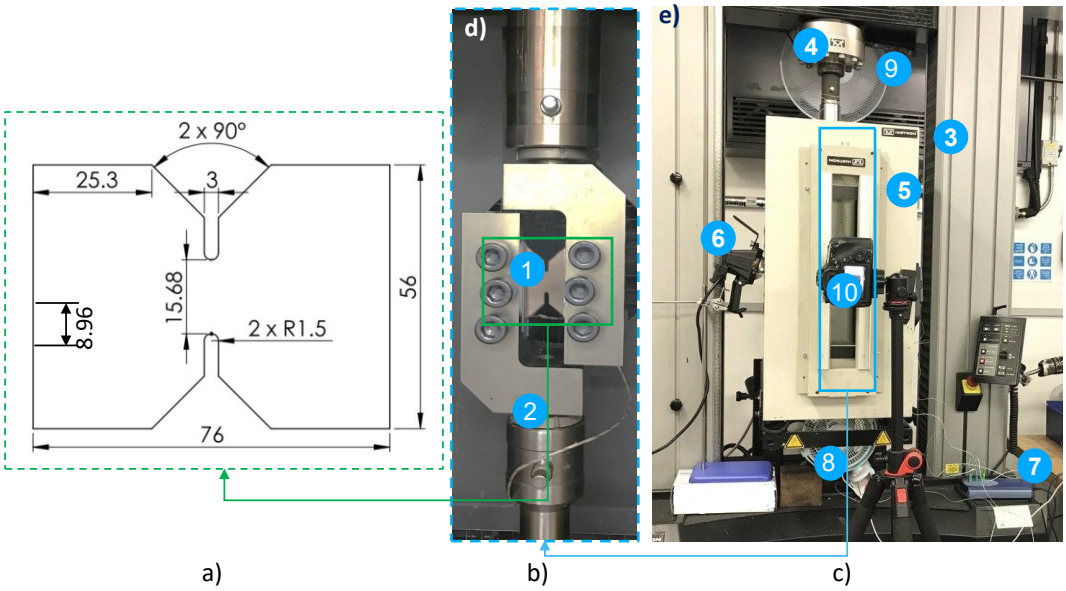


Figure 3-11. The experimental setup and specimen designs for VNR shear tests. a) Design and dimensions (unit: mm) of the specimen. b) An example picture of the specimen and grips used for the VNR shear test. c) Experimental setup: ①VNR shear test specimen; ②VNR shear test grips; ③Instron 5584 load frame; ④Crosshead embedded with a load cell; ⑤Environmental chamber; ⑥Light; ⑦Thermocouples and data-logger; ⑧⑨Fans; ⑩Camera

For strain measurement, the DIC method was used. The test images were captured at a resolution of approximately 225 pixels/mm2 and processed by a 2D DIC software, provided by GOM, Germany. The facet size and facet distance were defined as 16 × 16 pixels and 14 pixels, respectively, to create a surface component for strain calculation. For each facet, the software automatically calculated its strain state on the central point of the facet at each deformation

stage. An example strain field distribution on the VNR shear specimen is shown in Figure 3-12. The region of interest (ROI) was defined as the central section of the specimen at the length of the maximum recognisable length and a width of around 1 mm. The maximum principal strain provided by the DIC software as the experimental result is the arithmetic mean of the maximum principal strains of all facets within the ROI (note: all strains calculated by the software here are engineering strains).

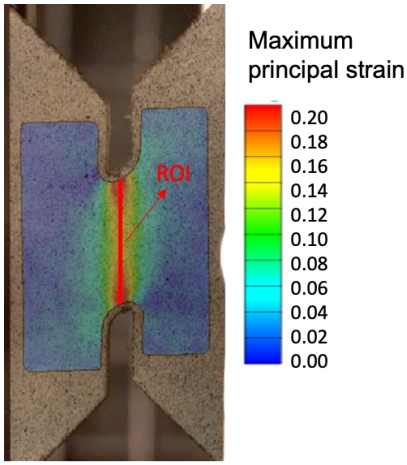


Figure 3-12. The DIC measurement methods for the VNR shear test

- Experimental results

Normally, for obtaining the effective stress-strain properties characterised by the VNR shear test, the standard post-processing process, i.e., treating the simple shear deformation as the combination of pure shear and rotation, is employed. In this research, the effective strain and effective stress were defined as the von-Mises effective strain ($\bar{\varepsilon}$) according to Eq. (3-6) and effective stress ($\bar{\sigma}$) according to (3-7) respectively. The detailed calculation process is listed in Table 3-3.

$$\bar{\varepsilon} = \frac{\sqrt{2}}{3}\sqrt{(\varepsilon_1 - \varepsilon_2)^2 + (\varepsilon_2 - \varepsilon_3)^2 + (\varepsilon_3 - \varepsilon_1)^2} \tag{3-6}$$

$$\bar{\sigma} = \frac{\sqrt{2}}{2}\left(\sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}\right) \tag{3-7}$$

Table 3-3. The detailed effective stress-strain calculation for the VNR shear tests

Variables	Calculation
Measured variables	Vertical Load (F) Initial cross-section area (A_0) Mean value of maximum principal strains (engineering strain) within the ROI (ε_{ROI}^{eng}) captured by GOM software
Ture strain	$\varepsilon_t = ln(1 + \varepsilon_{ROI}^{eng})$
Principal strains	$\varepsilon_1 = \varepsilon^t$ $\varepsilon_2 = 0$ $\varepsilon_3 = -\varepsilon^t$
Effective strain	$\bar{\varepsilon} = \frac{2}{\sqrt{3}}\varepsilon^t$
True stress	$\sigma_t = \frac{F}{A_o} * (1 + \varepsilon_{ROI}^{eng})$
Principal stresses	$\sigma_1 = \sigma^t$ $\sigma_2 = 0$ $\sigma_3 = -\sigma^t$
Effective stress	$\bar{\sigma} = \sqrt{3}\sigma^t$

(Note: all strains in the table refer to true strains unless particularly specified as an engineering strain).

The effective stress-strain curves obtained from the uniaxial tensile and the VNR shear tests on the PEEK specimens at 180 ℃ and 0.1 /s are shown in Figure 3-13. As can be observed, there is an obvious difference between the effective stress-strain curves obtained from the tensile and VNR shear characterisation. Various error sources can cause such a difference. Because the potential experimental error sources (such as the machining and measurement deviation) have been minimised with considerable efforts in this study (and confirmed through repeatability studies), and the reliability of the tensile test data has been confirmed through a systematic study in [106], the calculation errors in the VNR shear test are assumed to exist. To gain further insights into the problem, a detailed theoretical analysis on the material stress and strain states under shear deformations is presented in the next section.

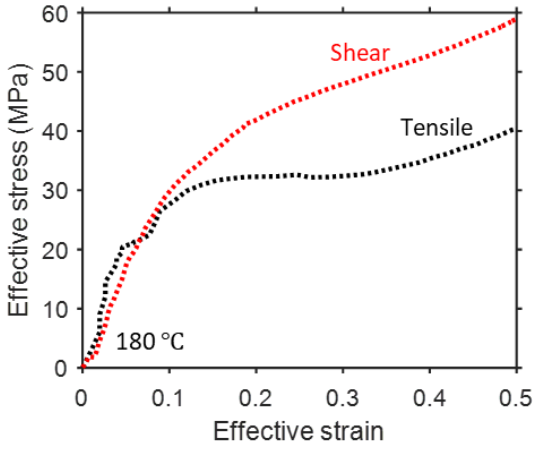


Figure 3-13. The effective stress-strain curves obtained from the tensile and the VNR shear tests on PEEK at a temperature of 180 °C and at a strain rate of 0.1 /s

3.4.3. A detailed explanation of VNR shear tests for stress-strain characterisation problem

1) Deformation mechanism of the VNR shear specimen

The deformation mechanism of the VNR shear specimen can be analysed from two perspectives: the stress and deformation state within an infinitesimal element; and the overall strain distribution within the entire ROI of the specimen.

Figure 3-14 illustrates the stress and deformation state within an infinitesimal element and the conventional approach to understand the VNR shear deformation. The stress state and deformation state of the infinitesimal element at the centre of the specimen are presented in Figure 3-14a. A vertical displacement on one side and a constraining force on the other side cause the central element to deform as simple shear deformation. To maintain the simple shear deformation, shear stress (σ_{12}) and normal stresses (σ_{11} and σ_{22}) exist within the analysed element. During the experiment, the vertical load F is the measurable variable that accounts for the vertical stress components σ_{12} and σ_{22}. The term $\gamma_{m,s}$ represents the maximum shear strain of an analysed element in simple shear deformation. To simplify the calculation for small deformations, simple shear is normally simplified to pure shear with a rigid-body rotation, shown in Figure 3-14b. The term $\gamma_{m,p}$ denotes the maximum shear strain of an analysed element in pure shear. In pure shear deformation, there is only a shear stress σ_{12} in the analysed unit

element. Thus, the measured load F accounts only for the shear stress. It is evident that the effective stress-strain is simpler to calculate from the pure shear deformation as the stress state is less complex.

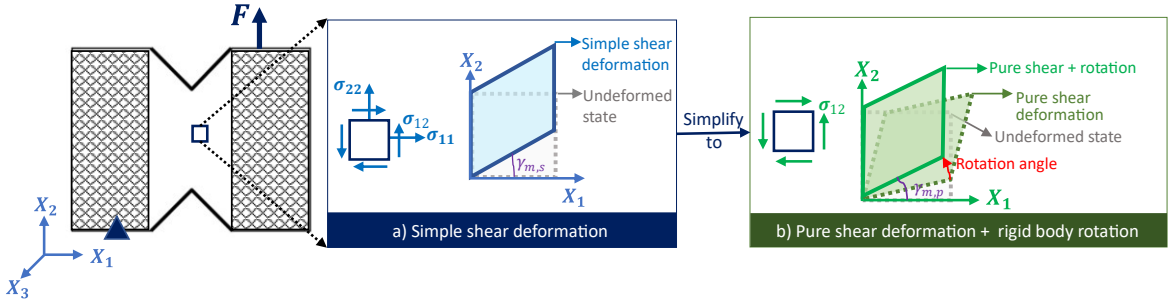


Figure 3-14. Deformation mechanism of the VNR shear specimen. a) Simple shear deformation; b) Simplification of the deformation mechanism at small deformation: pure shear deformation + rigid body rotation. (Terms $\sigma_{11}, \sigma_{22}, \sigma_{12}$ are the stress components of the analysed element. Terms $\gamma_{m,s}$ and $\gamma_{m,p}$ are the maximum shear strains under simple shear and pure shear deformation, respectively.)

To distinguish between simple shear and pure shear, a detailed analysis is presented in Figure 3-15. The pure shear is shown in Figure 3-15a, while for simple shear deformation, which involves both stretch and rotation, the polar decomposition is analysed to isolate and comprehend the deformation mechanism, as depicted in Figure 3-15b.

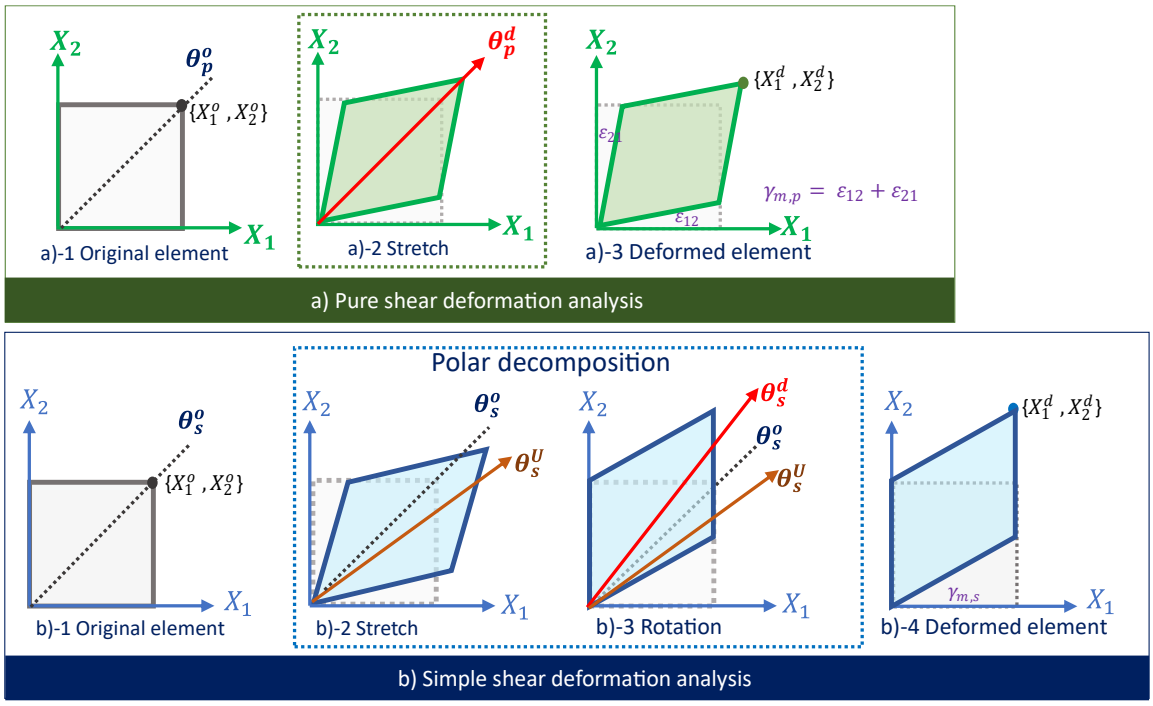


Figure 3-15. Deformation mechanisms of a) Pure shear and b) Simple shear deformation. (The terms X_i^o, X_i^U, X_i^d $(i = 1,2,3)$ are the coordinates of the original, stretched, and deformed configurations, respectively. The arrows, $\boldsymbol{\theta_p^o}$,$\boldsymbol{\theta_p^d}$ represent the maximum principal strain directions of pure shear under original and deformed configuration, respectively; $\boldsymbol{\theta_s^o}$, $\boldsymbol{\theta_s^U}$, $\boldsymbol{\theta_s^d}$ are the maximum principal strain directions of simple shear under the original, stretched, and deformed configurations, respectively. Note: $\boldsymbol{\theta_p^o}$, $\boldsymbol{\theta_s^o}$ are two virtual directions for the purpose of angle calculations.)

To analyse the deformation mechanism, the central element is taken as the analysed element, and the coordinates of the original, stretched, and deformed configurations are represented by X_i^o, X_i^U, X_i^d $(i = 1,2,3)$ respectively, according to the geometric changes.

In the case of pure shear deformation, the in-plane coordinates of the deformed elements, X_1^d, X_2^d can be calculated by comparing Figure 3-15a)-1 and a)-3 and represented by Eq. (3-8):

$$X_1^d = X_1^o + X_2^o \cdot \tan\left(\frac{\gamma_{\mathrm{m,p}}}{2}\right); \quad X_2^d = X_1^o \cdot \tan\left(\frac{\gamma_{\mathrm{m,p}}}{2}\right) + X_2^o \qquad (3\text{-}8)$$

Based on the incompressible assumption[106] on solid-state polymers, X_3^d can be calculated as below:

$$X_3^d = (1 - \tan^2(\frac{\gamma_{m,p}}{2})) \cdot X_3^o \tag{3-9}$$

Therefore, the relationship between the original configuration $\boldsymbol{X^o}$ and the deformed configuration $\boldsymbol{X^d}$, can be represented as Eq.(3-10).

$$\boldsymbol{X^d} = \begin{bmatrix} 1 & \tan\left(\frac{\gamma_{m,p}}{2}\right) & 0 \\ \tan\left(\frac{\gamma_{m,p}}{2}\right) & 1 & 0 \\ 0 & 0 & \frac{1}{1-\tan^2\left(\frac{\gamma_{m,p}}{2}\right)} \end{bmatrix} \cdot \boldsymbol{X^o} \tag{3-10}$$

And the deformation gradient of pure shear, F_p can be calculated accordingly:

$$\boldsymbol{F_p} = \frac{\partial \boldsymbol{X^d}}{\partial \boldsymbol{X^o}} = \begin{bmatrix} 1 & \tan\left(\frac{\gamma_{m,p}}{2}\right) & 0 \\ \tan\left(\frac{\gamma_{m,p}}{2}\right) & 1 & 0 \\ 0 & 0 & \frac{1}{1-\tan^2\left(\frac{\gamma_{m,p}}{2}\right)} \end{bmatrix} \tag{3-11}$$

As shown in Eq. (3-11), the deformation gradient tensor of pure shear, $\boldsymbol{F_p}$, is symmetrical, indicating that there is no rotation but only stretch in pure shear deformation. Therefore, the angle between the maximum principal strain and the X_1 axis remains 45° ($\boldsymbol{\theta_p^o}$, $\boldsymbol{\theta_p^d}$ in Figure 3-15a)-2) all the time.

Similarly, for simple shear deformation, by comparing Figure 3-15b)-1 and b)-4 the deformed configuration can be calculated as Eq. (3-12).

$$X_1^d = X_1^o;\ X_2^d = \tan(\gamma_{m,s}) \cdot X_1^o + X_2^o;\ X_3^d = X_3^o \tag{3-12}$$

the deformation gradient tensor, $\boldsymbol{F_s}$, can be expressed as:

$$\boldsymbol{F_s} = \frac{\partial \boldsymbol{X^d}}{\partial \boldsymbol{X^o}} = \begin{bmatrix} 1 & 0 & 0 \\ \tan(\gamma_{m,s}) & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3-13}$$

Eq.(3-13) shows that the deformation gradient tensor of simple shear, $\boldsymbol{F_s}$, is asymmetric, which indicates that rotation exists during such deformation. After polar decomposition with $\boldsymbol{F_s}$, the

rotation tensor $\boldsymbol{R}$ and stretch tensor $\boldsymbol{U}$, in a relation of $\boldsymbol{F_S} = \boldsymbol{RU}$, can be obtained as below (To simplify the expression, making $b = \tan(\gamma_{m,s})/2$):

$$\boldsymbol{R} = \begin{bmatrix} \frac{1}{\sqrt{1+b^2}} & \frac{-b}{\sqrt{1+b^2}} & 0 \\ \frac{b}{\sqrt{1+b^2}} & \frac{1}{\sqrt{1+b^2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad \boldsymbol{U} = \begin{bmatrix} \frac{1+2b^2}{\sqrt{1+b^2}} & \frac{b}{\sqrt{1+b^2}} & 0 \\ \frac{b}{\sqrt{1+b^2}} & \frac{1}{\sqrt{1+b^2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3-14}$$

The stretch process is illustrated in Figure 3-15b)-2, the eigenvectors of $\boldsymbol{U}$ are the directions principal strains, and of which the first eigenvector, $\widehat{\boldsymbol{v}}_{\boldsymbol{1}}$, is corresponding to $\boldsymbol{\theta_S^o}$ or $\boldsymbol{\theta_S^d}$ in the original and deformed configurations, respectively, as illustrated in Figure 3-15b. $\widehat{\boldsymbol{v}}_{\boldsymbol{1}}$ can be expressed as Eq. (3-15).

$$\widehat{\boldsymbol{v}}_{\boldsymbol{1}} = \begin{bmatrix} \sqrt{1+b^2}+b \\ 1 \\ 0 \end{bmatrix} \tag{3-15}$$

The direction of the maximum principal strain, $\widehat{\boldsymbol{v}}_{\boldsymbol{1}}$, can be understood as below:

In the original configuration, as shown in Figure 3-15b)-1, $b = 0$, therefore, $\widehat{\boldsymbol{v}}_{\boldsymbol{1}} = [1\ 1\ 0]^T$, which means $\widehat{\boldsymbol{v}}_{\boldsymbol{1}}$ is at a direction of 45° with the X_1 axis, shown as $\boldsymbol{\theta_S^o}$. It is worth mentioning that $\boldsymbol{\theta_S^o}$ is listed in the figure only for the purpose of calculating the rotation angle of the principal strain and no actual physical meaning is represented.

In the stretched configuration, as shown in Figure 3-15b)-2, the angle between the maximum principal strain and the X_1 axis, i.e., the angle between $\boldsymbol{\theta_S^U}$ and X_1 axis, can be calculated according to $\widehat{\boldsymbol{v}}_{\boldsymbol{1}}$:

$$\tan(\boldsymbol{\theta_S^U}) = \frac{1}{\sqrt{1+b^2}+b} = \sqrt{1+b^2}-b \tag{3-16}$$

Accordingly, from Figure 3-15b)-1 to b)-2, the change of the direction of maximum principal strain can be calculated:

$$\tan(\boldsymbol{\theta_S^{oU}}) = \tan(\boldsymbol{\theta_S^U} - \boldsymbol{\theta_S^o}) = \frac{\tan(\boldsymbol{\theta_S^U}) - \tan(\boldsymbol{\theta_S^o})}{1 + \tan(\boldsymbol{\theta_S^U}) \cdot \tan(\boldsymbol{\theta_S^o})} = \frac{1-\sqrt{1+b^2}}{b} \tag{3-17}$$

The rotation process is illustrated in Figure 3-15b)-3. The counterclockwise rotational angle of the analysed element (i.e., the rigid body rotation) $\boldsymbol{\theta_s^{Ud}} = \boldsymbol{\theta_s^{d}} - \boldsymbol{\theta_s^{U}}$. According to the definition of rational tensor $\boldsymbol{R}$, in the form of $\boldsymbol{\theta_s^{Ud}}$, $\boldsymbol{R}$ can be represented as:

$$R = \begin{bmatrix} \cos(\boldsymbol{\theta_s^{Ud}}) & -\sin(\boldsymbol{\theta_s^{Ud}}) & 0 \\ \sin(\boldsymbol{\theta_s^{Ud}}) & \cos(\boldsymbol{\theta_s^{Ud}}) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3-18}$$

By comparing Eqs. (3-14) and (3-18), $\boldsymbol{\theta_s^{Ud}}$ can be calculated as below:

$$\tan(\boldsymbol{\theta_s^{Ud}}) = b \tag{3-19}$$

It is found from Eq. (3-19) that:

$$\tan\left(\frac{\boldsymbol{\theta_s^{Ud}}}{2}\right) = \frac{\sin(\boldsymbol{\theta_s^{Ud}})}{1+\cos(\boldsymbol{\theta_s^{Ud}})} = \frac{\sqrt{1+\mathrm{b}^2}-1}{\mathrm{b}} \tag{3-20}$$

Therefore, comparing Eqs. (3-17 and (3-20), the following expression can be obtained:

$$\boldsymbol{\theta_s^{oU}} = -\frac{1}{2}\boldsymbol{\theta_s^{Ud}} = -\arctan(\frac{\sqrt{1+\mathrm{b}^2}-1}{\mathrm{b}}) \tag{3-21}$$

Then the change of the maximum principal strain from the original configuration to the deformed configuration can be obtained accordingly:

$$\boldsymbol{\theta_s^{od}} = \boldsymbol{\theta_s^{oU}} + \boldsymbol{\theta_s^{Ud}} = \arctan(\frac{\sqrt{1+\mathrm{b}^2}-1}{\mathrm{b}}) \tag{3-22}$$

From Eqs. (3-16) to (3-22), it is proved that $\boldsymbol{\theta_s^{od}} = -\boldsymbol{\theta_s^{oU}} = \boldsymbol{\theta_s^{Ud}}/2$. This means the principal strain axes rotation angle is different with the element rotation angle, and both angles are related to the shear strain. This rotation of principal strain axes will eventually induce the rotation of principal stress axes. The simplification from simple shear to pure shear has failed to address the stress state changes and the relation between the vertical load *F* and the stress components (σ_{22}, σ_{12}). Therefore, the effective stress calculation is inaccurate when the principal axes keep changing, and the stress components are not available.

The above explanations mainly focus on deformation of the central infinitesimal element. However, when considering the overall deformed area, it is acknowledged that the material does not deform uniformly due to the stress concentration effect, as observed by several

researchers [175, 176, 178, 179]. This non-uniformity of deformation is also evident in experimental investigations, as depicted in Figure 3-12. Although the average stress along the ligament can be obtained through the measurement of vertical load F, this average stress does not accurately represent the real material behaviour. Thus, the non-uniformity factor introduces another source of error in the calculation of effective stress-strain.

2) Strain measurement methods used in the VNR shear test

The measurable variables and related characterisation methods are crucial from an experimental perspective, for evaluating experimental methods and analysing calculation errors in shear tests. Measurable variables typically include the vertical load F, which is a stress-related variable, and the strain or the strain field of the deformed material, depending on the strain measurement methods used. Two commonly used strain measurement methods are the strain gauge measurement and the DIC measurement, which are illustrated in Figure 3-16 and introduced below.

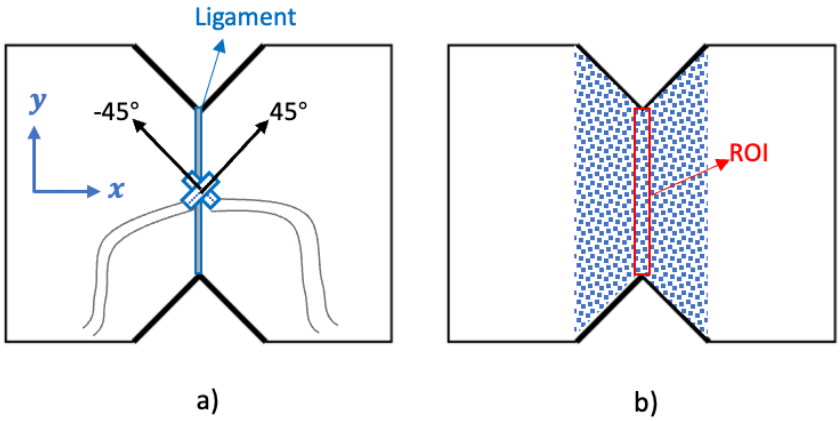


Figure 3-16. The shear strain measurement methods in the VNR shear test. a) Using strain gauge; b) Using DIC.

In accordance with the testing standard ASTM 7078 [83] of the VNR shear test, the shear strain can be measured using two strain gauges mounted along the +45° and -45° directions to the loading axis at the ligament centre, shown as Figure 3-16a. The strains along the +45° and -45° directions can be expressed using the rule of strain transformation as follows:

$$\varepsilon_{+45^\circ} = \frac{\varepsilon_x+\varepsilon_y}{2} + \frac{\varepsilon_x-\varepsilon_y}{2}\cos(2\times(+45^\circ)) + \frac{\gamma_{xy}}{2}\sin(2\times(+45^\circ)); \quad (3\text{-}23)$$

$$\varepsilon_{-45°} = \frac{\varepsilon_x+\varepsilon_y}{2} + \frac{\varepsilon_x-\varepsilon_y}{2}\cos(2\times(-45°)) + \frac{\gamma_{xy}}{2}\sin(2\times(-45°)), \quad (3\text{-}24)$$

where, $\varepsilon_{+45°}$ and $\varepsilon_{-45°}$ are strains measured from the +45° and -45° strain gauges, respectively. The strains of $\varepsilon_x, \varepsilon_y$ and γ_{xy} are the strain components of the analysed element at the specimen centre in the $x-y$ plane shown in Figure 3-16a. Subtracting Eq. (3-24) from Eq. (3-23), the maximum shear strain (γ_{xy}) of the analysed element can be obtained as:

$$\gamma_{xy} = \varepsilon_{+45°} - \varepsilon_{-45°} \quad (3\text{-}25)$$

Theoretical analysis showed that if the centre element undergoes pure shear deformation, the +45° and -45° directions are two principal axes of the deformed element (analysed in Section 3.1). Thus, the strain gauges mounted along these directions should measure opposite values. Taheri-Behrooz et al. [178] analysed the strains obtained by two strain gauges in the VNR shear test at a small deformation degree (the shear strain is less than 2 %), found that the two strain gauges gave the same absolute value but with opposite signs, indicating that it is acceptable to simplify the deformation mode as pure shear within small deformations. However, as the deformation increases, the absolute values of the two strain gauges begin to differ due to the rotation of principal axes.

The DIC method is for strain measurement is shown in Figure 3-16b and described in Section 2.2. To use this method, a speckle grayscale distribution is created on the target deformation area of the test specimen, and images of the speckled area are captured during deformation. Image processing algorithms are then applied to obtain the strain field, and the shear strain of the specimen is calculated by averaging the strain over a defined ROI. Under the pure shear assumption, the shear strain equals the maximum principal strain in magnitude, making the maximum principal strain a commonly used metric for effective strain calculation. The size and shape of ROI are defined differently in various studies. For example, Nunes et al. [154] defined the ROI as a small central zone on the ligament, while Filho et al. [180] divided the ligament into five evenly distributed subregions. Daiyan et al. [170] defined the ROI as a circle at the centre with a radius of 1-2 % of the ligament length (31 mm for standard specimen); Totry et al. [84] defined the ROI as a rectangular at 25 mm × 2 mm in the middle of the ligament.

In summary, the selection of strain measurement method in the VNR shear test can significantly affect the accuracy of the calculated material properties. The different methods, such as strain

gauge measurements and DIC, rely on data from various deformation regions on the specimen, leading to variations in the measured strain values.

Taking into account of the limitations and errors associated with strain measurement methods, it is challenging to establish a calculation method for obtaining accurate material properties solely based on experimental data, since potential errors may vary from the strain measurement method employed. To obtain a clear understanding of the stress-strain field of the VNR shear test specimen and evaluate the calculation error, numerical simulations were conducted in a commercial software, ABAQUS, as they provide the full field stress/strain data throughout the deformation process.

3.4.4. Numerical simulations

1) Temperature-dependent Johnson-Cook (JC) constitutive model

The material constitutive model for the studied materials was defined using the effective stress-strain curve obtained from the tensile tests at 180 – 300 °C. The model consists of an elastic part followed by a nonlinear plastic flow. Taking into account the effect of temperature on plastic deformation, a temperature-dependent Johnson-Cook (JC) plastic model [100, 113] was employed and calibrated using the tensile test data from section 3.3. The mathematic form of the temperature-dependent JC model between T_g and T_m is given below:

$$\sigma = [A + B(\bar{\varepsilon}^{pl})^n]\left(1 - \left(\frac{T - T_0}{T_m - T_0}\right)^m\right), T_0 < T < T_m \tag{3-26}$$

where the flow stress, σ, is a function of the plastic strain, $\bar{\varepsilon}^{pl}$, and the temperature, T. The terms T_0 and T_m are material constants representing the controlling temperature and the melting temperature, respectively. All temperatures in this model are in absolute temperatures. The terms A, B, n, m are the material constants to be determined.

The material constants were identified by first determining the yield stress and yield strain using the method described in Section 3.3.2. Once these properties were identified, an in-house MATLAB code was developed to determine the JC model constants. The nonlinear least-squares problem was solved using the lsqnonlin function in MATLAB.

Table 3-4 lists the specific material model parameters. The experimental flow stress-plastic strain curves and the fitted model are shown in Figure 3-17.

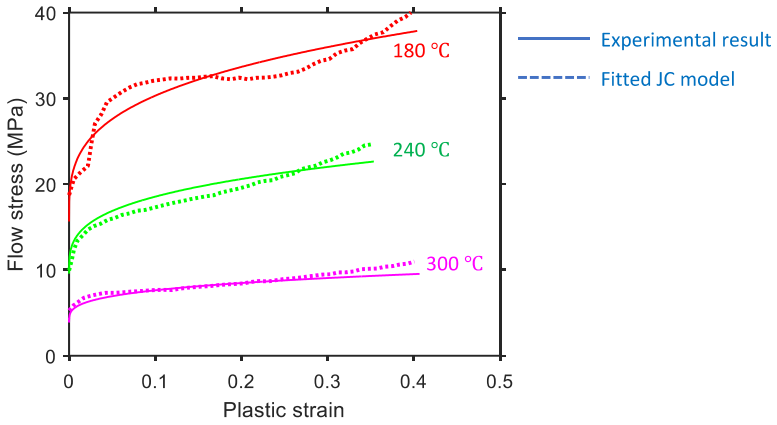


Figure 3-17. Flow stress comparison between the JC numerical modelling (solid lines) and the tensile test (dash lines) results of PEEK at different temperatures (180 – 300 °C) and at a strain rate of 0.1 /s.

2) FEA model set-up

The virtual specimen was modelled in ABAQUS as a deformable part with five integration points through the thickness of the specimens. The 4-node shell element with reduced integration (S4R) was used to define the mesh. Before finalising the model form, a mesh-sensitivity study was conducted to find the optimised mesh size. The numerical boundary condition was consistent with the experimental setup and is shown in Figure 3-18. Displacement control was applied to the sample during the simulation, with a total displacement was defined as 5 mm. The model was solved using the explicit solver in a Windows system with four CPUs. The total CPU time for completing one case was around 64 minutes.

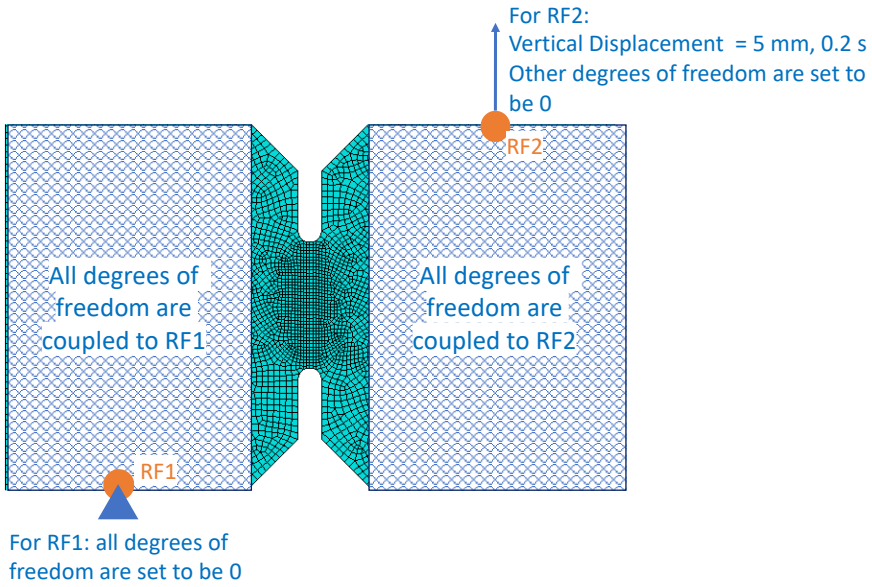


Figure 3-18. Schematic illustration of the boundary conditions of the FE model

Table 3-4. Input parameters fitted from tensile test results of PEEK at 180 – 300 ℃

Properties	Parameters
Density (tonne/mm^3)	$\rho = 1.3 \times 10^{-9}$
Modulus (MPa)	$E = 415.948$
Poisson's ratio	$\nu = 0.48$
JC model parameters	A=15.651,
	B=29.079,
	n=0.298,
	m =0.947,
	θ_0= 453K (180 ℃)
	θ_m = 616 K (343 ℃)
Predefined field temperature	453(180 ℃)

3) Model validation

The accuracy of the simulation results was verified by comparing the strain distribution and vertical load with the experimental data obtained from the VNR shear test on PEEK. The maximum principal strain distribution obtained from the numerical simulation, was compared with the DIC analysis results at four different deformation stages, as shown in Figure 3-19a. The deformation stages were selected based on a coordinate analysis of the experimental and simulation results, with an interval of approximately 0.7 mm. The position on the ligament of

the specimen was represented by a relative value from 0 to 1, where 0 corresponded to the lower end and 1 to the upper end.

At a specific stage of displacement, 1.98 mm, the strain fields measured from the experiment and simulation were compared and presented in Figure 3-19b and c, respectively. Furthermore, the vertical loads obtained from the experiment and simulation were compared and shown in Figure 3-19d. The comparison between the experimental and numerical results indicated that the strain fields of the numerical results (solid lines) and the experimental results (dotted lines) were in good agreement, and the vertical loads of both the experimental and numerical results were also consistent. These findings demonstrated the validity of the numerical simulation results.

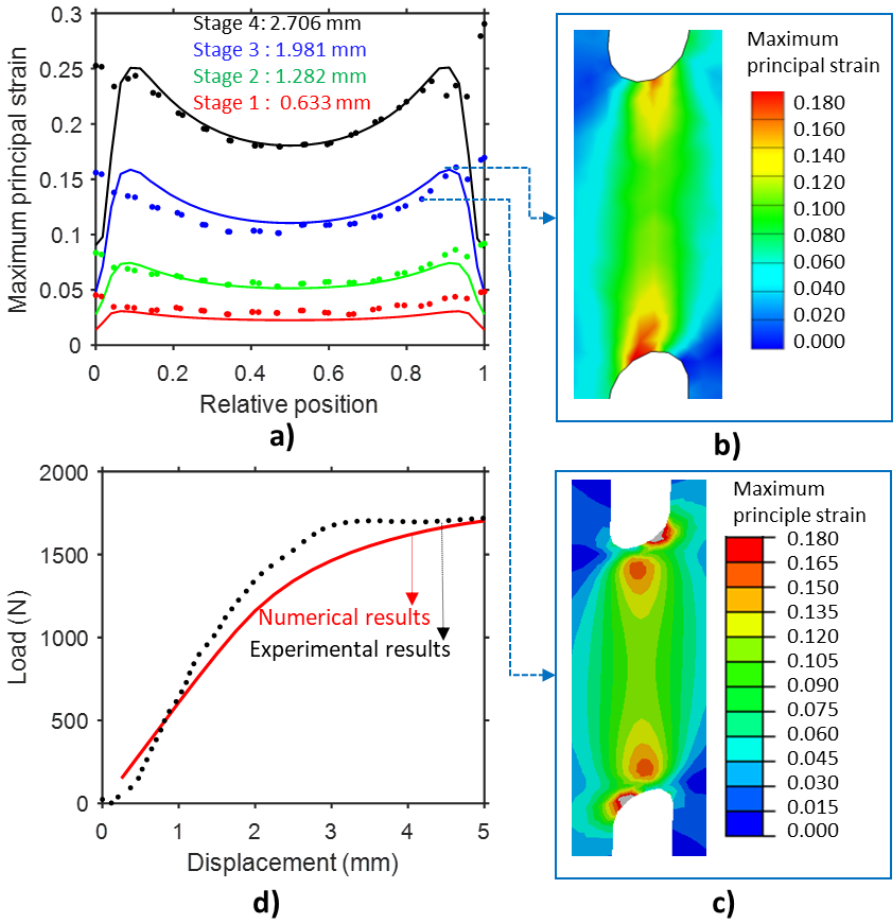


Figure 3-19. Validation of the numerical model by comparing a) strain distribution along the ligament at four displacement stages (Solid lines are the numerical results; Dot lines are the experimental results); strain fields of b) experimental and c) numerical results at Stage 3; d) vertical loads obtained from experimental and numerical results.

3.4.5. Analysis of the numerical simulation results

1) Analysis of the deformation state of tested materials

The forming limit diagram is a widely used tool for analysing the deformation behaviour of materials. Based on the strain field observed on the specimen in Figure 3-20a, it can be seen that the strain localisation occurs along the ligament. This suggests that at least two representative nodes should be selected for analysis: Node 1 is the strain-localised point near the upper end, and Node 2 is the central point of the ligament. The maximum and minimum principal strains of these two nodes at different deformation stages are presented in Figure 3-20b. The green reference line with a slope of -1 represents the strain state under pure shear deformation. It can be observed that Node 2 remains close to this reference line during the deformation process, indicating a nearly pure shear state. However, Node 1, deviates from the reference line (green line) when the minimum principal strain reaches approximately 0.25. This deviation suggests that Node 1 experiences additional deformation states beyond pure shear. Therefore, the central point (Node 2) is appropriate to be selected for analysing the actual material properties under shear loading.

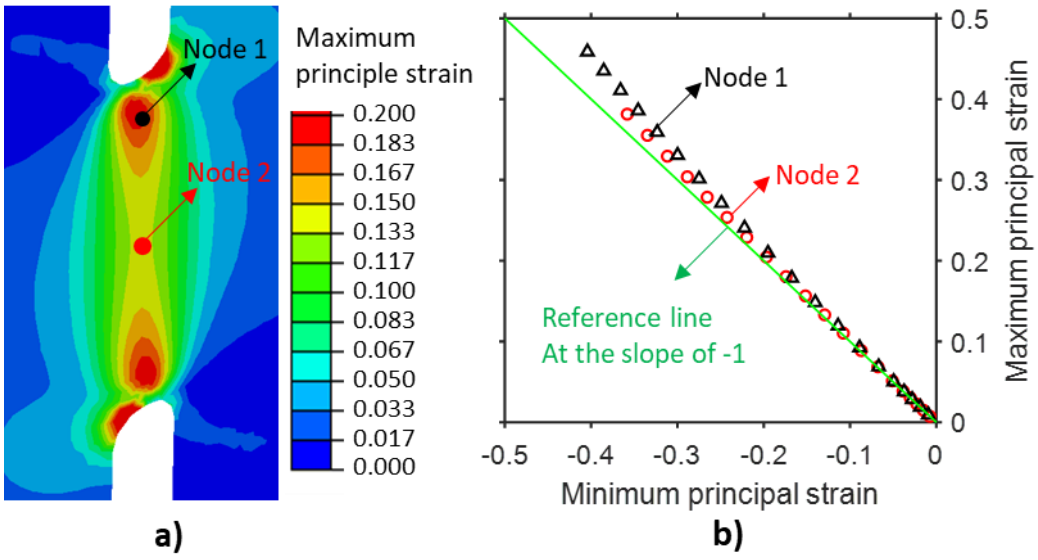


Figure 3-20. The deformation state of two representing nodes, Node 1 and Node 2, on the test specimen: a) illustration of the analysed nodes, and b) deformation state of the two analysed nodes in the forming limit diagram.

2) The analytical ground truth and error source

From the numerical simulation results, the strain and stress state of each node can be extracted for further analysis. Specifically, for the central point, three principal strains and three principal stresses were extracted against each deformation stage to obtain the effective stress-strain curves. The effective strains ($\bar{\varepsilon}_c$) and effective stresses ($\bar{\sigma}_c$) of the central point were then calculated using Eqs.(3-6) and (3-7) , respectively.

As the input constitutive JC model provides the ground truth for the material property, a comparison was made between the effective stress-strain curve of the central point and the input model data. Figure 3-21 shows that the effective stress-strain curve of the central point agrees well with the input constitutive JC model. Thus, the properties extracted from the central point of the ligament can be used as the ground truth property of the studied material. As mentioned previously, it is important to consider the potential errors associated with the measurement methods used for the material characterisation.

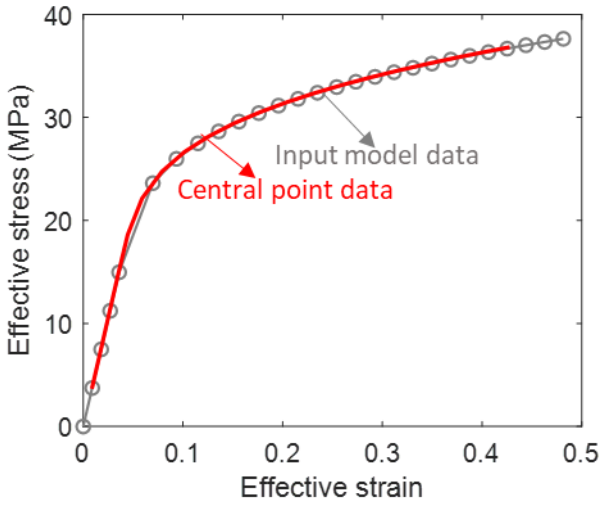


Figure 3-21. The ground truth validation: effective stress-strain curves of the central point (Node 2 in Figure 3-20) and the input constitutive model (the JC model).

In the case of experiments that rely on strain gauge measurement, as explained by Eq.(3-25), the measured strain corresponds to the maximum shear strain of the central point ($\gamma_{m,c}$), which is equivalent to the maximum principal strain under the assumption of pure shear. To obtain the effective strain (ε_{SG}) and effective stress (σ_{SG}) using this method, the steps listed in

Table 3-3 are followed:

$$\varepsilon_{SG} = \frac{2}{\sqrt{3}} \cdot \gamma_{m,c} \tag{3-27}$$

$$\sigma_{SG} = \sqrt{3}\frac{F}{A_0}\left(1 + \frac{2}{\sqrt{3}}(e^{\varepsilon_{SG}} - 1)\right) \tag{3-28}$$

where F is the measured vertical load. The term A_0 is the initial area of the cross-section along the ligament of the specimen. The term $\left(1 + 2(e^{\varepsilon_{SG}} - 1)/\sqrt{3}\right)$ in Eq. (3-28) is the process of converting the engineering stress to true stress.

In the case of experiments that rely on DIC measurement, the ROI is defined by the central section, which is a node set that composed of all the nodes along the central line of the ligament, same to the case that explained in Section 3.4.2. The effective strain (ε_{DIC}) and effective stress (σ_{DIC}) are calculated as follows:

$$\varepsilon_{DIC} = \frac{2}{\sqrt{3}}\bar{\varepsilon}_{ROI} \tag{3-29}$$

$$\sigma_{DIC} = \sqrt{3}\frac{F}{A_0}\left(1 + \frac{2}{\sqrt{3}}(e^{\varepsilon_{DIC}} - 1)\right) \tag{3-30}$$

where $\bar{\varepsilon}_{ROI}$ is the arithmetic mean of maximum principal strains of all nodes in the ROI. The term $\left(1 + 2(e^{\varepsilon_{DIC}} - 1)/\sqrt{3}\right)$ in Eq. (3-30) is also the process of converting the engineering stress to true stress.

The comparison of the effective strains and effective stresses obtained from virtual measurement by strain gauges and DIC with the ground truth data is presented in Figure 3-22. The results show that the measured strains from both strain gauges and DIC are in good agreement with the ground truth data. However, the calculated stresses from both measurement methods differ significantly from the ground truth data. As indicated in Figure 3-22b, the calculation errors become non-negligible when the effective strain exceeds 0.102, with errors of both measurement methods, i.e., ξ_{SG} and ξ_{DIC}, are observed as around 12.3 %. Therefore, a correction method is required to accurately calculate the effective stress at large deformations in VNR shear tests.

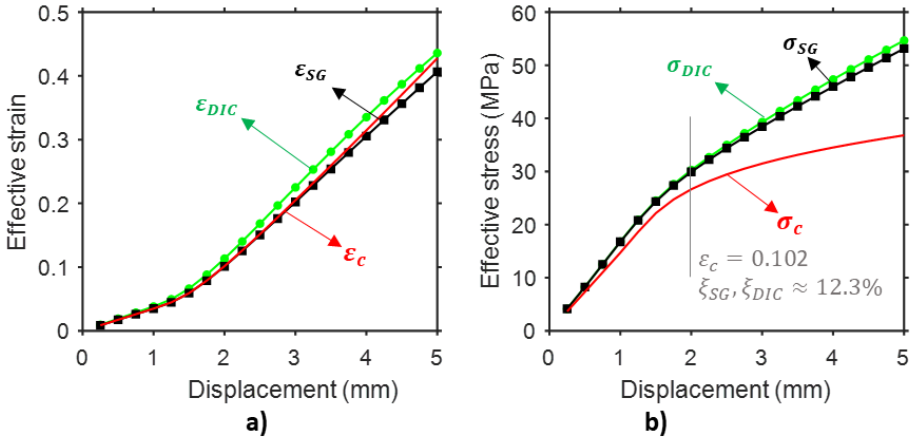


Figure 3-22. Comparison between the results of strain gauge and DIC measurements and the ground truth of material properties: a) effective strains and b) effective stresses.

3.4.6. A new two-step correction method

As explained in Section 3.4.3. , the errors of characterisation results by simple shear tests come from two main factors: the rotation factor and the nonuniformity factor. Furthermore, Figure 3-22 has indicated that the main error is introduced through the calculation of effective stress. To address this issue, this section proposes a two-step correction method to correct the two factors that caused the calculation error of effective stress characterised by the VNR shear test. The detailed process is shown in Figure 3-23. And the detailed calculation variables are listed in Table 3-5.

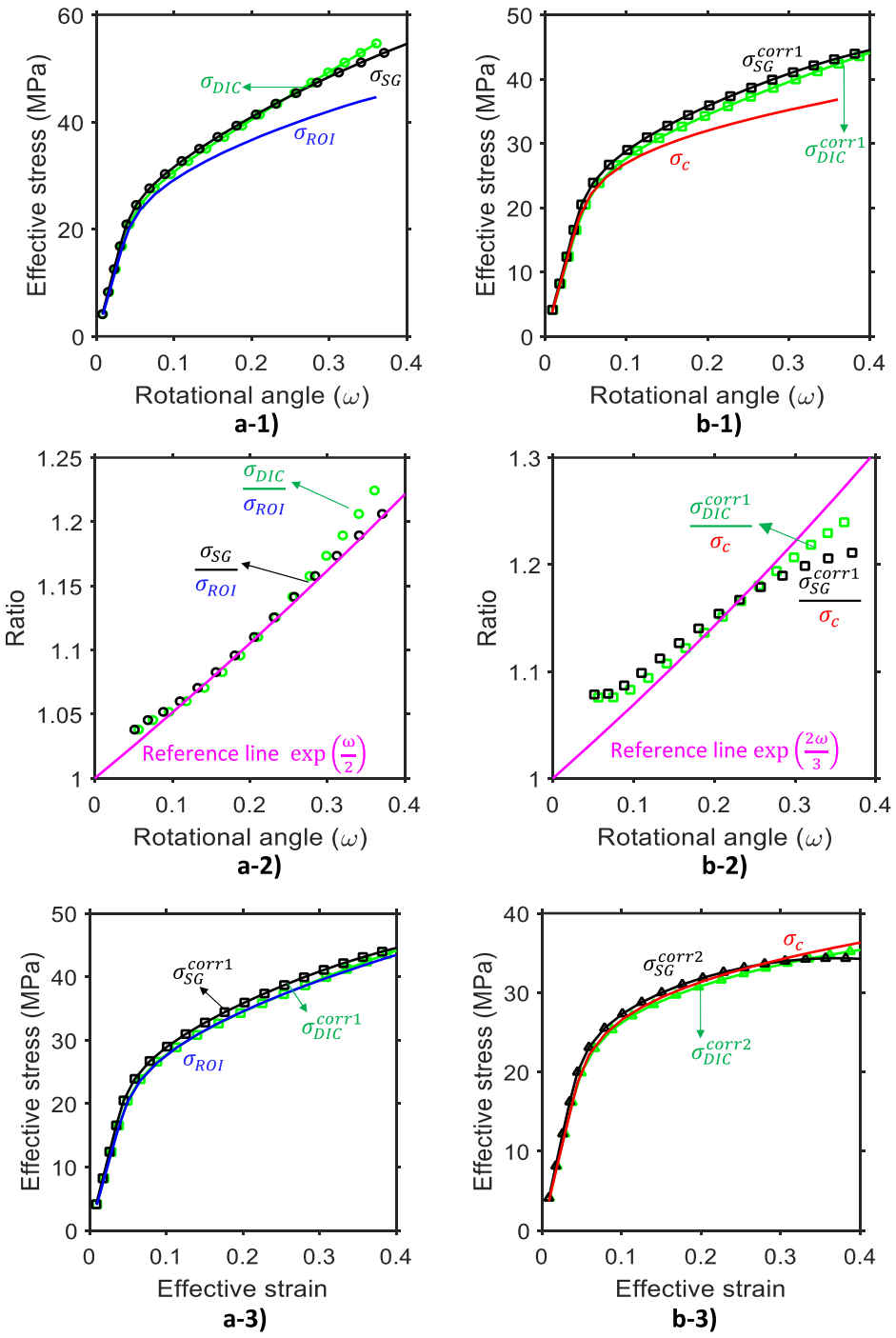


Figure 3-23. Two-step correction for the effective stress-strain curves. a) The first correction step: a)-1, The relationship between effective stresses and rotational angle; a)-2, The relationship between the calculation errors and rotational angle; a)-3, The first step correction effect; b) The second correction step: b)-1, The relationship between effective stresses and rotational angle; b)-2, The relationship between the calculation errors and rotational angle; b)-3, The second step correction effect.

The first step involves correcting the rotational factor without considering the non-uniformity factor. To achieve this, the target curve for the first correction step should account for the load contributions from all nodes in the ROI. Therefore, the first step aims to convert the initial calculations into the effective stress that takes into account every single node in the ROI by building a virtual state as the target curve. The effective strain (ε_{ROI}) of the target curve is defined by the averaged maximum principal strains ($\bar{\varepsilon}_{m,ROI}$) of all nodes in the ROI, while the effective stress (σ_{ROI}) of the target curve is defined by the averaged value of maximum principal stresses ($\bar{\sigma}_{m,ROI}$) of all the nodes in the ROI.

To achieve this goal, the first step is to establish the relationship between the errors and the rigid body rotational angles shown Figure 3-23a)-1 and a)-2. The rotational angle can be calculated geometrically from the maximum shear strain, and is also illustrated as $\boldsymbol{\theta_s^{od}}$ in Figure 3-15. The calculation of the rotation angle for the target curve, using strain gauges or DIC results is listed in Figure 3-23 as ω_{lg}, ω_{SG}, ω_{DIC}, respectively. The differences between ω_{SG} and ω_{DIC} arises from the available data of the two different measurement methods. For the strain gauge measurement, only one set of data is obtained ($\gamma_{m,c}$), whereas, for the DIC measurement, the strain data of all nodes inside the ROI ($\gamma_{m,ROI}$) are utilised. The calculation errors for both sets of data should be equal to the ratio of the initial calculated effective stresses (σ_{SG}, σ_{DIC}) and the target effective stresses (σ_{ROI}). The errors related to the rotational angle are illustrated in Figure 3-23a)-2. The correction factor regarding the rotation influence is proposed as the reference line. The results after the correction on the rotation factor are shown in Figure 3-23a)-3. It is obvious that the rotation correction factor works well for both measurement methods.

The second step involves correcting the non-uniformity error, with an objective of adjusting the effective stresses after the first step correction (σ_{SG}^{corr1} and σ_{DIC}^{corr1}) to match the ground truth of the material properties (σ_c and ε_c), represented by the target curve. This step is presented in Figure 3-23b)-1 and b)-2, where the rotational angle for the target curve (ω_c) is calculated using the maximum shear strain of the central point ($\gamma_{m,c}$). Similar to the first step correction, the non-uniformity correction factor is obtained by taking the ratio of the present effective stress (σ_{SG}^{corr1} and σ_{DIC}^{corr1}) and the target effective stress (σ_c). The correction factor corresponding to the non-uniform deformation is shown as the reference line in Figure 3-23b)-2, and is calculated via f_{SG}^2 or f_{DIC}^2. The results of the correction on the non-uniformity factor are shown in Figure

3-23b)-3, which demonstrates that the effective stress-strain curve of the VNR shear test can be corrected to the ground truth of the material properties using the two-step correction based on the experimentally measured data, regardless of the measurement methods.

Table 3-5. Details of the two-step correction method

Variables	The target curve	Virtual results of using strain gauges (SG)	Virtual results of using DIC
The first step correction (rotational factor)			
Rotational angle	$\omega_{tg} = \arctan\left(\frac{\tan(\bar{\gamma}_{m,ROI})}{2}\right)$; $\omega_{SG} = \arctan\left(\frac{\tan(\gamma_{m,c})}{2}\right)$; $\omega_{DIC} = \arctan\left(\frac{\tan(\bar{\gamma}_{m,ROI})}{2}\right)$ $\bar{\gamma}_{m-ROI}$ refers to the averaged maximum shear strain values of all the nodes in the ROI; and γ_{m-c} refers to the maximum shear strain values of the central point.		
Effective strain	$\varepsilon_{ROI} = \frac{2}{\sqrt{3}} \cdot \bar{\varepsilon}_{ROI}$	ε_{SG} (Eq. (3-27))	ε_{DIC} (Eq.(3-29))
Effective stress before correction	$\sigma_{ROI} = \sqrt{3} \cdot \bar{\sigma}_{ROI}$	σ_{SG} (Eq.(3-28))	σ_{DIC} (Eq.(3-30))
The first correct factor	N/A	$f_{SG}^{1} = e^{\frac{\omega_{SG}}{2}}$	$f_{DIC}^{1} = e^{\frac{\omega_{DIC}}{2}}$
Effective stress after the first step correction	N/A	$\sigma_{SG}^{corr1} = \frac{\sigma_{SG}}{f_{SG}^{1}}$	$\sigma_{DIC}^{corr1} = \frac{\sigma_{DIC}}{f_{DIC}^{1}}$
The second step correction (non-uniformity factor)			
Rotational angle	$\omega_c = \arctan\left(\frac{\tan(\gamma_{m,c})}{2}\right)$	ω_{SG}	ω_{DIC}
Effective strain	ε_c (Eq.(3-6))	ε_{SG}	ε_{DIC}
Effective stress before correction	σ_c (Eq.(3-7))	σ_{SG}^{corr1}	σ_{DIC}^{corr1}
The second correct factor	N/A	$f_{SG}^{2} = e^{\frac{2\cdot\omega_{SG}}{3}}$	$f_{DIC}^{2} = e^{\frac{2\cdot\omega_{DIC}}{3}}$
Effective stress after the second step correction	N/A	$\sigma_{SG}^{corr2} = \frac{\sigma_{SG}^{corr1}}{f_{SG}^{2}}$	$\sigma_{DIC}^{corr2} = \frac{\sigma_{DIC}^{corr1}}{f_{DIC}^{2}}$

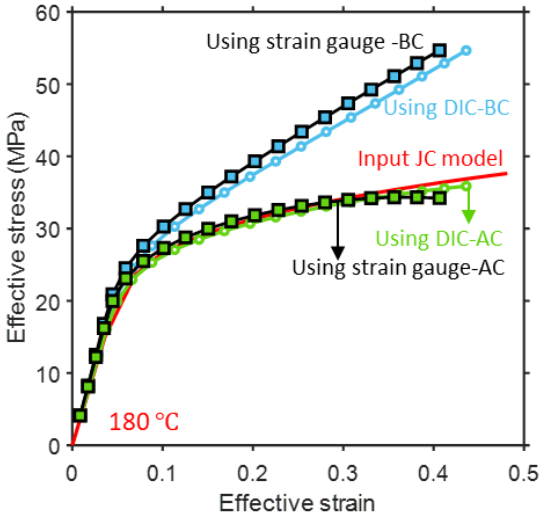
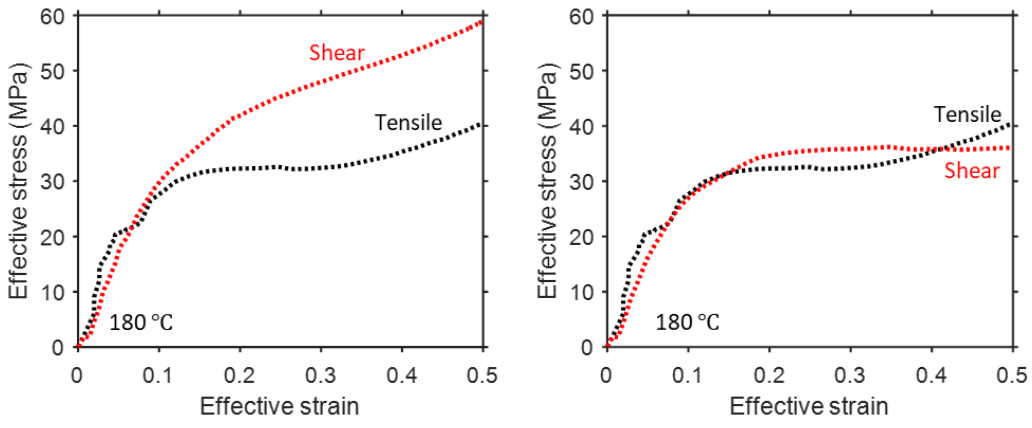


Figure 3-24. Comparison of the effective stress-strain curves of strain gauge and DIC methods before and after the two-step correction. (BC means before correction, and AC means after correction).

The above correction process is analysed and calculated according to numerical results. As the initial problem indicated in Figure 3-13, the effective stress-strain curves calculated from the tensile and the VNR shear test data are considerably different without the correction procedure. Therefore, by adopting the two-step correction method proposed above, the shear test results can be corrected. The effective stress-strain curves under tensile and the VNR shear tests on PEEK at 180 ℃ is shown in Figure 3-25. Compared to Figure 3-13, the characterised material properties after corrections show more reliability.



a) b)

Figure 3-25. The effective stress-strain curves of the VNR shear tests a) before and b) after correction on PEEK at a temperature of 180 ℃ and at a strain rate of 0.1 /s.

3.4.7. Validation of the developed correction method

To further validate the correction method, the model constants fitted from the corrected and uncorrected VNR shear test data were entered into ABAQUS, to show the detailed difference of the final material properties before and after correction. Figure 3-26 shows the JC model fitting results using the experimental data before and after correction. The fitted material parameters are listed in Table 3-6. The most notable difference is shown by the n parameter in JC model, representing the strain hardening exponent, which is also bolded. Without the correction, the strain hardening exponent, n, might be more than twice of the real value. The numerical simulation setup is the same as the previous setup.

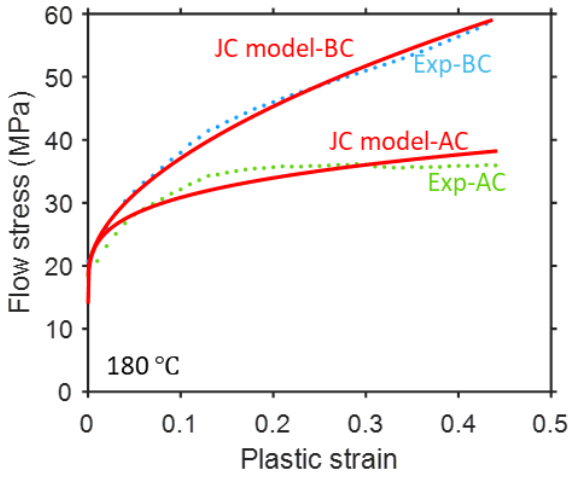


Figure 3-26. Flow stress-plastic strain curves obtained from the JC model predicted results and the VNR shear experimental results at a temperature of 180 °C and a strain rate of 0.1/s. (BC means before correction, AC means after correction. Exp refers to the post-processed experimental results)

Table 3-6. Input parameters fitted from the VNR test results of PEEK at 180 °C, before and after the correction

Properties	Parameters before correction	Parameters after correction
Density (tonne/mm3)	$\rho = 1.3 \times 10^{-9}$	$\rho = 1.3 \times 10^{-9}$
Modulus (MPa)	$E = 372.314$	$E = 312.559$
Poisson's ratio	$\nu = 0.48$	$\nu = 0.48$

JC model parameters	A=18.517, B=62.864, **n=0.530,** m =0.652, θ_0= 453K (180 ℃) θ_m = 616 K (343 ℃)	A=14.032, B=29.569, **n=0.245,** m =0.675, θ_0= 453K (180 ℃) θ_m = 616 K (343 ℃)
Predefined field temperature	453 K (180 ℃)	453 K (180 ℃)

The simulation results regarding these two sets of model input parameters were analysed and compared with the experimental results obtained from the DIC measurement. Similar to the previous analysis, the maximum principal strain distribution along the ligament was analysed at four displacement stages shown in Figure 3-27a and b, respectively. Under small deformation, i.e., when the displacement is lower than that defined in the first deformation stage (the effective strain is around 0.093 at this stage), the strain distributions are similar regardless of the model is fitted for uncorrected or corrected test data. However, with the deformation increases, the model output begins to deflect the experimental results with the model fitted using the uncorrected testing data. Besides, the results show that the extent of deflection increases with the increasing deformation. In contrast, with the model fitted from the corrected testing data, the predicted strain fields comply with the experimental results very well.

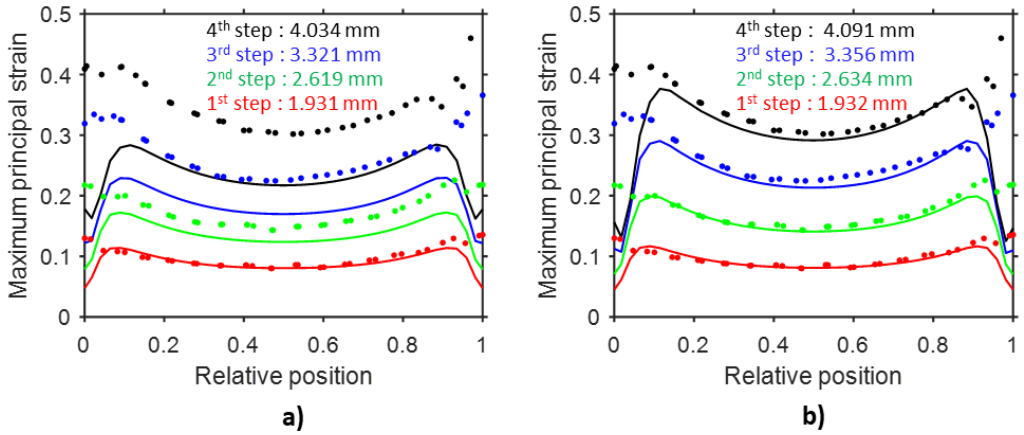


Figure 3-27. Comparison between experimental and numerical results using shear test results as model input at 180 ℃: strain field comparison at four displacement stages using a) model fitted using uncorrected shear test data and b) model fitted using corrected shear test data. (The solid lines are the numerical results, and the dot lines are the experimental results.)

The above analysis and validation are all based on the material properties of PEEK at 180 ℃. To further validate the universality of the two-step correction method on the post-processing of the VNR shear test results against other conditions and even on different materials, another two groups of simulation were conducted: 1) the PEEK material at 240 ℃; 2) the PA6 material at 160 ℃. The JC model parameters of PEEK are the same as listed in Table 3-4. The JC model constants of PA6 were also calibrated by the tensile test results, and listed in Table 3-7.

Table 3-7. Input parameters fitted from tensile test results of PA6 at 160 ℃

Properties	Parameters of PA6 at 160 ℃
Density (tonne/mm^3)	$\rho = 1.5 \times 10^{-9}$
Modulus (MPa)	$E = 180.13$
Poisson's ratio	$\nu = 0.48$
JC model parameters	A=11.489, B=14.531, n=0.729, m =1.079, θ_0= 433K (160 ℃) θ_m = 503 K (230 ℃)
Predefined field temperature	433 K (160 ℃)

Figure 3-28 presents a comparison of the effective stress-strain curves of the two validation simulations before and after correction. Similar to Figure 3-27, the effective stress-strain curves without correction always deflect from the ground truth at large deformation, regardless of the strain measurement methods. However, by adopting the two-step correction method during post-processing of the VNR shear test data, the correct material effective stress-strain properties can be obtained, regardless of different conditions or materials. As shown in Figure 3-28, the corrected effective stress-strain curves align well with the ground truth, indicating the effectiveness of the proposed correction method.

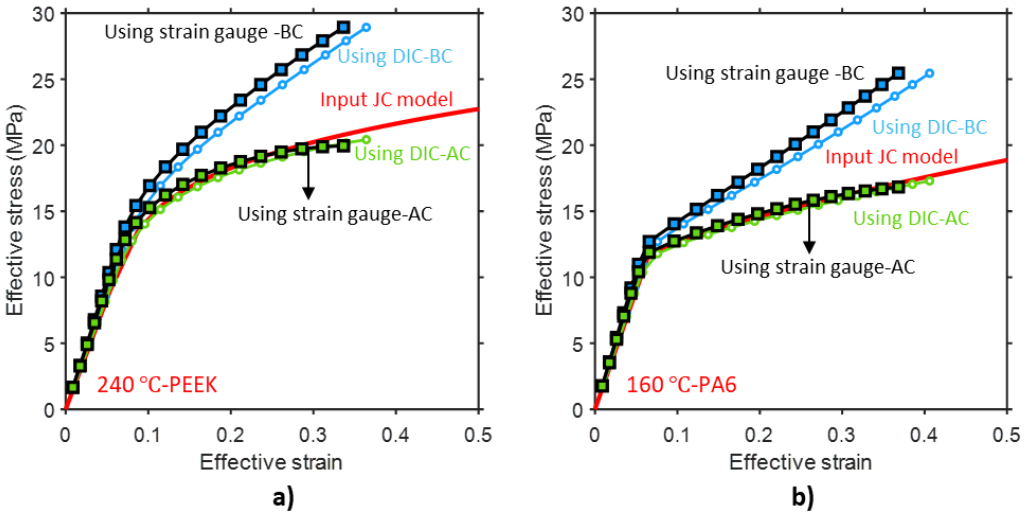


Figure 3-28. Verification of the correction method by a) changing the test condition of the same material (PEEK); and b) changing the material (PA6). (BC means before correction, and AC means after correction).

3.5. Summary

This chapter presents experimental studies conducted to investigate the thermomechanical properties of two commonly used semicrystalline thermoplastics, namely PA6 and PEEK, under solid-state stamp forming conditions. The objective is to gain a better understanding of their deformation properties. The effects of temperature and strain rate on the thermomechanical properties of these thermoplastics are thoroughly analysed. Additionally, an alternative method for characterising the mechanical properties of polymers, the VNR shear test method, is also investigated. This method is evaluated in terms of its experimental procedure, deformation mechanism, potential sources of errors, and correction methods. The correction method is also validated through numerical simulations.

Overall, the findings of this study contribute to our understanding of the deformation behaviours of semicrystalline thermoplastics under solid-state stamp forming conditions and provide valuable insights into the potential use of the VNR shear test method for characterising the mechanical properties of polymers.

CHAPTER 4 Investigation of the Thermomechanical Properties of UD FRTPs

*This Chapter is reproduced from publication **A2** in the Publication list.

4.1. Introduction

This chapter aims for fabricating UD FRTP laminates, specifically, UD CF/PA6 laminates from prepregs with satisfying quality and characterising the thermo-mechanical properties of CF/PA6 laminates. To achieve this, a hot press tool is designed and manufactured. To overcome the resin overflow during consolidation, three consolidation strategies are designed, and their produced laminates are presented and compared. Eventually the CF/PA6 laminates are fabricated with satisfying qualities. After the fabrication, the transverse tensile and longitudinal shear experiments are conducted for characterising the in-plane tensile and shear properties of UD CF/PA6, to provide a mesoscale in-plane deformation properties of UD CF/PA6 under the solid-state stamp forming conditions. After the experimental studies, the temperature and strain rate effects on the in-plane deformation properties are analysed. With the thermomechanical properties of UD CF/PA6 obtained, guidelines to the processing windows of the solid-state stamp forming process can be built and further numerical studies can be conducted.

4.2. Fabrication of UD FRTP laminates

4.2.1. Hot press and moulding rig design

For consolidating the UD CF/PA prepregs into laminates, a set of hot press that can provide sufficient heat while maintaining a certain level of pressure is required. A bespoken hot press was designed and manufactured for this purpose. Figure 4-1 shows a picture of the hot press that installed on a biaxial machine. The vertical axis in the biaxial machine was employed, to provide the vertical pressure when consolidating the laminate. As illustrated in Figure 4-1, there are two decks in the hot press, serve for heating and cooling functions, respectively. The heating function is provided by eight heating cartridges on each heating plate. Insulation cardboards are

inserted between the heating plates and their contacted plates, to minimise the heat that transferred to other parts of the system. For the CF/PA6 fabrication, the cooling deck is proved to be efficient enough by utilising the steel plates itself. A power control box is also designed and built as part of the heating system, as shown in Figure 4-1. The high voltage section distributes a three-phase inlet power to each single cartridge, with circuit breaker safety measures deployed. The low voltage section utilised an Arduino unit to provide feedback temperature control for a steady temperature profile. The temperature signal, measured by the thermocouples built in the heating plates, is fed to the Arduino and a self-developed Proportional-Integral-Derivative (PID) algorithm computes the duty cycle in real time.

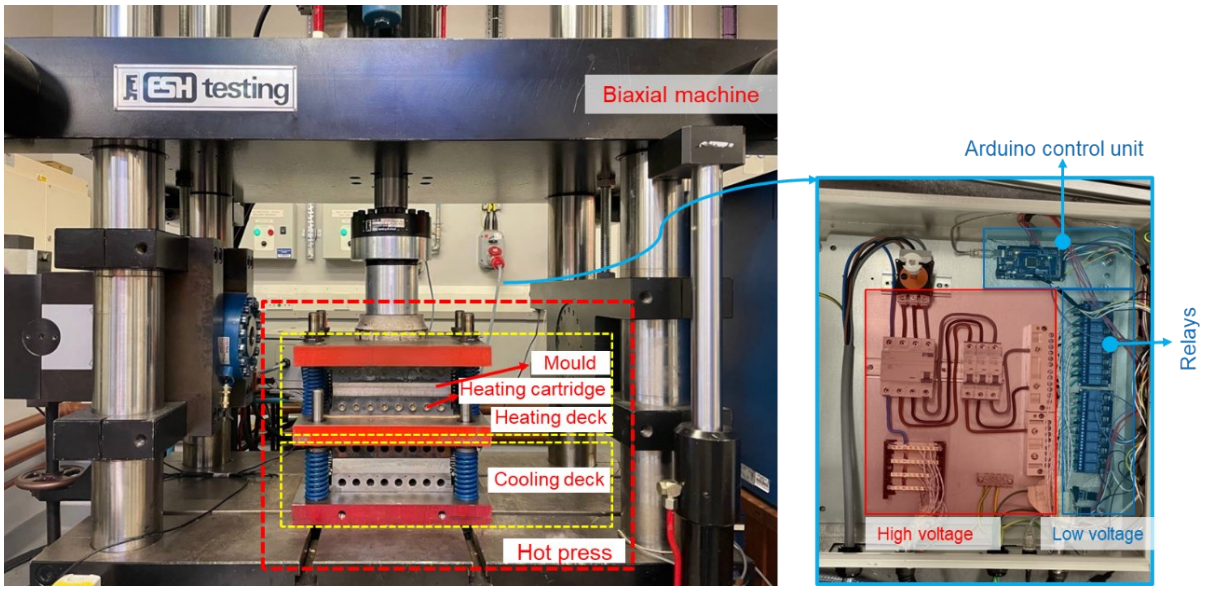


Figure 4-1. The hot press mounted on the biaxial machine, and its electrical unit for controlling the heating function

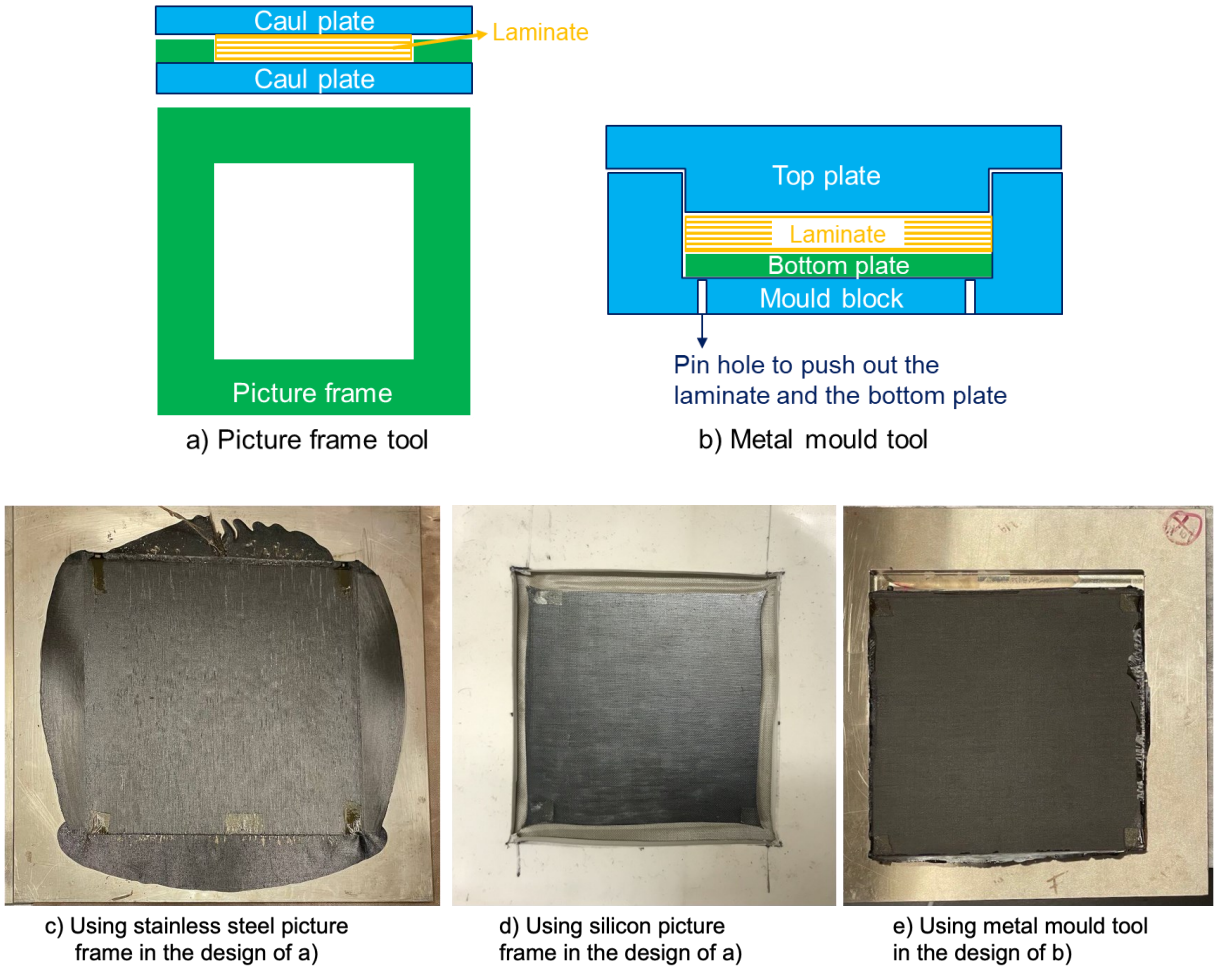


Figure 4-2. The two designs of consolidation tool: a) Picture frame tool and b) Metal mould tool. The example pictures of fabricated laminates via different consolidation tools (c-e).

For the laminate fabrication, a pair of mould is required to restrain the resin from overflowing. A few options were explored in this research, as illustrated in Figure 4-2. Many researchers [181], and the material supplier [182], recommended using picture frame tool (Figure 4-2a) to consolidate laminates. The frame, that normally made of steels, is supposed to constrain the resin from flowing away. However, in the preliminary consolidation trials using stainless steel picture frame, it was observed that, a considerable amount of overflow occurred as shown in Figure 4-2c. Later, to improve this overflow, the silicon picture frame was employed, borrowing its ability in elastic compression. The overflow issue was effectively solved by the silicon picture frame, yet the consolidated laminate showed distorted edges as shown in Figure 4-2d, which could impact the mechanical properties of the laminate.

Facing the problems occurred with the picture frame tool, another set of metal mould was proposed and manufactured, as illustrated in Figure 4-2b. In this newly designed metal mould tool, the prepregs are fully closed in a cavity that is designed according to the constant volume, so that even when the prepregs are under melt state, very small amount of resin can be squeezed out. Besides, it is assured that load can be effectively applied on the laminate during the whole consolidation process. Four pinholes are added on the mould block, for the purpose of pushing the bottom plate and laminate out after consolidation. After the improvement, the consolidated laminate is shown as Figure 4-2e, and it is obvious that there is only ignorable amount of overflow.

4.2.2. Laminate fabrication and quality assessment

The material studied in this thesis is Toray Cetex® TC910 carbon fibre reinforced UD tapes using PA6 matrix purchased from Toray Advanced Composites. The PA6 resin content by weight is 40 %. The T_g and T_m of the PA6 polymer are 60 ℃ and 233 ℃, respectively [183]. The UD tapes are consolidated at a configuration of $[0]_{24}$. The consolidation temperature is 250 ± 10 ℃. At the heating-up stage, the consolidation pressure is 2 bar, and the pressure is set to 4 bar for 20 minutes when the target temperature is reached. After 20 minutes of holding, the laminate is cooled down to 80 ℃ under 4 bar. The initial thickness of one layer of prepreg is 0.16 mm. After consolidation, the formed laminates are at a thickness of 3.3 mm.

In order to confirm the fabrication parameters used in this research, optical microscopic observation was conducted on several trial laminates. For the preparation of optical microscopic observation, the samples are firstly cut, grinded and polished via a semi-automatic polisher. The detailed grinding and polishing process are listed as Table 4-1. After the sample preparation, example pictures of microscopic observation of the CF/PA6 prepreg and consolidated CF/PA6 laminate at a configuration of $[0]_{24}$ are shown in Figure 4-3. As can be observed, the laminated CF/PA6 sample conserves the original microstructural features of the PA6, maintaining the layer boundaries. And meanwhile, there was no voids or delamination observed. This microstructural structural observation results proved the effectiveness of the laminate fabrication parameters.

Table 4-1. The grinding and polishing process for UD CF/PA6

Step	Sandpaper/cloth	Solution/ suspension	Load	RPM/direction	Time
Grinding-> wash with water-> dry with hot air					
1	P400	Water	7N	150/ Complementary	2.5 min
2	P600	Water	10N	150/ Complementary	1.5 min
3	P1200	Water	10N	150/ Complementary	1.5 min
4	P2500	Water	10N	150/ Complementary	1.5 min
Polishing -> wash with water -> dry with hot air -> ultrasonic wash -> wash with water -> dry with hot air					
6	Plano cloth	3 μm diamond sus.	5N	150/ Complementary	5 min
7	Plano cloth	0.3 alumina sus.	5N	150/ Complementary	5 min
8	Alpha cloth	0.04 μm OPUS sus.	5N	150/ Complementary	10 min

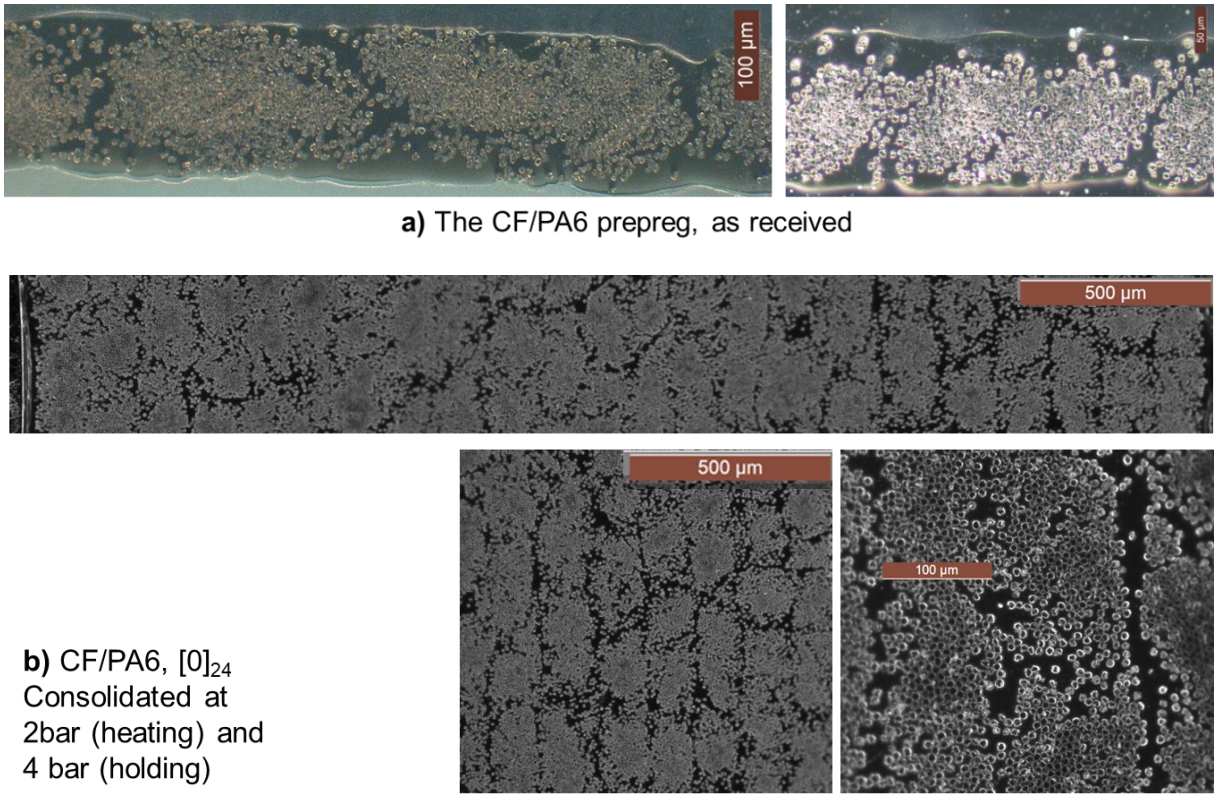


Figure 4-3. Microstructural observation of the laminate quality

Since the optical microscopic observation is a destructive quality checking method, after the fabrication parameter confirmation, a portable 'Prisma TOFD' ultrasonic C-scan device supplied by Sonatest Ltd. UK [184] was adopted to confirm the quality of all the laminates. All laminates used for the following studies are these defects-free ones.

4.3. Characterisation of the transverse tensile and longitudinal shear properties of UD FRTPs

As introduced in Section 2.4. the most significant intra-ply deformation mechanisms are transverse tension and longitudinal shear, and it is essential to characterise the thermomechanical properties for further understanding and analysis. It is believed that the VNR shear test can give good experimental results for UD FRTPs; and the traditional tensile test could also be used for characterising the transverse tensile properties on the condition of solving the problem of thermal expansion of metal fixtures. Both the VNR shear test and the transverse tensile test should be able to be conducted under solid-state stamp forming conditions.

4.3.1. Specimen design

The dimensions for the transverse tension and longitudinal shear specimen are show in Figure 4-4. The transverse tensile specimen was designed according to the aspect ratio of the dog-bone specimen described in ASTM 638 [185, 186]. With the consideration of the total length of the environmental chamber during test, the dimensions are scaled down accordingly. The detailed dimensions of the transverse tensile specimen are presented in Figure 4-4a with a 50.00 mm uniform region. The gauge length is defined as 50.00 mm as the deformation was observed to be evenly distributed without necking along the uniform area. As for the VNR shear specimen, the initial design was according to the original ASTM 7078 standard [83]. In the initial a few tests, however, it was difficult to concentrate stress along the ligament, resulting in poor repeatability. Therefore, a modification was made by adding two notches with a width of 1.5 mm (which was determined by the width of the water jet cutter), and the final specimen design is shown in Figure 4-4b. The ligament length of the VNR shear specimen after modification was 24.60 mm. All the experimental specimens were pre-dried at 70 °C for 7 hours to minimise deconsolidation during testing [112]. After the pre-drying, these specimens were stored in a dry environment and tested within 24 hours.

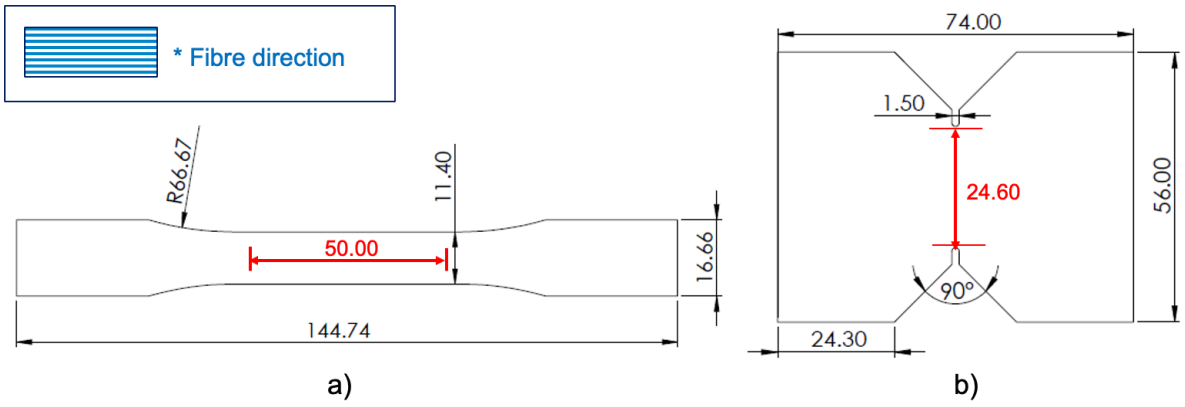


Figure 4-4. Design of experimental specimen for a) transverse tensile and b) VNR shear tests. (Unit: mm)

4.3.2. Experimental setup

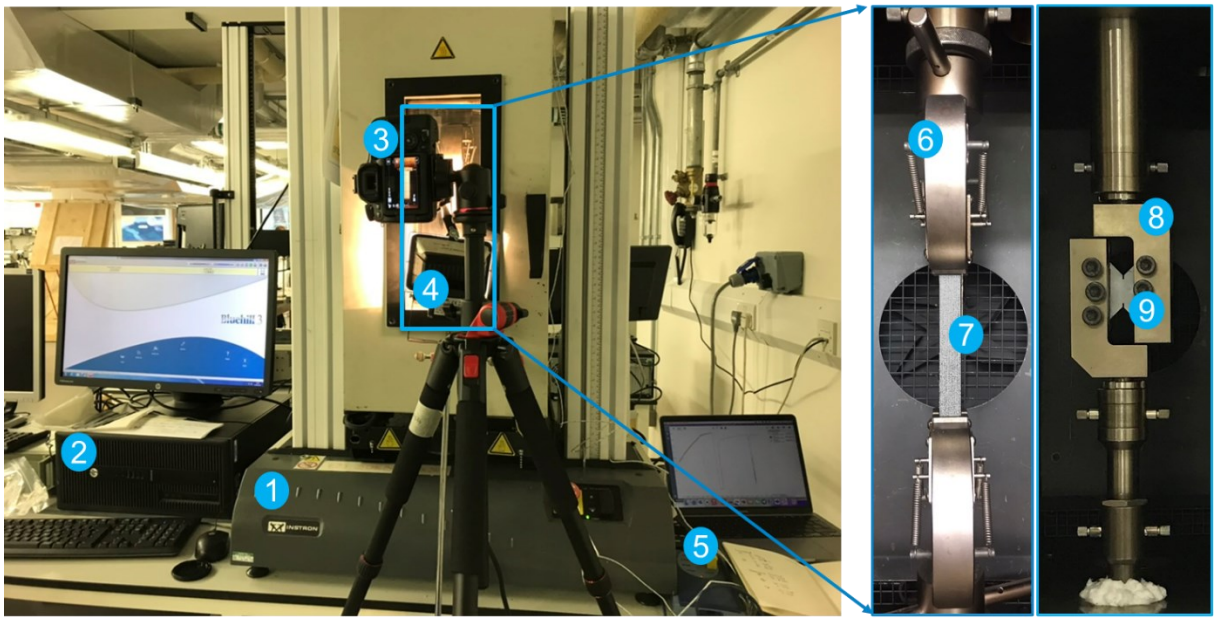


Figure 4-5. Experimental setup. ①Instron 3369 tensile testing machine with environmental chamber; ②PC and BLUEHILL software; ③Camera; ④Lamp; ⑤Thermocouples and datalogger; ⑥Tensile test rig; ⑦Transverse tensile test specimen; ⑧VNR shear test rig; ⑨VNR shear test specimen.

Figure 4-5 shows the experimental setup on an Instron 3369 tensile testing machine controlled by the BLUEHILL software. The components and setup for this experiment is similar to the experimental setup in Section 3.2. . The test rig for the transverse tensile test was the Instron mechanical clamp, and for the VNR shear test was a pair of lightweight clamps bespoken with titanium alloy. For each test, a heating-up procedure was strictly followed to minimise the potential compression that comes from the thermal expansion of the specimen and tools. Firstly, the environmental chamber together with the metal clamps was heated up to the desired temperature without the specimen. Then, the specimen was set up by clamping only one side. After that, the specimen was heated up to the desired temperature and then soaked for 10 minutes to achieve a uniform temperature. Later, the door of the environmental chamber was opened, and the other side of the specimen was quickly tightened. Following that, the door of the environmental chamber was closed again to heat the whole system. After the specimen was heated up to the desired temperature and soaked for another 20 minutes, the experiment started. Through this process, the compression load was observed to be no more than 20 N.

The test conditions employed in this study were determined by the indications from previous studies on PA6 in Section 3.3. For each test condition, three tests were conducted to check the repeatability. The detailed test matrices for transverse tensile and VNR shear tests are listed in Table 4-2.

Table 4-2. Test matrices for transverse tensile and VNR shear tests

	Transverse tension				VNR shear		
Strain rate (/s) / Temperature (°C)	0.001	0.01	0.1	0.25	0.004	0.04	0.4
180		√				√	
200	√	√	√	√	√	√	√
210		√				√	
220		√				√	

4.3.3. Strain measurement and data postprocessing

Both the tensile and shear strains were measured using the DIC technique. Before the experiments, a random speckle pattern was created on each specimen with white paint for the background and black paint for the speckles. This speckled pattern can create a grayscale

distribution for the DIC process. The deformation images were processed by a 2D commercial software GOM, Germany. For each recorded deformation, a facet size of 16 x 16 pixels with a distance of 14 pixels was used in the software.

For the transverse tensile test specimen, the tensile stress, σ_t, and tensile strain, ε_t, were defined as Eqs. (4-1) and (4-2) .

$$\sigma_t = \frac{F}{A_0} * (1 + d/l_0) \tag{4-1}$$

$$\varepsilon_t = \ln(1 + d/l_0) \tag{4-2}$$

where F is the load measured by the load cell of Instron, d is the change of the length within the gauge length area measured by the GOM software, $l_0 = 50.00$ is the initial gauge length, A_0 is the initial cross-section area of the transverse tensile test specimen.

For the VNR shear test, the shear stress, σ_s, and shear strain, ε_s, were defined as Eqs. (4-3) and (4-4).

$$\sigma_s = \frac{F}{A_0} \tag{4-3}$$

$$\varepsilon_s = \bar{\gamma}_{xy}; \tag{4-4}$$

where F is also the load measured by the load cell of Instron. Terms A_0 is the initial cross-section area of the VNR shear test specimen. The shear strain field was analysed via DIC by GOM software. A section was constructed in the GOM software along the ligament of the specimen, acting as the ROI. The term $\bar{\varepsilon}_{12}$ is the arithmetic mean value of the in-plane shear strains ε_{12} of all facets across the ROI.

4.4. Experimental results and analysis

4.4.1. Transverse tensile properties of UD FRTPs

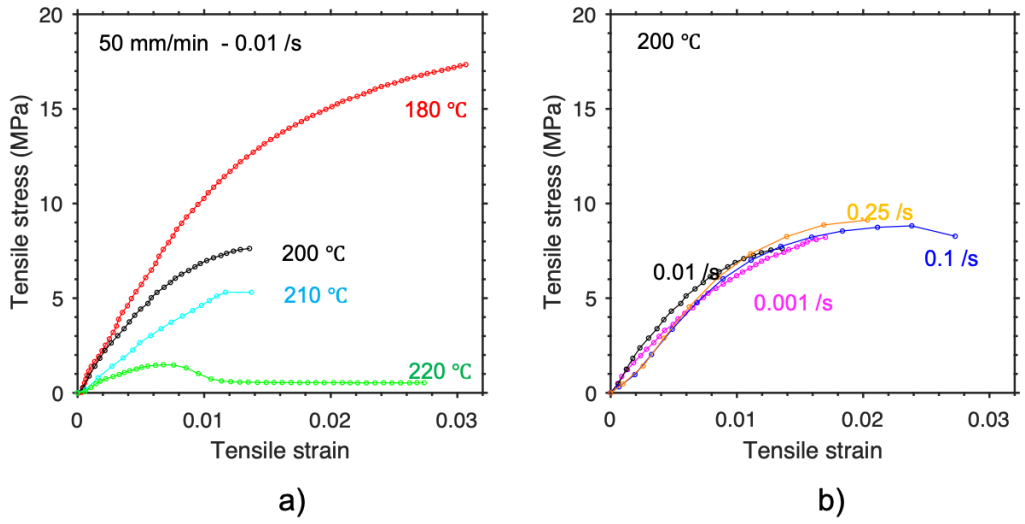


Figure 4-6. Experimental results of transverse tensile tests on UD CF/PA6 under a) temperature ranges in 180 – 220 °C, at a strain rate of 0.01 /s and b) a strain rate ranges in 0.001 – 0.25 /s, at a temperature of 200 °C.

The tensile stress-strain curves of transverse tensile tests are shown in Figure 4-6. It is evident that a linear elastic region exists at the beginning of the deformation. After the initial linear elastic deformation, nonlinear deformation regions appear, where strain hardening or softening phenomena occur. It is crucial to note that in this research, the term "strain hardening" refers to the increase in stress with increasing strain, whereas the term "strain softening" denotes a decrease in stress with increasing strain. Figure 4-6a demonstrates that temperature has a significant impact on the stress-strain behaviour (at a tensile strain rate of 0.01 /s). Specifically, an increase in temperature results in a notable decrease in stress at the same tensile strain level. Within the tested temperature region (180 – 220 °C), at lower temperatures (180/200 °C), there is strain hardening behaviour at the nonlinear deformation region. At higher temperatures (210/220 °C), however, strain softening phenomena are observed. At 220 °C, the stress level gradually reaches a plateau after the period of strain softening. On the other hand, Figure 4-6b indicates that the strain rate effects are marginal within the tested conditions (200 °C, 0.001 –

0.25 /s). Moreover, it is evident that the specimen fractures easily during the transverse tensile tests, as indicated by the relatively small strains at failure in Figure 4-6.

4.4.2. Longitudinal shear properties of UD FRTPs

Figure 4-7a illustrates the temperature effects on the longitudinal shear behaviour. Similar to transverse tensile deformation, the temperature effects (at a strain rate of 0.04 /s) significantly impact longitudinal shear behaviour, leading to a decrease in stress level with increasing temperature. An elastic deformation region exists at the beginning of deformation, followed by strain hardening or softening behaviour. At lower temperatures (180/200 °C), stress increases consistently with the increasing shear strain, indicating continuous strain hardening. At higher temperatures (210/220 °C), obvious strain softening behaviour occurs. Figure 4-7b shows that the strain rate effects at 200 °C with strain rate ranging from 0.004 to 0.4 /s are insignificant on the longitudinal shear properties.

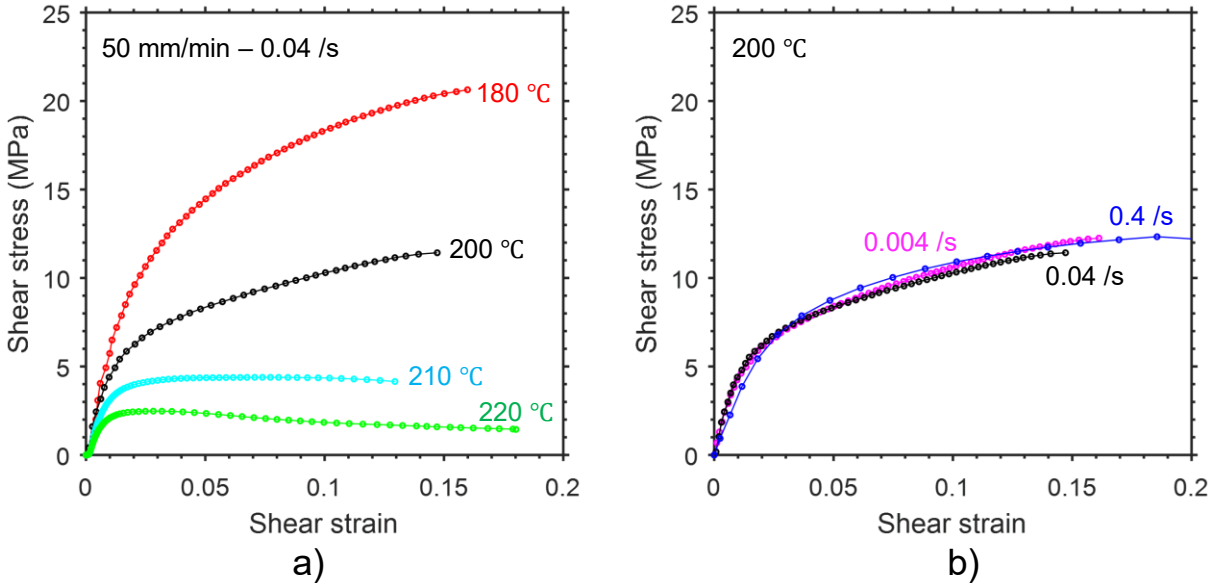


Figure 4-7. Experimental results of VNR shear tests on UD CF/PA6 under a) temperature ranges in 180 – 220 °C, and at a strain rate of 0.04 /s and b) strain rate ranges in 0.004 – 0.4 /s at a temperature of 200 °C.

There are noticeable strain softening effects observed at the starting temperature of 210 °C for both transverse tensile and VNR shear tests. Additionally, it is noteworthy that the material

exhibits a decreasing trend in strain-to-failure with increasing temperature at temperatures below or equal to 210 °C. However, this trend disappears when the temperature approaches T_m, i.e., 220 °C. To gain a further understanding of the strain hardening and softening behaviour, the transverse tensile and longitudinal shear properties are further analysed in the following sections.

4.4.3. Analysis of the intra-ply deformation properties

The strain hardening or softening behaviour can be inferred clearly by the tangent modulus evolution as indicated in Figure 4-8. The tangent modulus is defined as the derivative of the stress-strain curve. Therefore, according to the definition, after the initial elastic deformation, the positive tangent moduli indicate strain hardening, while the negative tangent moduli indicate strain softening. It is found that for both transverse tensile and VNR shear tests, at lower temperatures (180/200 °C), there is only strain hardening, although the tangent modulus representing the level of strain hardening gradually decreases during deformation; whereas at higher temperatures (210/220 °C), strain softening can be observed after certain deformation. Besides, strain softening occurs earlier at the higher temperature (220 °C), which indicates easier localised deformation.

To gain deeper insight into the material's strain hardening/softening responses and corresponding deformation mechanisms, further analysis was conducted and illustrated through Figure 4-9 and Figure 4-10. VNR shear results were selected for the analysis, because the material could achieve a much greater level of deformation under shear that is deemed the dominant deformation mode during forming. The nonlinear relationship between the shear strain evolution and the applied displacement onto the specimen, as shown in Figure 8a, can be used to infer localised deformation. The shear strains are similar for different test conditions until the displacement reaches approximately 0.4 mm, after which the strains in the ligament area increase more rapidly at higher temperatures, indicating greater levels of localised deformation. Figure 4-9b and c depict the shear strain fields across the specimen at displacements of 0.2 and 0.7 mm, respectively, for varying testing temperatures (180 – 220 °C). In Figure 4-9b, at a displacement of 0.2 mm, the shear strain fields show a similar distribution across different temperatures. However, in Figure 4-9c, the shear strain fields vary significantly at a displacement of 0.7 mm, with a higher level of strain localisation observed at the ligament area at a higher testing temperature. The analysis indicates that a greater strain softening in the

stress-strain response resulted in a higher level of strain localisation. During stamp forming processes, strain localisation can impede the continuous deformation of materials and induce premature material failure. Therefore, it is advisable to avoid forming temperatures of 210 ℃ or higher, to ensure good formability. Regarding forming speed, since the strain rate effects are negligible within the studied range, and with consideration of the forming machine capacity, a higher forming speed can be recommended to increase forming efficiency and reduce material heat loss when low-temperature tools are used.

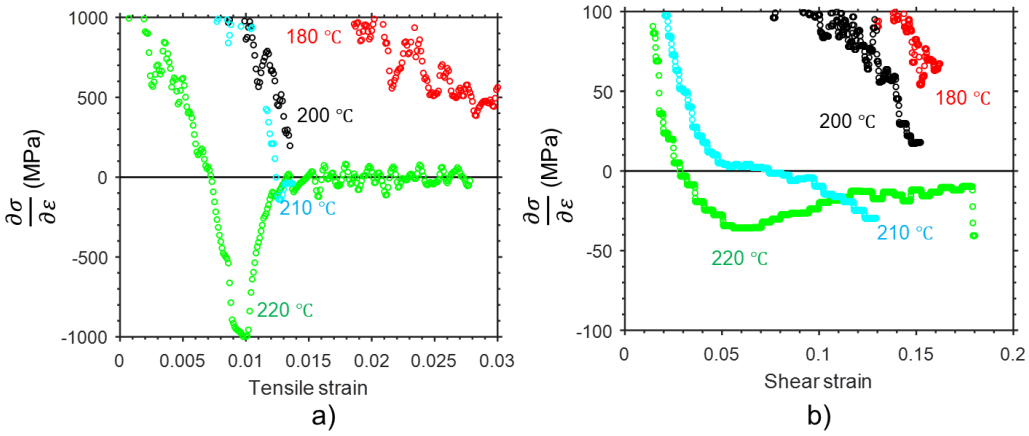


Figure 4-8. The tangent modulus versus strain at different temperatures (180 – 220 ℃) under a) transverse tensile at a strain rate of 0.01 /s, only the data within a range of ± 1000 are shown, and b) VNR shear tests at a strain rate of 0.04 /s, only the data within a range of ± 100 are shown.

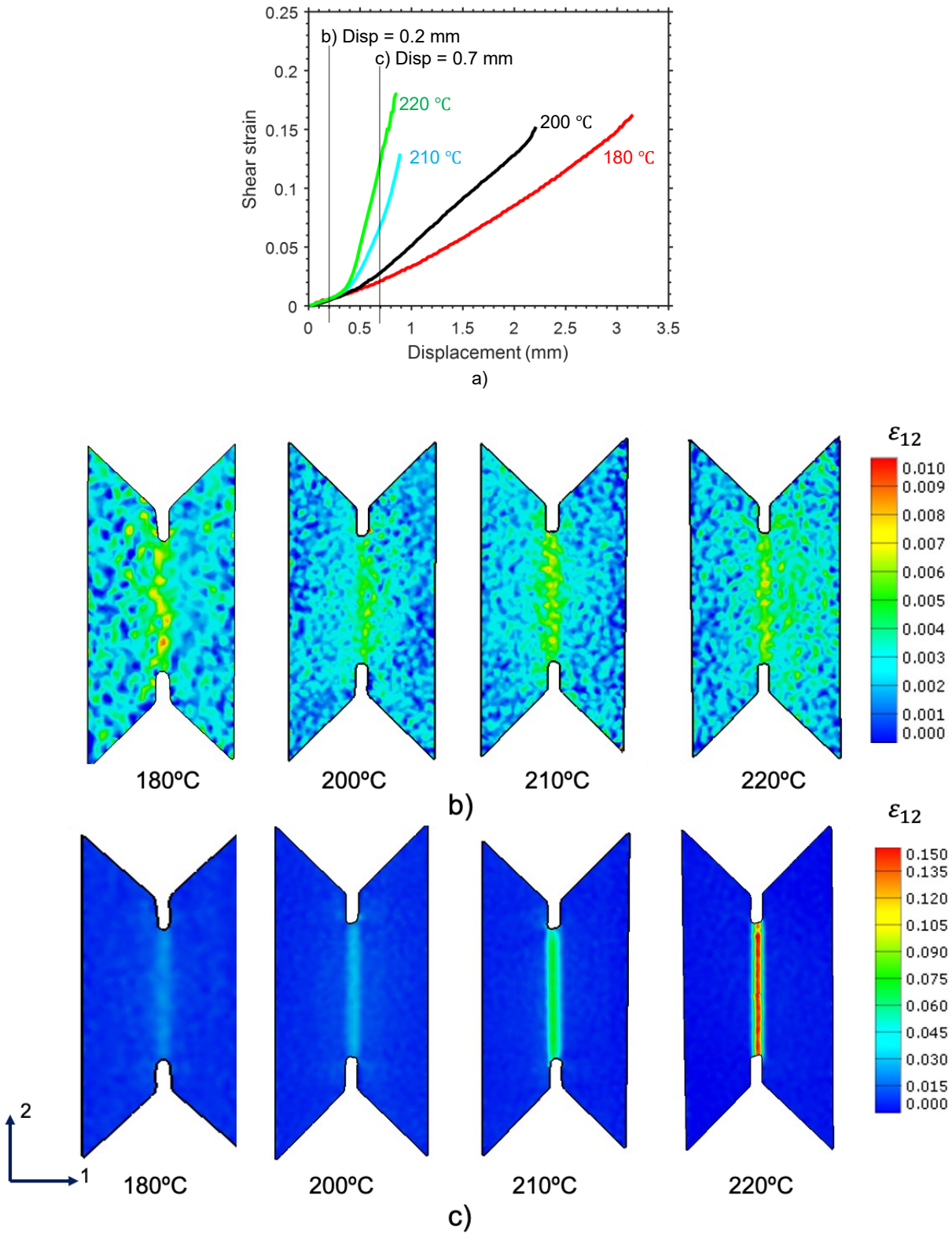


Figure 4-9. Localised deformation analysis of VNR shear tests under the temperature range of 180 – 220 °C. a) Shear strain evolution versus the crosshead displacement, and shear strain distribution at the displacement of b) 0.2 mm and c) 0.7 mm at different temperatures.

Figure 4-10 shows a comparison between the stress-strain properties of the UD CF/PA6 and PA6 polymer under different temperatures. The stress-strain curves of UD CF/PA6 are the longitudinal shear stress-strain curves, and the stress-strain curves of PA6 under uniaxial tension come from the previous study shown in Section 3.3. Note that although CF/PA6 was tested at 0.01 /s and PA6 was tested at 0.1 /s, the comparison can still provide valid indications because it has been proven that the strain rate effects on both materials are insignificant within the concerned temperature and strain rate ranges. As can be seen, the stress level of UD CF/PA6 is higher than PA6, at 180 and 200 °C, and for the early stage of deformation at 220 °C. This reflects the strengthening effect attributed to the fibre reinforcement. Additionally, in Figure 4-10, no significant softening behaviour is observed in PA6 under all tested temperatures (180 – 220 °C), while there is a noticeable strain softening observed after the initial deformation stage in UD CF/PA6 at 220 °C. When discussing the possible source of strain softening behaviour, all constituents in the material should be taken into consideration. Therefore, in the case of UD CF/PA6, the constituents that contribute to the total strength are PA6, reinforcing carbon fibres (CFs), and the interface between CFs and PA6. Since CFs can be considered rigid bodies and PA6 shows no strain softening behaviour, it is reasonable to conclude that the strain softening behaviour comes from the degradation of the fibre-matrix interface. These findings informed the development of the physically based constitutive model of UD FRTPs in Chapter 5.

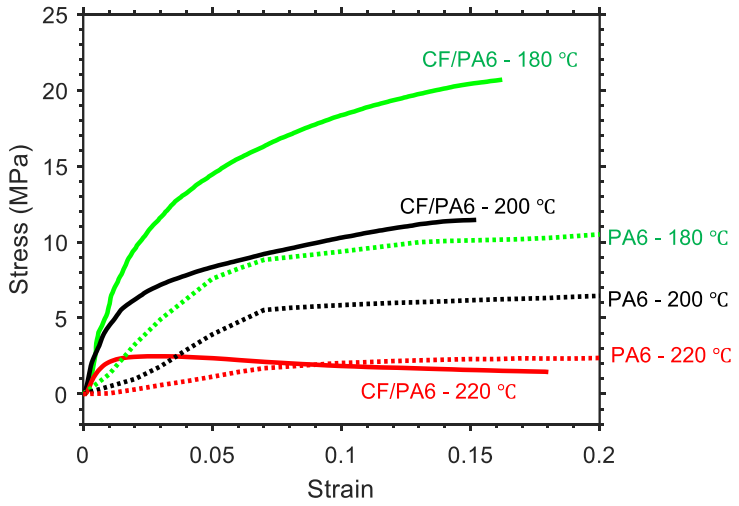


Figure 4-10. Stress-strain curves of UD CF/PA6 under longitudinal shear compared to PA6 under uniaxial tension, under various temperatures (180 – 220 °C).

4.5. Summary

This chapter has presented the experimental work on the design of hot press tool, the determination of UD CF/PA6 laminates fabrication method, the experimental design for characterising the thermomechanical properties of the UD CF/PA6. Besides, the temperature and strain rate effects on the thermomechanical properties of the UD CF/PA6 under solid-state stamp forming conditions are analysed and insights for the deformation mechanisms have gained.

CHAPTER 5 Development and Implementation of Constitutive Models

*This Chapter is reproduced from publication **A1, A2** and **A3** in the Publication list.

5.1. Introduction

This chapter proposed a new physically based viscoelastic-viscoplastic constitutive model with comprehensive consideration of the temperature and strain rate effects for thermoplastics. The developed constitutive model was subsequently employed to predict the stress-strain behaviour of these materials at different temperatures and strain rates for stamp forming applications. The comparison between the experimental and the predicted results by the constitutive model has shown a good prediction performance.

With the constitutive model for thermoplastic built, another set of physically based constitutive model was developed for the UD FRTPs, specifically, UD CF/PA6, to describe the intra-ply tensile and shear deformation. This constitutive model for UD CF/PA6 was built based on the constitutive model for PA6 with the same model form and material constants applied. This practice has effectively simplified the constitutive model for composites. Furthermore, the physics behind the deformation were taken into account, which gives insights to the deformation mechanism of the polymer-fibre interface properties inside composites.

After the constitutive models for UD CF/PA6 are developed, an FEA model was built and the solid-state stamp forming on UD CF/PA are simulated. Via conducting numerical studies on the forming process, the potential results for forming trial can be predicted.

5.2. Development of a constitutive model for thermoplastics

To build up a physically based viscoelastic-viscoplastic model for semicrystalline thermoplastics, rheological models are commonly used. The mechanical responses can be visually understood and expressed by combining simple responses of the basic elements, such as springs, dashpots, and friction elements, in the rheological models [187]. The stress-strain

relationship can be derived according to the stress and strain generated in the rheological models. In this section, a new physically-based constitutive model is developed based on the rheological model proposed by Polanco-Loria[121]. In the Polanco-Loria model, there are two sources of resistance, i.e., the intermolecular resistance and the network resistance, to overcome for material deformation. These two sources of resistance are assembled in parallel with each other, which is also mentioned in other studies [116, 188, 189]. The Polanco-Loria model is detailed below:

The intermolecular resistance (I). The intermolecular resistance is composed of an elastic part and a viscoplastic part in series. The elastic part is represented by a hyperelastic spring, while the viscoplastic part is represented by a dashpot and a friction element in parallel. The initial stiffness and flow stress are both results of the intermolecular interactions.

The network resistance (N): The resistance against network stretching is represented by a hyperelastic spring. The strain hardening during deformation is related to the network resistance, which is generated from molecular reorientation and alignment.

In this study, based on the experimental characterisation and analysis of the tensile stress-strain response of PA6 and PEEK, a new constitutive model which further develops both the rheological model form and the mathematical expressions is proposed, aiming to predict the thermomechanical behaviour of semicrystalline thermoplastics under solid-state stamp forming conditions in an accurate manner. Figure 5-1 shows the rheology scheme and the kinematics of the proposed model. Compared with the Polanco-Loria model, the new features in the rheological model form are that the hyperelastic springs in the intermolecular branch and the network branch are both replaced with a rate-dependent viscoelastic part, which is composed of a hyperelastic spring and a viscous dashpot [101]. In the mathematical expression of each rheological element, comprehensive temperature and strain rate effects are considered based on their physical meaning, which will be elaborated upon in detail in the following section.

Figure 5-1a shows the rheology scheme of the proposed model, which is composed of an intermolecular resistance (I) in parallel with a network resistance (N). The intermolecular resistance (I) branch is composed of a viscoelastic part and a viscoplastic part in series. The viscoelastic deformation is represented by a temperature-dependent Neo-Hookean hyperelastic spring in series with a temperature- and rate-dependent dashpot [102, 106, 121]. The hyperelastic spring characterises the initial elastic deformation due to the internal energy

change, and the dashpot in the viscoelastic part describes the viscosity contribution. The viscoplastic deformation is represented by a temperature- and rate-dependent dashpot in parallel with a friction element. It should be noted that the friction element is designed to represent the activation of plastic deformation, not as a physically microstructural response. When the stress in the friction element reaches the stress limit, the viscoplastic deformation begins. The stress contribution from viscoelastic deformation and viscoplastic deformation were analysed by Barba et al. [106]. Around 90 % permanent deformation was found in PEEK at temperatures around T_g, which also shows good agreement with the present model (detailed later). The network resistance is composed of a temperature-dependent eight-chains hyperelastic spring [117] in series with a temperature- and rate-dependent dashpot. The network resistance (N) branch describes the strain hardening behaviour in the later deformation stage caused by the reorientation and alignment of molecular chains.

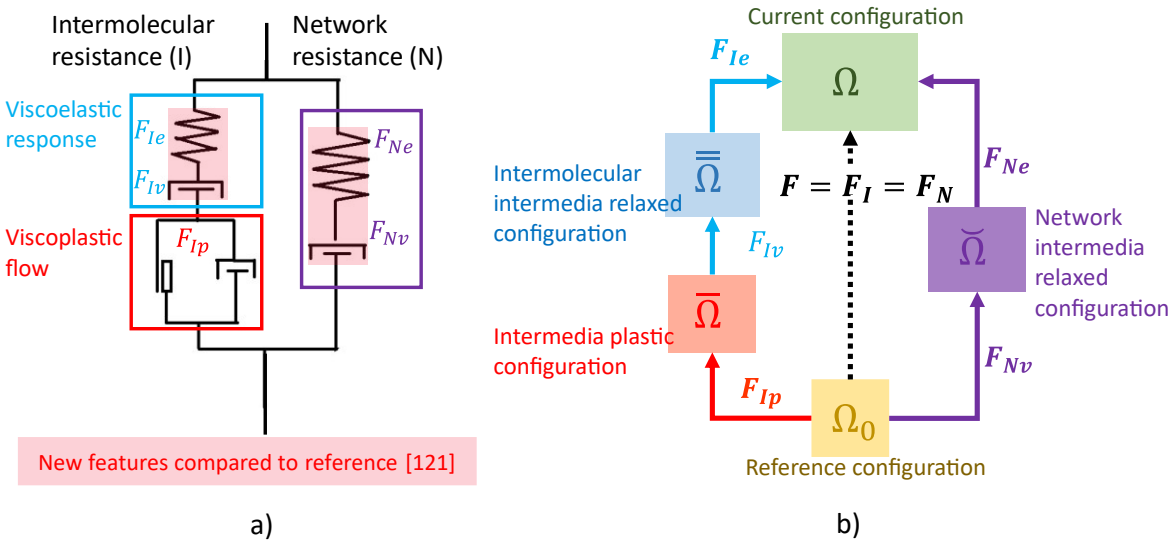


Figure 5-1. The proposed new constitutive model. (a) The rheology scheme; (b) The kinematics ($\boldsymbol{F}$ is the deformation gradient, $\boldsymbol{F_I}, \boldsymbol{F_N}$ represent the intermolecular deformation and the network deformation gradients, respectively. $\boldsymbol{F_{Ie}}, \boldsymbol{F_{Iv}}, \boldsymbol{F_{Ip}}$ represent the intermolecular elastic, viscous, and plastic deformation gradients, respectively.$\boldsymbol{F_{Ne}}, \boldsymbol{F_{Nv}}$ represent the network elastic and viscous deformation gradients, respectively.).

The kinematics analysis method of rheological models has been described in detail by Polanco-Loria et al. [121], Garcia-Gonzalez et al. [102, 115], and Barba et al. [106]. Similarly, Figure 5-1b shows the kinematics of this new model. Five configurations are established to define the

deformation kinematics of the new model. The initial reference configuration, $\boldsymbol{\Omega_0}$, deforms via the combination of three intermedia configurations to the current configuration, $\boldsymbol{\Omega}$. The intermedia plastic configuration, $\bar{\boldsymbol{\Omega}}$, and the intermolecular intermedia relaxed configuration, $\bar{\bar{\boldsymbol{\Omega}}}$, account for the plastic deformation and the viscous deformation in the intermolecular resistance, respectively. The network intermedia relaxed configuration, $\breve{\boldsymbol{\Omega}}$,accounts for the viscous deformation in the network resistance.

Setting $\boldsymbol{X}, \boldsymbol{x}$ as an arbitrary material point in the reference configuration $\boldsymbol{\Omega_0}$ and the current configuration $\boldsymbol{\Omega}$, respectively, the mapping relationship can be represented as $\boldsymbol{x} = \chi(\boldsymbol{X}, t)$, where t is time. Therefore, the deformation gradient, $\boldsymbol{F}$, can be derived as

$$\boldsymbol{F} = \nabla_{\boldsymbol{X}} \chi \tag{5-1}$$

where $\nabla_{\boldsymbol{X}}$ denotes the material gradient.

The deformation gradient, $\boldsymbol{F}$, is equivalent to both the intermolecular deformation gradient, $\boldsymbol{F_I}$, and the network deformation gradient, $\boldsymbol{F_N}$, according to the rheological model. Therefore,

$$\boldsymbol{F} = \boldsymbol{F_I} = \boldsymbol{F_N} \tag{5-2}$$

For the intermolecular side, the deformation gradient can be decomposed into the intermolecular elastic, $\boldsymbol{F_{Ie}}$, viscous $\boldsymbol{F_{Iv}}$, and plastic $\boldsymbol{F_{Ip}}$ deformation gradients. For the side associated with the network resistance, it is divided into network elastic $\boldsymbol{F_{Ne}}$, and viscous $\boldsymbol{F_{Nv}}$, deformation gradients. Therefore,

$$\boldsymbol{F_I} = \boldsymbol{F_{Ie}} \boldsymbol{F_{Iv}} \boldsymbol{F_{Ip}} \tag{5-3}$$

$$\boldsymbol{F_N} = \boldsymbol{F_{Ne}} \boldsymbol{F_{Nv}} \tag{5-4}$$

From the aforementioned rheology scheme, the Cauchy stress tensor (i.e., true stress tensor), $\boldsymbol{\sigma}$, in terms of the intermolecular resistance and the network resistance can be expressed as:

$$\boldsymbol{\sigma} = \boldsymbol{\sigma_I} + \boldsymbol{\sigma_N} \tag{5-5}$$

where, $\boldsymbol{\sigma_I}$ and $\boldsymbol{\sigma_N}$ denote the stresses contributed by the intermolecular resistance and the network resistance, respectively.

5.2.1. The intermolecular resistance (I)

The stress contributed by the intermolecular resistance is defined by a Neo-Hookean hyperelastic spring [190]. As Barba et al. [106] observed, polymers are nearly incompressible at temperatures above T_g; therefore, the incompressible Neo-Hookean material model is adopted. The Cauchy stress can be expressed as:

$$\boldsymbol{\sigma_I} = 2C_1(\theta)\boldsymbol{B_{Ie}} \tag{5-6}$$

where $\boldsymbol{B_{Ie}}$ is the intermolecular elastic left Cauchy-Green deformation tensor, calculated by $\boldsymbol{B_{Ie}} = \boldsymbol{F_{Ie}}\boldsymbol{F_{Ie}^T}$. With θ being the current temperature, the term $C_1(\theta)$ is a temperature-dependent parameter and is calculated by $C_1(\theta) = E(\theta)/(4(1+\nu))$, where ν is Poisson's ratio and $\nu = 0.48$ is used in this paper [106, 190]. The term $E(\theta)$ is the temperature dependent elastic modulus. From the research of Bouvard et al. [191] and Garcia-Gonzalez et al. [102], the decrease in elastic modulus $E(\theta)$ was modelled as a linear relationship with increasing temperature. However, according to the experimental observation presented in Figure 3-7b, the relationship between the elastic modulus and the temperature is nonlinear. Therefore, a nonlinear form of elastic modulus, $E(\theta)$, is proposed to describe the temperature dependence:

$$E(\theta) = E_{ref} + E_1\left(\theta - \theta_{ref}\right)^{m_E} \tag{5-7}$$

where E_{ref} is the elastic modulus at the reference temperature, θ_{ref}, and the terms E_1 and m_E are material constants. The reference temperature (θ_{ref}) in this model is set as the RT in Kelvin, 296 K. It should be noted that all temperatures referred to in the model development of this study are absolute temperatures.

For the uniaxial tensile deformation, the von-Mises effective stress form can be derived from the Cauchy stress tensor and expressed as $\sigma_I = 2C_1(\theta)(\lambda_{Ie}^2 - 1/\lambda_{Ie})$, where λ_{Ie} is the intermolecular elastic stretch ratio. According to the definition of the stretch ratio, in the form of effective (true) strain, λ_{Ie} can be expressed as $\lambda_{Ie} = \exp(\varepsilon_{Ie})$, where $\varepsilon_{Ie} = \int \dot{\varepsilon}_{Ie}\, dt$ refers to the effective (true) strain of the intermolecular elastic part, carried by the hyperelastic spring, and $\dot{\varepsilon}_{Ie}$ is the corresponding strain rate.

To calculate $\dot{\varepsilon}_{Ie}$, the deformation of each element in the intermolecular branch needs to be identified. There are viscoelastic and viscoplastic parts in the intermolecular branch, and the viscoplastic part comes into effect only when a temperature-dependent stress limit, $\sigma_{lim}(\theta)$, on

the friction element is reached. Therefore, the term $\dot{\varepsilon}_{Ie}$ is determined differently at two stages, i.e., before and after the stress limit is reached. In this research, the stress limit is defined as yield stress as shown in Figure 3-7a. The JC temperature softening model [102, 106, 115] is adopted in this work. The stress limit at current temperature is calculated by:

$$\sigma_{lim}(\theta) = \sigma_{ref}\left(1 - \left(\frac{\theta - \theta_{ref}}{\theta_m - \theta_{ref}}\right)^{m_\sigma}\right) \tag{5-8}$$

where, σ_{ref} is the stress limit at the reference temperature, m_σ is a material constant relating to σ_{lim}, and the term θ_m denotes the melting temperature of a studied material.

Before the stress limit is reached, i.e., $\sigma_I < \sigma_{lim}$, only the spring and the dashpot in the intermolecular viscoelastic part of the intermolecular branch share the total strain. Thus, the intermolecular elastic strain rate is given as:

$$\dot{\varepsilon}_{Ie} = \dot{\varepsilon} - \dot{\varepsilon}_{Iv} \tag{5-9}$$

where $\dot{\varepsilon}$ is the total effective (true) strain rate, $\dot{\varepsilon}_{Iv}$ is the intermolecular viscous strain rate in the intermolecular viscoelastic part, and $\dot{\varepsilon}_{Iv}$ is represented by a power-law flow rate [192]:

$$\dot{\varepsilon}_{Iv} = \dot{\gamma}_{01}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right)^{m_{rI}}\left(\frac{\sigma_I}{\eta_{Iv}(\theta)}\right)^{m_{Iv}} \tag{5-10}$$

where, $\dot{\gamma}_{01}$, m_{rI} and m_{Iv} are material constants controlling the temperature and strain rate sensitivity of the dashpot in the intermolecular viscoelastic part. The term $\dot{\varepsilon}_{ref}$ is the reference strain rate and is defined as the minimum strain rate of a studied strain rate range in this paper. The term $\eta_{Iv}(\theta)$ represents the temperature-dependent coefficient of viscosity of the dashpot in the intermolecular viscoelastic part. With the change in temperature from the glass transition temperature (θ_g) to the melting temperature (θ_m), the material evolves from a solid state to a liquid state, where the coefficient of viscosity decreases gradually from a large number to a small number. Therefore, the expression of $\eta_{Iv}(\theta)$ is shown as:

$$\eta_{Iv}(\theta) = \frac{\eta_{Iv0}}{\left(\frac{\theta - \theta_g}{\theta_m}\right)^{m_{\theta I}}} \tag{5-11}$$

where, η_{Iv0} and $m_{\theta I}$ are material constants relating to η_{Iv}.

After the stress limit is reached, i.e., $\sigma_I \geq \sigma_{lim}$, both the viscoelastic and viscoplastic parts share the total strain. Therefore, Eq. (5-9) is rewritten as:

$$\dot{\varepsilon}_{Ie} = \dot{\varepsilon} - \dot{\varepsilon}_{Iv} - \dot{\varepsilon}_{Ip} \tag{5-12}$$

where $\dot{\varepsilon}_{Ip}$ is the intermolecular plastic strain rate carried by the dashpot in the viscoplastic part, and $\dot{\varepsilon}_{Ip}$ can be expressed as an exponential flow rate according to references [121, 192]. In this paper, to address the strain rate and temperature effects on the viscoplastic flow, a revised equation form is proposed, given by:

$$\dot{\varepsilon}_{Ip} = \dot{\varepsilon}_{0p}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right)^{m_{Ip}}\left\{\exp\left(\frac{1}{C_p(\theta)}\left(\frac{\sigma_I}{\sigma_{lim}}-1\right)\right)-1\right\} \tag{5-13}$$

where $\dot{\varepsilon}_{0p}$ is a viscoplastic multiplier and m_{Ip} is a material constant showing the rate-dependence of the dashpot in the viscoplastic part. The term $C_p(\theta)$ is a temperature-dependent parameter that goes by the following form:

$$C_p(\theta) = \frac{C_{p1}}{\left(\frac{\theta}{\theta_m}\right)^{m_{cp}}} \tag{5-14}$$

where C_{p1} and m_{cp} are material constants relating to C_p.

5.2.2. The network resistance (N)

The stress contributed by the network resistance is defined by a hyperelastic spring which is modelled as an eight-chain model proposed by Arruda and Boyce [116, 189]. The eight-chain model form is expressed as:

$$\boldsymbol{\sigma_N} = \frac{C_R(\theta)}{3J_{Ne}}\frac{\bar{\lambda}_L}{\bar{\lambda}_{Ne}}\mathcal{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)\left(\boldsymbol{B^*_{Ne}} - \left(\bar{\lambda}_{Ne}\right)^2\boldsymbol{I}\right) \tag{5-15}$$

where, $C_R(\theta)$ is the temperature-dependent modulus of the network resistance, $\boldsymbol{I}$ is the identity tensor, and $J_{Ne} = \det(\boldsymbol{F_{Ne}})$ is the Jacobian determinant associated with the network elastic deformation gradient. The term $\boldsymbol{B^*_{Ne}}$ is the deviatoric network elastic left Cauchy-Green deformation tensor, and in the form of $\boldsymbol{B^*_{Ne}} = \boldsymbol{F^*_{Ne}}(\boldsymbol{F^*_{Ne}})^{\boldsymbol{T}}$, and $\boldsymbol{F^*_{Ne}} = (J_{Ne})^{-1/3}\boldsymbol{F_{Ne}}$ is the deviatoric part of $\boldsymbol{F_{Ne}}$. The term $\bar{\lambda}_{Ne}$ is the average network elastic stretch ratio, calculated from the form $\bar{\lambda}_{Ne} = \sqrt{\mathrm{tr}(\boldsymbol{B^*_{Ne}})/3}$; and in the form of effective (true) strain, $\bar{\lambda}_{Ne}$ can be expressed as $\bar{\lambda}_{Ne} = \sqrt{(\exp(2\varepsilon_{Ne}) + 2\exp(-\varepsilon_{Ne}))/3}$, with ε_{Ne} being the effective (true) strain

of network elastic deformation. The last term $\left(\boldsymbol{B}_{\boldsymbol{Ne}}^{*} - \left(\bar{\lambda}_{Ne}\right)^2 \boldsymbol{I}\right)$ in Eq. (5-15) could also be expressed in the form of effective (true) strain as $2\left((\bar{\lambda}_{Ne})^2 - 1/\bar{\lambda}_{Ne}\right)/3$. $\bar{\lambda}_L$ is a material constant associated with the number of "rigid links" between entanglements [116] and the stretching process locks when the number is reached. The term $\mathcal{L}^{-1}(\bar{\lambda}_{Ne}/\bar{\lambda}_L)$ is the inverse of Langevin function and the approximation form is expressed as:

$$\mathcal{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) = 3\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) + \frac{9}{5}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^3 + \frac{297}{175}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^5 + \frac{1539}{875}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^7 \quad (5\text{-}16)$$

From the strain energy function of Arruda-Boyce model [189], the eight-chain model could be regressed to the Neo-Hookean model if the higher order infinitesimal is ignored. Therefore, the modulus of the eight-chain model, i.e., the modulus of the network resistance, $C_R(\theta)$, can be inferred from the post-strain hardening modulus as shown by the experimental results in Figure 3-8b. Therefore, $C_R(\theta)$ is in a linear relationship with temperature, expressed as:

$$C_R(\theta) = C_{R_{ref}} + C_{r1}(\theta - \theta_{ref}) \quad (5\text{-}17)$$

where, $C_{R_{ref}}$ and C_{r1} are material constants relating to C_R.

To calculate the deformation carried by the hyperelastic spring, the strain rate of the network elastic deformation can be expressed as:

$$\dot{\varepsilon}_{Ne} = \dot{\varepsilon} - \dot{\varepsilon}_{Nv} \quad (5\text{-}18)$$

where $\dot{\varepsilon}_{Nv}$ is the strain rate of the viscous deformation in the network branch, and can be expressed in a similar form to Eq. (5-10), shown as:

$$\dot{\varepsilon}_{Nv} = \dot{\gamma}_{02}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right)^{m_{rN}}\left(\frac{\sigma_n}{\eta_{Nv}(\theta)}\right)^{m_{Nv}} \quad (5\text{-}19)$$

where σ_n is the von-Mises effective stress of the Cauchy stress tensor $\boldsymbol{\sigma_N}$, and $\dot{\gamma}_{02}$, m_{rN} and m_{Nv} are material constants controlling the temperature and strain rate sensitivity of the dashpot in the network resistance. The term $\eta_{Nv}(\theta)$ denotes the coefficient of viscosity of the dashpot in the network resistance, which is also temperature-dependent, and is expressed as:

$$\eta_{Nv}(\theta) = \frac{\eta_{Nv0}}{\left(\frac{\theta - \theta_g}{\theta_m}\right)^{m_{\theta N}}} \quad (5\text{-}20)$$

where η_{Nv0} and $m_{\theta N}$ are material constants relating to η_{Nv}.

5.2.3. The heat generation during deformation

As has been observed, there is adiabatic heat affecting stress levels during deformation at higher strain rates due to nonelastic deformation. Therefore, it is necessary to correct the total stress with consideration of the thermal effect caused by heat generation. In the form of total effective stress, σ, the total stress after thermal softening correction, σ_c, is calculated by the following form:

$$\sigma_c = \sigma\left(1 - \frac{\theta_c - \theta}{\theta_m - \theta}\right)^{m_h} \tag{5-21}$$

where m_h is the material constant representing the softening parameter. Term θ_c refers to the corrected temperature when the adiabatic heat comes into effect, and θ_c could be calculated from $\theta_c = \theta + \int \dot{\theta}\, dt$, where $\dot{\theta}$ is the temperature evolution during the deformation.

To estimate $\dot{\theta}$, the thermodynamics of the deformation process was explained in detail by Bouvard et al. [191] and Garcia-Gonzalez et al. [102]. In this research, to simplify the equation form, a ω factor is adopted to be a rate-dependent Taylor–Quinney coefficient to measure the ratio of plastic work converting into heat during plastic deformation [193, 194]. The thermoelastic coupling behaviour is ignored as the thermal expansion effect is relatively small compared with the intermolecular and network deformation [106]. In addition, to describe the material temperature evolution more accurately, the heat capacity $\bar{C}$ is also revised into a temperature-dependent parameter $\bar{C}(\theta)$ according to the studies of [105, 106, 195]. The temperature evolution during the deformation can be expressed as:

$$\dot{\theta} = \frac{\omega(\dot{\varepsilon})}{\bar{C}(\theta)}\left(\sigma_{Iv}\dot{\varepsilon}_{Iv} + \sigma_{Ip}\dot{\varepsilon}_{Ip} + \sigma_{Nv}\dot{\varepsilon}_{Nv}\right) \tag{5-22}$$

where $\sigma_{Iv}, \sigma_{Ip}, \sigma_{Nv}$ are intermolecular viscous stress, intermolecular plastic stress, and network viscous stress, respectively. The term $\omega(\dot{\varepsilon})$ is the ratio of energy that is converted to temperature change to the total plastic dissipation, and this ratio is a function of strain rate. According to the estimation by Garcia-Gonzalez et al. [102], the adiabatic heating effect is anticipated when the strain rate is higher than 0.01 /s. Further, according to the studies of Barba et al. [106], when the strain rate is higher than 1 /s, the deformation is under adiabatic

conditions, under which conditions $\omega = 1$. Therefore, $\omega(\dot{\varepsilon})$ of intermediate strain rates (from 0.01 /s to 1 /s) are estimated accordingly. The mathematical form of $\omega(\dot{\varepsilon})$ is defined as below:

$$\omega(\dot{\varepsilon}) = \begin{cases} 0 & ;\ if\ \dot{\varepsilon} \leq 0.01 \\ min(0.33 * (log_{10}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right),\ 1)\ ;\ if\ \dot{\varepsilon} > 0.01 \end{cases} \tag{5-23}$$

The temperature-dependent specific heat capacity $\bar{C}(\theta)$ is expressed as:

$$\bar{C}(\theta) = \bar{C}_{ref} + \bar{C}_1\left(\theta - \theta_{ref}\right) \tag{5-24}$$

where $\bar{C}_{ref}$ is $\bar{C}$ at the reference temperature, and $\bar{C}_1$ is a material constant relating to $\bar{C}$.

5.2.4. Discussion of the proposed constitutive model and associated deformation mechanisms

This section provides a more detailed explanation on the rationale behind the proposed constitutive model form and attempts to promote a better understanding on the deformation mechanisms by reflecting on the proposed constitutive model equations.

Figure 5-2 schematically illustrates the stress response and microstructural evolution at different deformation stages of semicrystalline thermoplastics under tensile deformation, and the correspondingly, the contributing part (i.e., rheological element) of the constitutive model at each stage. Note that, since microstructural investigation is not the focus of this study, the interpretation of the microstructural evolution below is extended discussion, based on literature studies and supported by references.

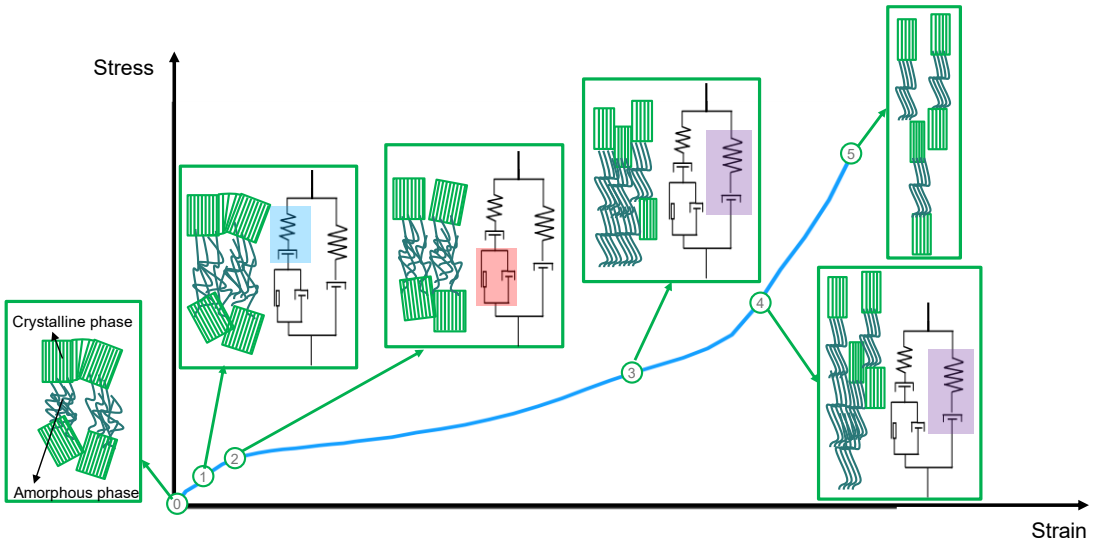


Figure 5-2. Schematic illustration of microstructural evolution and the governing parts of the constitutive model, corresponding to the stress-strain response at different stages, during the tensile deformation of semicrystalline thermoplastics [106, 196, 197].

3) Stage 0 shows the initial state of the semicrystalline thermoplastics without load applied. There are well-arranged crystals and amorphous chains connected with each other [156].

4) From stage 0 to stage 1, the viscoelastic deformation begins, which is mainly analogous to the viscoelastic part in the intermolecular branch of the constitutive model. During the viscoelastic deformation stage, the amorphous phase is the main deformation phase. The molecular chains in the amorphous region elongate and partially align reversibly. No chain unfolding or fracture in the crystalline region exists at this stage [198]. The stiffness, corresponding to the modulus of the hyperelastic spring in this viscoelastic part, is temperature-dependent during this stage (Eq. (5-7)), as the mobility of the molecular chains increases when the temperature increases. In the meantime, since there is a certain fraction of free volume [192], molecular relaxation occurs during deformation, shown as the viscous deformation represented by the dashpot in the intermolecular viscoelastic part. At a higher strain rate under the same temperature, there is less time for molecular relaxation, therefore greater stress is required for deformation (Eq. (5-10)). Further, the coefficient of viscosity decreases with increasing

temperature [192], indicating easier physical movement of the dashpot when the temperature increases (Eq. (5-11)).

5) From stage 1 to stage 2, when the stress limit is reached for activating the friction element in the viscoplastic part, the material yields. Accordingly, the viscoplastic part comes into effect. From a microstructural point of view, the stress limit is the stress barrier for the crystalline phase to overcome to accommodate further deformation, and this barrier is smaller at higher temperatures (Eq. (5-8)) [192]. After overcoming this stress barrier, the large crystal lamellae separate into smaller crystallites and reorientate towards the tensile axis [198]. The breakage and reorientation of crystallites results in the molecular chains moving more easily and therefore less stress is required [160]. Therefore, the material gradually shifts into a steady flow, shown as temperature- and rate-dependent viscous deformation. In the constitutive model, these dependencies are reflected through the dashpot in the viscoplastic part (Eq. (5-13) and (5-14)).

6) From stage 2 to stage 3, the network resistance against stretching starts to gradually take effect, and consequently the strain hardening process begins [199]. Although, in the constitutive model, the network stretch exists in the previous stages, the corresponding stress contribution to the whole system is almost negligible since its magnitude at this stage is relatively small. When the network stretch associated deformation takes the dominant role, the hyperelastic spring in the network branch results in the strain hardening response. In the microstructural aspect, the molecular chains in the amorphous region are reorientated and stretched. Meanwhile, there is transformation from crystallites to amorphous chains, which means the degree of crystallinity changes at the strain hardening stage [74, 160]. The molecular mobility at this stage is also temperature-dependent, as the above mentioned behaviours all require less energy at a higher temperature. Therefore, the strain hardening modulus is temperature-dependent (Eq. (5-17)), as proven by the experimental results. Further, there is also a molecular relaxation process [160], shown as rate-dependent viscous deformation, and the coefficient of viscosity deceases with increasing temperature. As a result, a temperature- and rate-dependent dashpot is also included (Eq.(5-19) and (5-20)).

7) From stage 3 to stage 4, the strain hardening continues with the continued deformation, since the amorphous molecular continues to be stretched and the network resistance

increases. As the molecular chains are reoriented and stretched reaching the limit of chain extensibility ($\bar{\lambda}_L$ in Eq. (5-15)) [200], a dramatic upturn in the strain hardening curve occurs. This strain hardening mechanism, from stage 2 onwards, is expressed by the eight-chains model of the hyperelastic spring [189].

8) At stage 5, the covalent bonds rupture and the molecular chains slide completely past each other, which results in the fracture of the thermoplastic materials.

5.2.5. Model calibration and performance

The material constants were identified based on the physical meaning of each rheological element and calibrated based on the experimental test results on PA6 and PEEK. The determination process is explained as follows:

(1) General properties: the material intrinsic properties, including Poisson's ratio ν, glass transition (θ_g) and melting (θ_m) temperatures, material constants relating to the specific heat capacity ($\bar{C}_{ref}$, $\bar{C}_1$), were determined by referencing to existing research papers [105, 106, 201]. The reference temperature (θ_{ref}) in this thesis was the RT in Kelvin, 296 K. The modulus at the reference temperature E_{ref} was referenced from the datasheet of the studied materials shown in Table 3-1.

(2) Material constants that were determined from analysis of experimental results: material constants relating to the elastic modulus, E_1, m_E , were identified according to the experimental results in Figure 3-7b. Material constants relating to the stress limit (σ_{ref}, m_σ) were identified by the experimental results in Figure 3-7a.

(3) Material constants determined from direct fitting to the stress-strain curves: material constants that show the rate-dependence ($\dot{\gamma}_0, m_{rI}, m_{Iv}, \dot{\varepsilon_{0p}}, m_{Ip}, \dot{\gamma_{02}}, m_{rN}, m_{Nv}$) and material constants that show temperature-dependence ($\eta_{Iv0}, m_{\theta I}$, $C_{p1}, m_{cp}, C_{R_{ref}}, C_{r1}, \eta_{Nv0}, m_{\theta N}$), the thermal softening constant m_h that defines the thermal softening behaviour caused by the adiabatic heat, the material constant associated with the number of "rigid links" between entanglements ($\bar{\lambda}_L$), were all determined by fitting the model to the true stress – true strain curves of the experimental results. It is noted that the true effective stresses from the experimental results correspond to σ_c in the constitutive model.

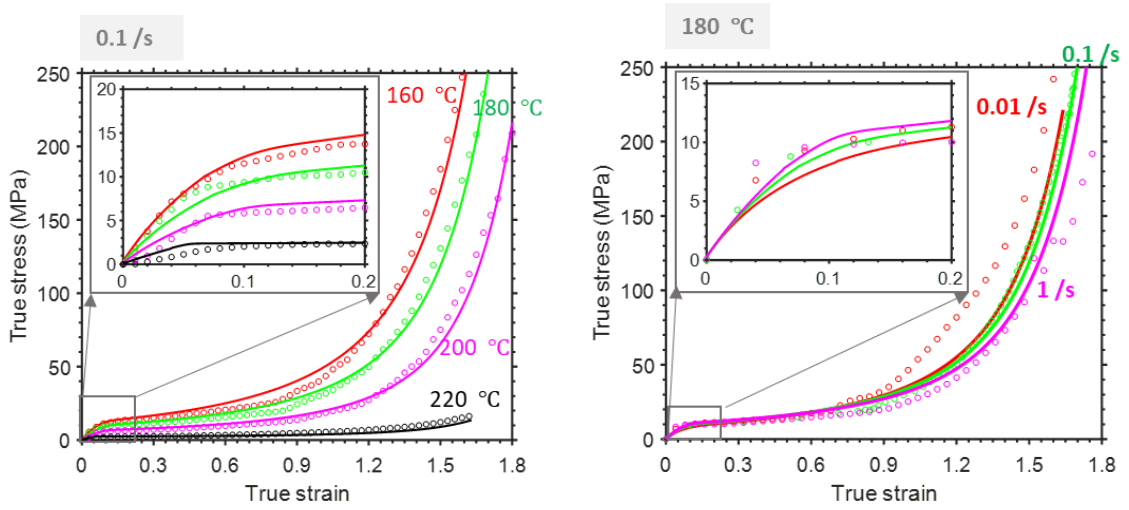
The material constants that need to be determined from direct fitting to the stress-strain curves were computed by employing a gradient-based optimisation program. The lsqnonlin algorithm in MATLAB was adopted to solve the nonlinear least-squares problems.

The material constants of the constitutive model for PA6 and PEEK materials are listed in Table 5-1.

Table 5-1. Material constants for PA6 and PEEK.

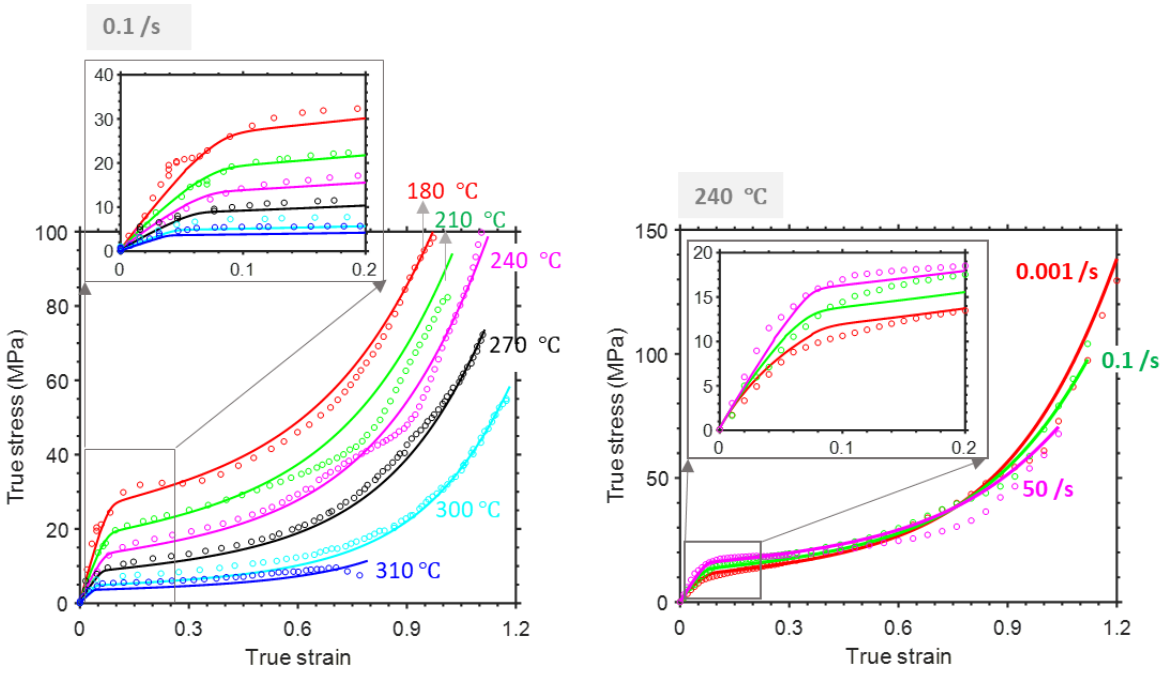
General properties										
	ν	$\bar{C}_{ref}$ (kJ/m^{3}K)	$\bar{C}_1$	θ_{ref} (K)	θ_g (K)	θ_m (K)	m_h			
PA6	0.48	3.00	0.01	296	320	503	0.5			
PEEK	0.48	2.56	0.01	296	416	616	0.5			
Material constants associated with the viscoelastic properties										
	E_{ref} (MPa)	E_1 (MPa)	m_E	σ_{ref} (MPa)	m_σ	$\dot{\gamma_{01}}$	m_{rI}	m_{Iv}	η_{Iv0}	$m_{\theta I}$
PA6	2400	-1041.2	0.1554	12.0	3.257	0.001	0.8	0.5	0.25	1.20
PEEK	3200	-1280.1	0.1572	108.0	0.240	0.002	0.8	0.5	3.20	1.26
Material constants associated with the viscoplastic properties										
	$\dot{\varepsilon_{0p}}$	m_{Ip}	C_{p1}	m_{cp}						
PA6	0.0010	0.5	0.05	3.0						
PEEK	0.0001	0.4	0.04	3.5						
Material constants associated with the network properties										
	$\dot{\gamma_{02}}$	m_{rN}	m_{Nv}	η_{Nv0}	$m_{\theta N}$	$\bar{\lambda}_L$	$C_{R_{ref}}$	C_{r1}		
PA6	0.001	0.1	0.10	490	0.01	2.60	23.958	-0.120		
PEEK	0.001	0.5	1.03	562	0.09	1.22	13.780	-0.045		

The comparison between the computed and experimental true stress – true strain curves, for PA6 and PEEK, are shown in Figure 5-3 and Figure 5-4, respectively. As can be seen, the model can accurately capture the deformation behaviour of PA6 and PEEK at the studied temperatures and strain rates. Specifically, Figure 5-3a and Figure 5-4a show the precise prediction of stress responses at the temperature approaching T_m (i.e., 220 ℃ for PA6 and 310 ℃ for PEEK), at which the dramatic strain hardening behaviour disappears. Additionally, Figure 5-3b and Figure 5-4b show the model's capabilities in capturing the changing trend of strain rate effects throughout deformation, by taking into account the adiabatic heat influence, which is correlated with the findings in section 3.3.



a) b)

Figure 5-3. Comparison between experimental (symbols) and modelling (solid lines) results at different temperatures and strain rates for PA6. a) PA6 at different temperatures (160 – 220 °C), at a strain rate of 0.1 /s; b) PA6 at different strain rates (0.01 – 1 /s), at a temperature of 180 °C.



a) b)

Figure 5-4. Comparison between experimental (symbols) and modelling (solid lines) results at different temperatures and strain rates for PEEK. a) PEEK at different temperatures (180 – 310 °C), at a strain rate of 0.1 /s; b) PEEK at different strain rates (0.001 – 50 /s), at a temperature of 240 °C.

5.3. Numerical implementation of the constitutive model for thermoplastics

With the constitutive model of thermoplastics obtained, a Fortran VUMAT subroutine code was developed and implemented in a commercial FEA software, ABAQUS, for explicit analysis to validate the constitutive model. Figure 5-5 shows the flowchart of the subroutine. The highlighted tensors, including $\boldsymbol{U}, \dot{\boldsymbol{U}}, \bar{\bar{\boldsymbol{M}}}, \hat{\boldsymbol{M}}, \boldsymbol{r}$ (subscripts are ignored) will be detailed later in the explanations of the subroutine.

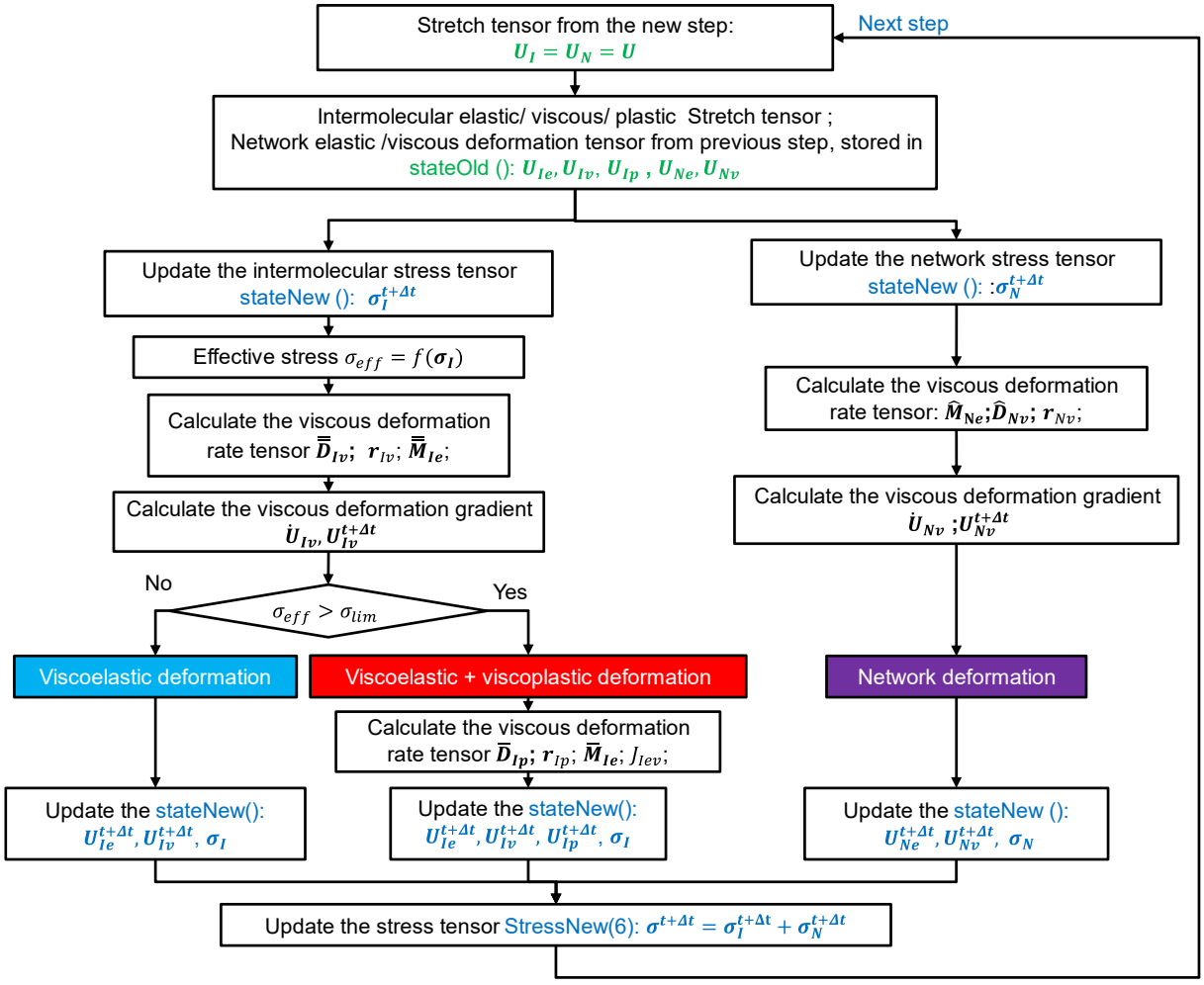


Figure 5-5. The flow chart of the ABAQUS subroutine

To account for large deformations in the hyperelastic constitutive model, it is advised to define the VUMAT in a corotational coordinate system where the basis system rotates with the material [114]. The stretch tensor $\boldsymbol{U}$, rather than the deformation gradient tensor, $\boldsymbol{F}$, is used for calculating the stress tensors from each step, based on $\boldsymbol{F} = \boldsymbol{R} \cdot \boldsymbol{U}$. The detailed calculation and

analysis are explained via the viscoelastic deformation, for example. The isolated kinematics are shown in Figure 5-6. According to the deformation mechanism, Eq (5-25) can be obtained.

$$\boldsymbol{F} = \boldsymbol{F_e F_v} = \boldsymbol{R_e U_e R_v U_v} \tag{5-25}$$

where, $\boldsymbol{R_e, R_v, U_e, U_v}$ are the rotation tensors for the elastic and viscous deformation, stretch tensors for the elastic and viscous deformation, respectively. To further decompose the stretch tensor, an assumption proposed by Mandel, the isoclinic concept [202] is employed. This concept assumes the rigid body rotation is included in elastic deformation only, which means $\boldsymbol{R_v = I}$. So that Eq (5-25) can be further revised to

$$\boldsymbol{F} = \boldsymbol{R_e U_e R_v U_v} = \boldsymbol{R U_e U_v} \tag{5-26}$$

where term $\boldsymbol{R}$ is the rotation tensor for the whole deformation, and only stretch tensors, $\boldsymbol{U_e}$ and $\boldsymbol{U_v}$ need to be considered while calculating the stress tensor.

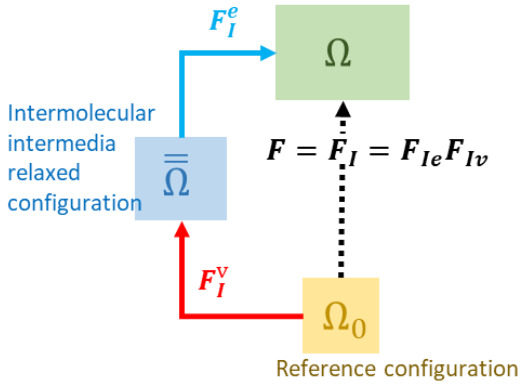


Figure 5-6. Kinematics of viscoelastic deformation

With the stretch tensor selected and the relationship built, previous kinematic relationships in Eqs (5-2)-(5-4) can be updated as below:

$$\begin{aligned} \boldsymbol{U} &= \boldsymbol{U_I} = \boldsymbol{U_N} \\ \boldsymbol{U_I} &= \boldsymbol{U_{Ie} U_{Iv} U_{Ip}} \\ \boldsymbol{U_N} &= \boldsymbol{U_{Ne} U_{Nv}} \end{aligned} \tag{5-27}$$

where $\boldsymbol{U}$ tensors with various subscriptions are the stretch tensors that can be obtained from the software each step, and updated during each calculation step, then stored in the state variables.

The stress tensors $\boldsymbol{\sigma_I}$ and $\boldsymbol{\sigma_N}$ are then calculated according to Eqs. (5-6) and (5-15), based on $\boldsymbol{U_{Ie}} = \boldsymbol{F_{Ie}}$ and $\boldsymbol{U_{Ne}} = \boldsymbol{F_{Ne}}$ respectively.

With the stress tensor for each configuration calculated, the next step is to update the stretch tensor for each configuration, i.e., $\boldsymbol{U_{Iv}}, \boldsymbol{U_{Ip}}$ and $\boldsymbol{U_{Nv}}$, via the deformation rate tensors, $\bar{\bar{\boldsymbol{D}}}_{\boldsymbol{Iv}}, \bar{\boldsymbol{D}}_{\boldsymbol{Ip}}, \breve{\boldsymbol{D}}_{\boldsymbol{Nv}}$, according to the definition $\boldsymbol{D} = \dot{\boldsymbol{F}}\boldsymbol{F}^{-1} = \dot{\boldsymbol{U}}\boldsymbol{U}^{-1}$. The equations for updating the stretch tensors are listed below:

$$\begin{aligned} \boldsymbol{U}_{\boldsymbol{Iv}}^{\mathbf{t+\Delta t}} &= \boldsymbol{U}_{\boldsymbol{Iv}}^{\mathbf{\Delta t}} + \dot{\boldsymbol{U}}_{\boldsymbol{Iv}} * \mathbf{\Delta t} = \boldsymbol{U}_{\boldsymbol{Iv}}^{\mathbf{\Delta t}} + \bar{\bar{\boldsymbol{D}}}_{\boldsymbol{Iv}}\boldsymbol{U}_{\boldsymbol{Iv}} * \mathbf{\Delta t} \\ \boldsymbol{U}_{\boldsymbol{Ip}}^{\mathbf{t+\Delta t}} &= \boldsymbol{U}_{\boldsymbol{Ip}}^{\mathbf{\Delta t}} + \dot{\boldsymbol{U}}_{\boldsymbol{Ip}} * \mathbf{\Delta t} = \boldsymbol{U}_{\boldsymbol{Ip}}^{\mathbf{\Delta t}} + \bar{\boldsymbol{D}}_{\boldsymbol{Ip}}\boldsymbol{U}_{\boldsymbol{Ip}} * \mathbf{\Delta t} \\ \boldsymbol{U}_{\boldsymbol{Nv}}^{\mathbf{t+\Delta t}} &= \boldsymbol{U}_{\boldsymbol{Nv}}^{\mathbf{\Delta t}} + \dot{\boldsymbol{U}}_{\boldsymbol{Nv}} * \mathbf{\Delta t} = \boldsymbol{U}_{\boldsymbol{Nv}}^{\mathbf{\Delta t}} + \breve{\boldsymbol{D}}_{\boldsymbol{Nv}}\boldsymbol{U}_{\boldsymbol{Nv}} * \mathbf{\Delta t} \end{aligned} \tag{5-28}$$

The equations for calculating $\bar{\bar{\boldsymbol{D}}}_{\boldsymbol{Iv}}, \bar{\boldsymbol{D}}_{\boldsymbol{Ip}}, \breve{\boldsymbol{D}}_{\boldsymbol{Nv}}$ refer to the J_2 plastic flow theory:

$$\begin{aligned} \bar{\bar{\boldsymbol{D}}}_{\boldsymbol{Iv}} &= \dot{\varepsilon}_{Iv}\boldsymbol{r}_{\boldsymbol{Iv}} \\ \bar{\boldsymbol{D}}_{\boldsymbol{Ip}} &= \dot{\varepsilon}_{Ip}\boldsymbol{r}_{\boldsymbol{Ip}} \\ \breve{\boldsymbol{D}}_{\boldsymbol{Nv}} &= \dot{\varepsilon}_{Nv}\boldsymbol{r}_{\boldsymbol{Nv}} \end{aligned} \tag{5-29}$$

where $\boldsymbol{r_{Iv}}, \boldsymbol{r_{Ip}}$ and $\boldsymbol{r_{Nv}}$ are the intermolecular viscous, intermolecular plastic and network viscous deformation potential tensors, respectively. According to Beytschko et al. [172], for the hyperelastic-plastic material model, the hyperelastic deformation potential is defined on the intermediate configuration $\bar{\bar{\boldsymbol{\Omega}}}$, rather than the deformed configuration $\boldsymbol{\Omega}$. Therefore, the second Piola-Kirchhoff stress is defined by the pull-back of the Kirchhoff stress. This pull-backed stress is the Mandel stress. The Mandel stresses on the intermolecular viscous, intermolecular plastic and network viscous configurations are defined according to Eq (5-30).

$$\begin{aligned} \boldsymbol{M}_{\boldsymbol{Iv}} &= \boldsymbol{J}_{\boldsymbol{Ie}}\boldsymbol{U}_{\boldsymbol{Ie}}^{\boldsymbol{T}}\boldsymbol{\sigma}_{\boldsymbol{Ie}}\boldsymbol{U}_{\boldsymbol{Ie}}^{-\boldsymbol{T}} \\ \boldsymbol{M}_{\boldsymbol{Ip}} &= \boldsymbol{J}_{\boldsymbol{Iev}}\boldsymbol{U}_{\boldsymbol{Iv}}^{\boldsymbol{T}}\boldsymbol{U}_{\boldsymbol{Ie}}^{\boldsymbol{T}}\boldsymbol{\sigma}_{\boldsymbol{Ie}}\boldsymbol{U}_{\boldsymbol{Ie}}^{-\boldsymbol{T}}\boldsymbol{U}_{\boldsymbol{Iv}}^{-\boldsymbol{T}} \\ \boldsymbol{M}_{\boldsymbol{Nv}} &= \boldsymbol{J}_{\boldsymbol{Ne}}\boldsymbol{U}_{\boldsymbol{Ne}}^{\boldsymbol{T}}\boldsymbol{\sigma}_{\boldsymbol{Ne}}\boldsymbol{U}_{\boldsymbol{Ne}}^{-\boldsymbol{T}} \end{aligned} \tag{5-30}$$

where $J = \det(\boldsymbol{F})$ is the determinant of the deformation gradient tensor. These deformation potential tensors are defined via $\boldsymbol{r} = \mathrm{dev}(\boldsymbol{M})/\sqrt{4/3J_2}$, where $\boldsymbol{M}$ is the Mandel stress tensor for

each configuration, and J_2 is the second invariant of the Mandel stress tensors and defined as $J_2 = \frac{1}{2}\mathrm{dev}(\boldsymbol{M}) : \mathrm{dev}(\boldsymbol{M})$. The colon operator, :, refers to the double dot product of tensors.

After all these stretch tensors are updated and stored in the state variables, the main function of the subroutine exits, and the next step starts.

To apply this VUMAT subroutine and validate the constitutive model, an ABAQUS explicit analysis FEA model was created based on the VNR shear test described in Section 3.4. The boundary conditions were set to mimic the experimental mechanism as shown in Figure 3-14, with one edge fixed and a vertical displacement added on the other edge. The simulation used solid elements (C3D8R).

Following the simulation, the strain distribution was compared to experimental results for the maximum principal strain and forming load, similar to Figure 3-19. Figure 5-7a presents the comparison of the maximum principal strain distribution at four deformation stages between experimental and simulation results. The horizontal axis 'relative position' indicates the position on the ligament of the specimen, with 0 referring to the lower end and 1 to the upper end. Figure 5-7b and c show the strain field measured from the experiment and simulation, respectively, at a displacement of 1.29 mm. Figure 5-7d compares the forming load in relation to displacement from experimental and FEA results. Overall, Figure 5-7 demonstrates both the reliability of the FEA model's prediction and the effectiveness of the subroutine in predicting polymer deformation.

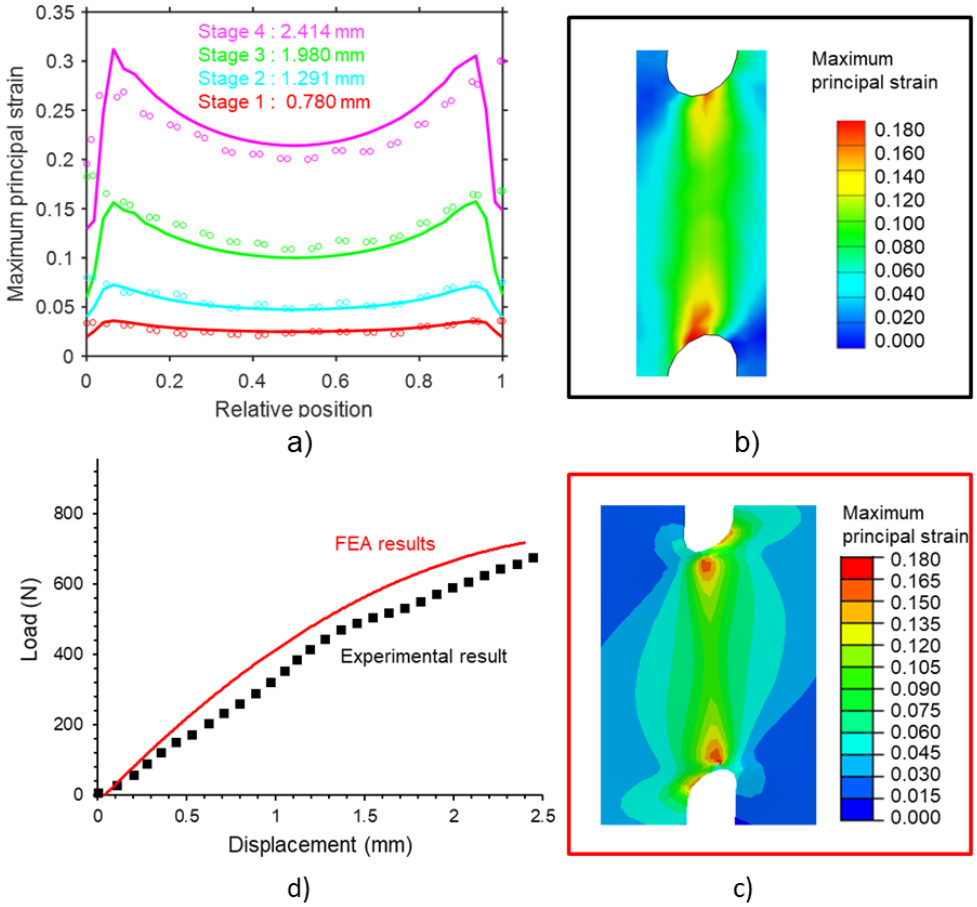


Figure 5-7. The validation of the ABAQUS subroutine via simulation of the VNR shear deformation. a) Comparison of the maximum principal strain distribution at four deformation stages from the simulation results (solid lines) and experimental results (circle lines); the maximum principal strain distribution of b) experimental and c) simulation results at deformation stage 2; d) the comparison of the load from the experimental and simulation results. (The studied case is the PEEK material at 240 °C).

5.4. Development of a constitutive model for UD FRTPs

A novel physically based constitutive model is presented. The new model comprises basic rheology elements to represent the viscous-elastic-plastic deformation properties of the composite material. The rheology of the model is shown in Figure 5-8, where the model for calculating the stress from deformation is composed of two parts: one part represents stress contributed by the polymer constituent reinforced by fibres (PrF), and the other part presents

stress contributed by the polymer-fibre interface (P-F). Note that the model does not consider the strain rate effects since they were found to be negligible in the concerned range.

The overall constitutive model is indicated as Eqs (5-31) and (5-32)

$$\boldsymbol{\sigma_C(F_C) = \sigma_{PrF}(F_{PrF}) + \sigma_{P-F}(F_{P-F})} \tag{5-31}$$

$$\boldsymbol{F_{PrF} = f(F_C); F_{P-F} = F_C} \tag{5-32}$$

where the form $\boldsymbol{\sigma(F)}$ denotes the stress response with respect to a given deformation of $\boldsymbol{F}$. $\boldsymbol{\sigma_C, \sigma_{PrF}}$, and $\boldsymbol{\sigma_{P-F}}$ are the stress tensors of the composite, the PrF part, and the P-F part, respectively. $\boldsymbol{F_C}$, $\boldsymbol{F_{PrF}}$ and $\boldsymbol{F_{P-F}}$ are the deformation gradient tensors corresponding to the composites, the PrF part, and the P-F part, respectively. The form $\boldsymbol{f(F_C)}$ represents that $\boldsymbol{F_{PrF}}$ is a function of $\boldsymbol{F_C}$, based on a hypothesis that there are mapping relationships between the overall deformations of the composite and the effective deformations of the polymer constituent. The investigation of the mapping relationships is detailed in the following subsection.

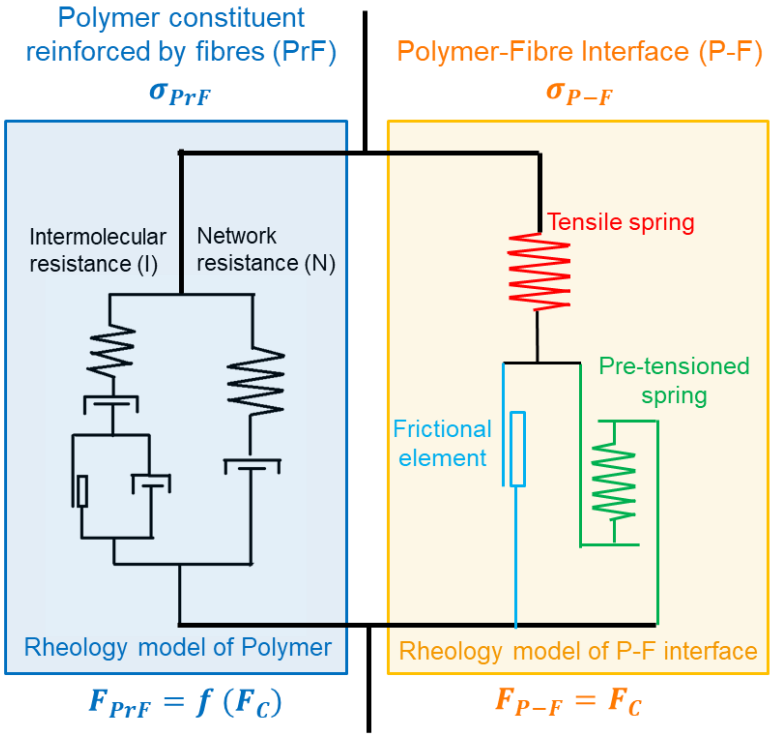


Figure 5-8. Rheology of the constitutive model form of UD CF/PA6.

5.4.1. The contribution of the polymer constituent reinforced by fibres (PrF)

One crucial question that needs to be addressed when determining the contribution of PrF is the extent of deformation that occurs in the polymers compared to the composite deformation. Given the difference in stiffness between the polymer and composite materials, the polymer constituent is likely to experience more deformation than the composite. As a result, it is necessary to establish a mapping relationship that describes the relative relationship between the deformation of the polymer constituent and the composite material. In order to calculate the specific deformation of the polymers within the composite, the representative volume element (RVE) technique is used, which involves creating an FEA model that comprises the basic microstructure of the composite. In a composite laminate, the actual fibres are randomly distributed. For the simplicity, however, the periodic square array arrangement of fibres are used [203]. In this case, the researchers have constructed a RVE model with nine unidirectional fibres that are uniformly distributed within the polymer matrix. The RVE model is then simulated using ABAQUS, a FEA software.

In this RVE model, the fibre volume fraction (V_{F_f}) was 46.87 %, corresponding to the datasheet of the UD CF/PA6 prepreg [183]. The fibre material was set to be orthotropic elastic, using the parameters for AS4 carbon fibres obtained from a reference [204]. For the matrix material (PA6), linear elasticity followed by the JC plasticity constitutive model were used, and the material constants were determined from the experimental results previously obtained in Section 3.3. The summary of material constants used for the RVE study are listed in Table 5-2.

Table 5-2. Material constants for carbon fibre and matrix material input in ABAQUS

Carbon fibre		Matrix (PA6)	
Orthotropic elasticity		Linear elasticity and Johnson cook plasticity	
E_1 (MPa)	214,000	E (MPa)	152.77 (180 °C); 70.02 (200 °C); 52.91 (210 °C); 27.79 (220 °C).
E_2 (MPa)	26,000	ν	0.48
E_3 (MPa)	26,000	A (MPa)	11.849
ν_{12}	0.28	B (MPa)	14.531
ν_{13}	0.28	n	0.729
ν_{23}	0.445	m	1.079

G_{12} (MPa)	112,000	θ_m (K)	503
G_{13} (MPa)	112,000	θ_0 (K)	433
G_{23} (MPa)	8,996.5		

The equation form of JC model is as below:

$\sigma = \left[A + B\left(\bar{\varepsilon}_{pl}\right)^n\right] \cdot \left(1 - \left(\frac{\theta-\theta_0}{\theta_m-\theta_0}\right)^m\right)$, where σ and $\bar{\varepsilon}_{pl}$ are the flow stress and plastic strain, respectively. The terms A, B, n, m are material constants for the Johnson-Cook model. Terms θ, θ_m and θ_0 are the current temperature, melting temperature and the controlling temperature, respectively, all at unit of Kelvin.

Two periodic boundary conditions (PBCs) were added on the RVE model, corresponding to the transverse tension and longitudinal shear deformation, respectively. The PBCs were implemented using an ABAQUS plugin called EasyPBC, developed by Omairey et al. [205]. The transverse tensile strain, ε_{22}^{RVE}, and longitudinal shear strain, ε_{12}^{RVE}, were obtained also according to the definition provided in reference [205].

From simulations on this RVE model, the strain in polymer elements, with respect to the overall composite deformation, can be obtained. Due to fibre reinforcement, the strain varies in polymer elements under both transverse tensile and longitudinal shear conditions, as shown in Figure 5-9b and d. An effective strain value representing the deformation of PrF, ε_{PrF}, is needed. This is determined by integrating the local strain corresponding to the material volumes, dv, and dividing the results by the total volume of the RVE, V_{RVE}. The mathematical expression of ε_{PrF} can be denoted by Eq. (5-33).

$$\varepsilon_{PrF} = \frac{\int \varepsilon_{equ-i}^{polymer} dv}{V_{RVE}} \tag{5-33}$$

where $\varepsilon_{equ-i}^{polymer}$ represents the equivalent strain of a local polymer element in RVE.

By introducing $\bar{\varepsilon}_{equ}^{polymer}$, representing the averaged equivalent strain of all polymer elements in the RVE, Eq. (5-33) can be simplified as follows:

$$\varepsilon_{PrF} = \frac{\bar{\varepsilon}_{equ}^{polymer} \cdot V_p}{V_{RVE}} = \bar{\varepsilon}_{equ}^{polymer} \cdot V_{F_p} \tag{5-34}$$

where V_p denotes the total volume of polymer elements inside the RVE, and V_{F_p} denotes the volume fraction of polymer constituent in the RVE.

The simulation results and mapping analysis of the RVE are presented in Figure 5-9. Specifically, Figure 5-9a and c depict the relationship between ε_{PrF} versus (vs.) ε_{22}^{RVE} and ε_{12}^{RVE}, respectively. Additionally, Figure 5-9b and d display examples of equivalent strain ($\varepsilon_{equ-i}^{polymer}$) fields for the polymer elements under transverse tension and longitudinal shear, respectively, both at 200 °C, and the corresponding deformation stages are indicated in the figures ($\varepsilon_{22}^{RVE} = 0.036, \varepsilon_{12}^{RVE} = 0.188$). To better illustrate the deformation of polymers, the fibres are excluded from the view. Notably, the equivalent strains of the polymer elements show a certain range under both deformation modes, with values ranging from 0.008 to 0.751 for transverse tension and 0.014 to 0.957 for longitudinal shear. From Figure 5-9a and c, it is evident that ε_{PrF} exhibits linear relationships with both ε_{22}^{RVE} and ε_{12}^{RVE}, independent of temperature across the studied range (180 to 220 °C). This linear relationship is demonstrated by the presence of trendlines in Figure 5-9a and c. Therefore, the constant ratios of ε_{PrF} to ε_{22}^{RVE} or ε_{12}^{RVE}, referred to as strain mapping factors, k^{map} be extracted as the slopes of their trendlines. It is clearly indicated that for transverse tension and longitudinal shear deformation, $k_{\varepsilon_{22}}^{map} = 1.76$ and $k_{\varepsilon_{12}}^{map} = 1.19$, respectively.

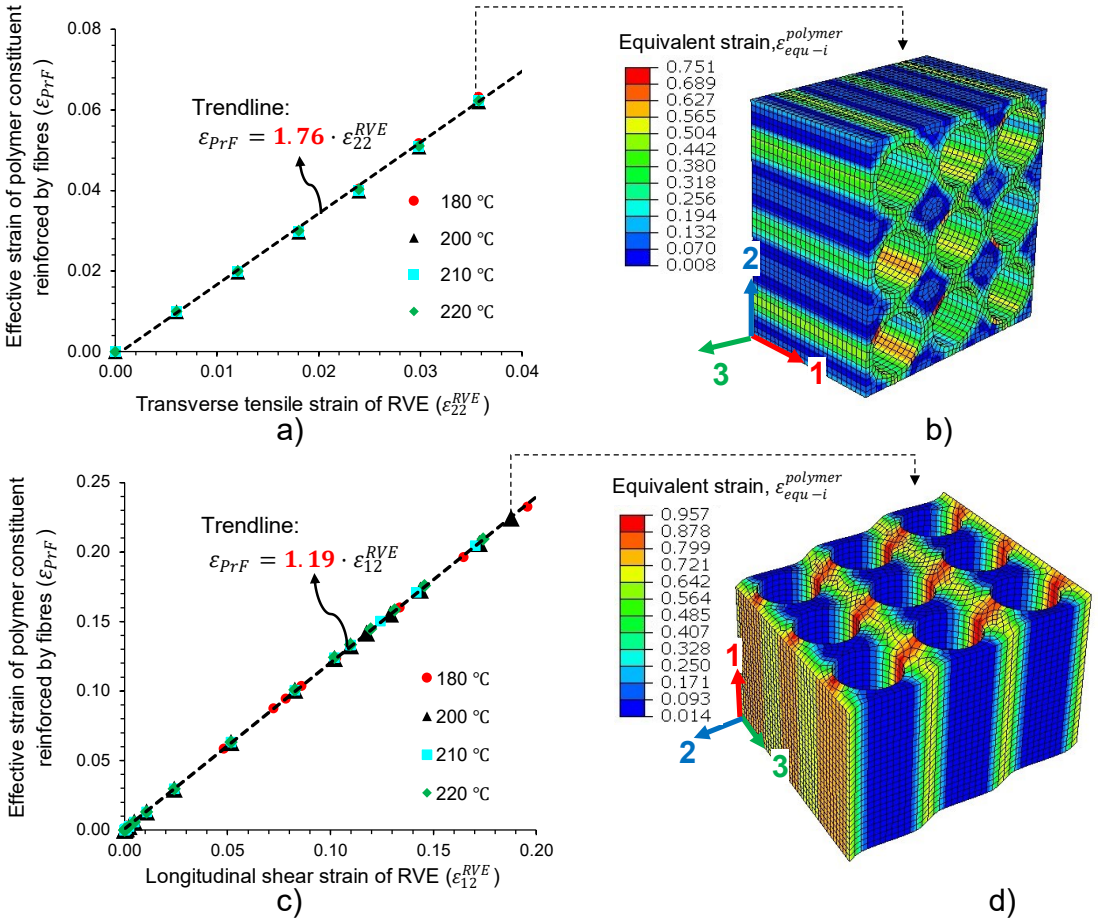


Figure 5-9. Analysis of the mapping relationships between the composite deformations and the effective deformations of PrF using RVE. a) ε_{PrF} vs. ε_{22}^{RVE} under transverse tensile at various temperatures (180 – 220 ℃), and b) the equivalent strain distribution in polymer elements when $\varepsilon_{22}^{RVE} = 0.036$, at 200 ℃. c) ε_{PrF} vs. ε_{12}^{RVE} under longitudinal shear at various temperatures (180 – 220 ℃) and d) the equivalent strain distribution in polymer elements when $\varepsilon_{12}^{RVE} = 0.188$, at 200 ℃.

With the polymer mapping relationship obtained, the stress contribution of fibre reinforced polymers can be calculated according to the polymer deformation. Therefore, Eqs. (5-31) and (5-32) can be revised as Eq. (5-35) in a one-dimensional form:

$$\sigma_C(\varepsilon_C) = \sigma_{PrF}(\varepsilon_{PrF}) + \sigma_{P-F}(\varepsilon_{P-T}) = \sigma_{PrF}(k^{map} \cdot \varepsilon_C) + \sigma_{P-F}(\varepsilon_C) \qquad (5\text{-}35)$$

where the form $\sigma(\varepsilon)$ denotes the stress response with respect to a given strain of ε. σ_C, σ_{PrF} and σ_{P-F} represent the stress responses of the composite, the PrF, and the P-F parts,

respectively; ε_C, ε_{PrF} and ε_{P-F} are composite strain and strains corresponding to the PrF and P-F parts, respectively. k^{map} represents $k_{\varepsilon_{22}}^{map}$ or $k_{\varepsilon_{12}}^{map}$, depending on whether the deformation is transverse tension or longitudinal shear. The key finding from the analysis validates the hypothesis on the mapping relationships between the composite deformations and the effective deformations of the PrF. This means that when the composite strain is ε_C , the contribution of the PrF equals to the stress generated in the polymer when the polymer strain is $k^{map} \cdot \varepsilon_C$.

The polymer constitutive model recently developed in the previous section can be adopted to calculate σ_{PrF}. The constitutive model is briefly introduced as below.

The polymer model comprises two branches: an intermolecular branch (I) that processes viscoelastic-viscoplastic behaviour to represent the intermolecular stretching within polymers, and a network branch (N) that processes viscoelastic behaviour to represent the network stretching. The constitutive model can be represented as Eqs. (5-36) and (5-37).

$$\boldsymbol{\sigma_P} = \boldsymbol{\sigma_I} + \boldsymbol{\sigma_N} \tag{5-36}$$

$$\boldsymbol{F_P} = \boldsymbol{F_I} = \boldsymbol{F_N};\quad \boldsymbol{F_I} = \boldsymbol{F_{Ie}} \cdot\ \boldsymbol{F_{Iv}} \cdot \boldsymbol{F_{Ip}}\ ;\quad \boldsymbol{F_N} = \boldsymbol{F_{Ne}} \cdot \boldsymbol{F_{Nv}} \tag{5-37}$$

Where $\boldsymbol{\sigma_P}$, $\boldsymbol{\sigma_I}$ and $\boldsymbol{\sigma_N}$ are the stress tensors of the polymer, the intermolecular branch, and the network branch, respectively. $\boldsymbol{F_P}$, $\boldsymbol{F_I}$ and $\boldsymbol{F_N}$ are the deformation gradient tensors of the polymer, the intermolecular branch, and the network branch, respectively. $\boldsymbol{F_{Ie}}, \boldsymbol{F_{Iv}}, \boldsymbol{F_{Ip}}$ are the elastic, viscous and plastic deformation gradient tensors in the intermolecular branch, respectively. And $\boldsymbol{F_{Ne}}$ and $\boldsymbol{F_{Nv}}$ are the elastic and viscous deformation gradient tensors in the network branch, respectively. In the intermolecular and network branches, the stress tensors ($\boldsymbol{\sigma_I}$ and $\boldsymbol{\sigma_N}$) are calculated from a Neo-Hookean hyperelastic spring and an Eight-chains hyperelastic spring, respectively. In this study, $\boldsymbol{\sigma_P}$ is equivalent to $\boldsymbol{\sigma_{PrF}}$, and $\boldsymbol{F_P}$ is equivalent to $\boldsymbol{F_{PrF}}$ which can be calculated from the composite deformation $\boldsymbol{F_C}$ and the mapping relationship k^{map}.

5.4.2. The contribution of the polymer-fibre interface (P-F)

The contribution of the P-F to the constitutive model is based on the adoption of the cohesive stress, that is commonly used to describe interface properties. Cohesive stress refers to the force required to separate individual atoms for a prescribed interatomic separation distance [206]. For this study, expected cohesive stress is defined to represent the P-F contribution. Figure 5-10a

shows the process of obtaining the expected cohesive stress (curve ③): by subtracting the polymer stress after mapping adjustment (curve ②) from the composite stress (curve ①) according to Eq.(5-35). The horizontal axis in Figure 5-10a is the composite (tensile/shear) strain instead of the separation distance typically used in cohesive theories [206] for the simplicity of later analysis. Theoretically, there will also be mapping relationships between the separation distance and composite deformation, but that would not be the focus of this study.

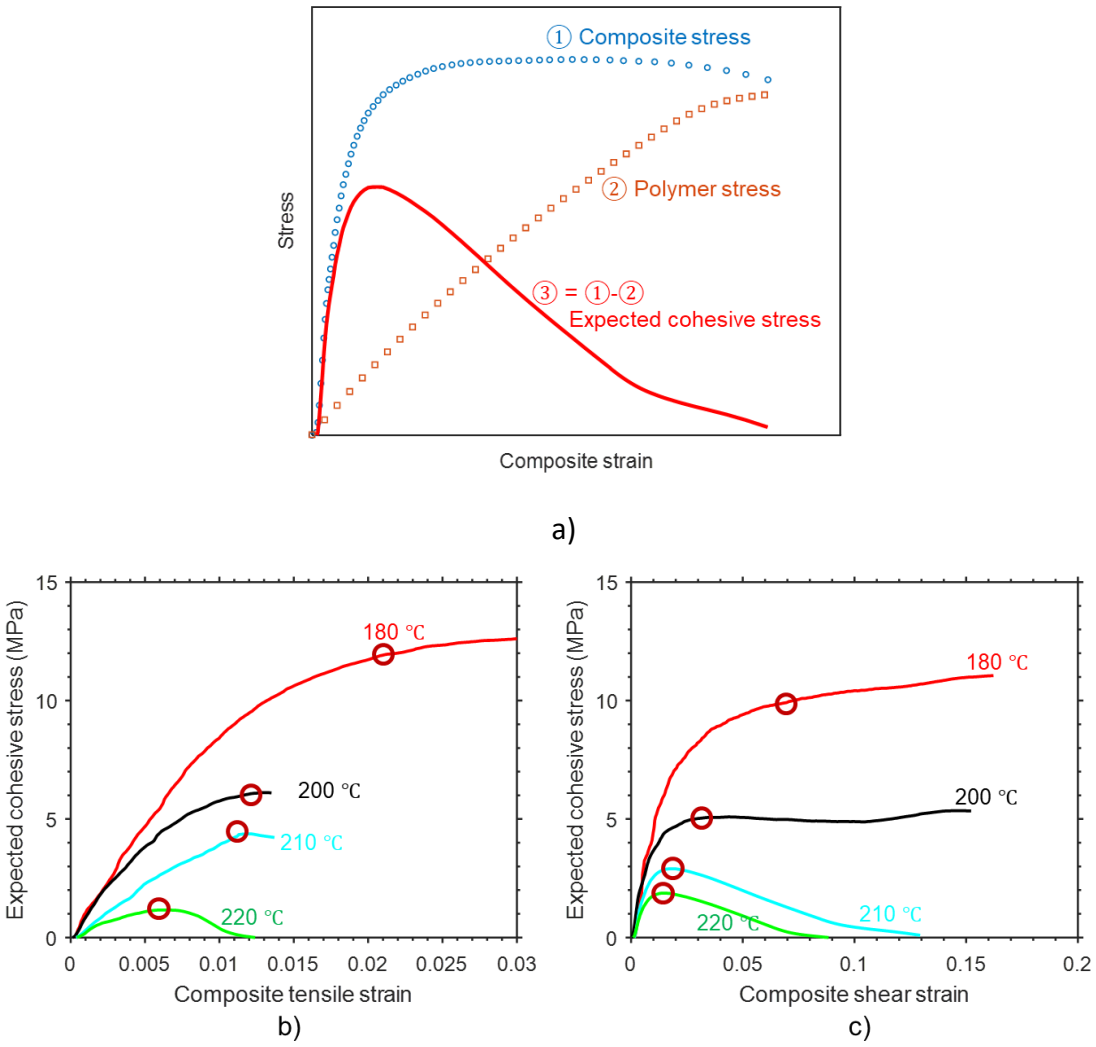


Figure 5-10. a) The calculation method for obtaining the expected cohesive stress curve. The Expected cohesive stress at different temperatures (180 – 220 ℃) under b) transverse tension and c) longitudinal shear deformation (circles representing the turning point of each curve).

Figure 5-10 b and c show the expected P-F cohesive properties behave as two different parts divided by the turning points, marked as the circles. For the higher temperatures (210 and 220 ℃), the turning points are simply defined by the peak in the curve (i.e., when the gradient turns 0). For the lower temperatures (180 and 200 ℃), due to lack of obvious softening, the turning points are determined by defining a criterion that the gradient is dropped to, which were 50 and 10 respectively for the tensile and shear curves. For each cohesive stress curve, the rapid increase occurs before the turning point, while the cohesive stress either stops increasing rapidly or starts decreasing after the turning point. The turning points in the cohesive stress-composite strain curve indicates the beginning of the P-F interface debonding. As can be inferred, the cohesive behaviour shows high temperature dependence. With the increase of temperature, there is noticeable reduction in the stress level and the strain value at the turning point.

Based on the analysis above, a rheology model was proposed for the P-F interface part comprising a tensile spring, a frictional element and a pre-tensioned spring arranged as shown in Figure 5-8. The frictional element is in parallel with the pre-tensioned spring and the two are in series with the tensile spring. At the initial stage of the deformation, only the tensile spring is activated, causing the stress to increase with the deformation. Once the stress in the frictional element reaches a certain value, the frictional element is activated and moves in conjunction with the pre-tensioned spring. According to the rheology form, it can be conjectured that with the further deformation, the tension in the pre-tensioned spring is reduced, causing a decrease in the total stress in this P-F interface part. Consequently, the stress in this P-F interface part changes. Depending on the stiffnesses of the tensile spring and compressive spring, the stress may continue to increase, remain constant or begin to decrease. Therefore, the proposed rheology model can effectively represent the cohesive stress characteristics of the P-F interface. The constitutive model can be represented as Eqs. (5-38) and (5-39).

$$\boldsymbol{\sigma_{P-F}} = \begin{cases} \boldsymbol{\sigma_{ts}} & \text{(before the friction element is activated)} \\ \boldsymbol{\sigma_{ts}} = \boldsymbol{\sigma_{pre-ts}} & \text{(after the friction element is activated)} \end{cases} \tag{5-38}$$

$$\boldsymbol{F_{P-F}} = \begin{cases} \boldsymbol{F_{ts}} & \text{(before the friction element is activated)} \\ \boldsymbol{F_{ts}} \cdot \boldsymbol{F_{pre-ts}} & \text{(after the friction element is activated)} \end{cases} \tag{5-39}$$

where $\boldsymbol{\sigma_{P-F}}$ and $\boldsymbol{F_{P-F}}$ refer to the Cauchy (true) stress tensor and deformation gradient tensor in the P-F. Meanwhile, $\boldsymbol{\sigma_{ts}}$, $\boldsymbol{\sigma_{pre-ts}}$ are the stress tensors in the tensile spring and pre-tensioned

spring, respectively. And $\boldsymbol{F_{ts}}, \boldsymbol{F_{pre-ts}}$ are the deformation gradient tensors in the tensile spring and pre-tensioned spring, respectively.

As for the cohesive model, a bilinear cohesive model is commonly used in microscale studies such as the RVE studies [207, 208]. The bilinear cohesive model is also embedded in ABAQUS [208]. However, in the macro-scale model, as indicated in Figure 5-10b and c, a more comprehensive model form is required to better capture the characteristics. Therefore, an exponential model [209] was selected to represent the macro-scale response of the cohesive behaviour, as schematically illustrated in Figure 5-11.

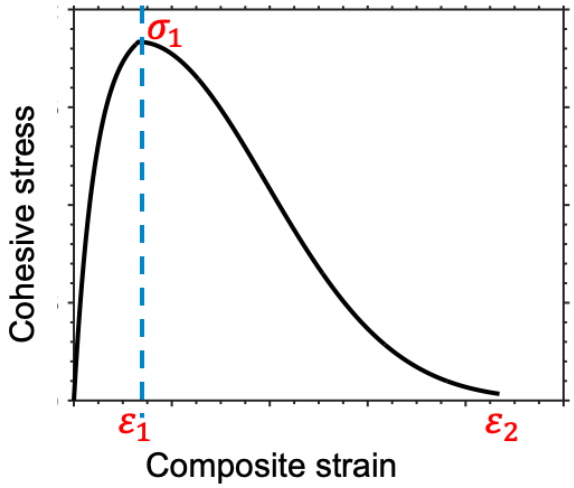


Figure 5-11. The cohesive model.

The controlling parameters for the exponential cohesive model are marked in Figure 5-11. The term ε_1 is the strain when the cohesive stress comes to a turning point, σ_1 is the stress corresponding to the turning point, and ε_2 is the strain when the cohesive model curve approaches zero, i.e., the complete debonding strain of cohesive elements. These controlling parameters, ε_1 , σ_1 and ε_2 are temperature dependent parameters. According to the expected cohesive stress curves shown in Figure 5-10b and c, the expected values of these controlling parameters, namely expected ε_1, expected σ_1 and expected ε_2 can be extracted and are shown in Figure 5-12. Note that for expected ε_2, Figure 5-10b and c show that there is no obvious failure strain under lower temperatures (180 and 200 °C), so it will be considered as infinity (∞) in this paper and represented by a relatively big number in the later fitting process. In Figure 5-12 only the values at 210 and 220 °C are shown.

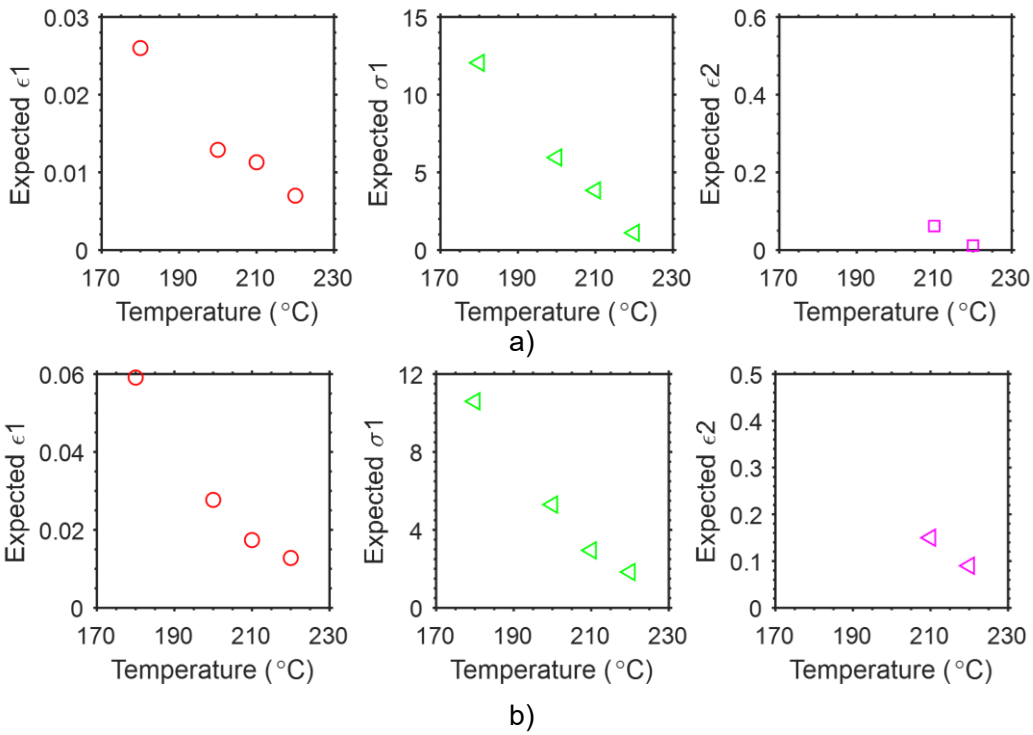


Figure 5-12. The expected controlling parameters for cohesive stress curves under a) transverse tension and b) longitudinal shear deformation.

The two-parts cohesive behaviour shown in Figure 5-11 can be represented by a piecewise-defined function: the first part is before reaching turning point, where the cohesive stress continuously increases with the increase of strain; the second part is the part after the turning point. The piecewise-defined function form is proposed as Eq.(5-40).

$$\sigma_{P-F}(\varepsilon_C) = \begin{cases} \left(-e^{\left(-\frac{3}{\varepsilon_1}\cdot\varepsilon_C+10\right)} + e^{10}\right)\cdot \sigma_1/(e^{10}-e^{7}) & (\varepsilon_C \le \varepsilon_1) \\ \sigma_1 \cdot e^{-\left((\varepsilon_C-\varepsilon_1)/(\varepsilon_2-\varepsilon_1)\cdot 2\right)^2} & (\varepsilon_C > \varepsilon_1) \end{cases} \tag{5-40}$$

where, the two formulas calculate the cohesive stresses of the first and second parts of the cohesive model, respectively. The constant values e^{10} and e^{7} are directly selected to represent considerable large numbers for obtaining the desired stress level. Other considerations are: for the continuity of the piecewise-defined function, $\sigma_{P-F}(\varepsilon_1) = \sigma_1$; $\sigma_{P-F}(\varepsilon_2)$ nears zero; marking the first-order derivative of the function as $\partial\sigma_{P-F}(\varepsilon_C)/\partial\varepsilon_C$, then $\partial\sigma_{P-F}(\varepsilon)/\partial\varepsilon$ is considerably large when $\varepsilon_C = 0$, and nears zero when $\varepsilon_C = \varepsilon_1$ and ε_2.

When it comes to the relationship between the controlling parameters, ε_1, σ_1 and ε_2, and temperatures, it can be expressed as Eqs. (5-41)-(5-43) according to Figure 5-12.

$$\varepsilon_1 = \varepsilon_{10} \cdot \mathrm{e}^{k_{\varepsilon 1} \cdot (\theta - \theta_m)} \tag{5-41}$$

$$\sigma_1 = \sigma_{10} + k_{\sigma 1} \cdot (\theta - \theta_m) \tag{5-42}$$

$$\varepsilon_2 = \varepsilon_{20} \cdot e^{k_{\varepsilon 2} \cdot (\theta - \theta_m)} + b_{\varepsilon 2} \tag{5-43}$$

where, $\varepsilon_{10}, k_{\varepsilon 1}, \sigma_{10}, k_{\sigma 1}, \varepsilon_{20}, k_{\varepsilon 2}$ and $b_{\varepsilon 2}$ are all cohesive-related material constants. θ and θ_m refer to the current temperature and the melting temperature of the matrix polymer, respectively. Note that all temperatures were converted to absolute temperatures in Kelvin (K) in the constitutive modelling.

By fitting the Eqs.(5-40)-(5-43) to Figure 5-10b and c, the cohesive-related constants are determined. The material constants together with the whole model are listed in next Section.

5.4.3. Model calibration and performance

The entire constitutive model and all model constants are summarised in Table 5-3 and Table 5-4.

Table 5-3. Summary of the unified constitutive model.

Composite model (C)

$$\sigma_C(\varepsilon_C) = \sigma_{PrF}(\varepsilon_{PrF}) + \sigma_{P-F}(\varepsilon_C)$$

Contribution of polymer constituent reinforced by Fibres (PrF)

$$\varepsilon_{PrF} = k^{map} \cdot \varepsilon_C$$

where k^{map} is the mapping relationship

For transverse tension: $k_{\varepsilon_{22}}^{map} = \bar{\varepsilon}_{equ}^{polymer} \cdot V_{F_p}/\varepsilon_{22}^{RVE}$

For longitudinal shear: $k_{\varepsilon_{12}}^{map} = \bar{\varepsilon}_{equ}^{polymer} \cdot V_{F_p}/\varepsilon_{12}^{RVE}$

Constitutive model of polymers:

$$\sigma_{PrF} = (\sigma_I + \sigma_N) \cdot Cor(\theta)$$

The intermolecular resistance (I)

Viscoelastic part ($\sigma_I < \sigma_{lim}$)

$$\sigma_I = 2 \cdot C_1(\theta) \cdot \left(\lambda_{Ie}^2 - \frac{1}{\lambda_{Ie}}\right);\ \lambda_{Ie} = \exp(\varepsilon_{Ie})$$

$$C_1(\theta) = (E_{ref} + E_1 \cdot (\theta - \theta_{ref})^{m_E})/(4 \cdot (1+\nu))$$

$$\dot{\varepsilon}_{Ie} = \dot{\varepsilon}_{PrF} - \dot{\varepsilon}_{Iv};\ \dot{\varepsilon}_{Iv} = \dot{\gamma}_{01} \cdot \left(\frac{\dot{\varepsilon}_{PrF}}{\dot{\varepsilon}_{ref}}\right)^{m_{rI}} \cdot \left(\frac{\sigma_I}{\eta_{Iv}(\theta)}\right)^{m_{Iv}}$$

$$\eta_{Iv}(\theta) = \eta_{Iv0}/\left(\frac{\theta-\theta_g}{\theta_m}\right)^{m_{\theta I}}$$

Viscoplastic part ($\sigma_I \geq \sigma_{lim}$)

$$\dot{\varepsilon}_{Ie} = \dot{\varepsilon}_{PrF} - \dot{\varepsilon}_{Iv} - \dot{\varepsilon}_{Ip};\ C_p(\theta) = C_{p1}/\left(\frac{\theta}{\theta_m}\right)^{m_{cp}}$$

$$\dot{\varepsilon}_{Ip} = \dot{\varepsilon}_{0p} \cdot \left(\frac{\dot{\varepsilon}_{PrF}}{\dot{\varepsilon}_{ref}}\right)^{m_{Ip}} \cdot \left\{\exp\left(\frac{1}{C_p(\theta)} \cdot \left(\frac{\sigma_I}{\sigma_{lim}} - 1\right)\right) - 1\right\}$$

Network resistance (N)

$$\sigma_N = \frac{C_R(\theta)}{3J_{Ne}} \cdot \frac{\bar{\lambda}_L}{\bar{\lambda}_{Ne}} \cdot \mathcal{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) \cdot \left(\frac{2}{3} \cdot \left((\bar{\lambda}_{Ne})^2 - 1/\bar{\lambda}_{Ne}\right)\right)$$

$$\bar{\lambda}_{Ne} = \sqrt{(\exp(2 \cdot \varepsilon_{Ne}) + 2\exp(-\varepsilon_{Ne}))/3}$$

$$\mathcal{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) = 3 \cdot \left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) + \frac{9}{5} \cdot \left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^3 + \frac{297}{175} \cdot \left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^5 + \frac{1539}{875} \cdot \left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^7$$

$$C_R(\theta) = C_{R_{ref}} + C_{r1} \cdot (\theta - \theta_{ref});\ \eta_{Nv}(\theta) = \frac{\eta_{Nv0}}{\left(\frac{\theta-\theta_g}{\theta_m}\right)^{m_{\theta N}}}$$

$$\dot{\varepsilon}_{Ne} = \dot{\varepsilon}_{PrF} - \dot{\varepsilon}_{Nv};\ \dot{\varepsilon}_{Nv} = \dot{\gamma}_{02} \cdot \left(\frac{\dot{\varepsilon}_{PrF}}{\dot{\varepsilon}_{ref}}\right)^{m_{rN}} \cdot \left(\frac{\sigma_n}{\eta_{Nv}(\theta)}\right)^{m_{Nv}}$$

Temperature evolution and thermal softening

$$Cor(\theta) = \left(1 - \frac{\theta_{cor}-\theta}{\theta_m-\theta}\right)^{m_h};\ \theta_{cor} = \theta + \int \dot{\theta}$$

$$\dot{\theta} = \frac{\omega(\dot{\varepsilon})}{\bar{C}(\theta)}\left(\sigma_{Iv} \cdot \dot{\varepsilon}_{Iv} + \sigma_{Ip} \cdot \dot{\varepsilon}_{Ip} + \sigma_{Nv} \cdot \dot{\varepsilon}_{Nv}\right)$$

$$\omega(\dot{\varepsilon}_{PrF}) = \begin{cases} 0 & ;\ if\ \dot{\varepsilon}_{PrF} \leq 0.01 \\ \min(0.33 \cdot (\log_{10}\left(\frac{\dot{\varepsilon}_{PrF}}{\dot{\varepsilon}_{ref}}\right), 1) & ;\ if\ \dot{\varepsilon}_{PrF} > 0.01 \end{cases}$$

$$\bar{C}(\theta) = \bar{C}_{ref} + \bar{C}_1 \cdot (\theta - \theta_{ref})$$

Contribution of Polymer-fibre interface (P-F)

$$\sigma_{P-F}(\varepsilon_C) = \begin{cases} \left(-e^{\left(-\frac{3}{\varepsilon_1}\cdot\varepsilon_C+10\right)} + e^{10}\right) \cdot \frac{\sigma_1}{(e^{10}-e^7)} & (\varepsilon_C \leq \varepsilon_1) \\ \sigma_1 \cdot e^{-\left(\frac{\varepsilon_C-\varepsilon_1}{\varepsilon_2-\varepsilon_1}\cdot 2\right)^2} & (\varepsilon_C > \varepsilon_1) \end{cases}$$

$$\varepsilon_1 = \varepsilon_{10} \cdot e^{k_{\varepsilon 1}\cdot(\theta-\theta_m)}$$

$$\sigma_1 = \sigma_{10} + k_{\sigma 1} \cdot (\theta - \theta_m)$$

$$\varepsilon_2 = \varepsilon_{20} \cdot e^{k_{\varepsilon 2}\cdot(\theta-\theta_m)} + b_{\varepsilon 2}$$

Table 5-4. Summary of all the material constants in the unified constitutive model

Polymer constituent (PA6) reinforced by fibres (PrF)										
General properties	ν	$\bar{C}_{ref}$ (kJ/m^{3}K)	$\bar{C}_1$	θ_{ref} (K)	θ_g (K)	θ_m (K)	m_h			
	0.48	3.00	0.01	296	320	503	0.5			
Viscoelastic properties	E_{ref} (MPa)	E_1 (MPa)	m_E	σ_{ref} (MPa)	m_σ	$\dot{\gamma_{01}}$	m_{rI}	m_{Iv}	η_{Iv0}	$m_{\theta I}$
	2400	-1041.2	0.1554	12.0	3.257	0.001	0.8	0.5	0.25	1.20
Viscoplastic properties	$\dot{\varepsilon_{0p}}$	m_{Ip}	C_{p1}	m_{cp}						
	0.0010	0.5	0.05	3.0						
Network properties	$\dot{\gamma_{02}}$	m_{rN}	m_{Nv}	η_{Nv0}	$m_{\theta N}$	$\bar{\lambda}_L$	$C_{R_{ref}}$		C_{r1}	
	0.001	0.1	0.10	490	0.01	2.60	23.958		-0.120	
Polymer contribution mapping parameter K_{map}										
Transverse tension : $k_{\varepsilon_{22}}^{map} = 1.76$					Longitudinal shear: $k_{\varepsilon_{12}}^{map} = 1.19$					
Polymer-fibre interface (P-F)										
	ε_{10}	$k_{\varepsilon 1}$	σ_{10} (MPa)	$k_{\sigma 1}$ (MPa/K)	ε_{20}	$k_{\varepsilon 2}$	$b_{\varepsilon 2}$			
Transverse tension	0.0045	-0.0328	-2.542	-0.272	6.0 x 10^{-8}	-0.55	0.011			
Longitudinal shear	0.0075	-0.0390	-1.710	-0.226	7.0 x 10^{-8}	-0.55	0.089			

Figure 5-13 and Figure 5-14 show the model prediction compared to experimental results for transverse tensile and VNR shear tests, respectively. The results show that the model can predict the transverse tensile and longitudinal shear properties with satisfying accuracy. The temperature effects are well reflected, and the strain hardening or softening features are well captured. The model takes the intra-ply tensile and shear deformation mechanisms into account, therefore, can be used later for numerical simulation of the solid-state stamping of sheet UD FRTPs.

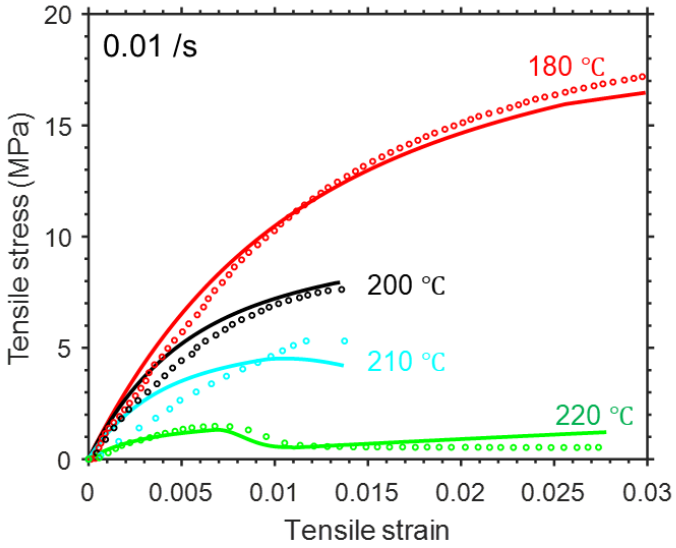


Figure 5-13. The comparison between the constitutive model predictions (solid lines) and the experimental results (circles) of transverse tensile tests.

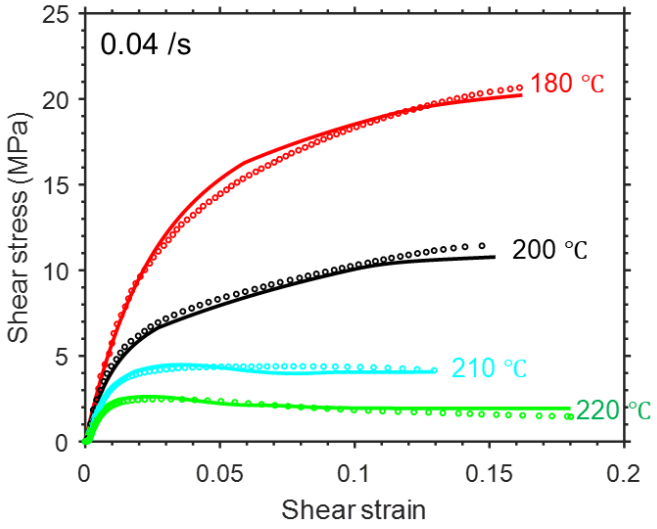


Figure 5-14. The comparison between the constitutive model predictions (solid lines) and the experimental results (circles) of VNR shear tests.

5.5. Numerical implementation of the UD FRTP model

The developed constitutive model was implemented into ABAQUS/Explicit using a user defined subroutine to simulate a bias-extension deformation. The validity and accuracy of this

constitutive model can be evaluated by comparing the FEA and experimental results of the bias-extension deformation.

The experimental setup of the bias-extension test is the same as the experimental setup of the transverse tension test. A bias-extension specimen with a dimension of 15 x 150 x 3.3 mm and a freely deforming length of 56 mm was tested. The configuration of the composite laminates was $[45_3/\text{-}45_3]_{2s}$. The bias-extension test was conducted at 200 °C with a crosshead speed of 50 mm/min. The strain field of the bias-extension specimen was also processed via the DIC method in GOM software with the same setting as stated in Section 3.4.

To validate the accuracy of the constitutive model, a FEA model was constructed with the same dimensions and carbon fibre configurations as the experimental specimen. The specimen at a thickness of 1.65 mm - half the thickness of the experimental specimen, was partitioned into four sections in the thickness direction, with material orientation of $[45/\text{-}45]_2$. For the boundary conditions for the specimen, a vertical displacement of 4 mm was applied to the top edge, while the bottom edge was fixed, and the back surface was set to be the symmetrical surface. The solid element (C3D8R) was selected for the specimen.

The shear strain field (ε_{12}) obtained from the FEA and experimental results at a displacement of 2.7 mm were compared and presented in Figure 5-15a. It is evident that the simulation using the developed constitutive model accurately predicts the shear strain field of the bias-extension deformation. Additionally, the averaged shear strains of the middle section marked by the white lines in Figure 5-15a were compared and presented in Figure 5-15b. The FEA results were calculated as the average value of all element values of the middle section, while the experimental results were computed as the average value of all the facets crossed by the middle section. Figure 5-15 demonstrates that the simulation using the developed constitutive model can effectively predict the in-plane deformation of UD CF/PA6.

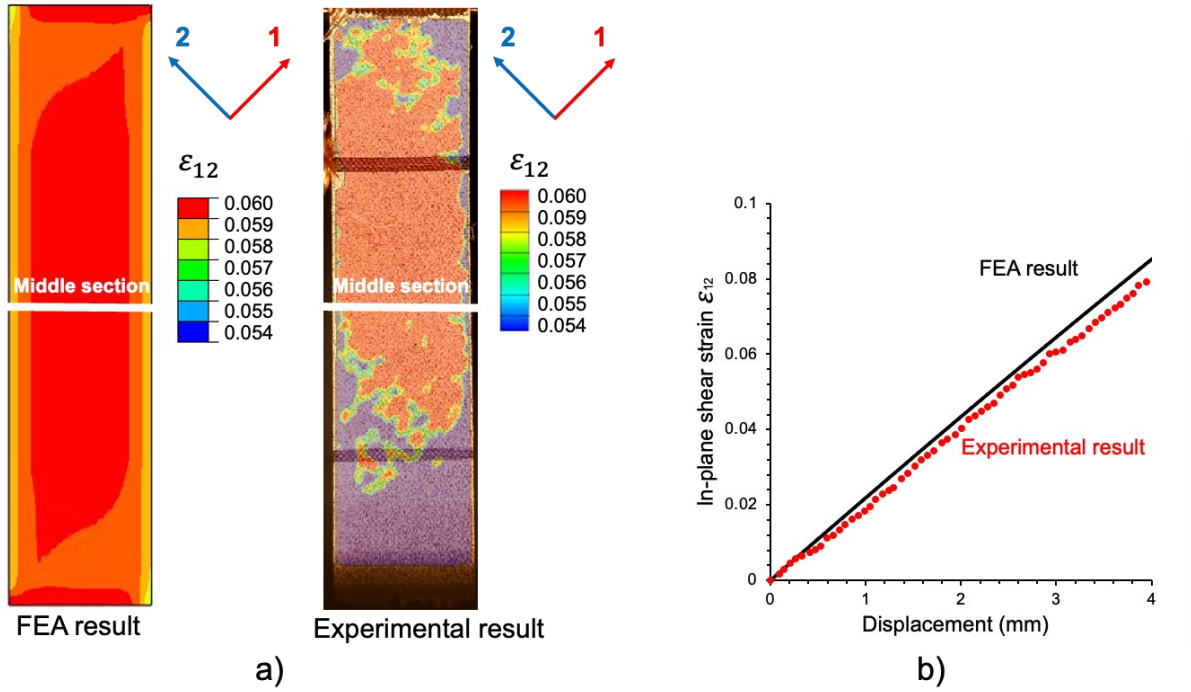


Figure 5-15. The comparison between the FEA and experimental results of a bias-extension test. a) Comparison of the in-plane shear strain distribution at the displacement of 2.7 mm (1 is the fibre direction); b) The comparison of the history of the averaged in-plane shear strain across the middle section of the specimen.

5.6. Finite element (FE) simulation and analysis of the solid-state stamp forming process

Using FEA simulation to predict the quality of the as-formed part is an essential step for providing guidance on forming process design. It is also a more efficient and economical way for these "trial and error" process during manufacturing technology development. Since Section 5.5. has successfully embedded the constitutive model into ABAQUS using a Fortran subroutine and checked its validity for predicting the in-plane deformation, FEA models to study the solid-state stamp forming process can be built to predict the potential results and provide indications on the forming trials.

5.6.1. Numerical model construction

Figure 5-16 illustrates the model assembly and loading conditions employed in the numerical simulation of the solid-state stamp forming process. Both the tool set up and material

configurations match the experiments, which will be discussed in Section 6.2. The model comprises forming tools and the workpiece. The forming tools, including punch, die and blank holder and spacers, were modelled as rigid body. The workpieces used in this study were cross-plied CF/PA6 laminates produced with UD tapes with various configurations. For example, for the laminate with a configuration of $[0_3/90_3]_{2s}$, it was modelled as 8 layers of shells, each of which represented a block of 3 plies of prepregs with the same fibre orientation. Hence, each layer of shell had a thickness of 0.4125mm, resulting in a total thickness of 3.3mm. The boundary conditions include a vertical displacement of 40.2 mm upward on the punch, a vertical pressure downward on the die, and fixed conditions on all degrees of freedom for the blank holder. To improve the simulation efficiency, only a quarter of the entire stamp forming set is modelled, assuming that the deformation of the cross-plied CF/PA6 laminate is symmetric about the X-Y plane and Y-Z plane.

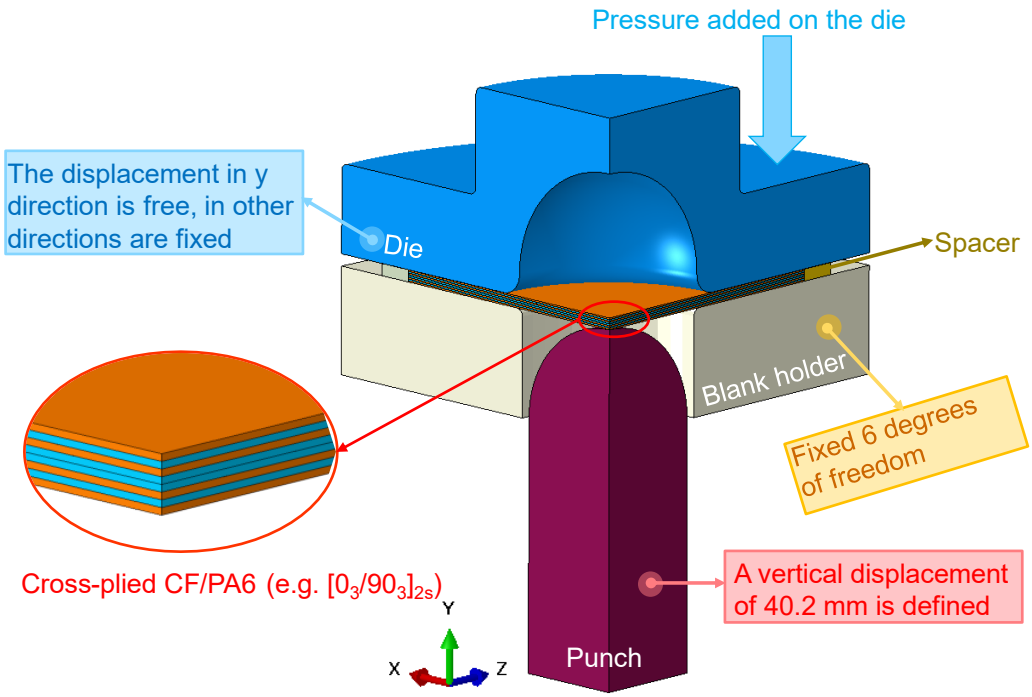


Figure 5-16. The model assembly and loading conditions of the FEA model.

To validate the bending decoupling method, two FEA models were constructed and evaluated: one utilising traditional solid element, and the other employing superimposed elements. Both simulations were conducted at a temperature of 200 ℃ with detailed material properties provided later in this study. The distribution of in-plane shear strain (ε_{12}) was analysed, as

shown in Figure 5-17. The results indicated that the simulation using superimposed elements successfully predicted the formation of wrinkles on the deformed part, while the solid elements failed to do so. Since accurately predicting wrinkles is crucial in capturing the key features of the as-formed part, it can be concluded that the bending decoupling approach is essential for effectively simulating the stamp forming process of CFRTPs.

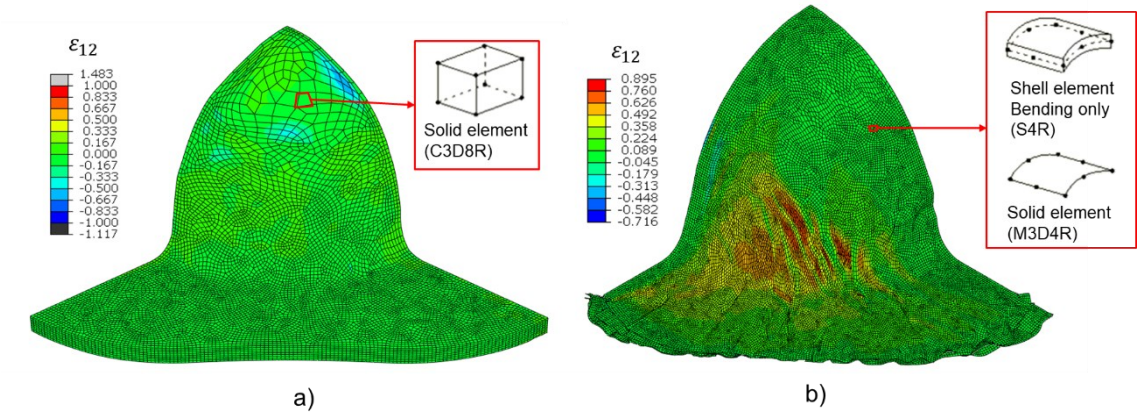


Figure 5-17. Comparison of the numerical simulation results a) with (using solid elements) and b) without bending decoupling (superimposed shell and membrane elements)

The FEA model's input, including material properties, is presented in Table 5-5. The density of the UD CF/PA6 was obtained from the material datasheet [183], while the membrane behaviour data were derived from the constitutive model and the corresponding VUMAT subroutine developed in the previous section. As the polymer properties dominate the transverse properties of the composite plies, the transverse bending rigidity was reasonably assumed to be the rigidity of pure PA6 characterised in Section 3.3. The inter-ply friction and tool-ply friction coefficients were determined based on [15]. To expedite the computation process, a mass scaling of 1e+05 was set. The default pressure on the die was set to be 2.77 MPa, corresponding to a blank holding force of 60 kN used in the experiments as described in the next section. Additionally, different values of blank holding force were also examined to study their effects.

Table 5-5. Model setup for numerical simulation

Item	Settings
UD CF/PA6	Density: 1.15e-09 tonne/mm3 Membrane(M3D4R): VUMAT subroutine Bending(S4R): temperature dependent linear elastic engineering constants $T = 453\ K$ (180 °C): $E22 = 115.57$ MPa $T = 463\ K$ (190 °C): $E22 = 93.55$ MPa $T = 473\ K$ (200 °C): $E22 = 72.61$ MPa $T = 483\ K$ (210 °C): $E22 = 52.65$ MPa $v12 = v13 = v23 = 0.32$
Tool section	Set as rigid section
Inter-ply friction coefficient	0.3812, maximum shear stress: 20 MPa
Tool-ply friction coefficient	0.1218
Initial temperatures	CFRTP sheet: 453 – 483 K (180 – 210 °C)
Mass scaling	1e+05
Blank holding force	0.92 MPa (20 kN), 2.77 MPa (60 kN, default), 6.92 MPa (150 kN)

5.6.2. Analysis of the numerical simulation results

This section explores various processing parameters to provide insights into the potential qualities of the parts and to identify the optimal stamp forming parameters.

1) The effects of blank shape

The initial blank shape plays a crucial role in determining the quality of stamp formed components, as it affects the extent of material stretching and the effectiveness of the applied blank holding force. Furthermore, the design of the blank shape can potentially affect the occurrence of wrinkles and cracks. In this study, three different blank shapes were designed and simulated to investigate the blank shape effects on the forming process, as shown in Figure 5-18. Figure 5-18 shows the initial blank shape design, represented by a black circular shape (Type A) with a diameter of 150 mm. Based on its simulation results of Type A, the excessive flange area was measured, and another two optimised blank shape were designed, shown as the red (Type B) and green (Type C) shapes in Figure 5, respectively. Compared to Type B, Type C overall has relative smaller dimensions.

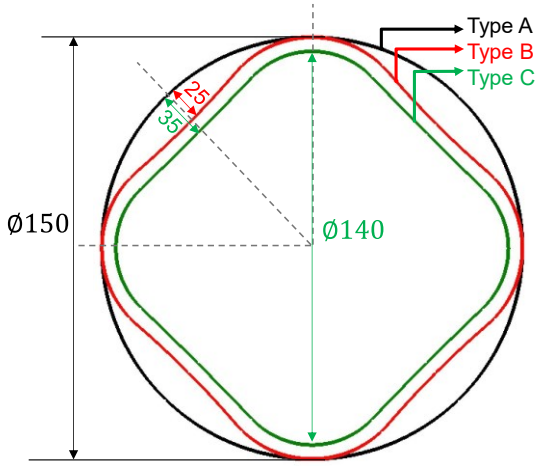


Figure 5-18. Design of blank shapes (unit: mm)

After conducting the FEA simulation, the distributions of in-plane shear strain (ε_{12}) obtained for the three blank shapes are compared and presented in Figure 5-19. For the easiness of presentation, only a quarter of the formed component is presented, corresponding to a quarter in the blank shape. The areas labelled ①② and ③, demonstrate the major differences between the three blank shapes, particularly in the flange areas. For Type A, an excessive amount of materials is observed in the flange area ①, which can potentially impede material stretching during forming. In contrast, the flange area of Type B is relatively small, resulting in an improved dome shape after forming. In terms of the in-plane shear strain (ε_{12}), Type B shows a reasonable distribution, whereas, Type A, shows some extreme values. When it comes to the simulation result of Type C, however, Type C exhibits an increasing number of wrinkles at position ③ due to the reduced overall dimension of the blank shape. The quality of the as-formed part can also be inferred from the distribution of shear strain. The scale bar shows that the maximum shear strain (1.381) for Type A is much higher than 1, indicating excessive element distortion. The shear strain distributions for Type B and Type C are more similar, indicating a comparable range of deformation. These findings confirm that Type B is the most optimal blank shape for forming the dome component.

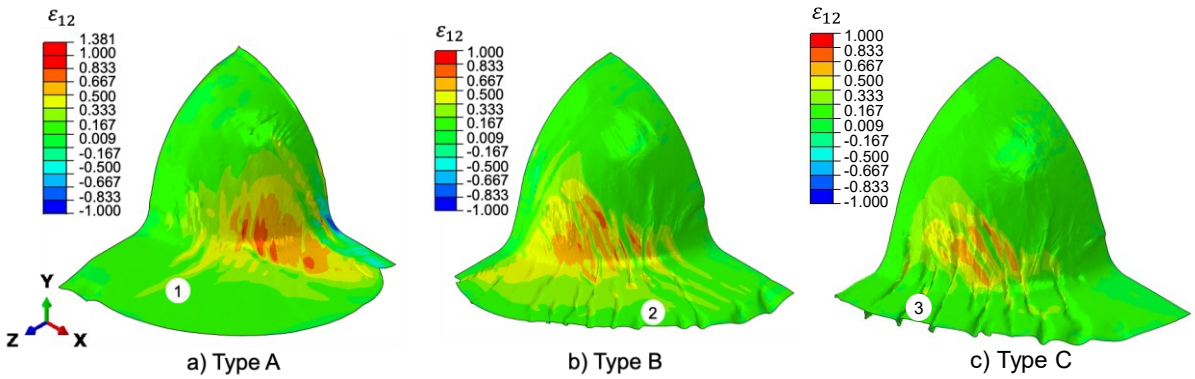


Figure 5-19. Comparison of the blank shape effects via the in-plane shear strain (ε_{12}) distributions from FEA simulation results. ① The excessive material on the flange area. ② the relative flat area on the flange. ③ the incompatible deformation between layers and wrinkles.

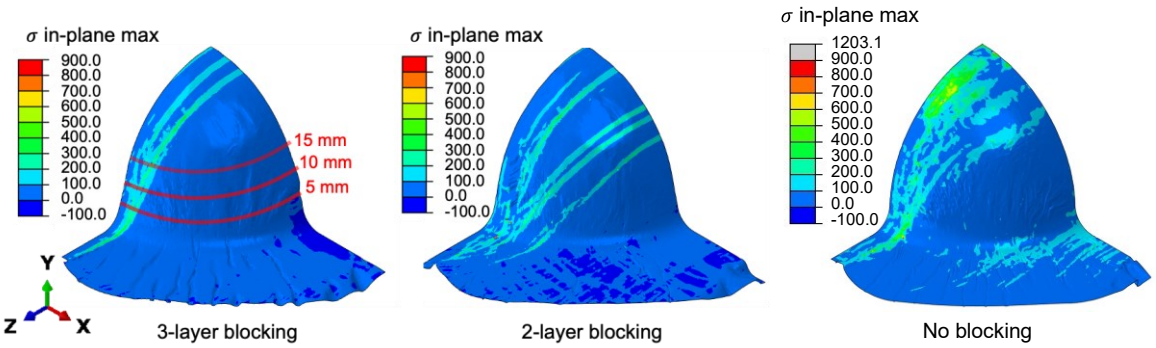
2) The effects of layer-blocking

The effects of layer blocking are investigated to determine the optimal carbon fibre configurations in cross-plied FRTPs. The strong in-plane strong anisotropy results in a lower transverse tensile strength compared to the fibre direction, increasing the risk of transverse cracking within one single layer. Layer blocking is employed to address this issue by consolidating a few layers together in the same direction during the forming process, potentially increasing the transverse tensile strength. However, if a single blocked layer becomes too thick, issues such as delamination and wrinkles may arise. To investigate this topic, except the layup mentioned before (i.e., $[0_3/90_3]_{2s}$), two additional layups: $[0_2/90_2]_{3s}$ and $[0/90]_{6s}$ with a single-ply thickness of 0.1375 mm, were simulated and analysed.

In order to investigate the effects of layer blocking on protecting against in-plane failure, the maximum stress failure criterion was employed, as it is a well-known method for predicting composite failure [210]. According to this criterion, failure occurs once the longitudinal tension, transverse tension or in-plane shear stresses exceed the corresponding yield stresses. Therefore, in order to visually represent the cumulative effects of these stresses, the maximum principal stress is employed as an indicator of the risk of fibre splits and cracks. It should be noted that due to the anisotropy of composites, there is no universal method for calculating principal stresses. Figure 5-20a shows the maximum principal stress distribution calculated by ABAQUS

for the three different configurations. It can be observed that the layer blocking cases (3-layer blocking and 2-layber blocking) exhibit a less pronounced distribution of maximum principal stress, with smaller magnitudes compared to the no-blocking case. Among the layer blocking configurations, the 3-layer blocking configuration demonstrates the least intensity of maximum principal stress distribution, indicating a lower risk for fibre splits or cracks. However, despite the advantage of layer blocking, there remains a risk of inter-ply failure due to differences in in-plane and inter-ply rigidities, which can lead to delamination. Figure 5-20b shows the section views of the as-formed part at the 5 mm, 10 mm and 15 mm from the bottom of the dome part for various layer-blocking configurations. It can be observed that while wrinkles were present in all configurations at 5 mm, the severity of wrinkles in the 3-layer blocking configuration is higher than that of the 2-layer blocking and no-blocking configurations. At higher positions in the cut view (10 mm and 15 mm), the no-blocking configuration shows almost no wrinkles between layers, whereas wrinkles are still present in the 3-layer and 2-layer configurations.

Based on the comparative results of three layer-blocking configurations analysed, some insights can be obtained regarding to the optimal configurations. The optimal configuration should take into account the precondition of avoiding crack failure. Additionally, it is observed that using less layer blocking can potentially reduce the severity of wrinkles and the risk of delamination.



a)

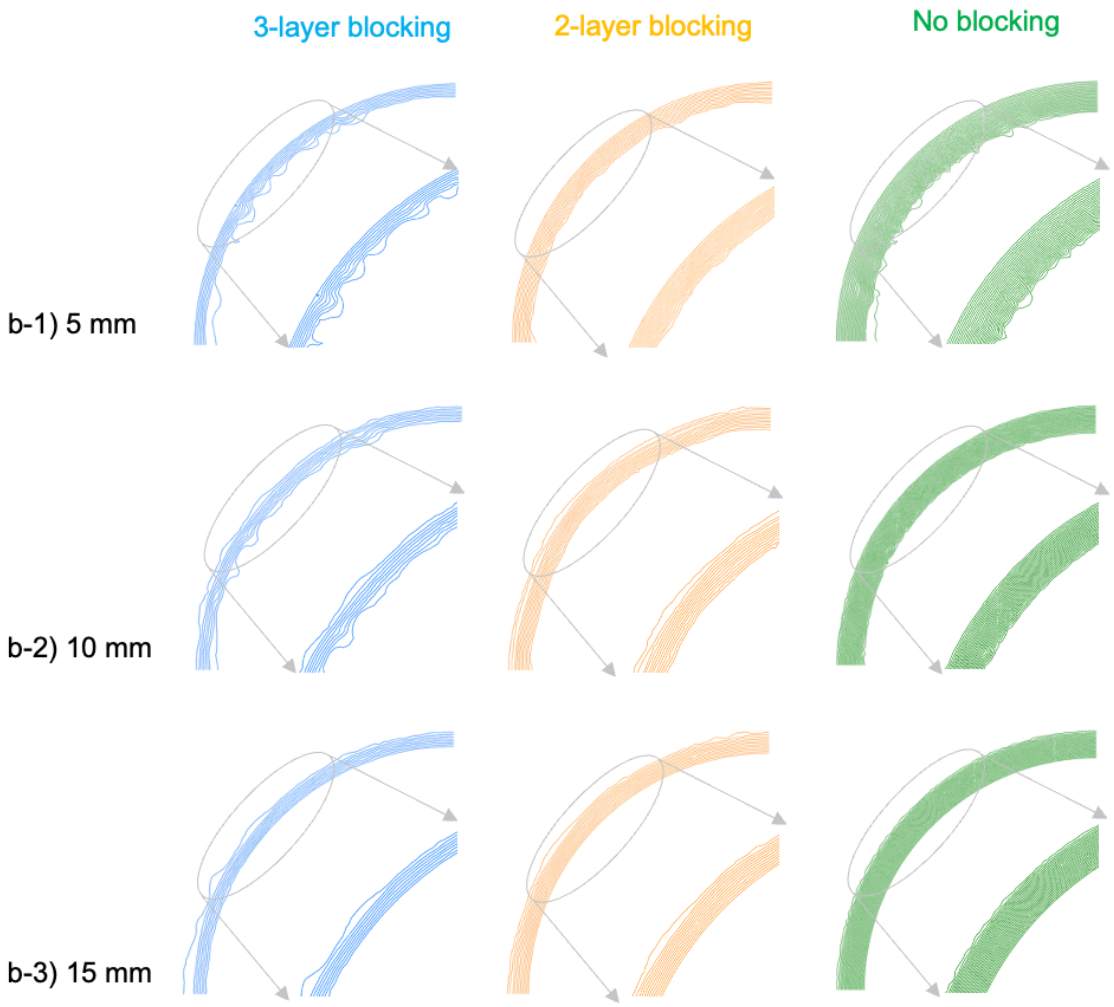


Figure 5-20. Analysis of the layer blocking effects reflected via a) the maximum in-plane stress (σ_{max}) and b) the cutting view of deformed components.

3) The effects of temperature

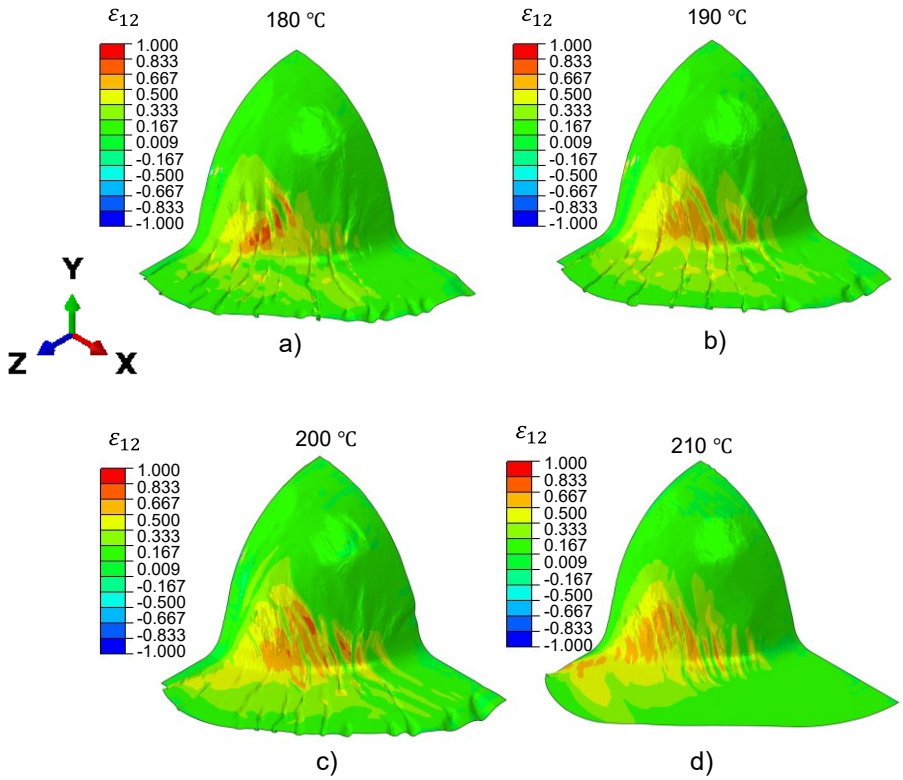


Figure 5-21. Analysis of the temperature effects reflected via the in-plane shear strain (ε_{12}) distribution at temperatures between 180 – 210 °C.

With the exploration of blank shape and layer-blocking effects, determining the processing parameters becomes the next crucial step. Among various processing parameters, temperature is identified as the most critical processing parameters that affects the quality of as-formed components, as concluded from the thermomechanical characterisation in Section 4.4. To assess the effects of temperature on the stamp formed parts, four simulations were conducted at temperature ranging from 180 to 210 °C. Figure 5-21 shows the distributions of in-plane shear strain for these simulations. It can be concluded that, with an increase in temperature, the area that exhibiting higher shear strain ($\varepsilon_{12} \geq 0.5$) expands. According to the maximum strain criterion [211], high strain values, particularly extreme ones, can lead to composite failure. Thus, components formed at lower temperature demonstrates a lower risk of failure. However, the simulation results reveal a noticeable trend of reduced wrinkles in components formed at higher temperatures. To assess the wrinkles of the deformed components, an indicator proposed

by Dörr [15], namely the modified mean curvature, $\bar{\kappa}$, was employed. The modified mean curvature is defined according to Eq. (5-44):

$$\bar{\kappa} = \frac{|SK1| + |SK2|}{2} \tag{5-44}$$

where, $SK1$ and $SK2$ refer to the curvature changes about material local 1- and 2-axes, respectively.

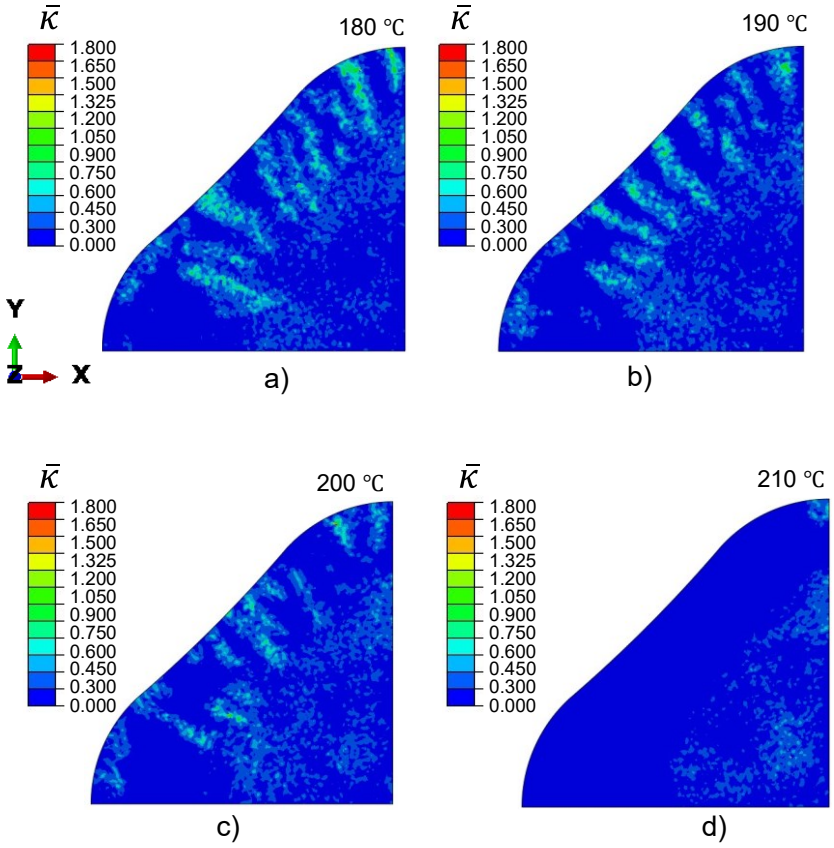


Figure 5-22. Analysis of the temperature effects reflected via the modified mean curvature ($\bar{\kappa}$) at temperatures between 180 – 210 °C.

The distribution of modified mean curvatures for the top blocked layer, simulated at various temperatures is plotted on the initial blank sheet, as shown in Figure 5-22. Notably, at lower temperatures (180/190 °C), several areas, particularly on the flange regions, exhibit high values of $\bar{\kappa}$, suggesting potential wrinkles in Figure 5-22. The wrinkles in the 200 °C case become less prominent and in the 210 °C case, they are almost eliminated. Consequently, it is reasonable to

infer that higher temperatures can help decrease wrinkle intensity. Based on the conclusions drawn from the thermomechanical characterisation studies, deformation at 210 °C and above should be avoided due to the strain softening characteristics. Thus, it is concluded that the optimal forming temperature for stamp forming UD CF/PA6 lies around 200 °C, as it balances the reduction of wrinkle intensity while avoiding strain softening effects.

5.7. Summary

This chapter has integrated experimental studies on polymers and UD CF/PA6 with the development of constitutive models for PA6 and CF/PA6, which have been implemented in the commercial FEA software, ABAQUS. The numerical simulations of the solid-state stamp forming process of CF/PA6 were conducted using the FEA model. Through an analysis of the simulation results, some preliminary conclusions have been drawn regarding the forming windows. These conclusions will be further validated and confirmed through additional experimental studies in the next chapter, following the guidelines developed in this chapter.

CHAPTER 6 Forming Experiments and Quality Assessment

*This Chapter is reproduced from publication **A3**. in the Publication list.

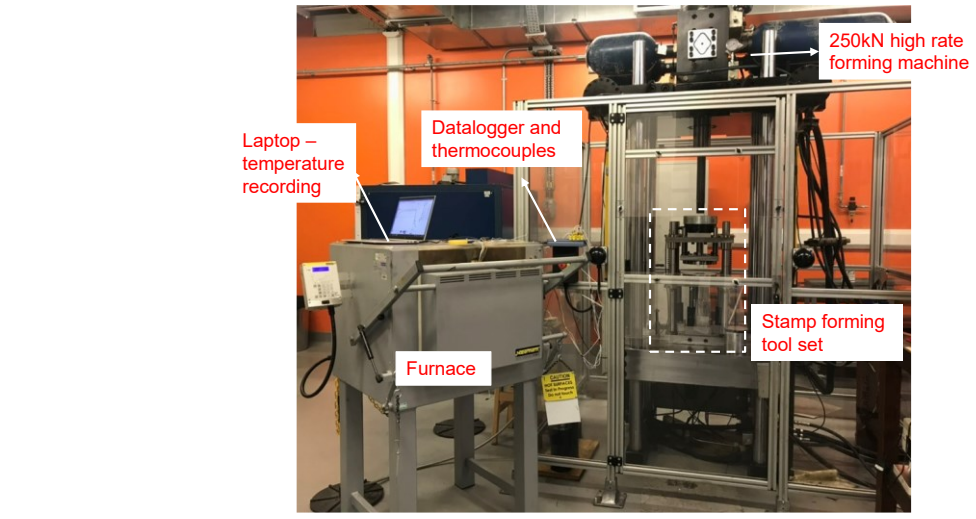
6.1. Introduction

This chapter is dedicated to the experimental trials conducted on UD CF/PA6 for solid-state stamp forming, with the primary objective of identifying the optimal processing windows and exploring the forming process in depth. The study specifically investigates and analyses the effects of the blank shape, layer blocking, and temperature. The results obtained are consistent with the simulation analysis presented in Section 5.6. Furthermore, to verify the accuracy of the FEA model, comparisons are made between the simulation results and the experimental results.

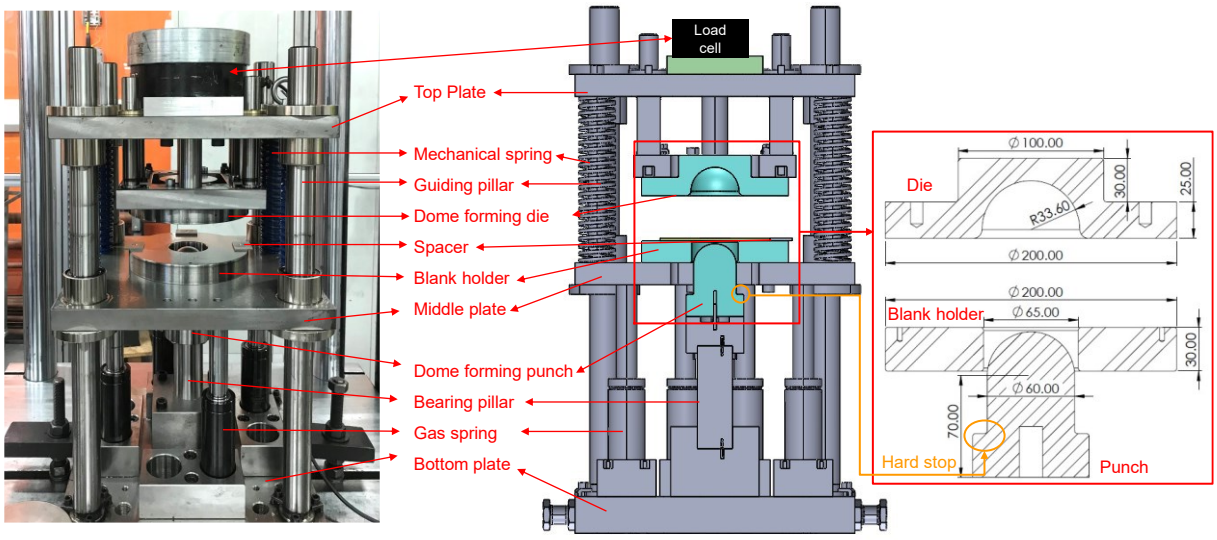
6.2. Experimental design for the solid-state stamp forming process

6.2.1. Setup of the stamp forming machine and tools

The UD CF/PA6 forming experiments were conducted using a high-rate forming machine with a capacity of 250 kN. The setup of the forming machine is illustrated in Figure 6-1a. The stamp forming tool set was mounted onto the machine, while a furnace was placed adjacent to the machine for heating up the UD CF/PA6 laminates. To ensure the workpiece reached the desired forming temperatures, the temperature of the workpiece was closely monitored and recorded during the heating and transferring stages. Thermocouples were used to measure the temperature, and a datalogger connected to a laptop was used for data collection and recording.



a)



b) c) d)

Figure 6-1. Experimental setup of the solid-state stamp forming process of CF/PA6 sheet. a) Stamp forming equipment setup b) Stamp forming tool set; c) Cross section view of the stamp forming tool set shown in a CAD model; d) Critical dimensions of the dome forming tools

Figure 6-1b and c show the image and dimensions of the stamp forming tool set and dome forming tools. The stamp forming tool set consists of two main parts: the tool frame and the dome forming tools. Both parts are custom-made from H13 tool steel and have undergone

appropriate heat treatment [4]. The tool frame is composed of two decks: the movable deck responsible for closing the stamp forming tools, and the stationary deck, which provides sufficient blank holding force via gas springs, providing blank holding force of 60 kN. The dome forming tools, including the punch, blank holder and die, were mounted in the forming frame. The diameter of the punch is 60 mm. To account for the thickness of the laminate, a clearance of 10 % was set when deciding the inner diameter of the dome shape die. Additionally, spacers with a thickness of 3.4 mm are used, providing a clearance of 2.94 % of the thickness, i.e., 0.1 mm, when the stamp forming tool is closed.

During the stamp forming process, the dome forming die moves downward with the top movable deck, exerting pressure on the laminate and deforming it into a dome shape. To prevent over-squeezing the materials, a step is designed as a 'hard stop' on the punch as a safety measure. When the blank holder touches the step on the punch, no further displacement can be achieved. The stamp forming tool was at room temperature during the forming process.

6.2.2. Stamp forming specimen preparation

Three blank shapes according to Figure 5-18, and two different blocking configurations, $[0_3/90_3]_{2s}$ and $[0/90]_{6s}$, investigated in Section 5.6. were set to validate simulation results. Laminates of both blocking configurations were fabricated and their quality was checked by ultrasonic C-scan. The specimens were cut using water jet cutting into blank shapes.

6.2.3. Solid-state stamping conditions and procedure

As previously mentioned, the forming temperature plays a crucial role in determining the quality of the components produced. It is noted that heat loss occurs during the transfer process from the furnace to the forming machine due to the relatively larger interior space of the furnace compared to the forming specimen, necessitating a higher actual set point for the furnace than the desired forming temperature. Consequently, precise calibration of the forming temperature is crucial.

[4] Hardened at 1025 ℃ for 30min; Quenched in air; Tempered twice at 610 ℃ for 2 hours with intermittent cooling; To hardness HRC 45-50.

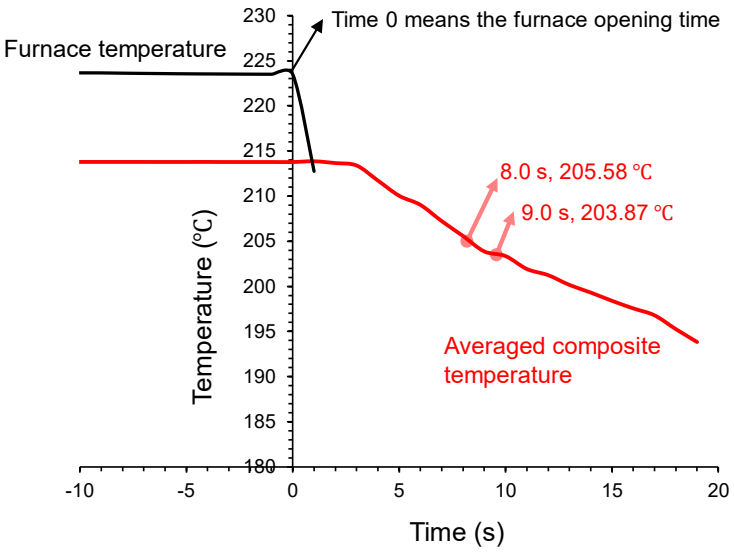


Figure 6-2. A temperature profile example of forming at 205 ℃.

In this experiment, a laminate sample was used as the calibration specimen. Four 2 mm diameter holes were drilled on the sample's side in the thickness direction, with depths of approximately 10 mm. Type K thermocouples with a diameter of 0.076 mm were inserted into the calibration sample. Figure 6-2 depicts the temperature profile of a 205 ℃ forming example, with the pink furnace temperatures assisting in identifying the furnace's opening time. The red solid line represents the average temperature of all four measurements. During the process, the transfer operation was recorded to verify the exact transfer time, which was found to be approximately 8 seconds through practice.

After the temperature is calibrated, the stamp forming conditions can be decided and are listed in Table 6-1. Based on the conclusions obtained from Section 4.4. the deformation rate was found to be a non-crucial parameter, and a large stamp forming speed was recommended. In this experimental setting, a stamp forming speed of 100 mm/s was set, resulting in a forming time of 0.4 s for each forming trial. The holding time was set to be 30s. The following topics are studied from the experimental matrices:

1) Trials 1-3 were conducted to compare the effects of the initial blank shape.
2) Trials 3 and 4 were utilised to compare the effects of layer blocking.

3) Trials 2 and 5-8 were conducted to assess the temperature effects on solid-state stamp forming.
4) Trials 2 and 8 were specifically employed to compare the quality of solid-state and melt-state stamp forming.

Table 6-1. Stamp forming condition matrices.

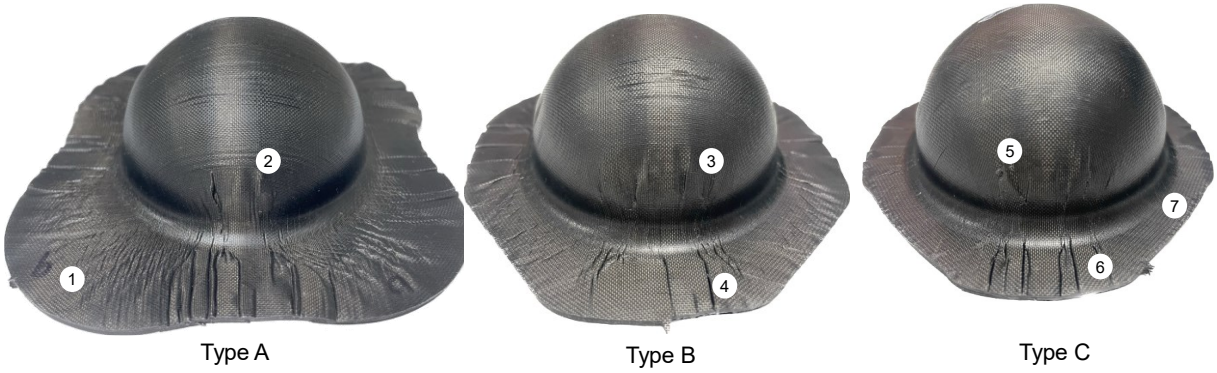
Forming trial number	Furnace setting temperature (°C)	Forming temperature (°C)	Blank sheet type	Composite configuration
1	230	208	Type A	$[0_3/90_3]_{2s}$
2	225	205	Type B	$[0_3/90_3]_{2s}$
3	225	205	Type C	$[0_3/90_3]_{2s}$
4	225	205	Type C	$[0/90]_{6s}$
5	200	180	Type B	$[0_3/90_3]_{2s}$
6	220	195	Type B	$[0_3/90_3]_{2s}$
7	240	215	Type B	$[0_3/90_3]_{2s}$
8(melt-state)	290	260	Type B	$[0_3/90_3]_{2s}$
Forming speed is 100 mm/s for each test. Holding time is 30 s for each test. Bank holding force is 60 kN for each test.				

6.3. Solid-state stamp forming trials and analysis

6.3.1. The effect of blank shape

The as-formed demonstrators with three blank shapes of are shown in Figure 6-3. It is evident that the flange area and wrinkles vary depending on the blank shape. The as-formed part resulting from Type A displays an excess of material in the flange region, as indicated by area ① in Figure 6-3a. Due to this surplus material, wrinkles form in regions where further material stretching is required, such as the side wall of the as-formed part, marked as area ②. In contrast, the as-formed part from shape Type B presents a more reasonable shape in the flange area, although wrinkles are still present in marked areas ③ and ④. The as-formed part in Figure 6-3c exhibits slightly worse wrinkles on the surface marked as area ⑤ and ⑥. This increasing trend of wrinkles may be ascribed to the less effective blank holding force, as the edge on area ⑦ is narrow. Based on this comparison, it is concluded that blank shape Type B yields the best-

formed quality with regard to wrinkles, which matches of the conclusion from the numerical analysis.



a) b) c)

Figure 6-3. The effect of blank shape. ① the excessive material on the flange area; ②③④⑤⑥ wrinkles ; ⑦ the limited area on the flange area.

6.3.2. The effect of layer blocking

As demonstrated in the numerical simulation, a dilemma arises when considering material configurations with or without layer blocking processes. When layer blocking is employed, inter-ply slippage increases, making it more susceptible to delamination. Conversely, without blocking, there is a higher likelihood of surface cracks due to the weakness of a thin layer. For the specific application of dome shape forming, two configurations are explored experimentally: $[0_3/90_3]_{2s}$ and $[0/90]_{6s}$. The configuration of $[0_3/90_3]_{2s}$, as outlined by Liu et al. [212], may create and amplify inter-ply failure. The experimental results of trial 3 and 4 show the blocking effects and are presented in Figure 6-4. Under the same forming conditions and setup, several cracks are observed on the surface of the unblocked component, whereas the layer-blocked component exhibits better quality.



a) b)

Figure 6-4. The effect of layer-blocking at a) three layers blocked (0.42 mm for a single layer) and b) unblocked (0.14 mm for a single layer) configurations.

6.3.3. The effect of forming temperature

The temperature effects on the stamp forming process were evaluated in Figure 6-5, where it was observed that stamp forming at 180 ℃, 215 ℃ and 260 ℃ resulted in unsatisfactory surface quality, whereas stamp forming at 195 ℃ and 205 ℃ produced acceptable surface quality. The details of each forming trial are listed as below:

1) The demonstrator formed at 180 ℃ exhibits large cracks on both the outer and inner surfaces. This is likely due to the high in-plane load induced by the strain hardening properties of the material. When wrinkles form during stamping, they create additional impedance, making it more difficult for the material to stretch. As a result, the in-plane stress can exceed the failure strength of the material, leading to cracks.

2) The demonstrator formed at 195 ℃ has a satisfying top surface quality with no visible fibre splits or cracks. However, there are wrinkles present on the dome part on both the outer and inner surfaces, which are marked as positions ④ and ⑤ on the as-formed part.

3) The demonstrator formed at 205 ℃ also exhibits satisfying surface quality, albeit with the presence of minor fibre splits on the external surface of the component, as shown from area ⑥. In addition, there are noticeable wrinkles occurring on the outer and inner surfaces of the flange and lower sidewall regions, marked respectively at positions ⑦ and ⑧.

4) The demonstrator formed at 215 ℃ shows some defects, such as the fibre waviness on the surface, marked as position ⑨, sever incompatible deformation between layers, marked as position ⑩, and wrinkles on the part, marked as ⑪ and ⑫.

5) The demonstrator formed at 240 ℃ was stamp formed under the melt state of the material. It is evident that, with the simple experimental setup employed, the quality of the part is subpar. There are notable instances of fibre waviness and splitting on the surface, as marked position ⑬, deformation incompatibility causing ply slippage, marked at position ⑭, and wrinkles marked as position ⑮. Furthermore, a holding time of 30 seconds is insufficient for the part to reach an adequate temperature drop, leading to unsatisfactory shape accuracy. The inner contour of the dome part exhibits distorted features, as indicated at position ⑯.

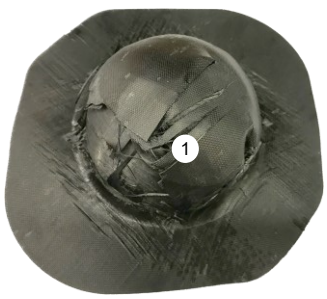
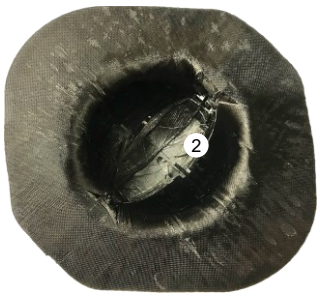
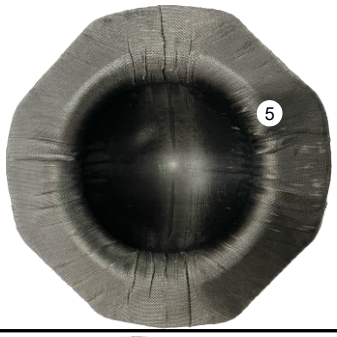
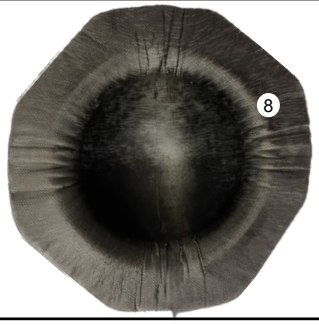
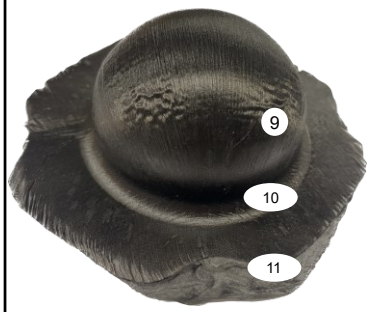
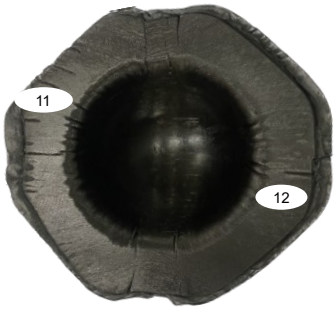
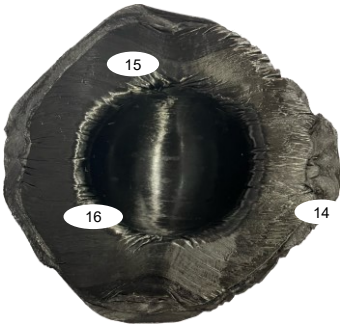
Forming temperature	Isometric view	Bottom view	Notes
180 ℃			1&2: large cracks
195 ℃			3&4&5: Wrinkles on flange and lower side wall regions
205 ℃			6: Local subtle split at outer layer 7&8: Winkles on flange and lower sidewall regions
215 ℃			9: Fibre waviness on surface 10&12: Wrinkles on surface 11: Excessive inter-ply slippage
260 ℃			32 13: Fibre waviness and splits on surface 14: Excessive inter-ply slippage and ply folding 15: Wrinkles on surface 16: Distorted inner contour of formed dome part

Figure 6-5. The effect of forming temperature (180 – 260 ℃).

Upon examining the demonstrator formed at temperatures ranging from 195 to 215 ℃, it was found that solid-state stamp forming yielded more promising results compared to melt-state forming. The experimental results under solid-state conditions showed two major trends with respect to demonstrator quality and temperature. Firstly, above a certain temperature (180 ℃ in this case), the risk of wrinkles formation decreased with increasing temperature due to a reduction in the material's in-plane rigidity and strain hardening. Secondly, an increase in temperature led to a decrease in transverse and bending rigidity, resulting in the formation of fibre splits and waviness, respectively. These two trends determine the temperature window for UD CF/PA6 stamp formed under solid state. Based on surface observations alone, it was found that parts formed at temperatures ranging from 195 to 208 ℃ (including the 208 ℃ part from forming trial 1) showed acceptable quality. A more detailed quality assessment will be provided in the following section.

6.4. Quality assessment of the formed components

Given the surface defects that have been observed and analysed, it is crucial to assess the internal quality of these components. Certain defects, like voids and delamination, have the potential to significantly impact the quality of the as-formed parts. Therefore, it is imperative to examine the inner quality of these demonstrators. As previously discussed, only the parts formed at temperatures between 195-208 ℃ with a three-layer-blocking configuration are necessary for further investigation.

To evaluate the quality of the as-formed part, the first approach is to check the microstructure of the as-formed part using optical microscope. For this purpose, the 208 ℃ formed part (trial 1) was cut and examined at three marked positions shown in Figure 6-6. As can be observed, the microstructure analysis of the checked positions revealed the presence of surface defects such as fibre splits in Section a-2 and wrinkles in Section c, which were also visually apparent. However, there were no inner defects detected, including voids and delamination. All three observed sections showed evidence of compaction within layers. Nevertheless, it should be noted that this method provides limited results for specific checked positions only. Therefore, a non-destructive quality check method, such as computed tomography (CT) scan, was adopted to obtain a comprehensive evaluation of the quality of the 195 ℃ and 208 ℃ formed parts, considering their size and layer thickness.

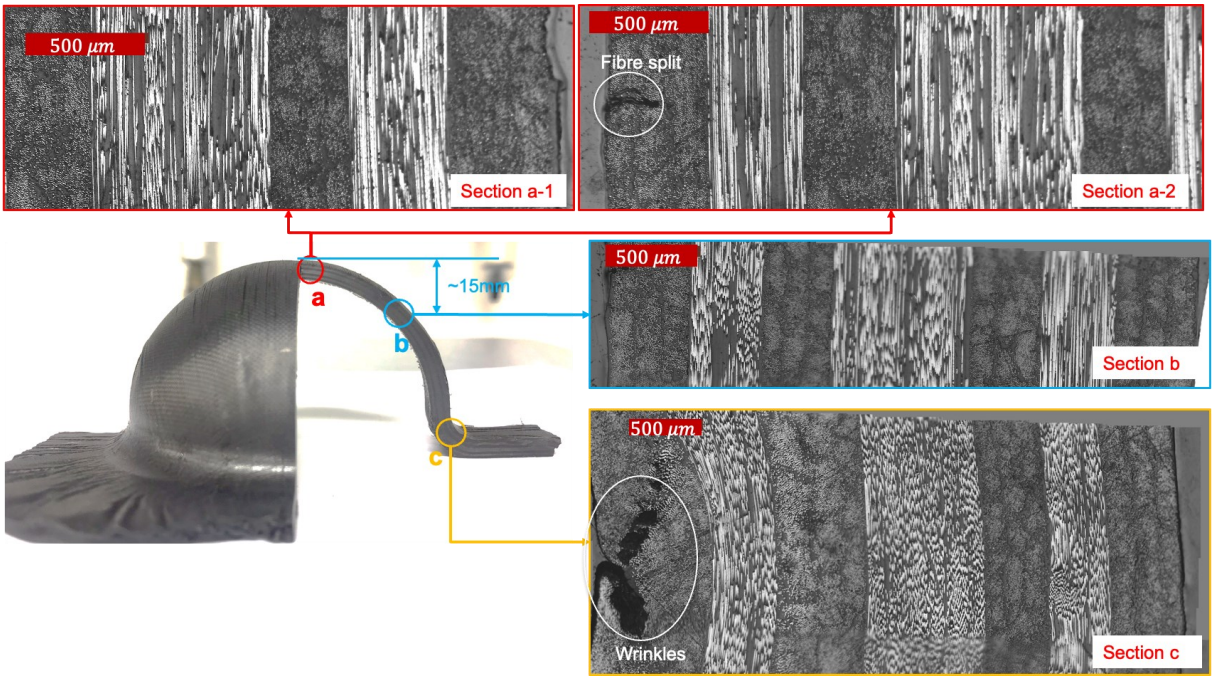
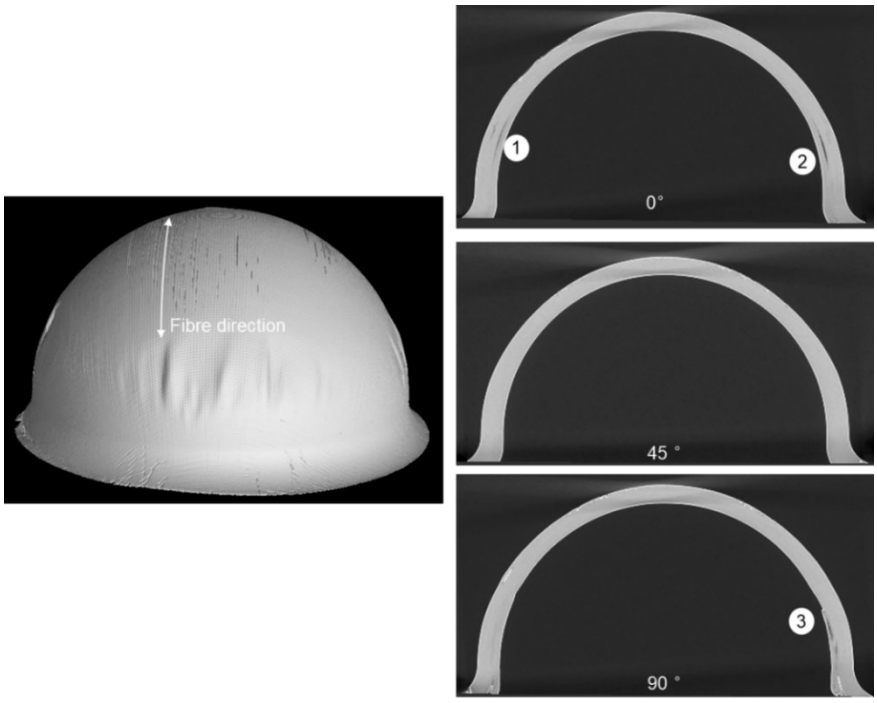


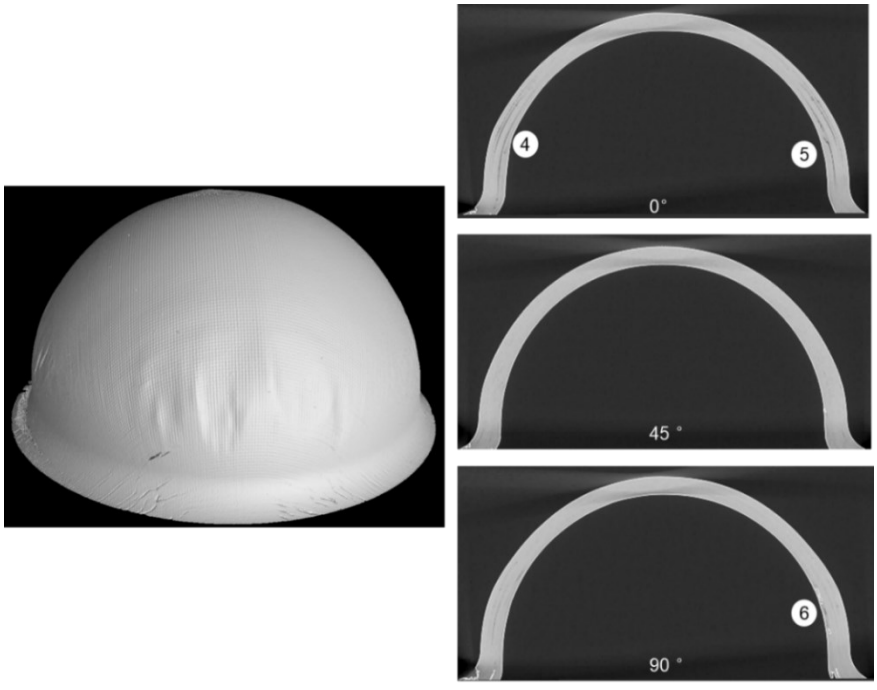
Figure 6-6. Optical microscope observation of the as-formed part trial 1 (forming temperature at 208 ℃)

A high-resolution 3D industrial CT scanner, DeskTom 150, provided by OR3D in the UK, was used to scan the 195 ℃ and 205 ℃ formed parts with a resolution of 40 μm. The commercial software myVGL, developed by Volume Graphics, was used for viewing the results. The CT scanning results are shown in Figure 6-7. As can be observed, although the top surface quality of the 195 ℃ part is better than the 205 ℃ part regarding fibre splits, more delamination inside the component is observed (positions ④⑤ and ⑥). In the 205 ℃ part, although there is delamination at positions ①② and ③, it is much less severe than in the 195 ℃ part. By comparing the positions of these inner defects with the positions of wrinkles on the as-formed parts, it is fair to say that all the inner defects are correlated to the wrinkles on the surface. Therefore, it is reasonable to assume that the intensity of the wrinkles reflects the overall quality of the as-formed parts. The possible reason for the difference in the intensity of these wrinkles between the parts deformed at 195 ℃ and 205 ℃ could be the strain hardening effects of the intra-ply deformation. Furthermore, it was observed that the wrinkles mostly happen on the 0/90 ° directions of the fibres. On the 45° section views for both the 195 ℃ and the 205 ℃ formed parts, no defects were observed, which provides the potential to avoid wrinkles by

altering the layer configurations. Based on the quality check conclusions, it is fair to say that 205 ℃ is the optimal forming temperature for the dome shape forming.



a)



b)

Figure 6-7. CT scanning results of a) forming trial 2 (forming temperature at 205 ℃) and b) forming trial 6 (forming temperature at 195 ℃)

6.5. Comparison of the numerical simulation and experimental results

From the analysis of the conclusions obtained from the experimental investigation, it can be concluded that the insights from the numerical simulation are valid. Admittedly, there are some physical data that necessitate further experimental characterisation, such as the bending properties of a single UD layer, the inter-ply, and tool-ply properties. Nevertheless, the numerical results from Section 5.6. are compared to the experimental results for the 205 °C parts to demonstrate the numerical simulation's capability. The comparison of the experimental and numerical simulations is illustrated in Figure 6-8. Compared to the as-formed part, the errors in length, width, and height are 3.48 %, 9.91 %, and 1.62 %, respectively. Additionally, the height between the top of the dome part and the highest position of wrinkles is 17.24 mm and 17.39 mm in the simulation results and the experimental results, respectively, resulting in an error of 0.86 %. The wrinkles on the top and bottom surfaces of the dome part were accurately predicted. This comparison reveals that, the numerical model can effectively predict the solid-state stamp forming process and provide a reliable guideline concerning processing windows and configurations.

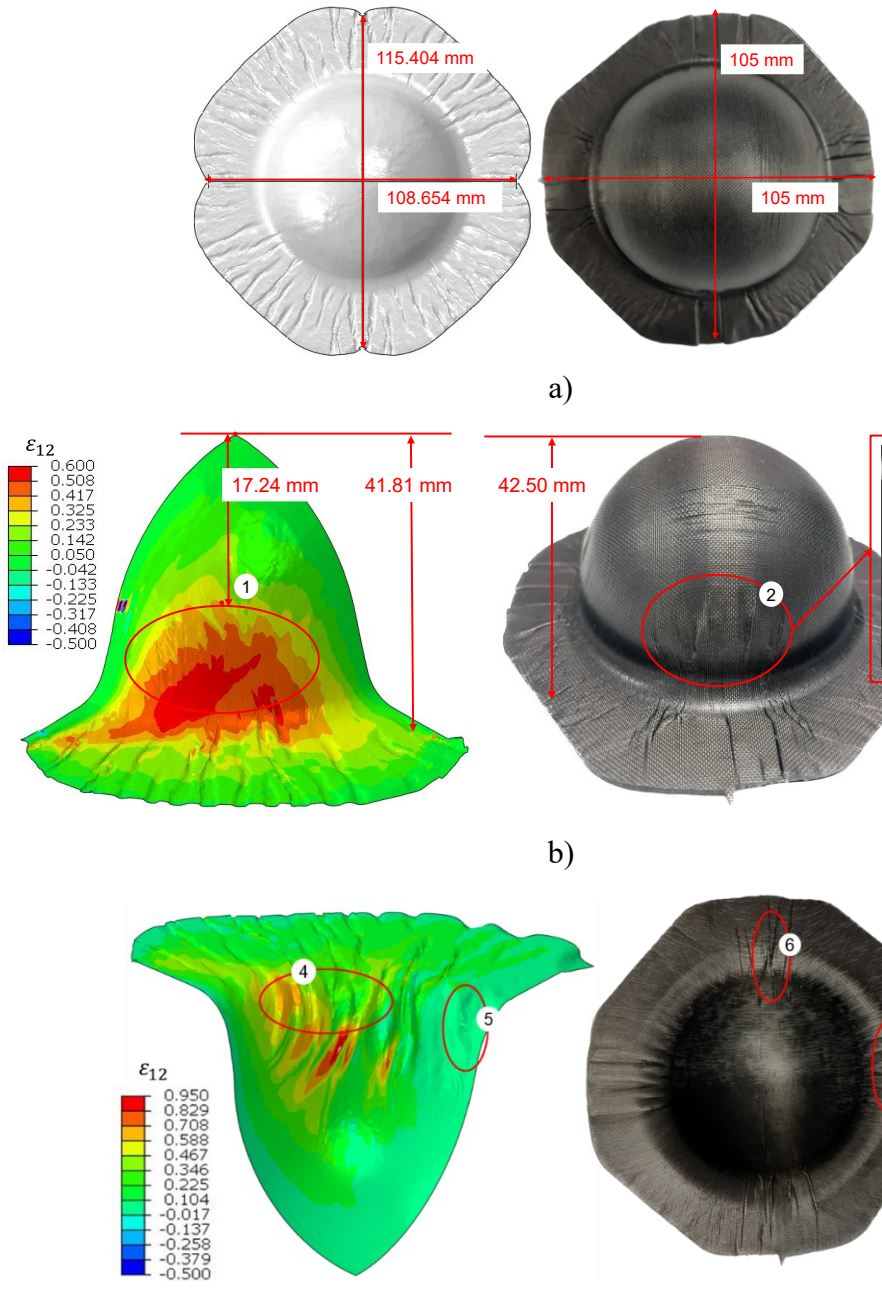


Figure 6-8. Comparison of the numerical results and the experimental results on a) as-formed part dimensions; b) comparison of wrinkles on the outer surface; c) comparison of wrinkles on the inner surface. ① Wrinkle area on the simulation results. ②Wrinkle area on the experimental results. ③Wrinkle area observed from CT scan on the experimental results. ④⑤Wrinkle area on the inner surface of the simulation results. ⑥⑦Wrinkle area on the inner surface of the experimental results.

6.6. Summary

This chapter has carried out solid-state stamp forming trials on CF/PA6 laminates with various layer configurations. The effects of the blank shape, layer blocking, and temperature on the solid-state stamp forming process have been examined. The formed component quality has been assessed through visual inspections of wrinkles, fibre split, cracks, fibre waviness, and shape accuracy. Microstructural observations and CT scanning have been employed to further confirm the inner quality. After quality checks and analysis, the optimal processing parameters have been identified. The numerical simulation results have been compared with the experimental results, demonstrating the ability of the constitutive and FEA model developed into predict the solid-state forming process.

CHAPTER 7 Conclusions and Suggestions for Future Work

7.1. Conclusions

Systematic studies on solid-state stamp forming of UD FRTPs have been conducted. The fundamental experimental studies involved characterising the thermomechanical properties and analysing the deformation mechanisms of deformable thermoplastics and UD FRTPs under specific conditions. To simulate the process, physically based constitutive models for thermoplastics and UD FRTPs were proposed and implemented in FEA software. Through experimental and numerical studies, valuable insights and guidelines have been provided for the solid-state stamp forming process, leading to the successful manufacture of demonstrators. The detailed conclusions for each section are further explained in four subsections.

7.1.1. Thermomechanical properties of thermoplastics

The thermomechanical behaviour of two representative semicrystalline thermoplastics, namely PA6 and PEEK, under solid-state stamp forming conditions (around 0.7 – 0.9 T_m at absolute temperature, and 0.001 – 50 /s) has been investigated through uniaxial tensile tests. The following main conclusions can be drawn:

- Temperature has significant effects on the thermomechanical behaviour of the thermoplastics in the studied temperature range. The elastic moduli, yield stresses, pre-strain hardening, and post-strain hardening moduli are all found to monotonically decrease with increasing temperature. At temperatures approaching the melting temperature, the post-strain hardening behaviour disappears.
- Strain rate effects on the thermomechanical properties of the thermoplastics are not obvious under the studied range. However, it is worth mentioning that adiabatic heat is found to be a non-negligible factor, which would cause material thermal softening during high-rate deformation.

Furthermore, an alternative experimental method, the VNR shear test, has been investigated. The main conclusions are as follows:

- The rotation factor and non-uniformity factor have been identified as significant sources of error in characterising material properties during simple shear deformation.
- A two-step correction method has been proposed and proven to be robust to rectify these error sources, with the correction factors showing an exponential relationship with the rotation angle calculated from the shear strain.

7.1.2. Thermomechanical properties of UD FRTPs

Having UD CF/PA6 laminates fabricated with good consolidation quality, the intra-ply transverse tensile and longitudinal shear properties of UD CF/PA6 have been systematically investigated to gain insights into the intra-ply deformation mechanisms under solid-state stamp forming conditions (180 – 220 ℃, 0.004 – 0.4 /s). The main conclusions are as follows:

- Temperature significantly impacts the intra-ply properties of UD CF/PA6 under examined conditions. Within the studied range, strain hardening follows initial elastic deformation at all temperatures, but at higher temperatures, strain hardening diminishes, leading to strain softening.
- The strain rate effects on the intra-ply deformation properties of UD CF/PA6 are marginal under the studied conditions and deformation modes.

7.1.3. Constitutive models and the implementation for numerical simulations

With the thermomechanical behaviour of thermoplastics, and UD FRTPs obtained, two new physically based viscoelastic-viscoplastic constitutive models were developed for thermoplastics and UD FRTPs, respectively. The main conclusions are as follows:

- The novel thermoplastics model integrates intermolecular and network responses, reflecting microstructural evolution. Experimental and model prediction results validate its accuracy in predicting PA6 and PEEK's viscoelastic-viscoplastic behaviours during solid-state stamping, considering temperature, strain rate, and adiabatic heat effects.
- The new model of UD FRTPs provides an in-depth understanding of the deformation mechanisms of UD FRTPs, especially the polymer-fibre (P-F) interface properties at temperatures close to the T_m of the polymer. The model accuracy is confirmed by the good agreement between the experimental result and model predictions, considering temperature effects.

With the constitutive models obtained, they have been implemented into FEA software, ABAQUS, for validating the model accuracy and effectiveness, the major conclusions are as follows:

- The comparison between the FEA and experimental bias-extension results of UD CF/PA6 demonstrates the effectiveness of implementing the UD FRTPs constitutive model into commercial FEA software. It suggests that the model can be effectively used to predict the intra-ply deformation behaviours of UD CF/PA6 for future solid-state stamp forming process simulations.
- The numerical simulations of the solid-state stamp forming process of dome-shaped UD CF/PA6 components have provided preliminary insights into the effects of key forming parameters, including the blank shape, layer-blocking configuration, and forming temperature, which confirms the effectiveness of the material modelling and process simulations.

7.1.4. Experimental investigations on the solid-state stamp forming process

Guided by the findings from the numerical simulations, solid-state stamp forming trials have been conducted on UD CF/PA6 laminates for producing dome-shaped demonstrator components. The major conclusions are as follows:

- The visual inspection, optical microscope observation, and CT scanning techniques provide a comprehensive assessment of the quality of the as-formed demonstrators. The findings from the analysis of the experimental results align with those derived from the numerical simulations. Through both numerical and experimental investigations, optimal stamp forming parameters, including blank shape design, layer-blocking configuration, and forming temperature, are identified for the solid-state stamp forming of the dome-shaped demonstrator.
- Furthermore, good agreement between the experimental results and previously obtained numerical simulation results confirms the FEA model's capability to effectively predict the solid-state forming process.

7.2. Suggestions for future work

- Characterisation of intra-ply bending and inter-ply properties of solid-state UD FRTPs

In order to improve the accuracy of FEA process simulation models, it is recommended to conduct further experimental studies on the intra-ply bending and inter-ply properties of UD FRTPs under solid-state stamp forming conditions. Although some experimental studies have already been conducted, there is still a lack of research specifically focused on UD FRTPs under solid-state deformation. For bending properties, it is suggested to characterise both the longitudinal and transverse directions. When it comes to inter-ply slip properties, considering the various possibilities of fibre arrangement, it is advisable to take into account the relative angles between layers.

- Deeper understanding of the wrinkle formation mechanism and quality assessment

Wrinkles formation is a persistent challenge in the manufacturing of FRTP components. While extensive research has been conducted on wrinkle formation in textile and woven FRTPs, the understanding of wrinkle formation in UD FRTPs, especially under solid-state conditions, is still limited. It is important to develop further theories and models to better comprehend this issue.

Furthermore, many researchers have noted that completely eliminating wrinkles can be a challenging task. As such, it's critical to establish standards that define an acceptable degree of wrinkles. To facilitate this, a comprehensive quality assessment of the demonstrators, considering their operational environment, is necessary. It's imperative to develop a specific set of criteria for the industrial application of FRTP components, to ascertain their acceptable use in various scenarios.

- Forming process simulation

There are two aspects to consider for improving the forming process simulation:

a. Optimising the Fortran code: The simulation process using user-defined subroutine is time-consuming, particularly with an explicit solver. Further research should focus on optimising the Fortran code to expedite the FEA simulation process and reduce computational resources.

b. Accounting for non-isothermal conditions: Currently, the FEA simulations in this study made simplification and primarily consider isothermal conditions. However, in reality, deformation is often accompanied by non-isothermal effects, especially in the area of flanges. To achieve more accurate predictions of the stamp forming process, it is recommended to incorporate

thermal analysis that accounts for the heat transfer process corresponding to the thermomechanical properties.

- Improvement of the stamp forming process

To enhance the stamp forming process, several areas can be addressed:

a. Enhancing the process stability could be facilitated using a gripper-based holder system with a spring mechanism. This system would offer superior control and stability during the forming process. Additionally, integrating automation into the holding and transfer system could yield a more precise control over transport time, ensuring improved accuracy and efficiency.

b. Exploring complex shapes: Further investigations into the formation of complex shapes are required, with an emphasis on post-forming assessment and evaluation. While this thesis has provided qualitative assessments, future studies should incorporate quantitative measures. These could include the development of theoretical approaches to quantify defects, establish damage tolerance thresholds, and design demonstrator components in a manner that facilitates post-forming mechanical testing.

c. Scaling up for near real-world demonstrator components: Manufacturing an industrial application demonstrator will allow for the assessment of the technology's robustness in a real-world setting or industrially relevant environment.

References

[1] IEA Statista, Distribution of carbon dioxide emissions produced by the transportation sector worldwide in 2021, by subsector. 2021, Statisa.

[2] Y. Wan and J. Takahashi, Development of Carbon Fiber-Reinforced Thermoplastics for Mass-Produced Automotive Applications in Japan. Journal of Composites Science, 2021. **5**(3): p. 86 DOI: https://doi.org/10.3390/jcs5030086.

[3] S. Melito. Thermoplastics vs. Thermoset Plastics: Material Properties Overview. 2022 [cited 2022 2022.09.11]; Available from: https://www.fictiv.com/articles/thermoplastic-vs-thermoset-plastic-mechanical-properties-overview.

[4] M. Valente, I. Rossitti, and M. Sambucci, Different Production Processes for Thermoplastic Composite Materials: Sustainability versus Mechanical Properties and Processes Parameter. Polymers (Basel), 2023. **15**(1) DOI: https://10.3390/polym15010242.

[5] A.F. Schug, Unidirectional Fibre Reinforced Thermoplastic Composites: A Forming Study, in Fraunhofer Research Institution for Casting, Composite and Processing Technology. 2019, Technischen Universität München.

[6] P. Mitschang, Manufacturing of Thermoplastic Fiber-Reinforced Polymer Composites, in Wiley Encyclopedia of Composites, L. Nicolais, Editor. 2011, John Wiley & Sons. p. 1-15 DOI: https://doi.org/10.1002/9781118097298.weoc130.

[7] U.K. Vaidya and K.K. Chawla, Processing of fibre reinforced thermoplastic composites. International Materials Reviews, 2008. **53**(4): p. 185-218 DOI: https://doi.org/10.1179/174328008x325223.

[8] S.P. Haanappel, Forming of UD fibre reinforced thermoplastics: a critical evaluation of intra-ply shear. 2013, University of Twente: Enschede, the Netherlands.

[9] F.C. Campbell, Structural composite materials. 2010: ASM international.

[10] P. Harrison, et al., Induction melt thermoforming of advanced multi-axial thermoplastic composite laminates. Journal of Manufacturing Processes, 2020. **60**: p. 673-683 DOI: https://doi.org/10.1016/j.jmapro.2020.10.026.

[11] Ruth Heuss, et al., Lightweight, heavy impact. 2012, McKinsey & Company.

[12] N. Shama Rao, et al., Carbon composites are becoming competitive and cost effective, in White paper. 2018.

[13] Statista Research Department, Lightweight automotive materials. 2019, Statista.

[14] Statista Research Department, Global organosheet & semi-finished thermoplastic UD-tape laminate market value 2030. 2023, Statista.

[15] D. Dörr, Simulation of the thermoforming process of UD fiber-reinforced thermoplastic tape laminates. 2021: Karlsruher Institut für Technologie (KIT).

[16] B. Zheng, et al., Formability and Failure Mechanisms of Woven CF/PEEK Composite Sheet in Solid-State Thermoforming. Polymers, 2019. **11**(6): p. 966 DOI: https://doi.org/10.3390/polym11060966.

[17] D.M. Bigg and J.R. Preston, Stamping of thermoplastic matrix composites. 1989. **10**(4): p. 261-268 DOI: https://doi.org/10.1002/pc.750100409.

[18] D. Leach, T. McCrea, and C. Chamberlain, Cost Effective Thermoplastic Composites in Aerostructures. Sampe Journal, 2020: p. 6-15.

[19] D. Brands, et al., Formability Experiments for Unidirectional Thermoplastic Composites. Key Engineering Materials, 2022. **926**: p. 1358-1371 DOI: https://doi.org/10.4028/p-x3g086.

[20] P. Boisse, et al., Advances in composite forming through 25 years of ESAFORM. International Journal of Material Forming, 2022. **15**(3): p. 39 DOI: https://doi.org/10.1007/s12289-022-01682-8.

[21] R. Guzman de Villoria, et al., In-plane strength enhancement of laminated composites via aligned carbon nanotube interlaminar reinforcement. Composites Science and Technology, 2016. **133**: p. 33-39 DOI: https://doi.org/10.1016/j.compscitech.2016.07.006.

[22] A.M. Almushaikeh, et al., Manufacturing of carbon fiber reinforced thermoplastics and its recovery of carbon fiber: A review. Polymer Testing, 2023. **122**: p. 108029 DOI: https://doi.org/10.1016/j.polymertesting.2023.108029.

[23] I. Fernández, F. Blas, and M. Frövel, Autoclave forming of thermoplastic composite parts. Journal of Materials Processing Technology, 2003. **143-144**: p. 266-269 DOI: https://doi.org/10.1016/S0924-0136(03)00309-1.

[24] Y. Song, et al., CAE method for compression molding of carbon fiber-reinforced thermoplastic composite using bulk materials. Composites Part A: Applied Science and Manufacturing, 2018. **114**: p. 388-397 DOI: https://doi.org/10.1016/j.compositesa.2018.09.002.

[25] K. Ahmed, et al., Effect of de-sizing on the structural and mechanical properties of carbon fiber reinforced polypropylene composites molded by the novel direct fiber feeding injection molding technology. Journal of Thermoplastic Composite Materials. **0**(0): p. 08927057221077872 DOI: https://doi.org/10.1177/08927057221077872.

[26] R. Nakao, H. Inoya, and H. Hamada, Mechanical Properties of Injection Molded Products Fabricated by Direct Fiber Feeding Injection Molding. Energy Procedia, 2016. **89**: p. 307-312 DOI: https://doi.org/10.1016/j.egypro.2016.05.039.

[27] V. Popineau, et al., Vacuum-Bag-Only (VBO) Molding of Flax Fiber-reinforced Thermoplastic Composites for Naval Shipyards. Applied Composite Materials, 2021. **28**(3): p. 791-808 DOI: https://doi.org/10.1007/s10443-021-09890-2.

[28] X. Fu, et al., High temperature vacuum assisted resin transfer molding of phenylethynyl terminated imide composites. Polymer Composites, 2011. **32**(1): p. 52-58 DOI: https://doi.org/10.1002/pc.21015.

[29] I. Baran, 16 - Pultrusion processes for composite manufacture, in Advances in Composites Manufacturing and Process Design, P. Boisse, Editor. 2015, Woodhead Publishing. p. 379-414 DOI: https://doi.org/10.1016/B978-1-78242-307-2.00016-6.

[30] M. Barile, et al., Thermoplastic Composites for Aerospace Applications, in Revolutionizing Aircraft Materials and Processes, S. Pantelakis and K. Tserpes, Editors. 2020, Springer International Publishing: Cham. p. 87-114 DOI: https://doi.org/10.1007/978-3-030-35346-9_4.

[31] K. Minchenkov, et al., Thermoplastic Pultrusion: A Review. Polymers, 2021. **13**(2): p. 180 DOI: https://doi.org/10.3390/polym13020180.

[32] B.t. composites. Processing methods of thermoplastic composites. Available from: https://thermoplasticcomposites.de/en/processing-2/.

[33] F. Li, et al., Enhanced mechanical properties of short carbon fiber reinforced polyethersulfone composites by graphene oxide coating. Polymer, 2015. **59**: p. 155-165 DOI: https://doi.org/10.1016/j.polymer.2014.12.067.

[34] P. Land, et al., Technology review of thermal forming techniques for use in composite component manufacture. 2016. **9**(1): p. 81-89 DOI: https://doi.org/10.4271/2015-01-2610.

[35] Y.D. Boon, et al., Recent Advances on the Design Automation for Performance-Optimized Fiber Reinforced Polymer Composite Components. Journal of Composites Science, 2020. **4**(2): p. 61 DOI: https://doi.org/10.3390/jcs4020061.

[36] Y.D. Boon, S.C. Joshi, and S.K. Bhudolia, Review: Filament Winding and Automated Fiber Placement with In Situ Consolidation for Fiber Reinforced Thermoplastic Polymer Composites. Polymers, 2021. **13**(12): p. 1951 DOI: https://doi.org/10.3390/polym13121951.

[37] G.D. Shrigandhi and B.S. Kothavale, Biodegradable composites for filament winding process. Materials Today: Proceedings, 2021. **42**: p. 2762-2768 DOI: https://doi.org/10.1016/j.matpr.2020.12.718.

[38] S. Das, The Cost of Automotive Polymer Composites: A Review and Assessment of DOE's Lightweight Materials Composites Research. United States, 2001 DOI: https://doi.org/10.2172/777656.

[39] R.A. Brooks, et al., A review on stamp forming of continuous fibre-reinforced thermoplastics. International Journal of Lightweight Materials and Manufacture, 2022. **5**(3): p. 411-430 DOI: https://doi.org/10.1016/j.ijlmm.2022.05.001.

[40] S.L. Wilkinson, Optimisation of the stamp forming process for thermoplastic composites, in Department of Aerospace Engineering. 2021, The University of Bristol: The University of Bristol.

[41] S.P. Haanappel, et al., Formability analyses of uni-directional and textile reinforced thermoplastics. Composites Part A: Applied Science and Manufacturing, 2014. **56**: p. 80-92 DOI: https://doi.org/10.1016/j.compositesa.2013.09.009.

[42] A. Burkhart and D. Cramer. Feasibility of continuous-fiber reinforced thermoplastic tailored blanks for automotive applications. in Proceedings of the 2005 Society of Plastics Engineers Automotive Composites Conferrence and Exposition. 2005.

[43] H. Lessard, et al., Influence of process parameters on the thermostamping of a [0/90] 12 carbon/polyether ether ketone laminate. Composites Part A: Applied Science and Manufacturing, 2015. **70**: p. 59-68 DOI: https://doi.org/10.1016/j.compositesa.2014.12.009.

[44] D. Dörr, F. Henning, and L. Kärger, Nonlinear hyperviscoelastic modelling of intra-ply deformation behaviour in finite element forming simulation of continuously fibre-reinforced thermoplastics. Composites Part A: Applied Science and Manufacturing, 2018. **109**: p. 585-596 DOI: https://doi.org/10.1016/j.compositesa.2018.03.037.

[45] T.K. Slange, Rapid Manufacturing of Tailored Thermoplastic Composites by Automated Lay-up and Stamp Forming, in A Study on the Consolidation Mechanisms. 2019, University of Twente: Enschede.

[46] T. Deng, et al., A hybrid lamination model for simulation of woven fabric reinforced thermoplastic composites solid-state thermo-stamping. Materials & Design, 2021. **200**: p. 109419 DOI: https://doi.org/10.1016/j.matdes.2020.109419.

[47] Y. Uriya, K. Ikeuch, and J. Yanagimoto, Cold and Warm V-bending Test for Carbon-fiber-reinforced Plastic Sheet. Procedia Engineering, 2014. **81**: p. 1633-1638 DOI: https://doi.org/10.1016/j.proeng.2014.10.203.

[48] Y. Uriya, K. Ikeuchi, and J. Yanagimoto, Enhanced formability of thin carbon fiber reinforced plastic sheets in cold/warm embossing with ductile dummy sheets of different thicknesses. International Journal of Material Forming, 2015. **8**(3): p. 415-421 DOI: https://doi.org/10.1007/s12289-014-1184-9.

[49] Y. Uriya and J. Yanagimoto, Erichsen cupping test on thermosetting CFRP sheets. International Journal of Material Forming, 2017. **10**(4): p. 527-534 DOI: https://doi.org/10.1007/s12289-016-1298-3.

[50] H. Wang, et al., Experimental and constitutive modelling studies of semicrystalline thermoplastics under solid-state stamp forming conditions. Polymer, 2021. **228**: p. 123939 DOI: https://doi.org/10.1016/j.polymer.2021.123939.

[51] U. Sachs, Friction and bending in thermoplastic composites forming processes. 2014, University of Twente: Enschede.

[52] J.S. Lightfoot, M.R. Wisnom, and K. Potter, A new mechanism for the formation of ply wrinkles due to shear between plies. Composites Part A: Applied Science and Manufacturing, 2013. **49**: p. 139-147 DOI: https://doi.org/10.1016/j.compositesa.2013.03.002.

[53] K. Shaker, Y. Nawab, and A. Saouab, Experimental and numerical investigation of reduction in shape distortion for angled composite parts. International Journal of Material Forming, 2020. **13**(6): p. 897-906 DOI: https://doi.org/10.1007/s12289-019-01510-6.

[54] T. Yoneyama, Teraoka, T., Masuzawa, K., Nishihara, Y., Nagashima, S., Yoshida, H., Press forming of carbon-fiber-reinforced thermoplastic sheets, Journal of Japan Society for Technology of Plasticity. Journal of Japan Society for Technology of Plasticity, 2012. **53**(613): p. 145–149 DOI: https://doi.org/10.9773/sosei.53.145.

[55] B. Khan, K.D. Potter, and M.R. Wisnom. Simulation of process induced defects in resin transfer moulded woven carbon fibre laminates and their effect on mechanical behaviour. in The 8th international conference on flow processes in composite materials (FPCM8). 2006. Douai, France.

[56] K. Potter, et al., Variability, fibre waviness and misalignment in the determination of the properties of composite materials and structures. Composites Part A: Applied Science and Manufacturing, 2008. **39**(9): p. 1343-1354 DOI: https://doi.org/10.1016/j.compositesa.2008.04.016.

[57] R. Sitohang, Effect of in-plane waviness defects on compressive failure of thermoplastic composite laminates. 2022, University of Twente: Enschede.

[58] R.D.R. Sitohang, et al., Effect of in-plane fiber waviness defects on the compressive properties of quasi-isotropic thermoplastic composites. Composite Structures, 2021. **272**: p. 114166 DOI: https://doi.org/10.1016/j.compstruct.2021.114166.

[59] E.T.M. Krämer, et al., Real-time observation of waviness formation during C/PEEK consolidation. Composites Part A: Applied Science and Manufacturing, 2020. **133**: p. 105872 DOI: https://doi.org/10.1016/j.compositesa.2020.105872.

[60] P. Boisse, et al., Bending and wrinkling of composite fiber preforms and prepregs. A review and new developments in the draping simulations. Composites Part B: Engineering, 2018. **141**: p. 234-249 DOI: https://doi.org/10.1016/j.compositesb.2017.12.061.

[61] P. Hallander, et al., An experimental study of mechanisms behind wrinkle development during forming of composite laminates. Composites Part A: Applied Science and Manufacturing, 2013. **50**: p. 54-64 DOI: https://doi.org/10.1016/j.compositesa.2013.03.013.

[62] P. Hallander, J. Sjölander, and M. Åkermo, Forming induced wrinkling of composite laminates with mixed ply material properties; an experimental study. Composites Part A: Applied Science and Manufacturing, 2015. **78**: p. 234-245 DOI: https://doi.org/10.1016/j.compositesa.2015.08.025.

[63] B. Engel and J. Brühmann, Mechanisms and sources of spring-in and spring-back of fiber reinforced thermoplastics. 2014.

[64] M. Zakaria, et al., Spring-back of Thick Uni-Directional Carbon Fibre Reinforced Composite Laminate for Aircraft Structure Application. Journal of Physics: Conference Series, 2018. **1005**(1): p. 012003 DOI: https://doi.org/10.1088/1742-6596/1005/1/012003.

[65] Q. Chen, et al., Intra/inter-ply shear behaviors of continuous fiber reinforced thermoplastic composites in thermoforming processes. Composite Structures, 2011. **93**(7): p. 1692-1703 DOI: https://doi.org/10.1016/j.compstruct.2011.01.002.

[66] S. Allaoui, C. Cellard, and G. Hivet, Effect of inter-ply sliding on the quality of multilayer interlock dry fabric preforms. Composites Part A: Applied Science and Manufacturing, 2015. **68**: p. 336-345 DOI: https://doi.org/10.1016/j.compositesa.2014.10.017.

[67] S.P. Haanappel and R. Akkerman, Shear characterisation of uni-directional fibre reinforced thermoplastic melts by means of torsion. Composites Part A: Applied Science and Manufacturing, 2014. **56**: p. 8-26 DOI: https://doi.org/10.1016/j.compositesa.2013.09.007.

[68] K. Potter, In-plane and out-of-plane deformation properties of unidirectional preimpregnated reinforcement. Composites Part A: Applied Science and Manufacturing, 2002. **33**(11): p. 1469-1477 DOI: https://doi.org/10.1016/S1359-835X(02)00138-0.

[69] H. Chen, et al., A focused review on the thermo-stamping process and simulation progresses of continuous fibre reinforced thermoplastic composites. Composites Part B: Engineering, 2021. **224**: p. 109196 DOI: https://doi.org/10.1016/j.compositesb.2021.109196.

[70] M.A. Khan, et al., Numerical and experimental analyses of woven composite reinforcement forming using a hypoelastic behaviour. Application to the double dome benchmark. Journal of Materials Processing Technology, 2010. **210**(2): p. 378-388 DOI: https://doi.org/10.1016/j.jmatprotec.2009.09.027.

[71] J. Weber and J. Schlimbach, Co-consolidation of CF/PEEK tape-preforms and CF/PEEK organo sheets to manufacture reinforcements in stamp-forming process. Advanced Manufacturing: Polymer & Composites Science, 2019. **5**(4): p. 172-183 DOI: https://doi.org/10.1080/20550340.2019.1673961.

[72] P. Harrison, R. Gomes, and N. Curado-Correia, Press forming a 0/90 cross-ply advanced thermoplastic composite using the double-dome benchmark geometry. Composites Part A: Applied Science and Manufacturing, 2013. **54**: p. 56-69 DOI: https://doi.org/10.1016/j.compositesa.2013.06.014.

[73] N. Reynolds, et al., Direct Processing of Structural Thermoplastic Composites Using Rapid Isothermal Stamp Forming. Applied Composite Materials, 2020. **27**(1): p. 107-115 DOI: https://doi.org/10.1007/s10443-020-09797-4.

[74] B. Zheng, et al., Experimental investigation and constitutive modeling of the deformation behavior of Poly-Ether-Ether-Ketone at elevated temperatures. Polymer Testing, 2017. **63**: p. 349-359 DOI: https://doi.org/10.1016/j.polymertesting.2017.08.020.

[75] M. Hou, Stamp forming of continuous glass fibre reinforced polypropylene. Composites Part A: Applied Science and Manufacturing, 1997. **28**(8): p. 695-702 DOI: https://doi.org/10.1016/S1359-835X(97)00013-4.

[76] P. Bussetta and N. Correia, Numerical forming of continuous fibre reinforced composite material: A review. Composites Part A: Applied Science and Manufacturing, 2018. **113**: p. 12-31 DOI: https://doi.org/10.1016/j.compositesa.2018.07.010.

[77] G.B. McGuinness and C.M. ÓBrádaigh, Characterisation of thermoplastic composite melts in rhombus-shear: the picture-frame experiment. Composites Part A: Applied Science and Manufacturing, 1998. **29**(1): p. 115-132 DOI: https://doi.org/10.1016/S1359-835X(97)00061-4.

[78] W. Tan and B.G. Falzon, Modelling the nonlinear behaviour and fracture process of AS4/PEKK thermoplastic composite under shear loading. Composites Science and Technology, 2016. **126**: p. 60-77 DOI: https://doi.org/10.1016/j.compscitech.2016.02.008.

[79] K. Potter, Bias extension measurements on cross-plied unidirectional prepreg. Composites Part A: Applied Science and Manufacturing, 2002. **33**(1): p. 63-73 DOI: https://doi.org/10.1016/S1359-835X(01)00057-4.

[80] P. Boisse, et al., The bias-extension test for the analysis of in-plane shear properties of textile composite reinforcements and prepregs: a review. International Journal of Material Forming, 2017. **10**(4): p. 473-492 DOI: https://doi.org/10.1007/s12289-016-1294-7.

[81] Y.R. Larberg, M. Åkermo, and M. Norrby, On the in-plane deformability of cross-plied unidirectional prepreg. Journal of Composite Materials, 2011. **46**(8): p. 929-939 DOI: https://doi.org/10.1177/0021998311412988.

[82] Y. Larberg and M. Åkermo, In-plane deformation of multi-layered unidirectional thermoset prepreg – Modelling and experimental verification. Composites Part A: Applied Science and Manufacturing, 2014. **56**: p. 203-212 DOI: https://doi.org/10.1016/j.compositesa.2013.10.005.

[83] A. International, ASTM D7078/D7078M-12 Standard Test Method for Shear Properties of Composite Materials by V-Notched Rail Shear Method. 2012, ASTM International: West Conshohocken, PA.

[84] E. Totry, et al., Effect of fiber, matrix and interface properties on the in-plane shear deformation of carbon-fiber reinforced composites. Composites Science and Technology, 2010. **70**(6): p. 970-980 DOI: https://doi.org/10.1016/j.compscitech.2010.02.014.

[85] E. Totry, et al., Mechanisms of shear deformation in fiber-reinforced polymers: experiments and simulations. International Journal of Fracture, 2009. **158**(2): p. 197-209 DOI: https://doi.org/10.1007/s10704-009-9353-4.

[86] J. Wang, A.C. Long, and M.J. Clifford, Experimental measurement and predictive modelling of bending behaviour for viscous unidirectional composite materials. 2010. **3**(2): p. 1253-1266 DOI: https://doi.org/10.1007/s12289-009-0670-y.

[87] A. Margossian, S. Bel, and R. Hinterhoelzl, On the characterisation of transverse tensile properties of molten unidirectional thermoplastic composite tapes for thermoforming simulations. Composites Part A: Applied Science and Manufacturing, 2016. **88**: p. 48-58 DOI: https://doi.org/10.1016/j.compositesa.2016.05.019.

[88] D. Dörr, et al., A coupled thermomechanical approach for finite element forming simulation of continuously fiber-reinforced semi-crystalline thermoplastics. Composites Part A: Applied Science and Manufacturing, 2019. **125**: p. 105508 DOI: https://doi.org/10.1016/j.compositesa.2019.105508.

[89] A.M. Roesner, L. Kärger, and F. Henning, Virtual characterization and macroscopic material modeling of a carbon fiber-reinforced PA6 UD composite. Journal of Composite Materials, 2016. **51**(21): p. 3075-3086 DOI: https://doi.org/10.1177/0021998316681186.

[90] T. Ishikawa, et al., Overview of automotive structural composites technology developments in Japan. Composites Science and Technology, 2018. **155**: p. 221-246 DOI: https://doi.org/10.1016/j.compscitech.2017.09.015.

[91] W.-T. He, et al., Structure and Properties Study of PA6 Nanocomposites Flame Retarded by Aluminium Salt of Diisobutylphosphinic Acid and Different Organic Montmorillonites. Polymers, 2018. **10**(3): p. 312 DOI: https://doi.org/10.3390/polym10030312.

[92] L. Minkova, et al., Interfacial tension of compatibilized blends of LDPE and PA6: the breaking thread method. 2003. **44**(26): p. 7925-7932 DOI: https://doi.org/10.1016/j.polymer.2003.09.054.

[93] D.J. Dijkstra, Guidelines for rheological characterization of polyamide melts (IUPAC Technical Report). Applied Chemistry, 2009. **81**(2): p. 339-349 DOI: https://doi.org/10.1351/PAC-REP-08-07-22.

[94] A. Arsad, et al., Mechanical and Rheological Characterization of PA6 and ABS Blends-With and Without Short Glass Fiber. 2011. **11**(13): p. 2313-2319 DOI: https://doi.org/10.3923/jas.2011.2313.2319

[95] L. Laiarinandrasana, et al., Structural versus microstructural evolution of semi-crystalline polymers during necking under tension: Influence of the skin-core effects, the relative humidity and the strain rate. Polymer Testing, 2016. **55**: p. 297-309 DOI: https://doi.org/10.1016/j.polymertesting.2016.09.012.

[96] E. Parodi, G.W.M. Peters, and L.E. Govaert, Structure–Properties Relations for Polyamide 6, Part 1: Influence of the Thermal History during Compression Moulding on Deformation and Failure Kinetics. 2018. **10**(7): p. 710.

[97] Á. Alvaredo, et al., Non-Isothermal Crystallization Behavior of PEEK/Graphene Nanoplatelets Composites from Melt and Glass States. 2019. **11**(1): p. 124 DOI: https://doi.org/10.3390/polym11010124.

[98] D.G. Papageorgiou, et al., Hybrid poly(ether ether ketone) composites reinforced with a combination of carbon fibres and graphene nanoplatelets. Composites Science and Technology, 2019. **175**: p. 60-68 DOI: https://doi.org/10.1016/j.compscitech.2019.03.006.

[99] N.D. Albérola, P. Mélé, and C. Bas, Tensile mechanical properties of PEEK films over a wide range of strain rates. II. 1997. **64**(6): p. 1053-1059 DOI: 10.1002/(sici)1097-4628(19970509)64:6<1053::Aid-app3>3.0.Co;2-k.

[100] F. Chen, et al., Hot tensile fracture characteristics and constitutive modelling of polyether-ether-ketone (PEEK). Polymer Testing, 2017. **63**: p. 168-179 DOI: https://doi.org/10.1016/j.polymertesting.2017.07.032.

[101] H.F. Brinson and L.C. Brinson, Polymer engineering science and viscoelasticity : an introduction. 2018, New York: Springer.

[102] D. Garcia-Gonzalez, R. Zaera, and A. Arias, A hyperelastic-thermoviscoplastic constitutive model for semi-crystalline polymers: Application to PEEK under dynamic loading conditions. International Journal of Plasticity, 2017. **88**: p. 27-52 DOI: https://doi.org/10.1016/j.ijplas.2016.09.011.

[103] V.G. Rostiashvili and T.A. Vilgis, Statistical Thermodynamics of Polymeric Networks, in Encyclopedia of Polymeric Nanomaterials, S. Kobayashi and K. Müllen, Editors. 2021, Springer Berlin Heidelberg: Berlin, Heidelberg. p. 1-18 DOI: https://doi.org/10.1007/978-3-642-36199-9_308-1.

[104] E. Parodi, G.W.M. Peters, and L.E. Govaert, Structure–Properties Relations for Polyamide 6, Part 1:Influence of the Thermal History during Compression Moulding on Deformation and Failure Kinetics. Polymers, 2018. **10**(7): p. 710 DOI: https://doi.org/10.3390/polym10070710.

[105] P.J. Rae, E.N. Brown, and E.B. Orler, The mechanical properties of poly(ether-ether-ketone) (PEEK) with emphasis on the large compressive strain response. Polymer, 2007. **48**(2): p. 598-615 DOI: https://doi.org/10.1016/j.polymer.2006.11.032.

[106] D. Barba, A. Arias, and D. Garcia-Gonzalez, Temperature and strain rate dependences on hardening and softening behaviours in semi-crystalline polymers: Application to PEEK. International Journal of Solids and Structures, 2020. **182-183**: p. 205-217 DOI: https://doi.org/10.1016/j.ijsolstr.2019.08.021.

[107] D. Brands, et al., In-Plane Shear Characterization of Unidirectional Fiber Reinforced Thermoplastic Tape Using the Bias Extension Method. Frontiers in Materials, 2022. **9** DOI: https://doi.org/10.3389/fmats.2022.863952.

[108] T.A. Martin, D. Bhattacharyya, and I.F. Collins, Bending of fibre-reinforced thermoplastic sheets. Composites Manufacturing, 1995. **6**(3): p. 177-187 DOI: https://doi.org/10.1016/0956-7143(95)95009-N.

[109] P. Harrison, T. Haylock, and A. Long. Measurement of the Transverse and Longitudinal Viscosities of Continuous Fibre Reinforced Composites. in 8th International ESAFORM Conference onMaterial Forming. 2005. Cluj-Napoca, Romania.

[110] Y. Larberg and M. Åkermo. In-plane properties of cross-plied unidirectional prepreg. in 16th ICCM International Conferences on Composite Materials. 2007. Kyoto, japan.

[111] Y. Larberg, Forming of stacked unidirectional prepreg materials, in Trita-AVE. 2012, KTH Royal Institute of Technology: Stockholm. p. vii, 31.

[112] Dennis Brands, et al., Intra-ply shear characterization of unidirectional fiber reinforced thermoplastic tape using the bias extension method, in 24th International Conference on Material Forming. 2021: Liège, Belgique.

[113] D. Garcia-Gonzalez, et al., Mechanical impact behavior of polyether–ether–ketone (PEEK). Composite Structures, 2015. **124**: p. 88-99 DOI: https://doi.org/10.1016/j.compstruct.2014.12.061.

[114] D. Systèmes, Abaqus User Subroutines Reference Manual. 2011, Dassault Systèmes: Dassault Systèmes.

[115] D. Garcia-Gonzalez, S. Garzon-Hernandez, and A. Arias, A new constitutive model for polymeric matrices: Application to biomedical materials. Composites Part B: Engineering, 2018. **139**: p. 117-129 DOI: https://doi.org/10.1016/j.compositesb.2017.11.045.

[116] E.M. Arruda and M.C. Boyce, Evolution of plastic anisotropy in amorphous polymers during finite straining. International Journal of Plasticity, 1993. **9**(6): p. 697-720 DOI: https://doi.org/10.1016/0749-6419(93)90034-N.

[117] M.C. Boyce, S. Socrate, and P.G. Llana, Constitutive model for the finite deformation stress–strain behavior of poly(ethylene terephthalate) above the glass transition. Polymer, 2000. **41**(6): p. 2183-2201 DOI: https://doi.org/10.1016/S0032-3861(99)00406-1.

[118] R. Xiao and C. Tian, A constitutive model for strain hardening behavior of predeformed amorphous polymers: Incorporating dissipative dynamics of molecular orientation. Journal of the Mechanics and Physics of Solids, 2019. **125**: p. 472-487 DOI: https://doi.org/10.1016/j.jmps.2019.01.008.

[119] H. Jiang and C. Jiang, Finite deformation constitutive model for macro-yield behavior of amorphous glassy polymers with a molecular entanglement-based internal-state variable. International Journal of Mechanical Sciences, 2019. **161-162**: p. 105064 DOI: https://doi.org/10.1016/j.ijmecsci.2019.105064.

[120] C. Jiang, et al., Constitutive modeling of the rate- and temperature-dependent macro-yield behavior of amorphous glassy polymers. International Journal of Mechanical Sciences, 2020. **179**: p. 105653 DOI: https://doi.org/10.1016/j.ijmecsci.2020.105653.

[121] M. Polanco-Loria, et al., Constitutive model for thermoplastics with structural applications. International Journal of Impact Engineering, 2010. **37**(12): p. 1207-1219 DOI: https://doi.org/10.1016/j.ijimpeng.2010.06.006.

[122] H. Abdul-Hameed, et al., Large-strain viscoelastic–viscoplastic constitutive modeling of semi-crystalline polymers and model identification by deterministic/evolutionary approach. Computational Materials Science, 2014. **90**: p. 241-252 DOI: https://doi.org/10.1016/j.commatsci.2014.03.043.

[123] S. Berezvai and A. Kossa, Performance of a parallel viscoelastic-viscoplastic model for a microcellular thermoplastic foam on wide temperature range. Polymer Testing, 2020. **84**: p. 106395 DOI: https://doi.org/10.1016/j.polymertesting.2020.106395.

[124] W.R. Yu, et al., Non-orthogonal constitutive equation for woven fabric reinforced thermoplastic composites. Composites Part A: Applied Science and Manufacturing, 2002. **33**(8): p. 1095-1105 DOI: https://doi.org/10.1016/S1359-835X(02)00053-2.

[125] W.R. Yu, et al., Analysis of flexible bending behavior of woven preform using non-orthogonal constitutive equation. Composites Part A: Applied Science and Manufacturing, 2005. **36**(6): p. 839-850 DOI: https://doi.org/10.1016/j.compositesa.2004.10.026.

[126] P. Xue, X. Peng, and J. Cao, A non-orthogonal constitutive model for characterizing woven composites. Composites Part A: Applied Science and Manufacturing, 2003. **34**(2): p. 183-193 DOI: https://doi.org/10.1016/S1359-835X(02)00052-0.

[127] X.Q. Peng and J. Cao, A continuum mechanics-based non-orthogonal constitutive model for woven composite fabrics. Composites Part A: Applied Science and Manufacturing, 2005. **36**(6): p. 859-874 DOI: https://doi.org/10.1016/j.compositesa.2004.08.008.

[128] X. Peng, G. Guo, and N. Zhao, An anisotropic hyperelastic constitutive model with shear interaction for cord–rubber composites. Composites Science and Technology, 2013. **78**: p. 69-74 DOI: https://doi.org/10.1016/j.compscitech.2013.02.005.

[129] Y. Gong, et al., An anisotropic hyperelastic constitutive model for thermoplastic woven composite prepregs. Composites Science and Technology, 2016. **128**: p. 17-24 DOI: https://doi.org/10.1016/j.compscitech.2016.03.005.

[130] H. Xiong, Simulation of forming, compaction and consolidation of thermoplastic composites based on solid shell elements

Simulations de la mise en forme, la compaction et la consolidation de composites thermoplastiques basées sur des éléments finis solides-coques. 2017, Université de Lyon.

[131] T.G. Rogers, Rheological characterization of anisotropic materials. Composites, 1989. **20**(1): p. 21-27 DOI: https://doi.org/10.1016/0010-4361(89)90677-0.

[132] D. Dörr, et al., A Benchmark Study of Finite Element Codes for Forming Simulation of Thermoplastic UD-Tapes. Procedia CIRP, 2017. **66**: p. 101-106 DOI: https://doi.org/10.1016/j.procir.2017.03.223.

[133] E.A. Lamers, Shape distortions in fabric reinforced composite products due to processing induced fibre reorientation. Universiteit Twente, 2004: p. 134 DOI: https://www.persistent-identifier.nl/urn:nbn:nl:ui:28-41422.

[134] F. Pascon, et al. Finite element analysis of the thermoforming process of thermoplastic composite parts. in NAFEMS World Congress. 2013. DOI: https://orbi.uliege.be/bitstream/2268/190041/1/Bruyneel_C90.pdf.

[135] A.A. Skordos, C. Monroy Aceves, and M.P.F. Sutcliffe, A simplified rate dependent model of forming and wrinkling of pre-impregnated woven composites. Composites Part A: Applied Science and Manufacturing, 2007. **38**(5): p. 1318-1330 DOI: https://doi.org/10.1016/j.compositesa.2006.11.005.

[136] A.A. Skordos and M.P.F. Sutcliffe, Stochastic simulation of woven composites forming. Composites Science and Technology, 2008. **68**(1): p. 283-296 DOI: https://doi.org/10.1016/j.compscitech.2007.01.035.

[137] M.V. d'Agostino, et al., Continuum and discrete models for structures including (quasi-) inextensible elasticae with a view to the design and modeling of composite reinforcements. International Journal of Solids and Structures, 2015. **59**: p. 1-17 DOI: https://doi.org/10.1016/j.ijsolstr.2014.12.014.

[138] P.P. Valentini, et al., Theoretical and experimental characterization of a FEM element assembly for the simulation of very compliant knitted mesh. International Journal for Numerical Methods in Engineering, 2016. **107**(5): p. 419-429 DOI: https://doi.org/10.1002/nme.5173.

[139] P. Harrison, W.-R. Yu, and A.C. Long, Rate dependent modelling of the forming behaviour of viscous textile composites. Composites Part A: Applied Science and Manufacturing, 2011. **42**(11): p. 1719-1726 DOI: https://doi.org/10.1016/j.compositesa.2011.07.026.

[140] A. Cherouat and J.L. Billoët, Mechanical and numerical modelling of composite manufacturing processes deep-drawing and laying-up of thin pre-impregnated woven fabrics. Journal of Materials Processing Technology, 2001. **118**(1): p. 460-471 DOI: https://doi.org/10.1016/S0924-0136(01)00987-6.

[141] P. Boisse, et al., Finite element simulations of textile composite forming including the biaxial fabric behaviour. Composites Part B: Engineering, 1997. **28**(4): p. 453-464 DOI: https://doi.org/10.1016/S1359-8368(96)00067-4.

[142] P. Boisse, et al., Simulation of wrinkling during textile composite reinforcement forming. Influence of tensile, in-plane shear and bending stiffnesses. Composites Science and Technology, 2011. **71**(5): p. 683-692 DOI: https://doi.org/10.1016/j.compscitech.2011.01.011.

[143] B. Liang, et al., Analysis of thermoplastic prepreg bending stiffness during manufacturing and of its influence on wrinkling simulations. Composites Part A: Applied Science and Manufacturing, 2014. **67**: p. 111-122 DOI: https://doi.org/10.1016/j.compositesa.2014.08.020.

[144] P. Wang, N. Hamila, and P. Boisse, Thermoforming simulation of multilayer composites with continuous fibres and thermoplastic matrix. Composites Part B: Engineering, 2013. **52**: p. 127-136 DOI: https://doi.org/10.1016/j.compositesb.2013.03.045.

[145] S.-W. Hsiao and N. Kikuchi, Numerical analysis and optimal design of composite thermoforming process. Computer Methods in Applied Mechanics and Engineering, 1999. **177**(1): p. 1-34 DOI: https://doi.org/10.1016/S0045-7825(98)00273-4.

[146] U. Sachs and R. Akkerman, Viscoelastic bending model for continuous fiber-reinforced thermoplastic composites in melt. Composites Part A: Applied Science and Manufacturing, 2017. **100**: p. 333-341 DOI: https://doi.org/10.1016/j.compositesa.2017.05.032.

[147] B. Chen, et al., A hypoelastic stress resultant shell approach for simulations of textile composite reinforcement forming. Composites Part A: Applied Science and Manufacturing, 2021. **149**: p. 106558 DOI: https://doi.org/10.1016/j.compositesa.2021.106558.

[148] O. Döbrich, et al., Decoupling the bending behavior and the membrane properties of finite shell elements for a correct description of the mechanical behavior of textiles with a laminate formulation. Journal of Industrial Textiles, 2014. **44**(1): p. 70-84 DOI: https://doi.org/10.1177/1528083713477442.

[149] D. Jauffrès, et al., Discrete mesoscopic modeling for the simulation of woven-fabric reinforcement forming. International Journal of Material Forming, 2010. **3**(2): p. 1205-1216 DOI: https://doi.org/10.1007/s12289-009-0646-y.

[150] M. Nishi, T. Hirashima, and T. Kurashiki. Textile composite reinforcement forming analysis considering out-of-plane bending stiffness and tension dependent in-plane shear behavior. in Proceedings of the 16th European Conference On Composite Materials, Seville, Spain. 2014.

[151] D. Dörr, et al., A viscoelastic approach for modeling bending behavior in finite element forming simulation of continuously fiber reinforced composites. Composites Part A: Applied Science and Manufacturing, 2017. **94**: p. 113-123 DOI: https://doi.org/10.1016/j.compositesa.2016.11.027.

[152] B. Schäfer, D. Dörr, and L. Kärger. Potential and challenges of a solid-shell element for the macroscopic forming simulation of engineering textiles. in ESAFORM 2021 - 24th International Conference on Material Forming. 2021. Online: ULiège Library DOI: https://doi.org/10.25518/esaform21.883.

[153] D.C. Moreira and L.C.S. Nunes, Comparison of simple and pure shear for an incompressible isotropic hyperelastic material under large deformation. Polymer Testing, 2013. **32**(2): p. 240-248 DOI: https://doi.org/10.1016/j.polymertesting.2012.11.005.

[154] L.C.S. Nunes, Modified slotted shear test for a thin sheet of solid polymer under large deformations. Polymer Testing, 2015. **45**: p. 20-30 DOI: https://doi.org/10.1016/j.polymertesting.2015.05.003.

[155] C. Thiel, et al., Shear, pure and simple. International Journal of Non-Linear Mechanics, 2019. **112**: p. 57-72 DOI: https://doi.org/10.1016/j.ijnonlinmec.2018.10.002.

[156] M.L. Di Lorenzo and C. Silvestre, Non-isothermal crystallization of polymers. Progress in Polymer Science, 1999. **24**(6): p. 917-950 DOI: https://doi.org/10.1016/S0079-6700(99)00019-2.

[157] H. Li, et al., Effects of thermal histories on carbon fiber/polyamide 6 microcomposite load transfer efficiency: Crystallization, modulus, and measurement. 2018. **39**(1): p. 102-109 DOI: https://doi.org/10.1002/pc.23907.

[158] ASTM D638-14 Standard Test Method for Tensile Properties of Plastics. 2014, ASTM International: West Conshohocken, PA.

[159] R. Artiaga and A.I. García Diez, Fundamentals of DMA. 2005.

[160] S. Jabbari-Farouji, et al., Plastic Deformation Mechanisms of Semicrystalline and Amorphous Polymers. ACS Macro Letters, 2015. **4**(2): p. 147-150 DOI: https://doi.org/10.1021/mz500754b.

[161] H.G.H. van Melick, L.E. Govaert, and H.E.H. Meijer, On the origin of strain hardening in glassy polymers. Polymer, 2003. **44**(8): p. 2493-2502 DOI: https://doi.org/10.1016/S0032-3861(03)00112-5.

[162] D. Rittel, On the conversion of plastic work to heat during high strain rate deformation of glassy polymers. Mechanics of Materials, 1999. **31**(2): p. 131-139 DOI: https://doi.org/10.1016/S0167-6636(98)00063-5.

[163] M.J. Kendall and C.R. Siviour, Experimentally simulating adiabatic conditions: Approximating high rate polymer behavior using low rate experiments with temperature profiles. Polymer, 2013. **54**(18): p. 5058-5063 DOI: https://doi.org/10.1016/j.polymer.2013.06.049.

[164] K. Zheng, et al., An experimental investigation of the drawability of AA6082 sheet under different elevated temperature forming processes. Journal of Materials Processing Technology, 2019. **273**: p. 116225 DOI: https://doi.org/10.1016/j.jmatprotec.2019.05.006.

[165] A. International, ASTM D4255 / D4255M-01(2007), Standard Test Method for In-Plane Shear Properties of Polymer Matrix Composite Materials by the Rail Shear Method. 2007, ASTM International: West Conshohocken, PA.

[166] I. De Baere, W. Van Paepegem, and J. Degrieck, Design of a modified three-rail shear test for shear fatigue of composites. Polymer Testing, 2008. **27**(3): p. 346-359 DOI: https://doi.org/10.1016/j.polymertesting.2007.12.006.

[167] A. International, ASTM D5379 / D5379M-19e1, Standard Test Method for Shear Properties of Composite Materials by the V-Notched Beam Method. 2019: West Conshohocken, PA.

[168] A. Maurel-Pantel, et al., A thermo-mechanical large deformation constitutive model for polymers based on material network description: Application to a semi-crystalline polyamide 66. International Journal of Plasticity, 2015. **67**: p. 102-126 DOI: https://doi.org/10.1016/j.ijplas.2014.10.004.

[169] D.F. Jones and L.R.G. Treloar, The properties of rubber in pure homogeneous strain. Journal of Physics D: Applied Physics, 1975. **8**(11): p. 1285-1304 DOI: https://doi.org/10.1088/0022-3727/8/11/007.

[170] H. Daiyan, et al., Shear Testing of Polypropylene Materials Analysed by Digital Image Correlation and Numerical Simulations. Experimental Mechanics, 2012. **52**(9): p. 1355-1369 DOI: https://doi.org/10.1007/s11340-012-9591-7.

[171] L.C.S. Nunes, Mechanical characterization of hyperelastic polydimethylsiloxane by simple shear test. Materials Science and Engineering: A, 2011. **528**(3): p. 1799-1804 DOI: https://doi.org/10.1016/j.msea.2010.11.025.

[172] T. Belytschko, W.K. Liu, and B. Moran, Nonlinear Finite Elements for Continua and Structures. 2000: John Wiley & Sons, Ltd.

[173] K.R. Rajagopal and A.S. Wineman, New universal relations for nonlinear isotropic elastic materials. Journal of Elasticity, 1987. **17**(1): p. 75-83 DOI: https://doi.org/10.1007/BF00042450.

[174] C. G'Sell and A. Dahoun, Evolution of microstructure in semi-crystalline polymers under large plastic deformation. Materials Science and Engineering: A, 1994. **175**(1): p. 183-199 DOI: https://doi.org/10.1016/0921-5093(94)91058-8.

[175] G. Hedner, R. Seldén, and P. Lagercrantz, Shear test for thermoplastic polymers. 1994. **34**(6): p. 513-518 DOI: https://doi.org/10.1002/pen.760340609.

[176] N. Melin, The modified Iosipescu shear test for orthotropic materials. 2008, KTH.

[177] J.H. Oh, et al., Interlaminar Shear Behavior of Thick Carbon/Epoxy Composite Materials. Journal of Composite Materials, 1999. **33**(22): p. 2080-2115 DOI: https://doi.org/10.1177/002199839903302202.

[178] F. Taheri-Behrooz and H.S. Moghaddam, Nonlinear numerical analysis of the V-notched rail shear test specimen. Polymer Testing, 2018. **65**: p. 44-53 DOI: https://doi.org/10.1016/j.polymertesting.2017.11.008.

[179] H. Ho, et al., Numerical analysis of the Iosipescu specimen for composite materials. Composites Science and Technology, 1993. **46**(2): p. 115-128 DOI: https://doi.org/10.1016/0266-3538(93)90167-F.

[180] J.C.A.D. Filho and L.C.S. Nunes, Simple shear response of a thin sheet of polytetrafluoroethylene under large deformations. Polymer Testing, 2019. **78**: p. 105942 DOI: https://doi.org/10.1016/j.polymertesting.2019.105942.

[181] H. Liu, et al., Experimental investigations on the effects of projectile hardness on the impact response of fibre-reinforced composite laminates. International Journal of Lightweight Materials and Manufacture, 2020. **3**(2): p. 77-87 DOI: https://doi.org/10.1016/j.ijlmm.2019.10.001.

[182] T.A. Composites, Toray Cetex® Stamp Forming Processing Guide—Thermoplastic UD Tapes. p. https://www.toraytac.com/resources/processing-guides.

[183] T.A. Composites. Datasheet of Toray Cetex® TC910. Available from: https://www.toraytac.com/product-explorer/products/r2Vf/Toray-Cetex-TC910.

[184] J. Liu, et al., The Impact Performance of Woven-Fabric Thermoplastic and Thermoset Composites Subjected to High-Velocity Soft- and Hard-Impact Loading. Applied Composite Materials, 2019. **26**(5): p. 1389-1410 DOI: https://doi.org/10.1007/s10443-019-09786-2.

[185] A. International, ASTM D638-14 Standard Test Method for Tensile Properties of Plastics. 2014, ASTM International: West Conshohocken, PA.

[186] I.D.G.A. Subagia and Y. Kim, Tensile behavior of hybrid epoxy composite laminate containing carbon and basalt fibers. Science and Engineering of Composite Materials, 2014. **21**(2): p. 211-217 DOI: https://doi.org/10.1515/secm-2013-0003.

[187] D.W. Holmes, J.G. Loughran, and H. Suehrcke, Constitutive model for large strain deformation of semicrystalline polymers. Mechanics of Time-Dependent Materials, 2006. **10**(4): p. 281-313 DOI: https://doi.org/10.1007/s11043-007-9023-8.

[188] R.N. Haward, G. Thackray, and T.M. Sugden, The use of a mathematical model to describe isothermal stress-strain curves in glassy thermoplastics. 1968. **302**(1471): p. 453-472 DOI: https://doi.org/10.1098/rspa.1968.0029.

[189] E.M. Arruda and M.C. Boyce, A three-dimensional constitutive model for the large stretch behavior of rubber elastic materials. Journal of the Mechanics and Physics of Solids, 1993. **41**(2): p. 389-412 DOI: https://doi.org/10.1016/0022-5096(93)90013-6.

[190] R.W. Ogden, Non-linear elastic deformations. 1997: Courier Corporation.

[191] J.L. Bouvard, et al., An internal state variable material model for predicting the time, thermomechanical, and stress state dependence of amorphous glassy polymers under large deformation. International Journal of Plasticity, 2013. **42**: p. 168-193 DOI: https://doi.org/10.1016/j.ijplas.2012.10.005.

[192] A.S. Argon, The Physics of Deformation and Fracture of Polymers. 2013, Cambridge: Cambridge University Press DOI: https://doi.org/10.1017/CBO9781139033046.

[193] L.-O. Fager and J.L. Bassani, Neck propagation in polymers with adiabatic heat generation. Mechanics of Materials, 1990. **9**(3): p. 183-194 DOI: https://doi.org/10.1016/0167-6636(90)90001-V.

[194] D. Rittel, L.H. Zhang, and S. Osovski, The dependence of the Taylor–Quinney coefficient on the dynamic loading mode. Journal of the Mechanics and Physics of Solids, 2017. **107**: p. 96-114 DOI: https://doi.org/10.1016/j.jmps.2017.06.016.

[195] W.N. dos Santos, J.A. de Sousa, and R. Gregorio, Thermal conductivity behaviour of polymers around glass transition and crystalline melting temperatures. Polymer Testing, 2013. **32**(5): p. 987-994 DOI: https://doi.org/10.1016/j.polymertesting.2013.05.007.

[196] Y. Zhang, et al., Quantification of strain-induced damage in semi-crystalline polymers: a review. Journal of Materials Science, 2019. **54**(1): p. 62-82 DOI: https://doi.org/10.1007/s10853-018-2859-2.

[197] W.D. Callister and D.G. Rethwisch, Materials science and engineering : an introduction. 2009, Hoboken, NJ: John Wiley.

[198] L. Lin and A.S. Argon, Structure and plastic deformation of polyethylene. Journal of Materials Science, 1994. **29**(2): p. 294-323 DOI: https://doi.org/10.1007/bf01162485.

[199] Jatin, V. Sudarkodi, and S. Basu, Investigations into the origins of plastic flow and strain hardening in amorphous glassy polymers. International Journal of Plasticity, 2014. **56**: p. 139-155 DOI: https://doi.org/10.1016/j.ijplas.2013.11.007.

[200] M.C. Boyce, D.M. Parks, and A.S. Argon, Large inelastic deformation of glassy polymers. part I: rate dependent constitutive model. Mechanics of Materials, 1988. **7**(1): p. 15-33 DOI: https://doi.org/10.1016/0167-6636(88)90003-8.

[201] X. Tardif, et al., Experimental study of crystallization of PolyEtherEtherKetone (PEEK) over a large temperature range using a nano-calorimeter. Polymer Testing, 2014. **36**: p. 10-19 DOI: https://doi.org/10.1016/j.polymertesting.2014.03.013.

[202] K. Hashiguchi, Chapter 8 - Multiplicative decomposition of deformation gradient tensor, in Nonlinear Continuum Mechanics for Finite Elasticity-Plasticity, K. Hashiguchi, Editor. 2020, Elsevier. p. 255-272 DOI: https://doi.org/10.1016/B978-0-12-819428-7.00008-0.

[203] C.T. Sun and R.S. Vaidya, Prediction of composite properties from a representative volume element. Composites Science and Technology, 1996. **56**(2): p. 171-179 DOI: https://doi.org/10.1016/0266-3538(95)00141-7.

[204] E. Totry, C. González, and J. Llorca, Prediction of the failure locus of C/PEEK composites under transverse compression and longitudinal shear through computational micromechanics. Composites Science and Technology, 2008. **68**(15): p. 3128-3136 DOI: https://doi.org/10.1016/j.compscitech.2008.07.011.

[205] S.L. Omairey, P.D. Dunning, and S. Sriramula, Development of an ABAQUS plugin tool for periodic RVE homogenisation. Engineering with Computers, 2019. **35**(2): p. 567-577 DOI: https://doi.org/10.1007/s00366-018-0616-4.

[206] J. Militký, M. Venkataraman, and R. Mishra, 12 - The chemistry, manufacture, and tensile behavior of polyamide fibers, in Handbook of Properties of Textile and Technical Fibres (Second Edition), A.R. Bunsell, Editor. 2018, Woodhead Publishing. p. 367-419 DOI: https://doi.org/10.1016/B978-0-08-101272-7.00012-2.

[207] L. Yang, et al., Micromechanical modelling and simulation of unidirectional fibre-reinforced composite under shear loading. Journal of Reinforced Plastics and Composites, 2014. **34**(1): p. 72-83 DOI: https://doi.org/10.1177/0731684414562873.

[208] L. Yang, et al., Microscopic failure mechanisms of fiber-reinforced polymer composites under transverse tension and compression. Composites Science and Technology, 2012. **72**(15): p. 1818-1825 DOI: https://doi.org/10.1016/j.compscitech.2012.08.001.

[209] F. Dogan, et al., Delamination of impacted composite structures by cohesive zone interface elements and tiebreak contact. Open Engineering, 2012. **2**(4): p. 612-626 DOI: https://doi.org/10.2478/s13531-012-0018-0.

[210] N.L. Feng, S.D. Malingam, and S. Irulappasamy, 4 - Bolted joint behavior of hybrid composites, in Failure Analysis in Biocomposites, Fibre-Reinforced Composites and Hybrid Composites, M. Jawaid, M. Thariq, and N. Saba, Editors. 2019, Woodhead Publishing. p. 79-95 DOI: https://doi.org/10.1016/B978-0-08-102293-1.00004-8.

[211] S. Li, The Maximum Stress Failure Criterion and the Maximum Strain Failure Criterion: Their Unification and Rationalization. Journal of composite Science, 2020. **4**(4): p. 157 DOI: https://doi.org/10.3390/jcs4040157.

[212] H. Liu, et al., A three-dimensional elastic-plastic damage model for predicting the impact behaviour of fibre-reinforced polymer-matrix composites. Composites Part B: Engineering, 2020. **201**: p. 108389 DOI: https://doi.org/10.1016/j.compositesb.2020.108389.

List of publications

A: Fist author publications:

A1. H. Wang, H. Liu, Z. Ding, N. Li, Experimental and constitutive modelling studies of semicrystalline thermoplastics under solid-state stamp forming conditions. Polymer, 2021. 228: p. 123939. https://doi.org/10.1016/j.polymer.2021.123939

A2. H. Wang, Z. Ding, X. Chen, H. Liu, N. Li, Experimental characterisation and constitutive modelling of the intra-ply tensile and shear properties of unidirectional fibre reinforced thermoplastics (UD FRTPs) under solid-state stamp forming conditions. Submitted to Composites Part A: Applied Science and Manufacturing, 2023. Under review.

A3. H. Wang, Z. Ding, Z. Shao, H. Liu, N. Li, Numerical and Experimental Investigations of Solid-State Stamp Forming of Unidirectional Carbon Fibre Reinforced Thermoplastics (UD CFRTPs), submitted to Journal of Material Processing Technologies, 2023. Under review.

A4. H. Wang, Z. Ding, Z. Shao, H. Liu, N. Li, Error analysis of stress-strain characterisation using V-Notched Rail shear test for sheet materials subject to large simple shear deformation. Submitted to International Journal of Mechanical Sciences, 2023. Under review.

B: Co-authored publications:

B1. Z. Ding, H. Wang, J. Luo, N. Li., A review on forming technologies of fibre metal laminates. International Journal of Lightweight Materials and Manufacture, 2021. 4(1): p. 110-126.

B2. R.A. Brooks, H. Wang, Z. Ding, et al., A review on stamp forming of continuous fibre-reinforced thermoplastics. International Journal of Lightweight Materials and Manufacture, 2022. 5(3): p. 411-430.

B3. Z. Ding, H. R. Attar, H. Wang, et al., Integrating Convolutional Neural Network and constitutive Model for Rapid Prediction of Stress-Strain Curves in Fibre Reinforced Polymers: A Generalisable Approach, submitted to Composite Science and Technology, 2023. Under review.

C: Conference presentation

C1: H. Wang, Z. Ding, H. Liu, N. Li, 2021, Error analysis and correction of stress-strain characterisation on sheet polymers using the V-Notched Rail (VNR) shear test method, Global conference on POLYMERS, PLASTICS AND COMPOSITES, Budapest, Hungary, 12 – 13, August, 2021.